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Dressing of Interaction Vertices in Multi-Reference Coupled Cluster Theory

Ph.D. thesis

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List of abbreviations

- **APSG**: Antisymmetrized Product of Strongly Orthogonal Geminals
- **CAS**: Complete Active Space
- **CC**: Coupled Cluster
  - **-S**: Singles
  - **-D**: Doubles
  - **-T**: Triples
  - **m-**: mosaic
  - **r-**: ring
- **CI**: Configuration Interaction
- **FDP**: Frequency-Dependent Polarizability
- **GNO**: Generalized Normal Ordering
- **GWT**: Generalized Wick Theorem
- **HF**: Hartree-Fock
- **GVB**: Generalized Valence Bond
- **ic**: internally contracted
- **id**: interaction dressing
- **JM**: Jeziorski-Monkhorst
- **KS**: Kohn-Sham
- **MBPT**: Many-Body Perturbation Theory
- **MR**: Multi-Reference
- **NSO**: Natural Spin Orbital

- **PES**: Potential Energy Surface
- **RPA**: Random Phase Approximation
  - **E-**: Extended
- **RDM**: Reduced Density Matrix
- **SVD**: Singular Value Decomposition
- **SAPT**: Symmetry-Adapted Perturbation Theory
- **SOSEX**: Second-Order Screened Exchange
- **UGA**: Unitary Group Approach

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Chapter 1

Introduction

An important goal of electronic structure theory is to find bound energy eigenstates of atomic or molecular systems. In a non-relativistic setting (and in the framework of the Born-Oppenheimer approximation [1]), the first step toward this goal can be reached by solving the electronic time-independent Schrödinger equation:

$$H|\Psi_\kappa\rangle = E_\kappa|\Psi_\kappa\rangle \ .$$

Electronic states and the corresponding energy levels are indexed by κ. The real space coordinate representation of the electronic Hamiltonian H reads (in $\hbar = m_e = e^2/(4\pi\epsilon_0) = 1$ atomic units)

$$H = \sum_i \left(-\frac{1}{2}\nabla_i^2 - \sum_A \frac{Z_A}{|\boldsymbol{r}_i - \boldsymbol{R}_A|} \right) + \sum_{i<j} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|} \ ,$$

where $\boldsymbol{r}_i$ and $\boldsymbol{R}_A$ denote electronic and nuclear coordinates, and Z_A are atomic numbers.
Knowing the electronic energy levels as functions of nuclear coordinates (i.e. $E_\kappa(\boldsymbol{R})$), one could take the next step and find the bound states of nuclei; this would completely determine the bound states of the molecule on the Born-Oppenheimer level. In many cases, one is interested only in the electronic ground state of the system; excited states will not be discussed here at all.

Finding the exact solution of the Schrödinger equation is not possible in the general N-electron case, and one must resort to approximations: analytical solutions only exist for very special cases, and exact numerical solutions (obtained by e.g. Configuration Interaction, that is, by representing and diagonalizing H as a matrix in the space of N-electron basis functions) usually come at extremely high computational cost.
The mean-field approximation (as done in the Hartree-Fock, or HF model) with perturbative or Coupled Cluster (CC) corrections built on it often provides very accurate approximations to the exact ground state as long as correlation between electrons is weak and the underlying quasi-particle picture remains valid. However, these models all tend to fail whenever the strong correlation (entanglement) of electrons destroys the quasi-particle picture: the most common examples of such phenomena include molecules considerably distorted from equilibrium structure (containing dissociating covalent bonds) and open-shell systems. In

these cases, a better first approximation is needed for the wave function than the one given by mean-field approximation. Such functions are generally called Multi-Reference (MR) functions, and the generalized perturbative or CC methods built on them are called MR electronic structure methods. In the past decades, MR methods have given invaluable insight into the structure of strongly correlated systems, yet, many problems still remain unresolved.

This thesis is the summary of my doctoral research on the so-called internally contracted MRCC (ic-MRCC) method. A new formulation of the theory is presented utilizing the concept of dressing Hamiltonian interaction vertices by density cumulants. These modified vertices formally resemble their original "bare" counterparts (the Fockian and the electron-electron interaction vertices), hence they can be similarly used as building blocks in the Feynman diagrammatic construction of CC equations. The transparent structure of the formulae makes theoretical investigations (like investigation of dispersion interactions) rather simple. The obtained ic-MRCC methods retain many nice features of HF-based CC, such as formal simplicity, extensivity and tolerable computational cost. At the same time, the MR character of the underlying wave function is encoded in the modified vertices through their cumulant dressing, making them more suitable for strongly correlated scenarios. Analytical and numerical investigation of this ic-MRCC formalism is the central theme of the thesis; performance of the method is examined with a so-called strongly orthogonal geminal wave function used as MR function.

Derivations in the thesis draw heavily on the MR generalization of certain mathematical techniques in many-body theory, such as cumulant expansions, normal ordering of operators and the Wick theorem. Despite their intuitive appeal and elegance, these MR-generalized tools (invented by Mukherjee and Kutzelnigg) have never achieved their due recognition or seen widespread usage in quantum chemistry, most likely because of the combinatorial hardship associated with them. This is unfortunate, because − in my opinion − these tools could be the perfect candidates to bridge the gap between the conceptual simplicity of mean-field-based methods and the numerical efficiency of MR methods. A secondary goal of this thesis is to present these MR tools in a unified and (mostly) self-contained manner, so that it can be used as an introduction to the subject. This goal might also serve as an excuse for the somewhat long and detailed chapter on the theoretical background.

The thesis contains seven chapters. After this introduction, Chapter 2 contains a review of the aforementioned MR tools of many-body theory, geminal wave functions and CC. Chapter 3 describes the main theoretical developments concerning the ic-MRCC with an emphasis on diagrammatic techniques in derivations and the concept of vertex dressing; a handful of approximation schemes are also introduced here. Description of dispersion interaction is the subject of Chapter 4, where emergence of the well-known London energy formula is demonstrated for certain approximations of the MR theory. A more technical issue on the linear dependence of elementary excitations used in MR calculations is addressed in Chapter 5. Finally, in Chapter 6, the numerical performance of these MRCC methods is tested on various strongly and weakly correlated molecular systems. The thesis is concluded with a short discussion and outlook to possible future projects in Chapter 7.

Chapter 2

Theoretical background

In this chapter we try to give a detailed introduction to the basic concepts of quantum chemistry, which will be needed for the rest of the thesis.
In Sec. 2.1 we review the necessary mathematical tools of many-body theory: the second quantized formalism, reduced density matrices and cumulants, normal ordering of operators and the generalized Wick theorem are all discussed simultaneously in the single determinantal and the multi-reference (MR) framework. In this way one can see how well-known concepts of single determinantal many-body theories can be naturally generalized in a multi-determinantal setting.
This is followed by a brief introduction to two rather different approaches to electronic structure calculations: a simple strong correlation model utilizing localized electron pairs (geminals) in Sec. 2.2, and a weak correlation model built on a single Slater determinant (Coupled Cluster) in Sec. 2.3. We discuss the advantages and disadvantages of both methods. A possible approach to combine weak and strong correlation models will be presented in the next chapter.

2.1 Some mathematical tools of many-body theory

2.1.1 Second quantized operators

Derivations in the many-body theory can become quite cumbersome if one has to deal explicitly with the fermionic nature of wave functions. This difficulty can be partially lifted by switching to the so-called second quantized representation. In the second quantized formulation of quantum theory not only observables, but even wave functions are represented in terms of operators: the proper antisymmetry with respect to interchanging particles is encoded in the algebra of these operators. Here we shall review the foundations of second quantization without going too much into details; a comprehensive treatment is given in Refs. [2–4], and a mathematically more rigorous one is given in Ref. [5]. Our other goal is to make the reader comfortable with the − in this context somewhat uncommon − tensor notation of second quantized operators. We shall strictly distinguish between upper and

lower indexed quantities as they are related to each other by hermitian conjugation. In the context of second quantization, this notation was introduced by Harris et al. [6] and further popularized by Kutzelnigg [7–10].

Let us consider a basis $\{\phi^p\}$ spanned by M orthogonal one-electron functions (spin orbitals); while M is in principle infinite, it is kept a large but finite number in all practical calculations. In real space representation we have $\phi^p(x) = \langle x|\phi^p\rangle$; here $x = (\boldsymbol{r}, \sigma)$ is used for brevity, with $\boldsymbol{r} \in \mathbb{R}^3$ and $\sigma = \alpha, \beta$ being the spatial and spin coordinates, respectively. Let $\mathcal{F}(M, N)$ denote the Hilbert space of all antisymmetric N-electron states built with these orbitals. The zero-particle sector contains only one state: $\mathcal{F}(M, 0) = \{|0\rangle\}$. This state is called the genuine vacuum; $\langle 0|0\rangle = 1$ by definition. Slater determinants constructed from orbitals $\phi^{p_1}, ..., \phi^{p_N}$ are denoted by $|\Phi^{p_1 \dots p_N}\rangle$ (or $\Phi^{p_1 \dots p_N}(x_1, ..., x_N)$ in real space); they form an orthonormal basis in $\mathcal{F}(M, N)$ ($\dim \mathcal{F}(M, N) = \binom{M}{N}$ being the number of linearly independent states), and therefore a general N-electron state can be expanded as

$$|\Psi_N\rangle = \frac{1}{N!} \sum_{p_1, \dots, p_N} C_{p_1 \dots p_N} |\Phi^{p_1 \dots p_N}\rangle \ . \tag{2.1.1}$$

Coefficients $C_{p_1 \dots p_N}$ are defined to be antisymmetric; the factor of $1/N!$ compensates for the overcounting stemming from the unrestricted summation.
The Fock-space $\mathcal{F}(M)$ is the space of all possible states for $N = 0, ..., M$. It is naturally given by

$$\mathcal{F}(M) = \bigoplus_{N=0}^{M} \mathcal{F}(M, N) \ ,$$

with dimension

$$\dim \mathcal{F}(M) = \sum_{N=0}^{M} \binom{M}{N} = 2^M \ .$$

We introduce annihilation operators a_p and creation operators $a^p = a_p^\dagger$ acting on the Fock space. While a^p places an electron on the p-th spin orbital, a_p deletes it. In other words, creation-annihilation operators map $\mathcal{F}(M, N)$ unto $\mathcal{F}(M, N \pm 1)$. As N cannot be negative, $a_p|0\rangle = |\emptyset\rangle$ must hold ($|\emptyset\rangle$ being the zero vector). Owing to their fermionic nature, these operators must obey anticommutation relations, thus forming a Grassmann algebra:

$$\{a_q, a_p\} = 0 \quad , \quad \{a^q, a^p\} = 0 \quad , \quad \{a_q, a^p\} = \delta_q^p \ . \tag{2.1.2}$$

We shall also introduce multi-particle creation and annihilation operators $a^{p_1 \dots p_K} = a^{p_1} ... a^{p_K}$ and $a_{p_1 \dots p_K} = a_{p_K} ... a_{p_1}$, respectively (so that $a^{p_1 \dots p_K} = a^\dagger_{p_1 \dots p_K}$). A Slater determinant is simply represented as

$$|\Phi^{p_1 \dots p_N}\rangle = a^{p_1 \dots p_N}|0\rangle \ . \tag{2.1.3}$$

Multiplying a creation operator by an annihilation operator from the right gives

$$a^{p_1 \dots p_K}_{q_1 \dots q_L} = a^{p_1 \dots p_K} a_{q_1 \dots q_L} \ .$$

For general K, L, operators like this provide a mapping $\mathcal{F}(M, N) \to \mathcal{F}(M, N + K - L)$, and can be used to describe excitation, ionization, or electron attachment processes (for

$K = L$, $K < L$ and $K > L$, respectively). Case $K = L$ signals a particle number conserving operator, as the number of electrons remains the same after its action. It changes, however, the distribution of electrons among the orbitals, hence the name: excitation operator. As operators of physical quantities are particle number conserving, they can be suitably expressed in terms of excitation operators. The second quantized form of a general operator acting on K electrons reads

$$X_K = \frac{1}{(K!)^2} \sum_{\substack{p_1,\ldots,p_K \\ q_1,\ldots,q_K}} \tilde{\chi}^{q_1\ldots q_K}_{p_1\ldots p_K} a^{p_1\ldots p_K}_{q_1\ldots q_K} \ , \tag{2.1.4}$$

where $\tilde{\chi}^{q_1\ldots q_K}_{p_1\ldots p_K}$ is a tensor antisymmetric for permutations either in its upper or lower indices:

$$\tilde{\chi}^{q_1\ldots q_K}_{p_1\ldots p_K} = \int \mathrm{d}x_1 \ldots \int \mathrm{d}x_K \Phi_{p_1\ldots p_k}(x_1, \ldots, x_k) \hat{X}_K(x_1, \ldots, x_K) \Phi^{q_1\ldots q_k}(x_1, \ldots, x_k) \ , \tag{2.1.5}$$

$\hat{X}_K(x_1, \ldots, x_K)$ being the analogous first quantized operator. For brevity $\mathrm{d}x = \mathrm{d}^3 r \mathrm{d}\sigma$ is used. A K-particle (either operator or c-number valued) quantity is said to have rank K; terms "K-particle" and "K-rank" shall be used interchangeably. We shall often use a condensed notation for such quantities, indicating only their rank in the subscript; for example, $\tilde{\chi}_K$ stands for $\tilde{\chi}^{q_1\ldots q_K}_{p_1\ldots p_K}$.

The most important operator for our investigations is the Hamiltonian:

$$H = \sum_{p,q} h^q_p a^p_q + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}^{qs}_{pr} a^{pr}_{qs} \ , \tag{2.1.6}$$

where

$$h^q_p = \int \mathrm{d}x \phi_p(x) \left(-\frac{1}{2}\nabla^2 - \sum_A \frac{Z_A}{|\boldsymbol{r} - \boldsymbol{R}_A|} \right) \phi^q(x) \tag{2.1.7}$$

and

$$v^{qs}_{pr} = \int \mathrm{d}x \int \mathrm{d}x' \phi_p(x) \phi_r(x') \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} \phi^s(x') \phi^q(x) \tag{2.1.8}$$

are the one- and two-particle matrix elements, respectively. According to Eq. (2.1.5), $\tilde{v}^{qs}_{pr} = v^{qs}_{pr} - v^{qs}_{rp}$ is the antisymmetrized integral.

Another important one-particle operator is the particle number operator

$$\hat{N} = \sum_p a^p_p \ , \tag{2.1.9}$$

which "reads off" the number of particles in a wave function: $\hat{N}|\Psi_N\rangle = N|\Psi_N\rangle$. This follows from the fact that

$$a^p_p |\Phi^{p_1\ldots p_N}\rangle = \nu^{p_1\ldots p_N}_p |\Phi^{p_1\ldots p_N}\rangle \ . \tag{2.1.10}$$

The so-called occupation number of $|\Phi^{p_1\ldots p_N}\rangle$ is $\nu^{p_1\ldots p_N}_p = 1$ if $p \in \{p_1, \ldots, p_N\}$ and $\nu^{p_1\ldots p_N}_p = 0$ otherwise, simply meaning that a_p annihilates any determinant not containing ϕ^p.

It is sometimes useful to think of a determinant as an M-component vector of the aforementioned occupation numbers:

$$|\Phi\rangle = |\nu_1, \nu_2, \ldots, \nu_M\rangle \ , \tag{2.1.11}$$

orbital indices of the determinant having been suppressed for simplicity. This representation will be used in Sec. 2.1.4.

Some further results of second quantization (real space formulation and relation with first quantization) are left to Appendix A.

2.1.2 Density matrices

The many-body wave function $\Psi(x_1, ..., x_N)$ gives the complete probabilistic description of our N-electron system, that is, the probability of finding N electrons at $x_1, ..., x_N$. This is much more information than what is actually needed in chemical applications. Density matrices provide a framework to extract the sufficient amount of information from the wave function leading to a simpler and more compact description. Here we review the most important basics of density matrices; the summary is mostly based on Refs. [11, 12], other references shall be marked explicitly.

Let us consider a normalized state $|\Psi\rangle$ with corresponding density operator $\rho = |\Psi\rangle\langle\Psi|$. The K-particle reduced density matrix (K-RDM) is then defined as

$$\gamma^{p_1...p_K}_{q_1...q_K} = \mathcal{N}_K \,\mathrm{Tr}[\rho a^{p_1...p_K}_{q_1...q_K}] = \mathcal{N}_K \langle a^{p_1...p_K}_{q_1...q_K} \rangle \ , \tag{2.1.12}$$

or as $\gamma_K = \mathcal{N}_K \langle a_K \rangle$ in a condensed notation. Here $\mathcal{N}_K$ is a normalization factor, and the notation $\langle ... \rangle$ is reserved for expectation value taken with the reference state. A reduced probabilistic description of the many-body system is encoded in the K-RDM in the sense that diagonal elements $\gamma^{p_1...p_K}_{p_1...p_K}$ are proportional to the probability of finding electrons occupying spin orbitals $\phi^{p_1}, ..., \phi^{p_K}$. Similarly, for real space density matrices

$$\gamma(x_1, ..., x_K; x'_1, ..., x'_K) = \sum_{\substack{p_1,...,p_K \\ q_1,...,q_K}} \gamma^{p_1...p_K}_{q_1...q_K} \phi_{p_1}(x'_1)...\phi_{p_K}(x'_K)\phi^{q_1}(x_1)...\phi^{q_K}(x_K) \ , \tag{2.1.13}$$

diagonal elements $\rho(x_1, ..., x_K) = \gamma(x_1, ..., x_K; x_1, ..., x_K)$ are proportional to the probability of finding electrons at $x_1, ..., x_K$; the relationship between the two representations of the K-RDM is outlined in Appendix A.
In order to avoid confusion, note that different conventions are used for the normalization of the K-RDM in the literature. Common choices include that of McWeeny ($\mathcal{N}_K = 1$), Löwdin ($\mathcal{N}_K = 1/K!$) and Coleman ($\mathcal{N}_K = (N-K)!/N!$). We shall use the McWeeny normalization throughout this thesis; from now on $\mathcal{N}_K = 1$ is understood everywhere.

Reduced density matrices have a particularly simple form if $|\Psi\rangle$ is just a single Slater determinant with orbitals provided by e.g. a Hartree-Fock (HF) or Kohn-Sham (KS) calculation. In this case the 1-RDM is

$$\gamma^p_q = \delta^p_q \nu_q \ , \tag{2.1.14}$$

with occupation numbers $\nu_q = 0, 1$ introduced in Eq. (2.1.10). The orthogonality of determinants is sufficient for easily evaluating the general single determinantal K-RDM as well:

$$\gamma^{p_1...p_K}_{q_1...q_K} = \mathcal{A}_{q_1...q_K} \left(\delta^{p_1}_{q_1}...\delta^{p_K}_{q_K}\right) \nu_{q_1}...\nu_{q_K} = \mathcal{A}_{q_1...q_K} \left(\gamma^{p_1}_{q_1}...\gamma^{p_K}_{q_K}\right) \ , \tag{2.1.15}$$

with the antisymmetrizer defined in Appendix A. This general K-RDM can be expressed in terms of the 1-RDM, reflecting the fact that the determinant wave function describes non-interacting (quasi-)particles. In order to arrive at a more compact notation, we introduce the wedge product of antisymmetric tensors $\chi^{p_1\dots p_K}_{q_1\dots q_K}$ and $\xi^{r_1\dots r_L}_{s_1\dots s_L}$ (following the normalization convention of Ref. [13]) as

$$(\chi_K \wedge \xi_L)^{p_1\dots p_K r_1\dots r_L}_{q_1\dots q_K s_1\dots s_L} = \mathcal{A}^{p_1\dots p_K r_1\dots r_L}\mathcal{A}_{q_1\dots q_K s_1\dots s_L}\left(\chi^{p_1\dots p_K}_{q_1\dots q_K}\xi^{r_1\dots r_L}_{s_1\dots s_L}\right) \ .$$

As a consequence of all tensors having an even number of indices, the wedge product is commutative. Using this notation, Eq. (2.1.15) can now be written in condensed notation as

$$\gamma_K = \frac{1}{K!}\underbrace{\gamma_1 \wedge \dots \wedge \gamma_1}_{K \text{ times}} \ . \tag{2.1.16}$$

The factor of $1/K!$ is introduced against overcounting in the wedge product, similarly to the case of Eq. (2.1.1).

In general, $|\Psi\rangle$ is a Multi-Reference (MR) wave function: it consists of multiple determinants (cf. Eq. (2.1.1)), each having a different set of occupied orbitals. The general form of density matrices is usually very complicated, although for simple MR wave functions tractable expressions can still be obtained (see Sec. 2.2.2). These complications can be partially lifted by working with the $\{\hat{\phi}^p\}$ eigenbasis of the 1-RDM, called natural spin orbital (NSO) basis:

$$\hat{\phi}^p = \sum_t U^p_t \phi^t \ .$$

Columns of unitary matrix U are chosen to be eigenvectors of γ_1:

$$\sum_r \gamma^r_s U^p_r = n^p U^p_s \ ,$$

and so $\{\hat{\phi}^p\}$ are eigenvectors of γ_1 in the integral kernel sense (cf. Eq. (2.1.13)):

$$\int \mathrm{d}x' \gamma(x;x')\hat{\phi}^p(x') = \sum_{r,s}\gamma^r_s U^p_r \phi^s(x) = n^p\hat{\phi}^p(x) \ .$$

This way we can keep using the diagonal form of the 1-RDM as in Eq. (2.1.14), the only difference being the appearance of fractional eigenvalues $n_p \in [0,1]$ instead of ν_p. These natural occupation numbers can be interpreted as average occupations of orbitals in the MR wave function with probabilities $P_{p_1\dots p_N} = C^{p_1\dots p_N}C_{p_1\dots p_N}$, as Eqs. (2.1.1) and (2.1.10) give

$$n_p = \gamma^p_p = \frac{1}{N!}\sum_{p_1,\dots,p_N} C^{p_1\dots p_N}C_{p_1\dots p_N}\nu^{p_1\dots p_N}_p = \frac{1}{N!}\sum_{p_1,\dots,p_N} P_{p_1\dots p_N}\nu^{p_1\dots p_N}_p \ .$$

Case $n_p = 1$ or $n_p = 0$ corresponds to an orbital that is occupied or unoccupied (virtual) in all of the determinants, while $0 < n_p < 1$ corresponds to a so-called active orbital found only in some of the determinants. An active orbital having $n_p \approx 1$ means that the number or weight of determinants containing it is high, while $n_p \approx 0$ signals a low number or weight

of host determinants. Natural orbitals have many other appealing features as well (e.g. providing the best approximation to the determinantal expansion of wave functions [14,15]), but these shall not be discussed here.
Since occupation numbers are real, writing them with upper or lower indices is immaterial and is just a question of aesthetics: $n_p = n^p$. Usually we shall write them in a way so that it matches the other indices in the formulae. From now on, NSO basis will generally be assumed and the hat from $\{\hat{\phi}^p\}$ will be dropped.

Useful partial trace relations of the K-RDM can be obtained by contracting one pair of upper-lower indices:

$$\sum_r \gamma^{p_1\dots p_{K-1}r}_{q_1\dots q_{K-1}r} = \langle a^{p_1\dots p_{K-1}} \hat{N} a_{q_1\dots q_{K-1}} \rangle = (N-K+1)\gamma^{p_1\dots p_{K-1}}_{q_1\dots q_{K-1}} \ , \tag{2.1.17}$$

where we used that $a_{q_1\dots q_{K-1}}|\Psi\rangle$ is an eigenfunction of the particle number operator defined in Eq. (2.1.9). Using Eq. (2.1.17) successively gives the trace formula of density matrices:

$$\mathrm{Tr}[\gamma_K] = \sum_{p_1,\dots,p_K} \gamma^{p_1\dots p_K}_{p_1\dots p_K} = \frac{N!}{(N-K)!} \ . \tag{2.1.18}$$

The importance of density matrices lies in the fact that the expectation value of a K-particle operator (cf. Eq. (2.1.4)) can be readily expressed with them as

$$\langle X_K \rangle = \frac{1}{(K!)^2} \sum_{\substack{p_1,\dots,p_K \\ q_1,\dots,q_K}} \tilde{\chi}^{q_1\dots q_K}_{p_1\dots p_K} \langle a^{p_1\dots p_K}_{q_1\dots q_K} \rangle = \frac{1}{(K!)^2} \sum_{\substack{p_1,\dots,p_K \\ q_1,\dots,q_K}} \tilde{\chi}^{q_1\dots q_K}_{p_1\dots p_K} \gamma^{p_1\dots p_K}_{q_1\dots q_K} = \frac{1}{(K!)^2} \mathrm{Tr}[\tilde{\chi}_K \gamma_K] \ .$$

Since in chemical applications one usually has to deal with one- and two-particle quantities only, knowledge of γ_1 and γ_2 already provides sufficient information on our many-body system; furthermore, γ_1 is completely determined by γ_2 because of Eq. (2.1.17). The expectation value of the Hamiltonian is given by

$$\langle H \rangle = \mathrm{Tr}[h\gamma_1] + \frac{1}{4}\mathrm{Tr}[\tilde{v}\gamma_2] = \sum_p h^p_p n_p + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}^{qs}_{pr} \gamma^{pr}_{qs} \ . \tag{2.1.19}$$

For a single Slater determinant this just falls back to the well-known formula

$$\langle H \rangle = \sum_k h^k_k + \frac{1}{2} \sum_{k,l} \tilde{v}^{kl}_{kl} \ ,$$

where indices k, l run through the occupied orbitals of the determinant.

Finally, another K-RDM-like quantity must be introduced:

$$\eta^{p_1\dots p_K}_{q_1\dots q_K} = \langle a_{q_1\dots q_K} a^{p_1\dots p_K} \rangle \ , \tag{2.1.20}$$

with all creation operators standing to the right from annihilation operators. As discussed earlier, diagonal elements of γ_K measure the occupancy of orbitals. On the contrary, diagonal elements of η_K measure the "unoccupancy" of orbitals, or vacancies, since $a^{p_1\dots p_K}$ annihilates any determinant $|\Phi^{r_1\dots r_N}\rangle$ if $\{p_1,\dots,p_K\} \cap \{r_1,\dots,r_N\} \neq \emptyset$. For this reason, η_K is often

referred to as the K-hole density matrix [16], but we shall try to avoid this nomenclature as it would conflict with the notion of particles and holes defined in Sec. 2.1.5.
In Secs. 2.1.5 and 2.1.6 a special role will be attributed to η_1:

$$\eta_q^p = \langle a_q a^p \rangle = \delta_q^p - \gamma_q^p = \delta_q^p (1 - n_q) = \delta_q^p \bar{n}_q \ , \tag{2.1.21}$$

where in the last two steps we used the NSO basis and introduced the "vacancy occupation number" $\bar{n}_q = (1 - n_q)$. The relation between γ and η can be generalized to rank K by successive use of the anticommutation relations and keeping track of the parity of permutations:

$$\gamma_K = \sum_{K'=0}^{K} \frac{(-1)^{K'}}{(K-K')!} \underbrace{\delta \wedge \ldots \wedge \delta}_{K-K' \text{ times}} \wedge \eta_{K'} \quad \Leftrightarrow \quad \eta_K = \sum_{K'=0}^{K} \frac{(-1)^{K'}}{(K-K')!} \underbrace{\delta \wedge \ldots \wedge \delta}_{K-K' \text{ times}} \wedge \gamma_{K'} \ , \tag{2.1.22}$$

The roles of γ_K and η_K are symmetric, reflecting the fact that a general state of the Fock space can be equally well described by specifying the N occupied or $M - N$ unoccupied orbitals of each determinant. They have very similar partial trace relations, too:

$$\sum_r \eta_{q_1 \ldots q_{K-1} r}^{p_1 \ldots p_{K-1} r} = \langle a_{q_1 \ldots q_{K-1}} (M - \hat{N}) a^{p_1 \ldots p_{K-1}} \rangle = (M - N - K + 1) \eta_{q_1 \ldots q_{K-1}}^{p_1 \ldots p_{K-1}} \ . \tag{2.1.23}$$

We shall return to this equivalence in Secs. 2.1.3 and 2.1.4 in the context of density cumulants.

2.1.3 Density cumulants

Although density matrices provide a sufficient probabilistic description of the N-electron system, it is often practical to use quantities more closely related to correlation effects: among other appealing properties, this is the main motivation to use density cumulants. A formal definition of cumulants in many-body theory can be given using a generating function [13, 17–19] (mirroring their original formulation in statistical physics [20]); we shall come back to this in Sec. 2.1.5. However, in order to highlight their relation to correlation, we shall now introduce cumulants in a more heuristic way, through an analogy with classical probability theory.

Let Ω_1 and Ω_2 be two elements of an event algebra with associated probabilites $P(\Omega_1)$ and $P(\Omega_2)$, respectively. If these events were independent, their joint probability would simply be $P(\Omega_1 \Omega_2) = P(\Omega_1) P(\Omega_2)$. However, if the events are related to each other (i.e. they are correlated) the joint probability cannot be simply expressed as a product, and a "correction term" must be introduced:

$$P(\Omega_1 \Omega_2) = P(\Omega_1) P(\Omega_2) + \mathcal{P}(\Omega_1 \Omega_2) \ .$$

The quantity $\mathcal{P}(\Omega_1 \Omega_2)$ measures the deviation from the case of indepent events, and therefore accounts for the genuine correlation between Ω_1 and Ω_2. Note that $\mathcal{P}(\Omega_1 \Omega_2)$ cannot be strictly interpreted as a probability as it can also be negative (meaning that correlation

makes $\Omega_1\Omega_2$ less likely). For a single event Ω_1, it is natural to define $P(\Omega_1) = \mathcal{P}(\Omega_1)$. Therefore, we have

$$P(\Omega_1\Omega_2) = \mathcal{P}(\Omega_1)\mathcal{P}(\Omega_2) + \mathcal{P}(\Omega_1\Omega_2) \ .$$

For three events Ω_1, Ω_2 and Ω_3, the joint probability can be partitioned in a similar manner:

$$\begin{aligned} P(\Omega_1\Omega_2\Omega_3) = &\mathcal{P}(\Omega_1)\mathcal{P}(\Omega_2)\mathcal{P}(\Omega_3) + \mathcal{P}(\Omega_1\Omega_2)\mathcal{P}(\Omega_3) + \mathcal{P}(\Omega_2\Omega_3)\mathcal{P}(\Omega_1) \\ &+ \mathcal{P}(\Omega_3\Omega_1)\mathcal{P}(\Omega_2) + \mathcal{P}(\Omega_1\Omega_2\Omega_3) \ . \end{aligned}$$

The first term refers to the case of three indepent events, the next three terms correspond to correlation between two events (the third one being independent), and the last term represents the fully correlated case. Following this recipe, one could partition any K-event joint probability to a sum of products of K'-event "connected probabilites" (for $K' \leq K$) by considering all possible partitions of K.

In many-body theory the joint probability of finding K electrons at K points of either orbital or real space is described by γ_K through Eqs. (2.1.12) or (2.1.13). As a joint classical probability having a purely product form describes independent events, a density matrix having a purely wedge product form (like the one in Eq. (2.1.12)) describes a system of non-interacting (quasi-)particles. Pursuing this analogy, we find that in a general K-RDM, correction terms must be introduced to describe correlation effects not covered by the 1-RDM wedge product part. The antisymmetry of the K-RDM dictates the correction terms to be also written in terms of wedge products. Then we have in condensed notation

$$\gamma_1 = \lambda_1 \ ,$$

$$\gamma_2 = \lambda_2 + \frac{1}{2}\lambda_1 \wedge \lambda_1 \ , \tag{2.1.24}$$

$$\gamma_3 = \lambda_3 + \lambda_2 \wedge \lambda_1 + \frac{1}{6}\lambda_1 \wedge \lambda_1 \wedge \lambda_1 \ , \tag{2.1.25}$$

$$\gamma_4 = \lambda_4 + \lambda_3 \wedge \lambda_1 + \frac{1}{2}\lambda_2 \wedge \lambda_2 + \frac{1}{2}\lambda_2 \wedge \lambda_1 \wedge \lambda_1 + \frac{1}{24}\lambda_1 \wedge \lambda_1 \wedge \lambda_1 \wedge \lambda_1 \ , \tag{2.1.26}$$

and in general

$$\gamma_K = \sum_{\nu=1}^{K} \sum_{K_1,\ldots,K_\nu \in \mathbb{P}_{K,\nu}} \frac{1}{w_{K_1\ldots K_\nu}} \bigwedge_{\mu=1}^{\nu} \lambda_{K_\mu} \ . \tag{2.1.27}$$

Here $\mathbb{P}_{K,\nu} = \{(K_1,\ldots K_\nu) \in \mathbb{Z}_+^\nu | K_1 \geq \ldots \geq K_\nu \, ; \, K_1 + \ldots + K_\nu = K\}$ is the set of ν-element partitions of K; for example $\mathbb{P}_{6,4} = \{(3,1,1,1),(2,2,1,1)\}$. Like in the classical analogy, Eq. (2.1.27) serves as the implicit definition of "connected K-RDM" λ_K: these are called K-particle density cumulants. The combinatorial weight factor compensates for overcounting similarly to Eq. (2.1.16): it is given by $w_{K_1\ldots K_\nu} = n_1!\ldots n_m!$, the $n_i!$ being the number of identical numbers in the partition (with $n_1 + \ldots + n_m = \nu$). For example, $w_{2,2,1,1} = (2!)^2$.
For future reference, we write out explicitly the expansion for $K = 1, 2$:

$$\gamma^p_q = \lambda^p_q \quad , \quad \gamma^{pq}_{rs} = \lambda^{pq}_{rs} + \lambda^p_q\lambda^r_s - \lambda^p_s\lambda^r_q \ ,$$

or, with a different notation (showing cumulants as formal expectation values with $\langle(...)\rangle_c$):

$$\langle a^p a_q\rangle = \langle a^p a_q\rangle_c \quad , \quad \langle a^p a^r a_s a_q\rangle = \langle a^p a^r a_s a_q\rangle_c + \langle a^p a_q\rangle_c\langle a^r a_s\rangle_c - \langle a^p a_s\rangle_c\langle a^r a_q\rangle_c \ . \tag{2.1.28}$$

Note that – as Eq. (2.1.27) is a non-linear relation – the coefficients of the expansion depend on the normalization convention used for γ_K. See for example Ref. [17] for the same formulae with Löwdin normalization.

Eq. (2.1.27) can be inverted to give λ_K in terms of density matrices; the first four cumulants are

$$\begin{aligned}
\lambda_1 &= \gamma_1 \\
\lambda_2 &= \gamma_2 - \frac{1}{2}\gamma_1 \wedge \gamma_1 \ , \\
\lambda_3 &= \gamma_3 - \gamma_2 \wedge \gamma_1 + \frac{1}{3}\gamma_1 \wedge \gamma_1 \wedge \gamma_1 \ , \\
\lambda_4 &= \gamma_4 - \gamma_3 \wedge \gamma_1 - \frac{1}{2}\gamma_2 \wedge \gamma_2 + \gamma_2 \wedge \gamma_1 \wedge \gamma_1 - \frac{1}{4}\gamma_1 \wedge \gamma_1 \wedge \gamma_1 \wedge \gamma_1 \ .
\end{aligned} \tag{2.1.29}$$

Transforming the above formulae with $\{\phi^p(x)\}$ (cf. Eq. (2.1.13)) yields the real space representation of cumulants, similarly to the one of RDM-s. For e.g. $K = 2$, we have

$$\lambda(x_1, x_2; x_1', x_2') = \gamma(x_1, x_2; x_1', x_2') - \gamma(x_1; x_1')\gamma(x_2; x_2') + \gamma(x_1; x_2')\gamma(x_2; x_1') \ .$$

The diagonal part $\lambda(x_1, x_2) = \lambda(x_1, x_2; x_1, x_2)$ can be interpreted as the correlation contribution to the pair density:

$$\lambda(x_1, x_2) = \rho(x_1, x_2) - \rho(x_1)\rho(x_2) + |\gamma(x_1; x_2)|^2 \ . \tag{2.1.30}$$

In the above sense, electron correlation is encoded in the correction terms to the γ_1 wedge product part of γ_K. Note, however, that there are other approaches to define electron correlation in the literature. According to the statistical definition, electrons are correlated if the probability of finding an electron at x_1 is not completely independent of finding another electron at x_2. In this sense, almost every properly antisymmetric wave function (even a single determinant) describes correlation (the so-called Fermi correlation) due to $\rho(x, x) = 0$, i.e. the unability to detect same spin electrons in the same point of space. Another definition given by Löwdin (to be briefly mentioned in Sec. 2.1.5) refers to all effects not covered by the HF approximation as correlation effects. The intrinsic definition of correlation based on cumulant expansion differs from both approaches, as no reference is made to the underlying wave function, and Fermi correlation is left out.
More details on the various definitions of electron correlation can be found in Ref. [21].

Partial trace relations of cumulants can be found from Eqs. (2.1.17) and (2.1.27). They are, however, more involved than those of of density matrices; see Ref. [22] for a more complete list. Here only the simplest cases are shown:

$$\sum_r \lambda^{pr}_{qr} = \delta^p_q(n_q^2 - n_q) \ , \tag{2.1.31}$$

$$\sum_t \lambda^{prt}_{qst} = \lambda^{pr}_{qs}(n^p + n^r + n_q + n_s - 2) \ ,$$

$$\begin{aligned}\sum_v \lambda^{prtv}_{qsuv} &= \lambda^{prt}_{qsu}(n^p + n^r + n^t + n_q + n_s + n_u - 3) \\ &+ \sum_v \Big(\lambda^{pv}_{qs}\lambda^{tr}_{uv} + \lambda^{pv}_{qu}\lambda^{rt}_{sv} + \lambda^{rv}_{sq}\lambda^{tp}_{uv} + \lambda^{rv}_{su}\lambda^{pt}_{qv} + \lambda^{tv}_{us}\lambda^{pr}_{qv} + \lambda^{tv}_{uq}\lambda^{rp}_{sv} \\ &+ \lambda^{pr}_{uv}\lambda^{tv}_{qs} + \lambda^{pt}_{sv}\lambda^{rv}_{qu} + \lambda^{tr}_{qv}\lambda^{pv}_{us} \Big) \ .\end{aligned}$$

From a practical point of view, these relations can serve as tests to check the numerical implementation of cumulants.

We need a final, slight generalization of the notion of cumulants. Cumulants have been introduced above as the "irreducible" components of density matrices. However, their role is more general: in fact, they serve as irreducible components in the expansion of any expectation value of creation-annihilation operator products. As a first example, we can check that γ_K and η_K have nearly identical cumulant expansions. Using Eqs. (2.1.24), (2.1.25), (2.1.26) and (2.1.22), we find

$$\eta_2 = \lambda_2 + \frac{1}{2}\eta_1 \wedge \eta_1 \ ,$$

$$\eta_3 = -\lambda_3 + \lambda_2 \wedge \eta_1 + \frac{1}{6}\eta_1 \wedge \eta_1 \wedge \eta_1 \ ,$$

$$\eta_4 = \lambda_4 - \lambda_3 \wedge \eta_1 + \frac{1}{2}\lambda_2 \wedge \lambda_2 + \frac{1}{2}\lambda_2 \wedge \eta_1 \wedge \eta_1 + \frac{1}{24}\eta_1 \wedge \eta_1 \wedge \eta_1 \wedge \eta_1 \ .$$

In the general case, cumulant expansion of η_K can be formally obtained from that of γ_K by replacing $\lambda_1 = \gamma_1$ with η_1, and $\lambda_{2K'-1}$ with $-\lambda_{2K'-1}$ for $K' \geq 2$.
If we want to adopt the notation introduced in Eq. (2.1.28), we must write for e.g. η_1 and η_2

$$\langle a_q a^p \rangle = \langle a_q a^p \rangle_c \ \ , \ \ \langle a_s a_q a^p a^r \rangle = \langle a_s a_q a^p a^r \rangle_c + \langle a_q a^p \rangle_c \langle a_s a^r \rangle_c - \langle a_s a^p \rangle_c \langle a_q a^r \rangle_c \ . \quad (2.1.32)$$

Then – due to γ_2 and η_2 having the same cumulant part – we find $\langle a_s a_q a^p a^r \rangle_c = \langle a^p a^r a_s a_q \rangle_c = \lambda^{pr}_{qs}$. With the help of anticommutation relations one can show that any product of a^p, a^r, a_q and a_s has the same cumulant part in its expansion. For example:

$$\langle a^p a_s a^r a_q \rangle = \langle a^p a_s a^r a_q \rangle_c + \langle a^p a_s \rangle_c \langle a^r a_q \rangle_c + \langle a^p a_q \rangle_c \langle a_s a^r \rangle_c \ , \quad (2.1.33)$$

with $\langle a^p a_s a^r a_q \rangle_c = -\langle a^p a^r a_s a_q \rangle_c = -\lambda^{pr}_{qs}$.
Generally, we can say that operators anticommute with each other within the "cumulant expectation value" $\langle a^{p_1 \cdots p_K}_{q_1 \ldots q_K} \rangle_c$ for $K \geq 2$. This property will have important consequences in the MR generalization of the Wick theorem (see Secs. 2.1.5 and 2.1.6).

2.1.4 Vanishing and non-vanishing of cumulants

Despite its nice features, cumulant expansion of the K-RDM has two drawbacks which limit its general application. One is the rapidly increasing combinatorial complexity of the expansion: a simple excercise in combinatorics shows that the expansion of the K-RDM contains

$$R_K = \sum_{\nu=1}^{K} \sum_{K_1,\ldots,K_\nu \in \mathbb{P}_{K,\nu}} \frac{1}{w_{K_1\ldots K_\nu}} \left(\frac{K!}{K_1!\ldots K_\nu!} \right)^2$$

unique terms, which is already 131 for e.g. the 4-RDM. The other problem is the number of elements in high rank cumulants: λ_K seemingly has M^{2K} entries, which quickly grows intractable even in a smaller basis set. Although antisymmetry and hermiticity properties reduce the number of independent elements, it still grows too rapidly with K and M. Similar problems arise in the context of the Wick theorem (see Sec. 2.1.6). As a way out, expansions in terms of cumulants are truncated, usually by neglecting higher rank cumulants. To understand the validity of such approximations, it is essential to know under what circumstances cumulant elements vanish, or at least become small.

We begin with some exact results. Let us assume that the N electron wave function can be written as the product of two independent $N_\mathbb{A}$- and $N_\mathbb{B}$-electron "groups" (with $N_\mathbb{A}+N_\mathbb{B} = N$), that is, as $|\Psi\rangle = |\Psi_\mathbb{A}\rangle \otimes |\Psi_\mathbb{B}\rangle$ (we neglect antisymmetry to keep things simple). Orbitals are grouped accordingly into two disjoint sets. The number of nonzero intergroup density matrix elements (containing orbitals from both groups) is now seriously reduced as creation-annihilation operators can only act on the associated Fock space of either $\mathbb{A}$ or $\mathbb{B}$. For example:

$$\gamma_{s_\mathbb{B}}^{p_\mathbb{A}} = \langle\Psi_\mathbb{B}| \otimes \langle\Psi_\mathbb{A}| \left(a^{p_\mathbb{A}} a_{s_\mathbb{B}}\right) |\Psi_\mathbb{A}\rangle \otimes |\Psi_\mathbb{B}\rangle = \langle\Psi_\mathbb{A}|a^{p_\mathbb{A}}|\Psi_\mathbb{A}\rangle\langle\Psi_\mathbb{B}|a_{s_\mathbb{B}}|\Psi_\mathbb{B}\rangle = 0 \ .$$

The only kind of nonzero intergroup element of γ_2 is $\gamma_{q_\mathbb{A} s_\mathbb{B}}^{p_\mathbb{A} r_\mathbb{B}} = \gamma_{q_\mathbb{A}}^{p_\mathbb{A}} \gamma_{s_\mathbb{B}}^{r_\mathbb{B}}$. More generally, all K-RDM elements factor into lower rank elements over the separate groups: this property of density matrices is called multiplicative separability. We immediately find that the intergroup λ_2 vanishes:

$$\lambda_{q_\mathbb{A} s_\mathbb{B}}^{p_\mathbb{A} r_\mathbb{B}} = \gamma_{q_\mathbb{A} s_\mathbb{B}}^{p_\mathbb{A} r_\mathbb{B}} - \gamma_{q_\mathbb{A}}^{p_\mathbb{A}} \gamma_{s_\mathbb{B}}^{r_\mathbb{B}} + \gamma_{s_\mathbb{B}}^{p_\mathbb{A}} \gamma_{q_\mathbb{A}}^{r_\mathbb{B}} = 0 \ .$$

From this it is quite obvious (but can also be proven formally [13, 19]) that intergroup cumulants vanish for all K. This means that − unlike density matrices − cumulants are additively separable over independent groups of the wave function (so $\lambda_K = \lambda_K^\mathbb{A} \oplus \lambda_K^\mathbb{B}$).
An important result can be readily obtained as a special case: an occupied orbital (found in all determinants of $|\Psi\rangle$) can always be factored out to form a one-electron group as $|\Psi\rangle = |\Psi'\rangle \otimes |\phi\rangle$. The fact that there is only one element in it, combined with the antisymmetry property of cumulants implies that cumulants vanish if at least one of their indices belongs to an occupied orbital. Similarly, cumulants also vanish if one of their indices is virtual (missing in all determinants of $|\Psi\rangle$). This means that only active orbitals can contribute to cumulants, and at most M_a^{2K} elements are to be considered for λ_K (M_a being the number of active orbitals); this also sets an upper bound of $K_{max} = M_a$ for the rank of cumulants,

as all higher rank cumulants vanish due to repeated indices and antisymmetry. Of course, all cumulants are zero in the limit of $|\Psi\rangle$ being purely a product of one-electron groups (i.e. a single determinant).

Another kind of exact cancellation concerns odd-rank cumulants only. As it was discussed in Sec. 2.1.2, a state of the Fock space can be equally well described by specifying either the occupied or the unoccupied states in determinants; in other words, γ_K and η_K have equivalent roles. This relationship was further deepened in Sec. 2.1.3 by noticing that these quantities had almost identical cumulant expansions. It follows from comparing the expansions of γ_K and η_K that odd rank cumulants vanish if $\gamma_K = \eta_K$ for all K. This seems very restrictive, but it makes sense in a special case. Let us consider a "half-filled" system with M_a active orbitals and $N_a = M_a/2$ electrons, whose wave function transforms as $|\Psi\rangle \to \pm|\Psi\rangle$ under flipping occupied and unoccupied states in each determinant (e.g. $|1001\rangle \to |0110\rangle$). This transformation leaves density operator $\rho = |\Psi\rangle\langle\Psi|$ invariant, and enforces a pairing of determinants (for example, $|1001\rangle$ and $|0110\rangle$ can only appear as $|1001\rangle \pm |0110\rangle$). Such a system can describe e.g. the degenerate case of dissociated covalent bonds. It is easy to show that in such a case $\gamma^{p_1\dots p_{N_a}}_{q_1\dots q_{N_a}} = \eta^{q_1\dots q_{N_a}}_{p_1\dots p_{N_a}} = \left(\eta^{p_1\dots p_{N_a}}_{q_1\dots q_{N_a}}\right)^*$, hence $\gamma_{N_a} = \eta_{N_a}$ if $|\Psi\rangle$ is taken to be real (as is done in most practical calculations). Comparison of Eqs. (2.1.17) and (2.1.23) shows that $\gamma_{N_a} = \eta_{N_a}$ automatically ensures the equality of all lower rank elements. We can then conclude that all odd-rank cumulants are zero if their indices belong to the active space of completely dissociated covalent bonds [13].

Not much more can be said about the exact vanishing of cumulants without further specifying the reference wave function. We shall now discuss under what circumstances cumulants can be considered negligible.

Negligibility of cumulants is certainly justified in systems well described by a single Slater determinant (e.g. closed-shell systems around equilibrium geometry), where a wedge product of K γ_1-s is a good approximation of γ_K. In this case cumulants rapidly become less and less important with increasing rank, as is supported by perturbative arguments [18]. But the situation is more complicated in systems with a strong MR character. Dissociation of multiple covalent bonds leads to a system of infinitely separated high spin fragments which are, however, still coupled to each other to form the correct spin state of the whole system. Such an MR wave function can produce density matrices which are vastly different from the wedge product, and one cannot expect the cumulants to be small in general. Odd-rank cumulants are an exception: not only are they negligible around equilibrium geometry, but they vanish in the dissociation limit, and therefore are assumed to have little effect in the intermediate range, too. The importance of high rank cumulants in the strongly correlated regime was demonstrated analitically for simple model systems [23], and numerically for realistic atomic, molecular systems [13, 24].

However, spin coupling does not occur if at least one of the two subsystems (denoted as $\mathbb{A}$ and $\mathbb{B}$) is a singlet. If they are separated by a distance large enough that exchange effects can be neglected, then the wave function is a product: $|\Psi\rangle \approx |\Psi_\mathbb{A}\rangle \otimes |\Psi_\mathbb{B}\rangle$, leading again to product-like intersystem γ_K and vanishing intersystem λ_K. This property will be used in Sec. 4.2, when investigating dispersion interactions between closed-shell systems.

Finally, a surprising feature of cumulants must be discussed. When rank K exceeds the

number of electrons in our system, K-particle density matrices naturally vanish by their definition (cf. Eq. (2.1.12)), but K-particle cumulants do not. Instead, they just become factorized and completely expressed by lower rank density matrices or cumulants. This non-vanishing appears to contradict the usual physical or probabilistic interpretation of cumulants, as K-particle correlation does not seem to make any sense for $K > N$. Kutzelnigg and Mukherjee coined the term Exclusion Principle Violating (EPV) cumulants [25] for these strange cumulants following a loose analogy with perturbation theory [26, 27].
The meaning of EPV cumulants has remained a somewhat controversial topic in the literature. Kong claimed that EPV cumulants were unphysical due to their disconnected nature, and proposed an ad-hoc modification of Eq. (2.1.27) that ensured their cancellation [28]; doing so, however, would lead to unwanted side effects (see Sec. 3.2). Contrary to this, Hanauer and Köhn argued that EPV cumulants are no less physical than ordinary cumulants based on the symmetric role of occupied and unoccupied states in Fock space [13]. Cumulants treat occupied and unoccupied states on an equal footing, as seen by e.g. λ_K being the cumulant of both γ_K and η_K, or any other expectation value with K upper-lower indices (up to a possible sign factor). Note that η_K has nonzero value for $K \leq M - N$ only. From this point of view, existence of λ_K for $K > N$ is no less meaningful than for $K > M - N$, in which case η_K identically vanishes instead of γ_K (assuming an active space of finite size). Cumulants simply provide a measure for correlation between occupancy of active orbitals, and this property is independent of the particle number.

All of the above general results will be demonstrated explicitly for cumulants of disjoint two-electron group functions (so-called strongly orthogonal geminals) in Sec. 2.2.2.

2.1.5 Normal ordering of operators

In various branches of physics, it is useful to decompose a physical quantity X as a sum of its expectation value (provided by some suitable averaging) and a (presumably small) fluctuation

$$X = \langle X \rangle + \delta X \ , \tag{2.1.34}$$

so that $\langle \delta X \rangle = 0$ and $\langle (\delta X)^2 \rangle = \langle X^2 \rangle - \langle X \rangle^2$. In the many-body problem, this average-fluctuation separation is implemented on the level of second quantized operators by their so-called normal ordering.

In order to build some intuition about normal ordering, first we define it for a system described by a single Slater determinant [26, 27, 29]. We shall refer to this determinant $|\Phi\rangle$ as the Fermi vacuum, and we shall distinguish between spin orbitals occupied (indexed by $i, j, k, l, ...$) and unoccupied (indexed by $a, b, c, d, ...$) in it. Following a useful interpretation originally rooted in quantum field theory, we formally relate the Fermi vacuum to the genuine vacuum $|0\rangle$ by calling occupied/unoccupied states hole/particle states. We define hole creation-annihilation operators b^i and b_i accordingly: annihilation of an electron from an occupied orbital is equivalent to the creation of a hole in the Fermi vacuum and vice versa: $a_i = b^i$ and $a^i = b_i$. Hole operators satisfy the same Grassmann algebra as particle operators, and particle and hole operators anticommute with each other. Evidently, $a_a|\Phi\rangle = |\emptyset\rangle$ and

$b_i|\Phi\rangle = |\emptyset\rangle$, meaning that − in the particle-hole sense − the Fermi vacuum acts similarly to the genuine one. Excitation $a_i^a|\Phi\rangle = a^a b^i|\Phi\rangle$ can be formally interpreted as the polarization of the vacuum by formation of a (virtual) particle-hole pair.
A product of creation-annihilation operators $\mathcal{S}$ is said to be normal ordered with respect to the Fermi vacuum if − in the particle-hole sense − all annihilation operators are standing right to the creation operators; normal ordering is the corresponding rearrangement of operators, denoted as $:\mathcal{S}:$. All operators anticommute within normal ordering, and the sign of the final product is determined by the parity of the permutation of operators compared to the original order. A few simple examples include

$$:a^i a^a: = :b_i a^a: = -a^a b_i = -a^a a^i \ , \tag{2.1.35}$$

$$\begin{gathered} :a^i a_k a_l: = :b_i b^k b^l: = b^k b^l b_i = a_k a_l a^i \ , \\ :a_a a^i a^j a^b a^k: = :a_a b_i b_j a^b b_k: = -a^b a_a b_i b_j b_k = -a^b a_a a^i a^j a^k \ , \\ :a^a a^b a_j a_i: = :a^a a^b b^j b^i: = a^a a^b b^j b^i = a^a a^b a_j a_i \ . \end{gathered} \tag{2.1.36}$$

A normal ordered operator always satisfies

$$\langle :\mathcal{S}: \rangle = 0 \ , \tag{2.1.37}$$

since $:\mathcal{S}:$ destroys the Fermi vacuum from either left ($:\mathcal{S}:\ |\Phi\rangle = |\emptyset\rangle$) or right ($\langle\Phi|\ :\mathcal{S}: = \langle\emptyset|$).

As long as our reference wave function is a single Slater determinant, the above prescription completely determines normal ordering. But generally our system is described by a linear combination of determinants. This makes a complete definition of normal ordering seemingly hard, since different orbitals are occupied and unoccupied in the various determinants, and thus no strict particle-hole classification is possible. A way out of this problem can be found by examining the relationship between "ordinary operators" and their normal ordered counterparts first in the single determinantal case: this leads to the so-called Wick theorem. It turns out that a new, equivalent definition of normal ordering can be found via the Wick theorem that can be readily generalized for MR wave functions.
In what follows, we will use general second quantized operators α_p which can be either creation or annihilation operators; they are not distinguished by upper or lower indices. Indices $p, q, r, s, ...$ refer to general orbitals with unspecified particle-hole character. We define the contraction between operators α_p and α_q as

$$\overbrace{\alpha_p \alpha_q} = \alpha_p \alpha_q - :\alpha_p \alpha_q: \ .$$

It follows from the definition of normal ordering that the only two nonzero contractions are

$$\overbrace{a^i a_k} = \delta_k^i \quad , \quad \overbrace{a_a a^c} = \delta_a^c \ .$$

Considering Eqs. (2.1.14) and (2.1.21), a contraction can be generally written as

$$\overbrace{\alpha_p \alpha_q} = \langle \alpha_p \alpha_q \rangle \ . \tag{2.1.38}$$

Note that the contraction can be nonzero only if α_p and α_q are not in normal order.
Now let us consider the product $\alpha_{p_1}\alpha_{p_2}...\alpha_{p_K}$ and its normal ordered counterpart $:\alpha_{p_1}\alpha_{p_2}...\alpha_{p_K}:$. The Wick theorem states that this product can be expressed as

$$\alpha_{p_1}\alpha_{p_2}...\alpha_{p_K} =:\alpha_{p_1}\alpha_{p_2}...\alpha_{p_K}: + \sum_{1\times} :\overbracket{\alpha_{p_1}\alpha_{p_2}...}\alpha_{p_K}: + \sum_{2\times} :\overbracket{\alpha_{p_1}\overbracket{\alpha_{p_2}...}}\alpha_{p_K}: + ... \ ,$$

that is, as a sum generated by contracting one, two, etc. pairs of indices of the normal ordered operator in every allowed way. For example:

$$\begin{aligned}
\alpha_p\alpha_q\alpha_r\alpha_s = &:\alpha_p\alpha_q\alpha_r\alpha_s: \\
&+ :\overbracket{\alpha_p\alpha_q}\alpha_r\alpha_s: + :\overbracket{\alpha_p\alpha_q\alpha_r}\alpha_s: + :\overbracket{\alpha_p\alpha_q\alpha_r\alpha_s}: \\
&+ :\alpha_p\overbracket{\alpha_q\alpha_r}\alpha_s: + :\alpha_p\overbracket{\alpha_q\alpha_r\alpha_s}: + :\alpha_p\alpha_q\overbracket{\alpha_r\alpha_s}: \\
&+ :\overbracket{\alpha_p\alpha_q}\overbracket{\alpha_r\alpha_s}: + :\overbracket{\alpha_p\overbracket{\alpha_q\alpha_r}\alpha_s}: + :\overbracket{\alpha_p\alpha_q\alpha_r\alpha_s}: \ .
\end{aligned} \tag{2.1.39}$$

The proof of the Wick theorem is just roughly outlined; see Ref. [27] for a more detailed justification. As a base case, we immediately see that the theorem holds if $\alpha_{p_1}\alpha_{p_2}...\alpha_{p_K}$ is in normal order, since then all contractions vanish. If one pair of operators – say α_{p_1} and α_{p_2} for the sake of simplicity – is not in normal order, and they do not anticommute with each other, then their contraction is nonzero, and we can write

$$\begin{aligned}
\alpha_{p_2}\alpha_{p_1}\alpha_{p_3}...\alpha_{p_K} &= \{\alpha_{p_2},\alpha_{p_1}\}\alpha_{p_3}...\alpha_{p_K} - \alpha_{p_1}\alpha_{p_2}\alpha_{p_3}...\alpha_{p_K} \\
&= \{\alpha_{p_2},\alpha_{p_1}\}\ :\alpha_{p_3}...\alpha_{p_K}: - :\alpha_{p_1}\alpha_{p_2}\alpha_{p_3}...\alpha_{p_K}: \\
&=:\overbracket{\alpha_{p_2}\alpha_{p_1}}\alpha_{p_3}...\alpha_{p_K}: + :\alpha_{p_2}\alpha_{p_1}\alpha_{p_3}...\alpha_{p_K}: \ .
\end{aligned}$$

All other contractions are zero due to the other operators being in normal order, hence the theorem holds. One could continue this way to show that all operator pairs out of the normal order give rise to contractions, and thus prove the Wick theorem.

Based on the findings of Sec. 2.1.3 (cf. Eqs. (2.1.28), (2.1.32), and (2.1.33), now without cumulants) and Eq. (2.1.38), multiple contractions in the Wick expansion can always be grouped into a single expectation value. The example of Eq. (2.1.39) then takes the form

$$\begin{aligned}
\alpha_p\alpha_q\alpha_r\alpha_s = &:\alpha_p\alpha_q\alpha_r\alpha_s: \\
&+ \langle\alpha_p\alpha_q\rangle\ :\alpha_r\alpha_s: - \langle\alpha_p\alpha_r\rangle\ :\alpha_q\alpha_s: + \langle\alpha_p\alpha_s\rangle\ :\alpha_q\alpha_r: \\
&+ \langle\alpha_q\alpha_r\rangle\ :\alpha_p\alpha_s: - \langle\alpha_q\alpha_s\rangle\ :\alpha_p\alpha_r: + \langle\alpha_r\alpha_s\rangle\ :\alpha_p\alpha_q: \\
&+ \langle\alpha_p\alpha_q\alpha_r\alpha_s\rangle \ .
\end{aligned}$$

The signs are simply given by $(-1)^\pi$, π being the parity of index permutation compared to the original order (i.e. the number of transpositions required to reorder the operators).
An important special case of the Wick theorem arises for operator $a^{p_1\cdots p_K}_{q_1\cdots q_L}$ where all annihilation operators in the product stand to the right from creation operators: in this case each contraction forms a one-particle density matrix element, and K' contractions are

combined into K'-particle density matrix elements (cf. Eq. (2.1.16)). The first few examples include

$$a_q^p = :a_q^p: + \gamma_q^p \ , \tag{2.1.40}$$
$$a_q^{pr} = :a_q^{pr}: + \gamma_q^r :a^p: - \gamma_q^p :a^r: \ ,$$

and

$$a_{qs}^{pr} = :a_{qs}^{pr}: + \gamma_q^p :a_s^r: - \gamma_s^p :a_q^r: - \gamma_q^r :a_s^p: + \gamma_s^p :a_q^p: + \gamma_{qs}^{pr} \ . \tag{2.1.41}$$

Excitation operators (i.e. case $K = L$) have a particularly simple and compact form in the condensed notation:

$$a_K = \sum_{K'=0}^{K} \gamma_{K'} \wedge :a_{K-K'}: \ . \tag{2.1.42}$$

For convenience, $a_0 = 1$ is understood. Since permutation $p_1q_1p_2q_2...p_Kq_K$ has the same parity as $p_1p_2...p_Kq_K...q_2q_1$, the sign of a term in Eq. (2.1.42) can be easily determined by counting the parity of upper and lower index permutations (π_U and π_L) independently. This is an equivalent but simpler way to find the sign, since counting the parity of two K-long strings is less cumbersome than doing the same thing for a $2K$-long string.

It is important to note that the Wick theorem (like Eq. (2.1.42) for excitation operators) is not simply a consequence of the original particle-hole-based normal ordering definition, but is completely equivalent to it. We could have used the Wick "theorem" to define normal ordered operators and we could have derived the rearrangement relations (such as Eqs. (2.1.35) and (2.1.36)) from it. However, the Wick theorem has the advantage that it does not make any explicit reference to particles, holes, or the structure of the underlying wave function: any MR wave function could be used as long as the arising expectation values $\langle \alpha_{p_1}...\alpha_{p_{K'}} \rangle$ are properly calculated. Therefore, we can actually take these relations as the definition of normal ordered operators with respect to an arbitrary MR wave function, and thus obtain a general tool with all the desirable properties of ordinary normal ordering retained (such as Eq. (2.1.37)). In the literature this approach − originating from Kutzelnigg and Mukherjee − is known as the Multi-Reference Generalized Normal Ordering (MRGNO) [9, 30–32].

The MRGNO can be easily formulated in terms of contractions instead of expectation values by means of the cumulant expansion. The only difference is that, in addition to the one-particle contraction of Eq. (2.1.38), we must have general K-particle contractions (that is, cumulants) as well:

$$\overbrace{\alpha_{p_1}\alpha_{p_2}...\alpha_{p_{2K}}} = \langle \alpha_{p_1}\alpha_{p_2}...\alpha_{p_{2K}} \rangle_c \ . \tag{2.1.43}$$

Then the MRGNO expansion of an operator is obtained by considering all possible K-particle contractions of its normal ordered counterpart. The MRGNO version of Eq. (2.1.39) is for example appended with $\overbrace{\alpha_p\alpha_q\alpha_r\alpha_s} = \langle \alpha_p\alpha_q\alpha_r\alpha_s \rangle_c$, while for a_3 we would find

$$a_3 = :a_3: + \lambda_1 \wedge :a_2: + \left(\lambda_2 + \frac{1}{2}\lambda_1 \wedge \lambda_1\right) \wedge :a_1: + \lambda_3 + \lambda_2 \wedge \lambda_1 + \frac{1}{6}\lambda_1 \wedge \lambda_1 \wedge \lambda_1 \ .$$

Some comments on MRGNO are in order. MRGNO is linear as long as all terms are ordered with respect to the same reference fuction: $:(c_1\mathcal{S}_1 + c_2\mathcal{S}_2): = c_1 :\mathcal{S}_1: + c_2 :\mathcal{S}_2:$.

By convention, normal ordering leaves c-number expressions unchanged. Actually, we already assumed this in Eq. (2.1.42), when we took $:a_0:=:1:=1$.
A word of caution: while normal ordering is perfectly well defined as a prescription for arranging elements in a product, it must not be thought of or applied as an operator acting on some vector space. Operations like "normal ordering both sides of an equation" can quickly lead to contradictions and should be avoided. For example, by normal ordering both sides of $\{a_a, a^b\} = \delta_a^b$ and using the fact that operators within normal order anticommute, one could "prove" $0 = \delta_a^b$.

There are alternative or further generalized extensions of normal ordering for MR wave functions, such as the quasi-normal ordering of Kong et al. [33], or the generalized antisymmetric ordered product of Sinha et al. [34]. When applied to a single determinant, the traditional normal ordering is recovered by these different orderings, but they are not equivalent in general. For example, the quasi-normal ordering does not satisfy Eq. (2.1.37) for an MR wave function. We shall not discuss these alternative orderings here in any more detail.

Our original motivation for normal ordering was to separate physical quantities to an expectation value and a zero-average fluctuation part, similarly to Eq. (2.1.34). Indeed, for a one-particle operator X_1 ($K = 1$ in Eq. (2.1.4)) we immediately have (cf. Eq. (2.1.40))

$$X_1 = \mathrm{Tr}[\chi_1\gamma_1] + \sum_{p,q} \chi_p^q \, :a_q^p: = \sum_p \chi_p^p n_p + \sum_{p,q} \chi_p^q \, :a_q^p: \ .$$

Eq. (2.1.37) ensures that the average of the fluctuation term vanishes, as desired. For the Hamiltonian, using Eqs. (2.1.40) and (2.1.41) leads to

$$H = \langle H \rangle + \mathcal{H} \ , \tag{2.1.44}$$

where

$$\mathcal{H} = \sum_{p,q} f_p^q \, :a_q^p: + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \, :a_{qs}^{pr}: \ , \tag{2.1.45}$$

We introduced the generalized Fockian matrix element

$$f_p^q = h_p^q + \sum_r \tilde{v}_{pr}^{qr} n_r \ . \tag{2.1.46}$$

For a single Slater determinant, this reduces to the well-known Fockian of HF theory. The expectation value $\langle H \rangle$ was calculated in Eq. (2.1.19). Eq. (2.1.44) implies that $\mathcal{H}$ is the operator of the energy not covered by the (either single, or multi-determinantal) reference wave function; for a HF reference, it is the operator of correlation energy according to the definition of Löwdin. One could naively expect the appearance of h_p^q in $\mathcal{H}$. The fact that the Fockian f_p^q appears instead implies a so-called "dressing" or "screening" effect: the original ("bare") one-particle part of $\mathcal{H}$ is modified by the partially averaged electron-electron interaction leading to an effective one-particle quantity. This one-particle quantity incorporates some of the physics of the interacting N-electron system, while keeping a simple and compact form. Similar considerations will play an important role in later chapters.

Finally, we mention that normal ordering can actually be used to introduce the cumulant expansion of RDM-s in a more formal manner, by a generating function [19]:

$$1+ :\mathcal{C}:=:\exp(\mathcal{T}): \ , \tag{2.1.47}$$

where

$$\mathcal{C} = \sum_{p,q} \gamma_q^p a_p^q + \frac{1}{4} \sum_{p,q,r,s} \gamma_{qs}^{pr} a_{pr}^{qs} + \frac{1}{36} \sum_{p,q,r,s,t,u} \gamma_{qsu}^{prt} a_{prt}^{qsu} + \dots \ , \tag{2.1.48}$$

$$\mathcal{T} = \sum_{p,q} \lambda_q^p a_p^q + \frac{1}{4} \sum_{p,q,r,s} \lambda_{qs}^{pr} a_{pr}^{qs} + \frac{1}{36} \sum_{p,q,r,s,t,u} \lambda_{qsu}^{prt} a_{prt}^{qsu} + \dots \ . \tag{2.1.49}$$

Operator $\exp(\mathcal{T})$ is defined through its Taylor series. After expanding it and using the fact that creation and annihilation operators anticommute within normal ordering, we can match same rank operators on both sides of Eq. (2.1.47) and obtain Eq. (2.1.27). The inverse of Eq. (2.1.47) is

$$:\mathcal{T}:=:\ln(1+\mathcal{C}): \ . \tag{2.1.50}$$

Similar relations will be used later in the so-called cluster expansion of the wave function (see Sec. 2.3.1).

2.1.6 The generalized Wick theorem

In Sec. 2.1.5 we saw how normal ordering of operators can be used to partition physical quantities to a mean and a fluctuation part. In order to make MRGNO a really useful tool when dealing with the many-body problem, the multiplication rules of normal ordered operators must be worked out. This set of rules is collectively known as the generalized Wick theorem: it states that a product of normal ordered operators $:S_1:$ and $:S_2:$ is given by a Wick-like expansion generated by all K-body contractions of the "merged" operator $:S_1S_2:$, written symbolically as

$$:S_1::S_2:=:S_1S_2: + :\overbrace{S_1 S_2}: \ .$$

Only contractions connecting the two operators are needed, as self-contractions like $\overbrace{S_1}$ vanish. Similarly to normal ordering, the generalized Wick theorem with respect to a single Slater determinant (the Fermi vacuum) is well known and widely used [26, 27], while its MR generalization (introduced by Mukherjee and Kutzelnigg [30–32]) is somewhat less recognized. The theorem can be proven by induction [19, 33, 34]. However, we will only demonstrate its validity for simple cases through explicit calculation.

The first few examples are trivial consequences of normal ordering (cf. Eqs. (2.1.40) and (2.1.5)):

$$:a^p::a_q:= a^p a_q =:a_q^p: +\gamma_q^p \quad , \quad :a_q::a^p:= a_q a^p = - :a_q^p: + \left(\delta_q^p - \gamma_q^p\right) = - :a_q^p: +\eta_q^p \ ,$$

and

$$:a_q^p::a_s:= -a_{qs}^p - \gamma_q^p \ :a_s:= - :a_{qs}^p: -\gamma_s^p \ :a_q: \ ,$$

$$:a^p_q::a^r:= -a^{pr}_q + \delta^r_q\,:a^p: -\gamma^p_q\,:a^r:= -\,:a^{pr}_q: + \left(\delta^r_q - \gamma^r_q\right):a^p:= -\,:a^{pr}_q: +\eta^r_q\,:a^p:\ . \tag{2.1.51}$$

The product of one-particle operators is

$$\begin{aligned}
:a^p_q::a^r_s: &= a^{pr}_{qs} + \delta^r_q a^p_s - \gamma^p_q a^r_s - \gamma^r_s a^p_q + \gamma^p_q\gamma^r_s \\
&=:a^{pr}_{qs}: + \left(\delta^r_q - \gamma^r_q\right):a^p_s: -\gamma^p_s\,:a^r_q: +\gamma^p_s\left(\delta^r_q - \gamma^r_q\right) + \lambda^{pr}_{qs} \\
&=:a^{pr}_{qs}: +\eta^r_q\,:a^p_s: -\gamma^p_s\,:a^r_q: +\gamma^p_s\eta^r_q + \lambda^{pr}_{qs}\ ,
\end{aligned} \tag{2.1.52}$$

and using this with Eq. (2.1.41), after some algebra, we find for $:a_2::a_1:$

$$\begin{aligned}
:a^{pr}_{qs}::a^t_u:= {}& a^{prt}_{qsu} + \delta^t_q a^{pr}_{us} + \delta^t_s a^{pr}_{qu} - \gamma^t_u a^{pr}_{qs} \\
&- \mathcal{A}^{pr}\mathcal{A}_{qs}\left(\gamma^p_q\,:a^r_s::a^t_u:\right) - \left(\lambda^{pr}_{qs} + \gamma^p_q\gamma^r_s - \gamma^p_s\gamma^r_q\right):a^t_u: \\
= {}& :a^{prt}_{qsu}: +\eta^t_q\,:a^{pr}_{us}: +\eta^t_s\,:a^{pr}_{qu}: -\gamma^p_u\,:a^{tr}_{qs}: -\gamma^r_u\,:a^{pt}_{qs}: \\
&+ \left(\gamma^r_u\eta^t_s + \lambda^{rt}_{su}\right):a^p_q: - \left(\gamma^r_u\eta^t_q + \lambda^{rt}_{qu}\right):a^p_s: -\lambda^{tr}_{qs}\,:a^p_u: \\
&- \left(\gamma^p_u\eta^t_s + \lambda^{pt}_{su}\right):a^r_q: + \left(\gamma^p_u\eta^t_q + \lambda^{pt}_{qu}\right):a^r_s: -\lambda^{pt}_{qs}\,:a^r_u: \\
&- \lambda^{pr}_{us}\,:a^t_q: -\lambda^{pr}_{qu}\,:a^t_s: \\
&+ \eta^t_q\lambda^{pr}_{us} + \eta^t_s\lambda^{pr}_{qu} - \gamma^p_u\lambda^{tr}_{qs} - \gamma^r_u\lambda^{pt}_{qs} + \lambda^{prt}_{qsu}\ .
\end{aligned} \tag{2.1.53}$$

We see that all of these products indeed result in Wick-like expansions with self-contractions cancelling.

As a general rule, we can say that product of two MRGNO operators $:S_1:$ and $:S_2:$ is evaluated by considering all possible contractions between them, each contraction producing a K-body cumulant (with $\gamma_1 = \lambda_1$ understood). The one exception under this rule is the pair-contraction between an annihilation operator of $:S_1:$ and a creation operator of $:S_2:$, which yields η_1.

The sign of the various terms is again given by $(-1)^\pi$, π being the parity of permutation. When dealing with products of excitation operators, there is a simpler way to determine the sign, by counting the parity of upper and lower indices (π_U and π_L) independently; the reason for this is similar to the one discussed for normal ordering. The only difference is that this method would count the index order of $\eta^p_q = \langle a_q a^p\rangle$ incorrectly as pq and not as qp. Because of this, an extra factor of $(-1)^{n_\eta}$ appears for the n_η η_1-s in the term: the sign is then given by $(-1)^{\pi_U+\pi_L-n_\eta}$. Signs of terms can be easily evaluated this way in e.g. Eqs. (2.1.52) and (2.1.53).

The above rules for evaluating products of normal ordered operators with respect to a general MR wave function are collectively known as the Multi-Reference Generalized Wick Theorem (MRGWT). If the reference function is a single determinant, all cumulants vanish leaving only γ_1-s and η_1-s, and the more common Wick theorem is recovered. Since multiplication of operators is associative and distributive with addition, product of an arbitrary number of normal ordered operators can be evaluated with the successive application of the MRGWT. If one is interested in the expectation value of a product, it is sufficient to evaluate only the fully contracted part of the product (with no residual operator) as all the other terms vanish by virtue of Eq. (2.1.37). Closing this section, we present the expression for $\langle :a_2::a_2:\rangle$, as we

will need it later:

$$\begin{aligned}\langle :a^{pr}_{qs}::a^{tv}_{uw}:\rangle =&\gamma^p_u\gamma^r_w\eta^t_q\eta^v_s - \gamma^p_w\gamma^r_u\eta^t_q\eta^v_s - \gamma^p_u\gamma^r_w\eta^t_s\eta^v_q + \gamma^p_w\gamma^r_u\eta^t_s\eta^v_q\\
&+ \lambda^{pt}_{qu}\gamma^r_w\eta^v_s - \lambda^{pt}_{qw}\gamma^r_u\eta^v_s - \lambda^{pv}_{qu}\gamma^r_w\eta^t_s + \lambda^{pv}_{qw}\gamma^r_u\eta^t_s\\
&- \lambda^{pt}_{su}\gamma^r_w\eta^v_q + \lambda^{pt}_{sw}\gamma^r_u\eta^v_q + \lambda^{pv}_{su}\gamma^r_w\eta^t_q - \lambda^{pv}_{sw}\gamma^r_u\eta^t_q\\
&- \lambda^{rt}_{qu}\gamma^p_w\eta^v_s + \lambda^{rt}_{qw}\gamma^p_u\eta^v_s + \lambda^{rv}_{qu}\gamma^p_w\eta^t_s - \lambda^{rv}_{qw}\gamma^p_u\eta^t_s\\
&+ \lambda^{rt}_{su}\gamma^p_w\eta^v_q - \lambda^{rt}_{sw}\gamma^p_u\eta^v_q - \lambda^{rv}_{su}\gamma^p_w\eta^t_q + \lambda^{rv}_{sw}\gamma^p_u\eta^t_q\\
&+ \lambda^{tv}_{qs}\gamma^p_u\gamma^r_w - \lambda^{tv}_{qs}\gamma^p_w\gamma^r_u + \lambda^{pr}_{uw}\eta^t_q\eta^v_s - \lambda^{pr}_{uw}\eta^t_s\eta^v_q + \lambda^{pr}_{uw}\lambda^{tv}_{qs}\\
&+ \lambda^{pt}_{qu}\lambda^{rv}_{sw} - \lambda^{pt}_{su}\lambda^{rv}_{qw} - \lambda^{rt}_{qu}\lambda^{pv}_{sw} + \lambda^{rt}_{su}\lambda^{pv}_{qw}\\
&- \lambda^{pt}_{qw}\lambda^{rv}_{su} - \lambda^{pv}_{qu}\lambda^{rt}_{sw} + \lambda^{pt}_{sw}\lambda^{rv}_{qu} + \lambda^{pv}_{su}\lambda^{rt}_{qw}\\
&- \lambda^{pr}_{qu}\lambda^{tv}_{sw} - \lambda^{pr}_{qw}\lambda^{tv}_{us} - \lambda^{pr}_{us}\lambda^{tv}_{qw} - \lambda^{pr}_{ws}\lambda^{tv}_{uq}\\
&- \lambda^{pt}_{qs}\lambda^{rv}_{uw} - \lambda^{pv}_{qs}\lambda^{tr}_{uw} - \lambda^{tr}_{qs}\lambda^{pv}_{uw} - \lambda^{pv}_{qs}\lambda^{tr}_{uw}\\
&+ \lambda^{prv}_{usw}\eta^t_q + \lambda^{prt}_{wsu}\eta^v_q + \lambda^{prv}_{quw}\eta^t_s + \lambda^{prt}_{qwu}\eta^v_s\\
&- \lambda^{trv}_{qsw}\gamma^p_u - \lambda^{trv}_{usq}\gamma^p_w - \lambda^{ptv}_{qsw}\gamma^r_u - \lambda^{pvt}_{qsu}\gamma^r_w + \lambda^{prtv}_{qsuw}\ . \end{aligned} \tag{2.1.54}$$

2.1.7 Spin-free formalism

Up to this point, our entire discussion has been based on spin orbitals, without explicitly referencing spin degrees of freedom. This is quite useful when developing theories as all quantities can be conveniently expressed in terms of antisymmetric tensors. But in the end, these theories must be implemented in a way that ensures the proper spin symmetry of the system: that is, our models are to be adapted to the desired spin state. In this subsection − contrary to earlier ones − latin indices correspond to spatial orbitals and spin indices are written out explicitly; a spin orbital index is then a composite index $p\sigma$ made up of spatial part $p = 1, ..., M/2$ and spin part $\sigma = \alpha, \beta$. Throughout the thesis we shall always point out if spatial orbitals are used in a section or subsection. When not stated otherwise explicitly, latin indices will keep referring to spin orbitals.

The spin state of N-electron systems is characterized by the eigenvalues of the third component and the squared magnitude of the spin operator ($\hat{S}_z$ and $\hat{S}^2$, respectively). The reason for this is that $\hat{S}_z$ and $\hat{S}^2$ can be simultaneously measured, meaning that $[\hat{S}_z, \hat{S}^2] = 0$. As a consequence, a common set of eigenvectors $\{|S, M_S\rangle\}$ (for $S = 0, 1/2, 1, ...,$ and $M_S = -S, -S+1, ..., S$) can be found:

$$\hat{S}_z|S, M_S\rangle = M_S|S, M_S\rangle \quad , \quad \hat{S}^2|S, M_S\rangle = S(S+1)|S, M_S\rangle \ .$$

See e.g. Ref. [2] for the second quantized form of spin operators. In this thesis we shall be concerned with singlet states (i.e. the case of $S = 0$, $M_S = 0$).

In a non-relativistic theory the Hamiltonian does not contain any spin-dependent interactions. This means that the matrix elements of Eqs. (2.1.7) and (2.1.8) are diagonal in their spin indices: $h^{q\sigma_1}_{p\sigma'_1} = h^q_p\delta^{\sigma_1}_{\sigma'_1}$ and $v^{q\sigma_1 s\sigma_2}_{p\sigma'_1 r\sigma'_2} = v^{qs}_{pr}\delta^{\sigma_1}_{\sigma'_1}\delta^{\sigma_2}_{\sigma'_2}$, where

$$h^q_p = \int \mathrm{d}^3r\phi_p(\boldsymbol{r})\left(-\frac{1}{2}\nabla^2 - \sum_A \frac{Z_A}{|\boldsymbol{r} - \boldsymbol{R}_A|}\right)\phi^q(\boldsymbol{r}) \ ,$$

and

$$v_{pr}^{qs} = \int \mathrm{d}^3 r \int \mathrm{d}^3 r' \phi_p(\boldsymbol{r}) \phi_r(\boldsymbol{r}') \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} \phi^s(\boldsymbol{r}') \phi^q(\boldsymbol{r}) \ . \tag{2.1.55}$$

Spin independence of H implies $[H, \hat{S}_z] = 0$ and $[H, \hat{S}^2] = 0$, meaning that energy eigenstates must be spin eigenstates as well. To make this relation more explicit, it is useful to consider excitation operators E_K with spin indices summed up:

$$E_{q_1 \ldots q_K}^{p_1 \ldots p_K} = \sum_{\sigma_1, \ldots, \sigma_K} a_{q_1 \sigma_1 \ldots q_K \sigma_K}^{p_1 \sigma_1 \ldots p_K \sigma_K} \ .$$

with the obvious properties $[E_K, \hat{S}_z] = 0$ and $[E_K, \hat{S}^2] = 0$. Note that these operators are not antisymmetric with respect to individual permutations of upper or lower indices, but symmetric under their simultaneous permutations. The Hamiltonian in terms of these spin-free operators reads

$$H = \sum_{p,q} h_p^q E_q^p + \frac{1}{2} \sum_{p,q,r,s} v_{pr}^{qs} E_{qs}^{pr} \ .$$

Spin-free counterparts of density matrices and cumulants are defined analogously:

$$\Gamma_{q_1 \ldots q_K}^{p_1 \ldots p_K} = \sum_{\sigma_1, \ldots, \sigma_K} \gamma_{q_1 \sigma_1 \ldots q_K \sigma_K}^{p_1 \sigma_1 \ldots p_K \sigma_K} = \langle E_{q_1 \ldots q_K}^{p_1 \ldots p_K} \rangle \ , \tag{2.1.56}$$

$$\Lambda_{q_1 \ldots q_K}^{p_1 \ldots p_K} = \sum_{\sigma_1, \ldots, \sigma_K} \lambda_{q_1 \sigma_1 \ldots q_K \sigma_K}^{p_1 \sigma_1 \ldots p_K \sigma_K} = \langle E_{q_1 \ldots q_K}^{p_1 \ldots p_K} \rangle_c \ . \tag{2.1.57}$$

In the basis of natural orbitals, the 1-RDM reads

$$\Gamma_q^p = \sum_\sigma \delta_q^p \delta_\sigma^\sigma n_{q\sigma} = 2 \delta_q^p n_q \ ,$$

where $n_q = n_{q\alpha} = n_{q\beta} \in [0, 1]$.
Excitation operators $\{E_K\}$ provide the starting point of the so-called Unitary Group Approach (UGA) [35–37] to spin-free quantum chemistry. The name originally refers to the fact that spin-free one-particle excitations form the basis of Lie-algebra $u(M/2)$ (thus generating Lie-group $U(M/2)$):

$$\left[E_q^p, E_s^r \right] = \sum_{t,u} C_{qst}^{pru} E_u^t \ ,$$

coefficients $C_{qst}^{pru} = \delta_q^r \delta_t^p \delta_s^u - \delta_s^p \delta_t^r \delta_q^u$ being the structure constants. Such group theoretical considerations may be used to arrive at efficient spin-free formulations of Configuration Interaction (CI).

Many-body theories written purely in terms of spin-free quantities have many desirable properties. The most important one is that a spin-free correction method built on some reference wave function $|\Phi_{S,M_S}\rangle$ will not contaminate the original spin: as a consequence of all operators commuting with $\hat{S}_z$ and $\hat{S}^2$, the corrected wave function will remain the very same spin eigenstate. Also, dealing with $M/2$ spatial orbitals instead of M spin orbitals in our equations may lead to a great reduction in computational efforts (roughly a factor of 2^K

for K nested summations).
The only question is whether one can easily formulate a theory in terms of spin-free quantities. This requires integrating out the spin degrees of freedom in the original spin orbital model, that is, summing up the spin indices of tensors (mostly density matrices or cumulants) appearing in it. This spin-summation is always straightforward for a single tensor, as seen in Eqs. (2.1.56) and (2.1.57). However, doing the same thing for a product with contracted spin indices on different tensors is highly non-trivial. One immediately faces this problem when trying to perform the spin-summation of terms in cumulant expansions or in results of MRGWT; for example, in order to arrive at spin-free versions of Eqs. (2.1.26) or (2.1.54), one should be able to express a sum like $\sum_{\sigma_1,\ldots,\sigma_4} \lambda^{p\sigma_1 r\sigma_2}_{u\sigma_3 w\sigma_4} \lambda^{t\sigma_3 v\sigma_4}_{q\sigma_1 s\sigma_2}$ as a function of Λ_2. Of course, every many-body model built on cumulant expansion or MRGWT inherits this difficulty.

The problem can be in principle solved as follows. The function $\Lambda_K = g_K(\lambda_K)$ mapping spin orbital cumulants unto spin-free ones is simply given by Eq. (2.1.57). The inverse relation $\lambda_K = g_K^{-1}(\Lambda_K)$ mapping spin-free cumulants unto spin orbital ones is, however, much less trivial. Knowledge of g_K^{-1} would allow us to reexpress spin orbital cumulants of any complicated product in terms of spin-free ones, making the summation straightforward [22, 38]. These inverse relations can be deduced from the invariance of singlet states and density matrices under the action of spin rotation group $SU(2)$. The somewhat technical derivations are left to Appendix C; here we just quote the nonzero results for $K = 1, 2$:

$$\lambda^{p\alpha}_{q\alpha} = \frac{1}{2}\Lambda^{p}_{q} \,, \tag{2.1.58}$$

$$\lambda^{p\alpha r\alpha}_{q\alpha s\alpha} = \frac{1}{6}\left(\Lambda^{pr}_{qs} - \Lambda^{pr}_{sq}\right) \quad , \quad \lambda^{p\alpha r\beta}_{q\alpha s\beta} = \frac{1}{6}\left(2\Lambda^{pr}_{qs} + \Lambda^{pr}_{sq}\right) \,. \tag{2.1.59}$$

All other nonzero elements can be obtained by permuting the indices or interchanging α and β spin labels. One then easily finds

$$\sum_{\sigma_1,\ldots,\sigma_4} \lambda^{p\sigma_1}_{u\sigma_3}\lambda^{r\sigma_2}_{w\sigma_4}\lambda^{t\sigma_3 v\sigma_4}_{q\sigma_1 s\sigma_2} = \frac{1}{4}\Lambda^{p}_{u}\Lambda^{r}_{w}\Lambda^{tv}_{qs} \,, \tag{2.1.60}$$

$$\sum_{\sigma_1,\ldots,\sigma_4} \lambda^{p\sigma_1 r\sigma_2}_{q\sigma_1 u\sigma_3}\lambda^{t\sigma_3 v\sigma_4}_{s\sigma_2 w\sigma_4} = \frac{1}{2}\Lambda^{pr}_{qu}\Lambda^{sv}_{uw} \,, \tag{2.1.61}$$

$$\sum_{\sigma_1,\ldots,\sigma_4} \lambda^{p\sigma_1 r\sigma_2}_{u\sigma_3 w\sigma_4}\lambda^{t\sigma_3 v\sigma_4}_{q\sigma_1 s\sigma_2} = \frac{1}{6}\left(2\Lambda^{pr}_{uw}\Lambda^{tv}_{qs} + \Lambda^{pr}_{wu}\Lambda^{tv}_{qs} + \Lambda^{pr}_{uw}\Lambda^{tv}_{sq} + 2\Lambda^{pr}_{wu}\Lambda^{tv}_{sq}\right) \,, \tag{2.1.62}$$

and so on. Knowing Eqs. (2.1.58) and (2.1.59) is sufficient for performing the spin-summation of any quantity up to rank four, like the terms in Eqs. (2.1.26) and (2.1.54).
While theoretically speaking, the problem of spin-summation is solved by finding g_K^{-1}, practically speaking, evaluating the sum for a general product containing many high rank tensors still poses a problem of rapidly increasing complexity. Fortunately, we will not need to handle more complicated cases then the ones given above.

2.2 Geminal wave functions

A wave function consisting of a single Slater determinant cannot properly describe strong correlation effects. It is well known [1] that a chemical bond treated on single determinantal level is described by a mixture of ionic and covalent contributions, no matter how stretched the bond is; ionic terms (with excess electrons localized on one of the atomic sites) do not vanish even in the dissociation limit, leading to a qualitatively wrong wave function and an anomalously high ground state energy. An MR wave function is needed to get at least qualitatively good results along the complete dissociation process.

If finding the best one-electron functions (as is done in the HF approximation) is not good enough for our goals, then − as the logical next step − we might try to determine the best two-electron functions or electron pairs. These two-electron functions are called geminals. In geminal models the whole N-electron wave function (N assumed to be even) is built from such two-electron blocks. In many cases this is in very good accordance with the chemist way of thinking; after all, most of our chemical intuition is gained from models in which molecules are formed by localized electron pairs as bonds. Such considerations − in the form of Valence Bond theory − go back to the early days of quantum chemistry (see Ref. [39] for historical perspectives and applications).
We will be concerned with so-called strongly orthogonal and singlet coupled geminal wave functions only. In the following section, basic properties of such wave functions with their corresponding density matrices and cumulants will be discussed. More details on this subject with an extensive list of references can be found in Refs. [40–43].

Since spin states of geminals must be taken care of, spatial and spin indices are written separately throughout this whole section; latin indices denote spatial orbitals accordingly.

2.2.1 General considerations

Let us consider sets of spin orbitals $\{\phi^{p\sigma}\}_P$; these could in principle be any kind of subsets of M spin orbitals. Furthermore, let the set contaning the spatial indices appearing in $\{\phi^{p\sigma}\}_P$ be denoted as $\mathbb{G}_P$.
The general form of a two-electron function built from orbitals of $\{\phi^{p\sigma}\}_P$ reads (cf. Eq. (2.1.1))

$$|\psi^P\rangle = \frac{1}{2} \sum_{p_1,p_2 \in \mathbb{G}_P} \sum_{\sigma_1,\sigma_2} c^P_{p_1\sigma_1 p_2\sigma_2} |\Phi^{p_1\sigma_1 p_2\sigma_2}\rangle \ . \tag{2.2.1}$$

Such functions are called geminals; their real space representation is simply $\psi^P(x,x') = \langle x,x'|\psi^P\rangle$ (cf. Appendix A). Of course, geminals are chosen to be orthonormal:

$$\langle \psi_Q|\psi^P\rangle = \int \mathrm{d}x \int \mathrm{d}x' \psi_Q(x,x')\psi^P(x,x') = \frac{1}{2} \sum_{p_1,p_2 \in \mathbb{G}_{PQ}} \sum_{\sigma_1,\sigma_2} c_Q^{p_1\sigma_1 p_2\sigma_2} c^P_{p_1\sigma_1 p_2\sigma_2} = \delta^P_Q \ ,$$

where $\mathbb{G}_{PQ} = \mathbb{G}_P \cap \mathbb{G}_Q$ represents the intersection of the two orbital sets the geminals are expanded on.

We can think of Eq. (2.2.1) as $|\psi^P\rangle = \psi^P|0\rangle$, that is, as action on the vacuum with a composite creation operator:

$$\psi^P = \frac{1}{2} \sum_{p_1,p_2 \in \mathbb{G}_P} \sum_{\sigma_1,\sigma_2} c^P_{p_1\sigma_1 p_2\sigma_2} a^{p_1\sigma_1 p_2\sigma_2} \ . \tag{2.2.2}$$

Similarly, the adjoint operator ψ_Q is interpreted as the annihilation operator of a composite quasiparticle. Because of their two-electron nature, one would anticipate boson-like commutation relations for these operators. This turns out to be only partially true:

$$[\psi_Q, \psi_P] = 0 \quad , \quad [\psi^Q, \psi^P] = 0 \quad , \quad [\psi_Q, \psi^P] = \mathcal{Q}^P_Q \ ,$$

where

$$\mathcal{Q}^P_Q = \delta^P_Q - \sum_{p_1,p_2,q_2 \in \mathbb{G}_{PQ}} \sum_{\sigma_1,\sigma_2,\sigma_2'} c^{p_1\sigma_1 q_2\sigma_2'}_Q c^P_{p_1\sigma_1 p_2\sigma_2} a^{p_2\sigma_2}_{q_2\sigma_2'} \ . \tag{2.2.3}$$

These relations are easily verified by using Eqs. (2.2.1) and (2.1.2). Instead of δ^P_Q, the rather complicated looking $\mathcal{Q}^P_Q$ arises in the commutator of ψ_Q and ψ^P: it is not diagonal, and – due to the one-particle operator in Eq. (2.2.3) – it is not even c-number valued. In general, this would make manipulation of geminal creation-annihilation operators rather difficult, and would seriously reduce their applicability in building N-electron wave functions. We need an extra condition to simplify the structure of $\mathcal{Q}^P_Q$. Note that we have not specified the sets $\{\mathbb{G}_P\}$ yet.

A diagonal $\mathcal{Q}^P_Q$ could already make things much simpler, since then at least different geminal creation-annihilation operators would commute with each other. This can be achieved by enforcing the so-called strong orthogonality condition on geminals. Strong orthogonality of geminals in either real or orbital space means

$$\int \mathrm{d}x' \psi_Q(x,x')\psi^P(x'',x') = 0 \quad \Leftrightarrow \quad \frac{1}{2} \sum_{p_1 \in \mathbb{G}_{PQ}} \sum_{\sigma_1} c^{p_1\sigma_1 q_2\sigma_2'}_Q c^P_{p_1\sigma_1 p_2\sigma_2} = 0 \quad \text{if } P \neq Q \ . \tag{2.2.4}$$

This is obviously a stronger condition that the usual orthogonality, and it immediately leads to $\mathcal{Q}^P_Q$ being diagonal. Eq. (2.2.4) can be easily satisfied if different geminals have no common orbitals assigned to them (i.e. by setting $\mathbb{G}_{PQ} = \emptyset$ for $P \neq Q$). The converse of the statement is also true: strong orthogonality implies the mutually disjoint distribution of orbitals among geminals. The equivalence of these two conditions is known as the Arai theorem [44].

Strongly orthogonal geminals are much easier to handle and use as building blocks. Acting on the vacuum with the product of $N/2$ (different) geminal creation operators yields a simple N-electron wave function Ansatz, the Antisymmetrized Product of Strongly Orthogonal Geminals (APSG) [45]:

$$|\Psi_{APSG}\rangle = \prod_{P=1}^{N/2} \psi^P|0\rangle \ .$$

The name makes more sense in real space representation:

$$\Psi_{APSG}(\boldsymbol{x}_N) = \langle \boldsymbol{x}_N|\Psi_{APSG}\rangle = \sqrt{\frac{2^{N/2}}{N!}} \mathcal{A}_{1\ldots N} \left(\prod_{P=1}^{N/2} \psi^P(x_{2P-1}, x_{2P}) \right) \ .$$

The antisymmetrizer (according to its definition in Appendix A) interchanges coordinate indices only between different geminals. Note that strongly orthogonal geminals in $|\Psi_{APSG}\rangle$ are independent two-electron groups in the same sense as was used in Sec. 2.1.4.

Two electrons can be coupled to form either a singlet or a triplet state, and coefficients of geminal wave functions should be chosen accordingly. We shall only deal with the singlet case, that is, we require $M_S = 0$ and $S = 0$ for all geminals. The former can be easily achieved by restricting the coefficients to have only one α and one β spin index. The latter can be ensured by choosing the geminal coefficient matrix to be symmetric with respect to interchanging spatial indices. Then we have (in accordance with Eq. (C.0.14)) $c^P_{p_1\sigma_1 p_2\sigma_2} = c^P_{p_1 p_2}\epsilon_{\sigma_1\sigma_2}$, where $c^P_{p_1p_2} = c^P_{p_2p_1}$, and $\epsilon_{\sigma_1\sigma_2} = \delta^\alpha_{\sigma_1}\delta^\beta_{\sigma_2} - \delta^\alpha_{\sigma_2}\delta^\beta_{\sigma_1}$ is the Levi-Civita symbol. Substituting in Eq. (2.2.2) yields

$$\psi^P = \sum_{p_1,p_2\in\mathbb{G}_P} c^P_{p_1p_2} a^{p_1\alpha p_2\beta} \ .$$

This expression can be further simplified by switching to the diagonal basis of the coefficient matrix:

$$\psi^P = \sum_{p\in\mathbb{G}_P} c^P_p a^{p\alpha p\beta} \ .$$

In fact, we will see in Sec. 2.2.2 that the orbitals spanning this basis coincide with the natural orbitals.

APSG belongs to the family of Multiconfigurational Self-Consistent Field (MCSCF) methods: orbitals and geminal coefficients must be simultaneously optimized in order to arrive at the lowest APSG energy (cf. Eq. (2.1.19); see Sec. 2.2.2 for the density matrices). At the end of the complicated variational optimization procedure one obtains unique geminals localized to certain areas within the molecule (between atoms or around a single atom), which represent bonding and lone electron pairs according to the usual chemical interpretation.

Loosely speaking, APSG separates electron correlation into an intrageminal and an intergeminal part. While intrageminal correlation is properly taken into account (corresponding to the fact that a variational problem is solved for each electron pair), intergeminal correlation is not included. This is due to the Ansatz being a product of strongly orthogonal geminals, meaning that interaction of electron pairs is still described only on a mean field level.

The values of geminal coefficients can be guessed in the limiting case of weak correlation (around equilibrium geometry of a bonding geminal or in a lone pair geminal) and strong correlation (in a dissociating bonding geminal). In the former case a mean field description of the electron pair is expected to be sufficient, the geminal is dominated by a single two-electron determinant with a coefficient close to unity, all other determinants having negligible weight. In the latter case two determinants (say the p-th and r-th) have equal weight in the geminal and all the others vanish. The two corresponding coefficients are then $c^P_p = -c^P_r = \frac{1}{\sqrt{2}}$; the sign difference can be guessed from the requirement that the wave function describes a purely covalent system with one electron localized on each of the two atomic sites [1,46] (see also Appendix B).

The number of spatial orbitals assigned to geminals is an adjustable parameter of the theory.

Obviously, a geminal containing only one orbital (i.e. $\psi^P = a^{p\alpha p\beta}$) accounts for treating the corresponding electron pair on a spin restricted mean field level. The simplest improvement over mean field treatment requires two orbitals:

$$\psi^P = c_p^P a^{p\alpha p\beta} + c_r^P a^{r\alpha r\beta} \ .$$

The special case of APSG wave function built with geminals containing at most two orbitals (i.e. $|\mathbb{G}_P| \leq 2$) is called Generalized Valence Bond (GVB) [47–49]. We shall use GVB in all practical calculations.

APSG has numerous advantages. It represents a conceptually and computationally simple improvement over mean field techniques and can accurately describe the dissociation of a single covalent bond. The APSG energy always represents an upper bound for the exact ground state energy, as orbitals and coefficients are determined variationally. Furthermore, it gives the exact variational solution for two-electron systems, provided that all orbitals are assigned to the geminal (i.e. $|\mathbb{G}| = M/2$). The APSG energy is also size consistent and extensive over singlet subsystems.

However, there are also disadvantages. While APSG recovers a large part of static electron correlation, it fails to incorporate much of dynamic correlation, which would be needed for accurate energies and molecular properties. This weakness of APSG is manifested e.g. in its inability to properly describe intermolecular (e.g. dispersion) interactions, as these would require a more refined treatment of interbond interactions than mean field. Several correction methods have been proposed to add dynamic correlation to APSG, including both perturbative [50, 51] and non-perturbative [42, 52–59] approaches.
Also, APSG relies heavily on the localized nature of chemical bonds in molecules (i.e. the existence of a single dominant Lewis structure), and therefore cannot be properly applied to e.g. delocalized π-electron systems.

There is a more subtle problem too, concerning the APSG wave function. While APSG yields the correct wave function for single bond dissociation (describing two infinitely separated, but spin coupled doublet subsystems), the same is not true for multiple bond dissociations [60,61]. A simple example is provided by the symmetric dissociation of the H_2O molecule: we know from the Hund rule that in the infinite separation limit the system should consist of a triplet oxygen atom and two triplet coupled hydrogen atoms, all coupled together to form a singlet state. In the APSG description, however, the system consists of two singlet O-H electron pairs, no matter how stretched the bonds are: essentially, spin coupling between electrons of different bonds is missing. One thus generally concludes that APSG yields a qualitatively wrong wave function for multiple bond dissociations. This error may seriously effect correction methods using APSG as starting point, and may generate unphysical local maxima or even singularities on potential energy surfaces [53, 54].

2.2.2 APSG density matrices and cumulants

We shall now calculate density matrices and cumulants of the APSG wave function up to rank four. Derivations are substantially facilitated by the strong orthogonality of geminals and the simple structure of the APSG Ansatz.

We start with the 1-RDM $\gamma^{p\sigma_1}_{q\sigma_2} = \langle a^{p\sigma_1}_{q\sigma_2}\rangle$. Because of strong orthogonality ϕ^p and ϕ^q must belong to the same geminal (i.e. $p, q \in \mathbb{G}_P$), otherwise the 1-RDM element vanishes. Then we have

$$\gamma^{p\sigma_1}_{q\sigma_2} = \delta^P_Q \langle 0|\psi_P a^{p\sigma_1}_{q\sigma_2}\psi^P \prod_{\substack{R=1\\R\neq P}}^{N/2} \psi_R\psi^R|0\rangle = \delta^P_Q \langle 0|\psi_P a^{p\sigma_1}_{q\sigma_2}\psi^P|0\rangle$$

Calculation of the 1-RDM therefore reduces to evaluating the one-geminal contribution:

$$\gamma^{p\sigma_1}_{q\sigma_2} = \delta^P_Q \langle 0|\psi_P a^{p\sigma_1}_{q\sigma_2}\psi^P|0\rangle = \delta^P_Q \sum_{p_1,p_2\in P} c^{p_1}_P c^P_{p_2} \langle 0|a_{p_1\beta p_1\alpha} a^{p\sigma_1}_{q\sigma_2} a^{p_2\alpha p_2\beta}|0\rangle = \delta^p_q \delta^{\sigma_1}_{\sigma_2} c^p_P c^P_p = \delta^p_q \delta^{\sigma_1}_{\sigma_2} n_q \ .$$

Diagonal form of the 1-RDM shows that the orbitals diagonalizing the geminal coefficient matrix are indeed natural orbitals.
Looking at the 2-RDM $\gamma^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} = \langle a^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4}\rangle$, we see that strong orthogonality leaves us with two non-vanishing options: either all four orbitals belong to the same geminal (i.e. $p, q, r, s \in \mathbb{G}_P$) or they belong to two different geminals (that is, $p, q \in \mathbb{G}_P$ and $r, s \in \mathbb{G}_R$, or $p, s \in \mathbb{G}_P$ and $r, q \in \mathbb{G}_R$ for $P \neq R$). Then

$$\begin{aligned}\gamma^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} &= \delta^{PR}_{QS}\langle 0|\psi_P a^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4}\psi^P|0\rangle + (\delta^P_Q\delta^R_S + \delta^P_S\delta^R_Q)(1-\delta_{QS})\langle 0|\psi_R\psi_P a^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4}\psi^P\psi^R|0\rangle \\ &= \omega^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} + (1-\delta_{QS})\left(\gamma^{p\sigma_1}_{q\sigma_2}\gamma^{r\sigma_3}_{s\sigma_4} - \gamma^{p\sigma_1}_{s\sigma_4}\gamma^{r\sigma_3}_{q\sigma_2}\right) \ ,\end{aligned}$$

where $\omega^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4}$ is the connected one-geminal contribution to the 2-RDM:

$$\omega^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} = \delta^{PR}_{QS}\langle 0|\psi_P a^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4}\psi^P|0\rangle = \delta^P_Q \delta^{pr}\delta_{qs}\epsilon^{\sigma_1\sigma_3}\epsilon_{\sigma_2\sigma_4} c^p_P c^P_q \ . \tag{2.2.5}$$

In condensed notation, we can write the 2-RDM as

$$\gamma_2 = \{\omega_2\} + \frac{1}{2}\{\gamma_1\}\wedge\{\gamma_1\} \ .$$

In this notation tensors within curly parantheses belong to the same geminal, while tensors in different parantheses belong to strictly different geminals. The two-particle cumulant is then

$$\begin{aligned}\lambda_2 &= \{\omega_2\} + \frac{1}{2}\{\gamma_1\}\wedge\{\gamma_1\} - \frac{1}{2}\gamma_1\wedge\gamma_1 \\ &= \{\omega_2\} + \frac{1}{2}\{\gamma_1\}\wedge\{\gamma_1\} - \frac{1}{2}\{\gamma_1\}\wedge\{\gamma_1\} - \frac{1}{2}\{\gamma_1\wedge\gamma_1\} \\ &= \{\omega_2\} - \frac{1}{2}\{\gamma_1\wedge\gamma_1\} \ .\end{aligned} \tag{2.2.6}$$

Here we separated the wedge product part into an intrageminal and an intergeminal part. The intergeminal part of the wedge product cancels with that of the 2-RDM, leading to λ_2 having only intrageminal contributions. Written out explicitly, λ_2 is

$$\lambda^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} = \omega^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} - \delta_{QS}\left(\gamma^{p\sigma_1}_{q\sigma_2}\gamma^{r\sigma_3}_{s\sigma_4} - \gamma^{p\sigma_1}_{s\sigma_4}\gamma^{r\sigma_3}_{q\sigma_2}\right) \ . \tag{2.2.7}$$

Calculation of the 3-RDM is similar to that of the 2-RDM. Note, however, that all indices now cannot belong to the same geminal as no more than two electrons may be deleted from

a geminal; then we can only have 4-2 or 2-2-2 type index distributions. This means that γ_3 can be completely expressed in terms of ω_2 and γ_1:

$$\gamma_3 = \{\omega_2\} \wedge \{\gamma_1\} + \frac{1}{6}\{\gamma_1\} \wedge \{\gamma_1\} \wedge \{\gamma_1\} \ .$$

Using Eq. (2.2.6) and

$$\frac{1}{6}\gamma_1 \wedge \gamma_1 \wedge \gamma_1 = \frac{1}{6}\{\gamma_1\} \wedge \{\gamma_1\} \wedge \{\gamma_1\} + \frac{1}{2}\{\gamma_1 \wedge \gamma_1\} \wedge \{\gamma_1\} + \frac{1}{6}\{\gamma_1 \wedge \gamma_1 \wedge \gamma_1\} \ ,$$

we find again that the corresponding cumulant has only intrageminal contributions:

$$\begin{aligned}\lambda_3 &= \{\omega_2\} \wedge \{\gamma_1\} + \frac{1}{6}\{\gamma_1\} \wedge \{\gamma_1\} \wedge \{\gamma_1\} - \lambda_2 \wedge \gamma_1 - \frac{1}{6}\gamma_1 \wedge \gamma_1 \wedge \gamma_1 \\ &= -\{\lambda_2 \wedge \gamma_1\} - \frac{1}{6}\{\gamma_1 \wedge \gamma_1 \wedge \gamma_1\} \ .\end{aligned}$$

The four-particle density matrix and cumulant can be obtained analogously:

$$\gamma_4 = \frac{1}{2}\{\omega_2\} \wedge \{\omega_2\} + \frac{1}{2}\{\omega_2\} \wedge \{\gamma_1\} \wedge \{\gamma_1\} + \frac{1}{24}\{\gamma_1\} \wedge \{\gamma_1\} \wedge \{\gamma_1\} \wedge \{\gamma_1\} \ ,$$

$$\lambda_4 = -\frac{1}{2}\{\lambda_2 \wedge \lambda_2\} + \frac{1}{2}\{\lambda_2 \wedge \gamma_1 \wedge \gamma_1\} + \frac{1}{8}\{\gamma_1 \wedge \gamma_1 \wedge \gamma_1 \wedge \gamma_1\} \ .$$

Cancellation of intergeminal contributions to cumulants is the manifestation of the general result in Sec. 2.1.4. This important property leads to the vanishing of sufficiently high rank cumulants. If $|\mathbb{G}_P|$ spatial orbitals (that is, $2|\mathbb{G}_P|$ spin orbitals) are assigned to the P-th geminal, than any possible cumulant of rank $K > 2|\mathbb{G}_P|$ would contain at least one repeated upper/lower index and vanish by virtue of antisymmetry. Therefore, the highest cumulant rank for APSG is $K_{max} = 2\max_P(|\mathbb{G}_P|)$, and $K_{max} = 4$ for GVB [18]. Note that this vanishing is independent of the particle number: while λ_3 or λ_4 in GVB can be nonzero for a two-electron system, λ_K with $K > 4$ is always zero.

Spin-free counterparts of the above formulae can be easily found up to rank three since the 1-RDM is diagonal in spin. We present the results for cumulants only:

$$\Gamma^p_q = 2\delta^p_q n_q \quad , \quad \Lambda^{pr}_{qs} = 2\delta^P_Q \delta^{pr} \delta_{qs} c^p_P c^P_q - \delta_{QS}\left(\Gamma^p_q \Gamma^r_s - \frac{1}{2}\Gamma^p_s \Gamma^r_q\right) \ , \tag{2.2.8}$$

and

$$\Lambda^{prt}_{qsu} = -\delta_{QS}\left(\Gamma^p_q \Lambda^{rt}_{su} - \frac{1}{2}\Gamma^p_s \Lambda^{rt}_{qu} - \frac{1}{2}\Gamma^p_u \Lambda^{rt}_{sq}\right) - \delta_{SU}\left(\Gamma^r_s \Lambda^{tp}_{uq} - \frac{1}{2}\Gamma^r_u \Lambda^{tp}_{sq} - \frac{1}{2}\Gamma^r_q \Lambda^{tp}_{us}\right) \tag{2.2.9}$$

$$-\delta_{UQ}\left(\Gamma^t_u \Lambda^{pr}_{qs} - \frac{1}{2}\Gamma^t_q \Lambda^{pr}_{us} - \frac{1}{2}\Gamma^t_s \Lambda^{pr}_{qu}\right) \tag{2.2.10}$$

$$-\delta_{QS}\delta_{SU}\delta_{UQ}\left(\Gamma^p_q \Gamma^r_s \Gamma^t_u - \frac{1}{2}\Gamma^p_q \Gamma^r_u \Gamma^t_s + \frac{1}{4}\Gamma^p_u \Gamma^r_q \Gamma^t_s - \frac{1}{2}\Gamma^p_u \Gamma^r_s \Gamma^t_q + \frac{1}{4}\Gamma^p_s \Gamma^r_u \Gamma^t_q - \frac{1}{2}\Gamma^p_s \Gamma^r_q \Gamma^t_u\right) \ . \tag{2.2.11}$$

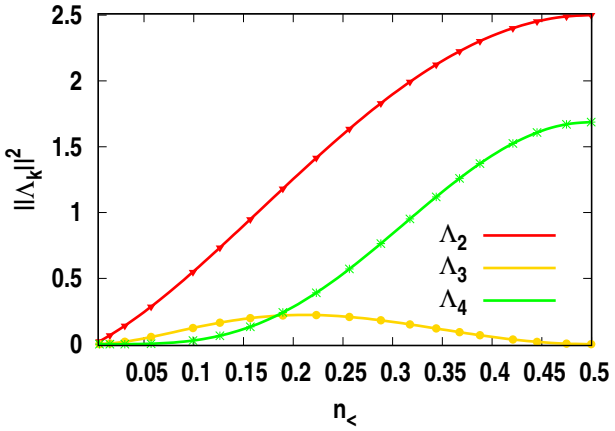


Figure 2.1: magnitude of spin-free cumulants in a single GVB geminal with increasing static correlation.

The case of the 4-RDM and its cumulant is less trivial, because products of rank two quantities appear; the spin-summation can be performed using Eqs. (2.1.60), (2.1.61) and (2.1.62). The results are too lengthy to type out, but they can be manipulated rather easily with symbolic algebra programs.

Asymptotic properties of cumulants can be studied in a dissociating geminal. In this limiting case only two orbitals (say the p-th and r-th) have nonzero coefficient: $c_p^P = -c_r^P = \frac{1}{\sqrt{2}}$ and so $n_p = n_r = \frac{1}{2}$. Both Γ_1 and Λ_2 tend to nonzero values:

$$\Gamma_p^p = \Gamma_r^r = 1 \quad , \quad \Lambda_{pp}^{pp} = \Lambda_{rr}^{rr} = \Lambda_{rp}^{pr} = \frac{1}{2} \quad , \quad \Lambda_{pr}^{pr} = \Lambda_{rr}^{pp} = \Lambda_{pp}^{rr} = -1 \ .$$

Using these values one can easily verify that Λ_3 vanishes in the dissociating limit, in accordance with the general result of Sec. 2.1.3. Nonvanishing elements of Λ_4 are

$$\Lambda_{pprr}^{pprr} = \Lambda_{rrpp}^{pprr} = -\frac{3}{2} \quad , \quad \Lambda_{prpr}^{pprr} = \frac{3}{4} \ .$$

As an illustration of cumulant magnitudes in a single GVB geminal, the squared norm of Λ_K is plotted in Fig. 2.1 for $K = 2, 3, 4$ as a function of the smaller of the two occupation numbers, $n_<$. The squared norm is defined as $||\Lambda_K||^2 = \mathrm{Tr}[\Lambda_K \Lambda_K]/K!$. Around equilibrium bond distance the geminal is dominated by a single determinant, $n_< \approx 0$ and cumulants are negligible, as expected. On the other hand, both Λ_2 and Λ_4 become quite large in the dissociating limit (where $n_< \approx 1/2$). The Λ_3 vanishes in both limiting cases, having only a tiny maximum around $n_< \approx 0.2$.

Real space counterparts of the spin-free cumulants are obtained simply by transforming them with $\{\phi^p(\boldsymbol{r})\}$, and exploiting the fact that intergeminal elements vanish. We find for $K = 1, 2$

$$\Gamma(\boldsymbol{r}_1; \boldsymbol{r}_1') = \sum_P \sum_{p,q \in \mathbb{G}_P} \Gamma_q^p \phi^q(\boldsymbol{r}_1) \phi_p(\boldsymbol{r}_1') = 2 \sum_p n_p \phi^p(\boldsymbol{r}_1) \phi_p(\boldsymbol{r}_1') \ ,$$

and

$$\Lambda(\boldsymbol{r}_1,\boldsymbol{r}_2;\boldsymbol{r}'_1,\boldsymbol{r}'_2)=\sum_P\sum_{p,q,r,s\in\mathbb{G}_P}\Lambda^{pr}_{qs}\phi^q(\boldsymbol{r}_1)\phi^s(\boldsymbol{r}_2)\phi_p(\boldsymbol{r}'_1)\phi_r(\boldsymbol{r}'_2)=\sum_P\Lambda^P(\boldsymbol{r}_1,\boldsymbol{r}_2;\boldsymbol{r}'_1,\boldsymbol{r}'_2)\ .$$

Orbitals assigned to geminals are localized around atoms or bonds, and so are cumulants. For spatially well separated electron pairs, $\Lambda^P(\boldsymbol{r}_1,\boldsymbol{r}_2;\boldsymbol{r}'_1,\boldsymbol{r}'_2)$ takes a non-negligible value only if points $\boldsymbol{r}_1,\boldsymbol{r}_2,\boldsymbol{r}'_1,\boldsymbol{r}'_2$ are all in the vicinity of the P-th pair. The above formulae again signal the fact that while intrabond correlation is taken care of by strongly orthogonal geminals, interbond correlation is not: finding an electron around a bond and finding a second electron around a different bond are essentially uncorrelated events.
As demonstrated in Appendix B, cumulants are essential in realizing the so-called left-right correlation in bonds, that is, the proper localization of electrons in the dissociating limit. Besides the advantages, cumulants naturally inherit the disadvantages of the APSG wave function, mostly the erroneous spin coupling in multiple bond dissociations. Looking at the example of symmetric H_2O dissociation, APSG cumulants do describe the correct localization of electrons on the oxygen atom, but they lack the coupling terms necessary for a triplet state. This weakness will have some troubling effects on APSG-based correction methods employing cumulants.

2.3 The Coupled Cluster model

In general, the exact wave function of the N-electron system is a combination of multiple determinants. However, in many important cases, the wave function is dominated by dynamic electron correlation, i.e. it is well approximated by a single determinant (built from e.g. HF orbitals), while other determinants give only minor contributions; closed-shell molecules around their equilibrium structure are the main examples. In such cases these small corrections can be easily introduced by means of many-body perturbation theory (MBPT). Assuming the convergence of the MBPT series, one expects more and more accurate results by calculating successively higher order contributions to the energy and the wave function. Unfortunately, with increasing order, the equations quickly become intractable and the computational cost increases dramatically.

The Coupled Cluster (CC) model provides a very efficient alternative approach to incorporate dynamic correlation. Introduced originally in the context of the nuclear many-body problem by Coester and Kümmel [62], CC was largely reformulated and improved on by Čížek [63,64]. Since then, following the works of many (e.g. Refs. [65, 66] just to name a few), CC has become perhaps the most successful tool in ab initio quantum chemistry. Instead of an exact calculation of the first few orders of MBPT, CC aims to take approximately all orders of MBPT into account. This approximate resummation can be seen most easily by formulating the theory in terms of Feynman diagrams, as will be outlined below.

The following summary is mostly based on Refs. [26, 27, 67, 68]; all other references will be shown explicitly.

2.3.1 General considerations

The exact wave function $|\Psi\rangle$ can be expanded in the basis of Slater determinants, as seen in Eq. (2.1.1). Since a single dominant determinant $|\Phi\rangle$ (set to be the Fermi vacuum) is assumed, it is useful to express all other determinants of the expansion relative to $|\Phi\rangle$ with the help of excitation operators:

$$|\Psi\rangle = |\Phi\rangle + \sum_{\substack{i\\a}} c_a^i a_i^a |\Phi\rangle + \frac{1}{4}\sum_{\substack{i,j\\a,b}} c_{ab}^{ij} a_{ij}^{ab} |\Phi\rangle + \frac{1}{36}\sum_{\substack{i,j,k\\a,b,c}} c_{abc}^{ijk} a_{ijk}^{abc} |\Phi\rangle + \dots \ . \tag{2.3.1}$$

Indices $i, j, k, l, \dots$ and $a, b, c, d, \dots$ denote the occupied and unoccupied (virtual) spin orbitals (or equivalently, hole and particle states) of $|\Phi\rangle$, respectively. Eq. (2.3.1) is interpreted as taking all possible polarizations of the Fermi vacuum into account by formation of particle-hole pairs (i.e. $a_{i_1\dots i_K}^{a_1\dots a_K} = a^{a_1\dots a_K} b^{i_K\dots i_1} = a^{a_1} b^{i_1} \dots a^{a_K} b^{i_K}$; see Sec. 2.1.5). The unit coefficient of $|\Phi\rangle$ reflects the so-called intermediate normalization of $|\Psi\rangle$ (fulfilling $\langle\Phi|\Psi\rangle = 1$). In principle, one could determine all coefficients variationally and thus obtain exact solutions to the Schrödinger equation, as done in CI. In practice, however, this is possible only for very small systems. Truncating the expansion after a finite excitation rank (e.g. $K = 2$) signficantly decreases the computational cost, but tends to yield poor results violating size consistency and extensivity.
The above problems can be solved by reparametrizing $|\Psi\rangle$. The expansion in Eq. (2.3.1) is written more compactly as

$$|\Psi\rangle = \hat{\Omega}|\Phi\rangle \ ,$$

$\hat{\Omega}$ being the so-called wave operator that maps the reference function unto the exact ground state. It simply reads $\hat{\Omega} = (1 + :C:)$, where

$$C = \sum_K C_K = \sum_K \frac{1}{(K!)^2} \sum_{\substack{i_1,\dots,i_K\\a_1,\dots,a_K}} c_{a_1\dots a_K}^{i_1\dots i_K} a_{i_1\dots i_K}^{a_1\dots a_K} = \sum_{\substack{i\\a}} c_a^i a_i^a + \frac{1}{4}\sum_{\substack{i,j\\a,b}} c_{ab}^{ij} a_{ij}^{ab} + \dots \ .$$

Since occupied and virtual orbitals form disjoint sets, putting operators in normal order is in fact immaterial (as $C = :C:$); its sole purpose is to emphasize the connection with MR theories in later chapters. In the spirit of cumulant expansion (cf. Eqs. (2.1.47), (2.1.48) and (2.1.49)) $\hat{\Omega}$ can be also written as

$$\hat{\Omega} = (1 + :C:) = :\exp(T): \ , \tag{2.3.2}$$

where

$$T = \sum_K T_K = \sum_K \frac{1}{(K!)^2} \sum_{\substack{i_1,\dots,i_K\\a_1,\dots,a_K}} t_{a_1\dots a_K}^{i_1\dots i_K} a_{i_1\dots i_K}^{a_1\dots a_K} = \sum_{\substack{i\\a}} t_a^i a_i^a + \frac{1}{4}\sum_{\substack{i,j\\a,b}} t_{ab}^{ij} a_{ij}^{ab} + \dots \ .$$

The t_K parameters are called CC amplitudes. After matching the operator ranks on both sides of Eq. (2.3.2), we find the following relations:

$$c_1 = t_1 \quad , \quad c_2 = t_2 + \frac{1}{2} t_1 \wedge t_1 \quad , \quad c_3 = t_3 + t_2 \wedge t_1 + \frac{1}{6} t_1 \wedge t_1 \wedge t_1 \ ,$$

and so on. Following the analogy with the cumulant expansion of density matrices, amplitudes can be thought of as "irreducible" components of CI coefficients. Due to the exponential form of Eq. (2.3.2), even a truncated T-expansion preserves size extensivity.

The next question is how to find amplitudes t_K. Substituting $|\Psi\rangle =:\exp(T):|\Phi\rangle$ into the Schrödinger equation gives

$$\mathcal{H} :\exp(T):|\Phi\rangle = \Delta E :\exp(T):|\Phi\rangle \ .$$

The normal product Hamiltonian was defined in Eq. (2.1.44), and the energy correction is $\Delta E = E_{exact} - \langle H\rangle$. Projecting the equation by $\langle\Phi|$ and exploiting the intermediate normalization gives a simple energy formula:

$$\Delta E = \langle \mathcal{H} :\exp(T):\rangle \ . \tag{2.3.3}$$

A similar projection by excited determinants $\{\langle\Phi| :a^{i_1\ldots i_K}_{a_1\ldots a_K}:\}$ $(K = 1, 2, ..., N)$ leads to a set of independent equations for each unknown amplitude:

$$\langle :a^{i_1\ldots i_K}_{a_1\ldots a_K}: \mathcal{H} :\exp(T):\rangle = \Delta E \langle :a^{i_1\ldots i_K}_{a_1\ldots a_K}::\exp(T):\rangle \ .$$

One can show that this amplitude equation can be simplified to

$$\langle :a^{i_1\ldots i_K}_{a_1\ldots a_K}: \mathcal{H} :\exp(T):\rangle_C = 0 \ , \tag{2.3.4}$$

where capital C in the subscript denotes the connected expectation value of the operator product: each operator must be connected to the rest of the product by a Wick-contraction of at least one pair of indices. The proof of this result is postponed until Sec. 3.1, where we will verify it in a more general MR setting.
Note that we could truncate the T-expansion anywhere, and we would still have the right amount of equations after projecting with the appropriate excited determinants. The most common choices for truncated expansions correspond to keeping only double excitations (CCD) or single and double excitations (CCSD). In this work, we shall be mostly concerned with these two approaches..
Eqs. (2.3.3) and (2.3.4) are the standard equations of CC. To proceed further, the expectation values of their operator products must be evaluated using the GWT. In general, this would be quite tedious; however, the work can be considerably reduced with the diagrammatic methods introduced in the next section.

2.3.2 Diagrammatic derivation of the Coupled Cluster equations

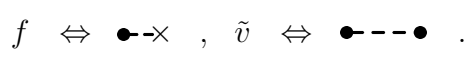
The operator products of Eqs. (2.3.3) and (2.3.4) are to be evaluated by going through all possible Wick-contractions in the operator product. Since $|\Phi\rangle$ is a single determinant, cumulants vanish and only pair contractions need to be considered (given by Eqs. (2.1.14) and (2.1.21)). For this reason, the exponential operator is always truncated after the $(K+2)$-th order (K being the rank of the excitation operator on the left, 2 being the highest rank

of the Hamiltonian). This makes the derivation of the energy formula relatively simple:

$$\begin{aligned}\Delta E &= \langle \mathcal{H} :T:\rangle + \frac{1}{2}\langle \mathcal{H} :T^2:\rangle \\ &= \sum_{p,q} f_p^q \langle :a_q^p: :T_1:\rangle + \frac{1}{4}\sum_{p,q,r,s} \tilde{v}_{pr}^{qs}\langle :a_{qs}^{pr}: :T_2:\rangle + \frac{1}{8}\sum_{p,q,r,s} \tilde{v}_{pr}^{qs}\langle :a_{qs}^{pr}: :T_1^2:\rangle \\ &= \sum_{p,q} f_p^q \sum_{\substack{k\\c}} t_c^k \langle :a_q^p: :a_k^c:\rangle + \frac{1}{16}\sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \sum_{\substack{k,l\\c,d}} \left(t_{cd}^{kl} + 2t_c^k t_d^l\right) \langle :a_{qs}^{pr}: :a_{kl}^{cd}:\rangle \\ &= \sum_{\substack{k\\c}} f_k^c t_c^k + \frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_{cd}^{kl} + \frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_c^k t_d^l \ . \end{aligned} \tag{2.3.5}$$

The last line is obtained using the expectation value of Eq. (2.1.52) and Eq. (2.1.54); antisymmetry of $\tilde{v}$ and t_2 was used to collect equivalent contractions into a single term.

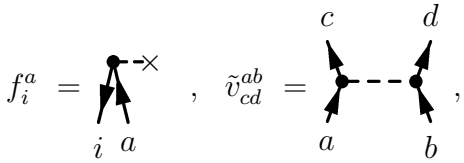
This purely algebraic approach becomes very cumbersome for the general amplitude equations. To make things easier, an intuitive graphical notation is used instead, originally introduced by Feynman in the context of quantum field theory (and later redesigned by Goldstone, Hugenholtz and others for MBPT [29, 69]). Let us represent the parameters of the Hamiltonian with the following dashed horizontal lines:



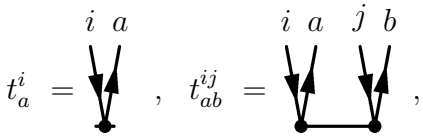
The number of dots indicates the rank of the quantity (i.e. the number of upper-lower index pairs attributed to it). The $\times$ symbol of the Fockian serves as a reminder that, although a one-particle quantity, f still carries an implicit two-particle contribution due to Eq. (2.1.46). Analogously, CC amplitudes shall be represented by continuous horizontal lines:



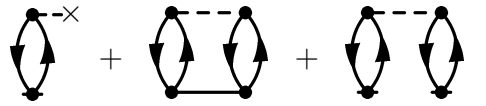
The above horizontal lines shall be referred to as vertices. Hole or particle indices (originating from γ- or η-type pair contractions) are represented by oriented lines entering and leaving the dots of vertices. By definition, down- and upgoing lines correspond to hole and particle indices, while incoming and outgoing lines denote upper and lower indices, respectively. Examples of Hamiltonian matrix elements include



while for the amplitudes (which only describe hole-particle transitions) we have



and so on. Contracting such quantities translates to simply connecting their oriented fermion lines to each other. For example, diagrams of the energy formula in Eq. (2.3.5) read

$\Delta E =$  . (2.3.6)

In the algebraic approach, all possible contractions must be considered when applying the GWT. Equivalently, in the diagrammatic approach, all possible connectivity patterns must be drawn: for example, three unique diagrams for the three unique terms of Eq. (2.3.5). Since all CC equations are linear in $\mathcal{H}$, each diagram must contain only one Hamiltonian vertex; this can be connected to an arbitrary number of amplitude vertices. Connecting a quantity to itself is forbidden: this reflects the fact that self-contractions are omitted in a normal ordered formalism. Similarly to the energy formula, one can also represent all unique terms of the amplitude equations with connected diagrams, the only difference being the appearance of $2K$ free (uncontracted) fermion lines belonging to the indices of the projecting determinants in Eq. (2.3.4). This highlights the usefulness of the diagrammatic approach: in this way, instead of calculating a large number of equivalent terms in GWT and then manually simplifying the results, one can immediately generate the unique contributions only.
The only question left is how to obtain the final form of expressions from these drawings. The so-called diagram rules of CC provide a one-to-one map to readily translate diagrams to algebraic expressions. The proof of these interpretation rules will be briefly discussed in Sec. 3.2.1; here we shall only quote them:

- A1: Write the product of algebraic expressions corresponding to interaction and amplitude vertices.
- A2: Assign a hole or particle index to each down- or upgoing line.
- A3: Sum over the indices of each internal line.
- A4: Assign a phase factor $(-1)^{H-L}$, with H and L standing for the number of hole lines and closed fermion loops (including quasi-loops connecting external lines), respectively.
- A5: Assign a symmetry factor of $1/2$ for each equivalent pair of internal lines (same particle-hole character, connecting the same two quantities) and $1/2$ for each equivalent pair of amplitudes.
- A6: Antisymmetrize all external upper/lower indices that are not manifestly antisymmetric.

These rules are sufficient to obtain Eq. (2.3.5) from Eq. (2.3.6). None of the diagrams produce a sign, since they all have the same number of hole lines and fermion loops. The second diagram contains a factor of $(1/2)^2$ due to a pair of equivalent hole lines and a pair of equivalent particle lines. In the third diagram there are no equivalent lines but a pair of equivalent t_1 amplitudes, hence the factor of $1/2$. Note that the vertices represent

antisymmetric quantities, and for this reason there are multiple equivalent ways to draw diagrams. For example, the second diagram could also be drawn as



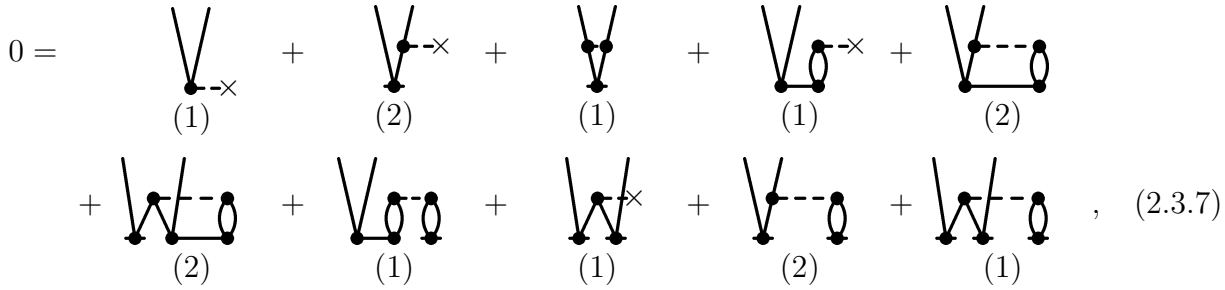
$$\Leftrightarrow \quad \frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_{cd}^{kl} \equiv -\frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_{dc}^{kl} \ .$$

The sign change is encoded in the changing number of fermion loops. Antisymmetry is naturally built into all of the diagrams considered here, therefore, direct and exchange processes do not need to be treated separately. Such diagrams are generally referred to as antisymmetrized Goldstone diagrams (to be distinguished from "ordinary" Goldstone diagrams of spin-free theories; see Sec. 3.3.2).

The above procedure can be applied to the amplitude equation of Eq. (2.3.4) as well. In the CCSD approximation $T = T_1 + T_2$ and only equations

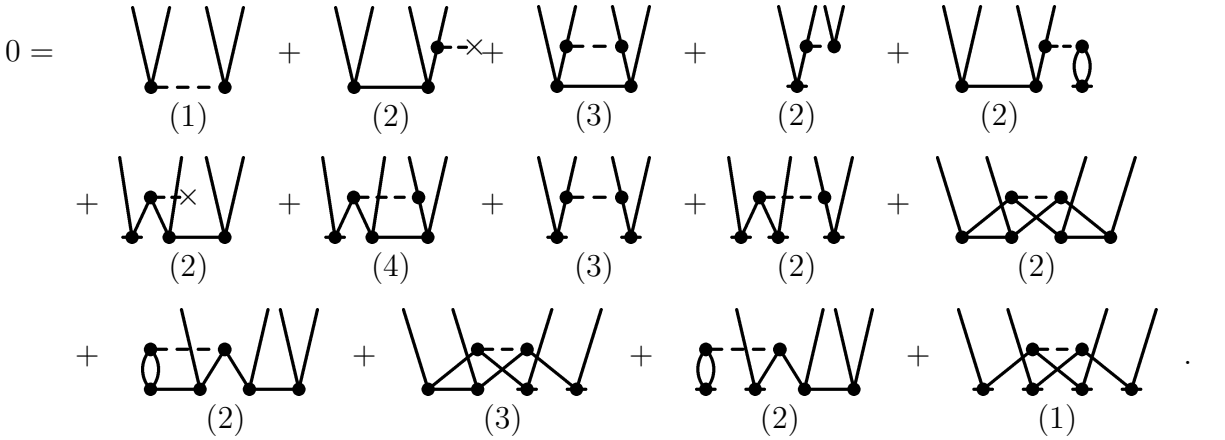
$$\langle :a_a^i: \mathcal{H} :\exp(T):\rangle_C = 0 \quad , \quad \langle :a_{ab}^{ij}: \mathcal{H} :\exp(T):\rangle_C = 0$$

are considered. The following diagrams are found for the t_1 equation:



(2.3.7)

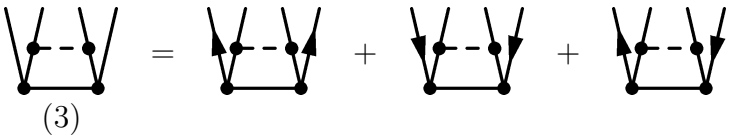
while for the t_2 equation we have



(2.3.8)

One could of course go beyond the CCSD approximation and derive equations for higher t_K in a similar manner.

To keep things simple, only the unoriented skeletons of diagrams are drawn above; the number of oriented diagrams represented by a single skeleton (i.e. the number of unique orientations one can assign to the skeleton) are shown in parantheses below each diagram. For example:



(2.3.9)

For the sake of readability, only the sufficient minimal number of oriented lines is drawn that already makes the orientation unambiguous. Again, due to antisymmetry, many of the above diagrams have multiple equivalent forms. For example, the third diagram of Eq. (2.3.9) could also be drawn as



(2.3.10)

The diagrams of Eqs. (2.3.7) and (2.3.8) are translated to the following set of algebraic equations:

$$
\begin{aligned}
0 =& f_a^i - \sum_k f_k^i t_a^k + \sum_c f_a^c t_c^i + \sum_{\substack{k\\c}} \tilde{v}_{ak}^{ic} t_c^k + \sum_{\substack{k\\c}} f_k^c t_{ca}^{ki} - \frac{1}{2}\sum_{\substack{k,l\\d}} \tilde{v}_{kl}^{id} t_{ad}^{kl} + \frac{1}{2}\sum_{\substack{k\\c,d}} \tilde{v}_{al}^{cd} t_{cd}^{il} \\
&- \frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_c^i t_{ad}^{kl} - \frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_a^k t_{cd}^{il} + \sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_c^k t_{ad}^{il} - \sum_{\substack{k\\c}} f_k^c t_a^k t_c^i \\
&- \sum_{\substack{k,l\\d}} \tilde{v}_{kl}^{id} t_a^k t_d^l + \sum_{\substack{l\\c,d}} \tilde{v}_{al}^{cd} t_c^i t_d^l - \sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_c^i t_a^k t_d^l \,,
\end{aligned}
\tag{2.3.11}
$$

and

$$
\begin{aligned}
0 =& \tilde{v}_{ab}^{ij} - \mathcal{A}^{ij} \sum_k f_k^i t_{ab}^{kj} + \mathcal{A}_{ab} \sum_c f_a^c t_{cb}^{ij} + \frac{1}{2}\sum_{k,l} \tilde{v}_{kl}^{ij} t_{ab}^{kl} + \frac{1}{2}\sum_{c,d} \tilde{v}_{ab}^{cd} t_{cd}^{ij} + \mathcal{A}^{ij}\mathcal{A}_{ab} \sum_{\substack{k\\c}} \tilde{v}_{ak}^{ic} t_{cb}^{kj} \\
&- \mathcal{A}_{ab} \sum_k \tilde{v}_{kb}^{ij} t_a^k + \mathcal{A}^{ij} \sum_c \tilde{v}_{ab}^{cj} t_c^i - \mathcal{A}^{ij} \sum_{\substack{k,l\\c}} \tilde{v}_{lc}^{jk} t_k^c t_{ab}^{il} + \mathcal{A}_{ab} \sum_{\substack{k\\c,d}} \tilde{v}_{bc}^{dk} t_k^c t_{ad}^{ij} \\
&- \mathcal{A}^{ij} \sum_{\substack{k\\c}} f_k^c t_c^i t_{ab}^{kj} - \mathcal{A}_{ab} \sum_{\substack{k\\c}} f_k^c t_a^k t_{cb}^{ij} \\
&+ \mathcal{A}^{ij} \frac{1}{2}\sum_{\substack{k,l\\c}} \tilde{v}_{kl}^{cj} t_c^i t_{ab}^{kl} - \mathcal{A}_{ab} \frac{1}{2}\sum_{\substack{k\\c,d}} \tilde{v}_{kb}^{cd} t_a^k t_{cd}^{ij} - \mathcal{A}^{ij}\mathcal{A}_{ab} \sum_{\substack{l\\c,d}} \tilde{v}_{kl}^{id} t_a^k t_{db}^{lj} + \mathcal{A}^{ij}\mathcal{A}_{ab} \sum_{\substack{l\\c,d}} \tilde{v}_{al}^{cd} t_c^i t_{db}^{lj} \\
&+ \mathcal{A}_{ab} \frac{1}{2}\sum_{k,l} \tilde{v}_{kl}^{ij} t_a^k t_b^l + \mathcal{A}^{ij} \frac{1}{2}\sum_{c,d} \tilde{v}_{ab}^{cd} t_c^i t_d^j - \mathcal{A}^{ij}\mathcal{A}_{ab} \sum_{\substack{l\\d}} \tilde{v}_{al}^{id} t_d^j t_b^l \\
&+ \mathcal{A}^{ij}\mathcal{A}_{ab} \frac{1}{2}\sum_{\substack{k,l\\c}} \tilde{v}_{kl}^{cj} t_c^i t_a^k t_b^l - \mathcal{A}^{ij}\mathcal{A}_{ab} \frac{1}{2}\sum_{\substack{k\\c,d}} \tilde{v}_{kb}^{cd} t_c^i t_a^k t_d^j + \frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_{cd}^{ij} t_{ab}^{kl} + \mathcal{A}^{ij}\mathcal{A}_{ab} \frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}_{kl}^{cd} t_{ac}^{ik} t_{db}^{lj}
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{A}^{ij}\frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{il}_{cd} t^{kj}_{ab} - \mathcal{A}_{ab}\frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{kl}_{ad} t^{ij}_{cb} \\
&+ \mathcal{A}_{ab}\frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{ij}_{cd} t^{k}_{a} t^{l}_{b} + \mathcal{A}^{ij}\frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{i}_{c} t^{j}_{d} t^{kl}_{ab} - \mathcal{A}^{ij}\mathcal{A}_{ab}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{ik}_{ac} t^{j}_{d} t^{l}_{b} \\
&- \mathcal{A}^{ij}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{i}_{c} t^{kj}_{ab} t^{l}_{d} - \mathcal{A}_{ab}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{k}_{a} t^{ij}_{cb} t^{l}_{d} + \mathcal{A}^{ij}\mathcal{A}_{ab}\frac{1}{4}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{i}_{c} t^{k}_{a} t^{j}_{d} t^{l}_{b} \ . \qquad (2.3.12)
\end{aligned}
$$

Note that some of the above expressions involving equivalent amplitudes can be further simplified by exploiting their hidden antisymmetry. For example:

$$
\mathcal{A}_{ab}\frac{1}{2}\sum_{k,l} \tilde{v}^{ij}_{kl} t^{k}_{a} t^{l}_{b} = \sum_{k,l} \tilde{v}^{ij}_{kl} t^{k}_{a} t^{l}_{b} \quad , \quad \mathcal{A}^{ij}\mathcal{A}_{ab}\frac{1}{2}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{ik}_{ac} t^{lj}_{db} = \mathcal{A}^{ij}\sum_{\substack{k,l\\c,d}} \tilde{v}^{cd}_{kl} t^{ik}_{ac} t^{lj}_{db} \ .
$$

We mention that the "quasi-loops" of sign rule A4 imply a pairing of external indices like ia or jb, and simply produce a global sign for each term of the respective amplitude equation. Solving Eqs. (2.3.11) and (2.3.12) for t_1 and t_2, we find the corrected wave function. Also, substituting these values back into Eq. (2.3.5) gives the energy correction.

The amplitude equations are non-linear, and in general can only be solved iteratively. To this end, let us denote them as $\Omega^{i_1\ldots i_K}_{a_1\ldots a_K}(t) = 0$, t standing for all of the amplitudes. Then the iterative form of the general amplitude equations is (iteration steps being $n = 1, 2, \ldots$)

$$
(t^{(n)})^{i_1\ldots i_K}_{a_1\ldots a_K} = (t^{(n-1)})^{i_1\ldots i_K}_{a_1\ldots a_K} + \frac{\Omega^{i_1\ldots i_K}_{a_1\ldots a_K}(t^{(n-1)})}{D^{i_1\ldots i_K}_{a_1\ldots a_K}} \ . \qquad (2.3.13)
$$

Here $D^{i_1\ldots i_K}_{a_1\ldots a_K}$ are (in principle) arbitrary parameters controlling the iteration. The most common, MBPT-like choice is to use the difference of diagonal Fockian matrix elements:

$$
D^{i_1\ldots i_K}_{a_1\ldots a_K} = f^{i_1}_{i_1} - f^{a_1}_{a_1} + \ldots + f^{i_K}_{i_K} - f^{a_K}_{a_K} \ . \qquad (2.3.14)
$$

The initial guess $t^{(0)}$ of the amplitudes is obtained by neglecting all terms of the equations, except for the inhomogeneous term and the diagonal part of the linear Fockian term. These zeroth-order equations are immediately solved by

$$
(t^{(0)})^{i}_{a} = \frac{f^{i}_{a}}{f^{i}_{i} - f^{a}_{a}} \quad , \quad (t^{(0)})^{ij}_{ab} = \frac{\tilde{v}^{ij}_{ab}}{f^{i}_{i} + f^{j}_{j} - f^{a}_{a} - f^{b}_{b}} \ ,
$$

and $t^{(0)}_{K} = 0$ for $K \geq 3$. Starting from this guess, with each update using Eqs. (2.3.13) and (2.3.14), the value of $t^{(n)}$ converges to the solution of the amplitude equations; the energy difference $\Delta E^{(n)} - \Delta E^{(n-1)}$ provides a useful numerical measure to monitor convergence.

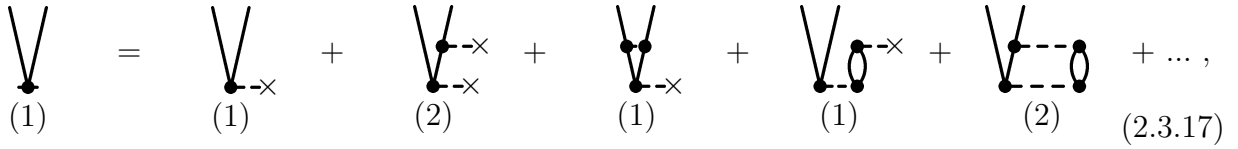
The iteration procedure outlined above can also be visualized with diagrams in order to establish the relationship between CC and MBPT. MBPT is based on the following $H = H^{(0)} + W$ partitioning of the Hamiltonian:

$$
H^{(0)} = \langle H \rangle + \sum_{p} f^{p}_{p} :a^{p}_{p}: \quad , \quad W = \sum_{p\neq q} f^{q}_{p} :a^{p}_{q}: + \frac{1}{4}\sum_{p,q,r,s} \tilde{v}^{qs}_{pr} :a^{pr}_{qs}: \ . \qquad (2.3.15)
$$

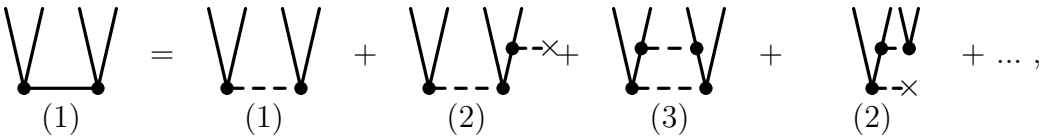
Substituting $t^{(0)}$ into the linear part of Eq. (2.3.5) yields the second-order MBPT energy:

$$E^{(2)} = \sum_{\substack{i \\ a}} \frac{f_a^i f_i^a}{f_i^i - f_a^a} + \frac{1}{4} \sum_{\substack{i,j \\ a,b}} \frac{\tilde{v}_{ij}^{ab} \tilde{v}_{ab}^{ij}}{f_i^i + f_j^j - f_a^a - f_b^b} \ . \tag{2.3.16}$$

One could similarly investigate the higher orders. Graphically, the CCSD iteration is represented by the following expansion of t_1 and t_2:

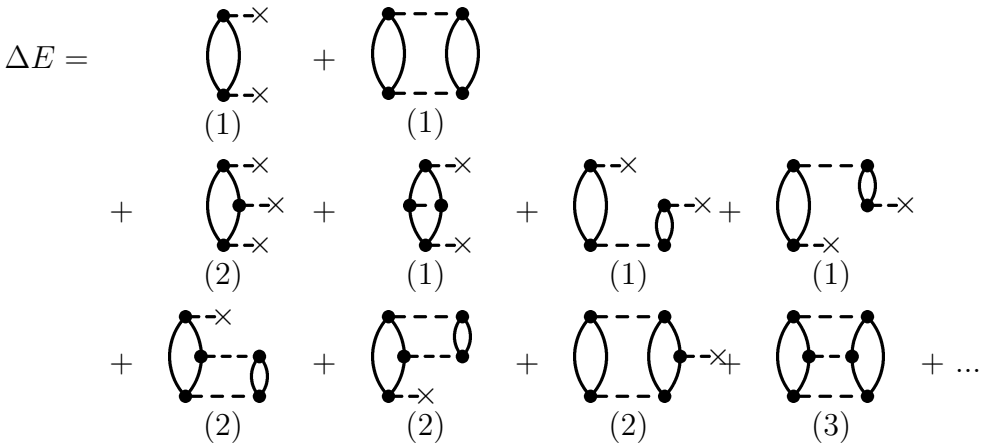


(2.3.17)



(2.3.18)

Substituting these into the energy formula leads to the following expansion:



(2.3.19)

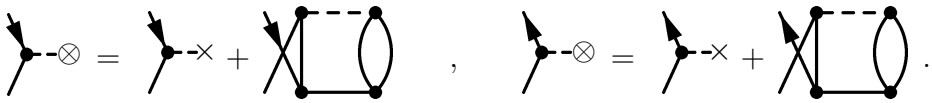
These diagrams can be interpreted just like the CC diagrams with two additional rules:

- P1: assign a factor of $1/D_{a_1 \ldots a_K}^{i_1 \ldots i_K}$ (see Eq. (2.3.14)) to all (fictitious) horizontal lines between interaction vertices crossing K pairs of fermion lines,
- P2: cancel all diagonal elements of the Fockian vertex f_p^q from the summation over internal lines (i.e. restrict the summation so that $p \neq q$ in the numerator).

The above diagrammatic expansion obtained from the CC iteration is just the MBPT expansion of the wave function, and the energy (for example, the first line of Eq. (2.3.19) is just Eq. (2.3.16)). It shows that all approximations of the general CC model correspond to some infinite-order summation of MBPT diagrams. It also establishes the interpretation of amplitudes as "renormalized" interaction vertices describing screened hole-particle scattering.

The CCSD approximation already recovers all first-order wave function diagrams and all second- and third-order energy diagrams shown above. The missing three-body and higher rank excitations start contributing to the wave function in second order and to the energy in fourth order. This is the reason why CCSD performs so well in all cases where good convergence of MBPT can be expected. When working with HF orbitals (meaning $f_a^i = 0$), contributions of one-body excitations to the first-order wave function and to the second- and third-order energy vanish; for this reason, HF-CCD and HF-CCSD results tend to be rather similar.

Many different connectivity patterns ("topologies") can be identified in the above diagrams; some of the most important ones are referred to as ring-, ladder- and mosaic-type diagrams. Ring diagrams describe purely hole-particle excitations at each intermediate state (i.e. at each fictitious horizontal line between Hamiltonian vertices): the third diagram of Eq. (2.3.9) (or equivalently the one in Eq. (2.3.10)) is a ring diagram linear in t. The first and second diagrams of Eq. (2.3.9) are called hole-hole and particle-particle ladder diagrams, as their two-electron vertex describes $hh \rightarrow hh$ and $pp \rightarrow pp$ scattering, respectively. The inhomogenous term of Eq. (2.3.8) (the "bare" hole-particle vertex) can be classified as both ring- and ladder-type. The last skeleton in the second line of Eq. (2.3.8) stands for the non-linear ring and ladder diagrams.
Some well-known approximations of CCD are based on keeping only certain classes of diagrams, such as ring-CCD, or ladder-CCD [70]. These approximations often have an interesting relationship with other electronic structure models; equivalence of (a certain type of) ring-CCD and the Random Phase Approximation (RPA) has been noticed by many [63, 71, 72].
The so-called mosaic diagrams are represented by the first skeleton of the third line in Eq. (2.3.8). Note that these diagrams can be united with the hole-hole and particle-particle Fockian diagrams (represented by the second skeleton of the first line) after introducing



The interpretation of this is that mosaic diagrams provide a screening effect to the "bare" Fockian vertices.
We shall revisit this concept and the role of the various diagram classes (especially ring diagrams and RPA) later in the framework of MRCC.

2.3.3 Advantages and disadvantages of Coupled Cluster

The CC model has many appealing features even from a formal point of view. Its exponential Ansatz combined with the manifestly connected nature of CC equations ensures size consistency and extensivity of the results as long as all Hamiltonian vertices between the non-interacting subsystems vanish. This property holds, no matter how the expansion of T is truncated.
A truncated CC approach with $T = T_1 + ... + T_K$ gives exact results for systems containing

$N \leq K$ electrons: CCSD is therefore exact for two-electron systems, such as the hydrogen molecule. This also explains why CCSD can often describe well the dissociation of single covalent bonds, while finite-order MBPT breaks down.
The CC energy is invariant under unitary mixing of orbitals in the occupied and virtual subspaces, giving e.g. the same value for canonical and localized HF orbitals. Such a transformation can be formally written as

$$\begin{bmatrix} \tilde{\phi}_h \\ \tilde{\phi}_p \end{bmatrix} = \begin{bmatrix} U_h & 0 \\ 0 & U_p \end{bmatrix} \begin{bmatrix} \phi_h \\ \phi_p \end{bmatrix} ,$$

U_h and U_p being unitary matrices mixing hole and particle states, respectively. The invariance follows simply from the fact that the energy is a scalar obtained from the contraction of interaction and amplitude vertices, both of which (also the amplitude equations) transforming as tensor components under the mixing.

Spin-adaptation of the CC equations can be readily achieved by the UGA (see Sec. 2.1.7). When dealing with closed-shell systems, there exists a simpler, diagrammatic approach for the CCSD equations. Both of these approaches will be discussed in Sec. 3.3.2 in the context of MRCC.

Approximations of the CC model often provide a very good tradeoff between accuracy and conceptual-computational simplicity. Evaluation of the most "expensive" diagram of the CCSD equations (the second one of Eq. (2.3.9)) requires roughly $\mathcal{O}(n_h^2 n_p^4) \sim \mathcal{O}(M^6)$ steps per iteration (n_h and n_p being the number of hole and particle states), making the computational cost of CCSD similar to that of MBPT(3). The cost of non-linear terms in the amplitude equation can be reduced to that of linear terms by introducing appropriate intermediate quantities; see later in Sec. 3.3.3.
Inclusion of three-body excitations (CCSDT) would make the wave function correct through second order and the energy through fourth order, leading to very accurate results. This would, however, come at the price of a roughly $\mathcal{O}(M^8)$ computational cost, which is already too much for general applications. For this reason, several approaches have been proposed for approximate, less expensive treatment of T_3, typically based on the analysis of MBPT diagrams. Common choices include the CCSD[T] [73], CCSD(T) [74], CC3 [75] and iCCSDn [76]. CCSD(T) has become perhaps the most popular one, often being referred to as "gold standard", offering chemical accuracy ($\sim$1 kcal mol^{-1} error) for an $\mathcal{O}(M^7)$ computational cost. Recently, CCSD(T) level investigation of extremely large systems (containing hundreds of atoms) has become possible [77–79].

Besides quantum chemistry, CC has also been investigated in the context of solid state theory [80–82]. Generalizations of CC to finite temperature [83–85], to nuclear three-body Hamiltonians [86, 87] and to relativistic settings [88, 89] are also known.

While CC is remarkably successful as a weak correlation model, it tends to fail in the regime of strong correlation, where the Fermi vacuum is qualitatively different from the exact ground state (multiple covalent bond dissociation being a common example). In such cases finding the converged solution to the amplitude equation proves to be hard (often impossible), and even when found, the results are usually of poor quality. These

problems can often be cured by alternative CC-like approaches preserving the concept of Fermi vacuum, such as the Method of Moments CC of Piecuch et al. [90], the Quadratic CC of Van Voorhis and Head-Gordon [91], or the Extended CC of Arponen et al. [92–97]. Variational and unitary reformulations of CC [98–100] can be more robust against the breakdown of reference function. Also, different flavours of variational CC can be related to other electronic structure methods: variational CCS is equivalent to HF [101], while APSG corresponds to an approximate variational CCSD [41].
In many cases, dissociation energy curves can be nicely cured by a judicious exclusion of spurious exchange contributions in the CC equations, as is done in the Distinguishable Cluster Approximation of Kats et al. [102–105].

Despite the often impressive success of modified single determinantal CC models in describing strong correlation, the ultimate solution to the problem appears to be the generalization of the formalism to an inherently MR wave function. In the following section the literature of MRCC approaches is briefly reviewed.

2.3.4 Overwiew of Multi-Reference Coupled Cluster approaches

In order to account for the missing static correlation plaguing single determinantal CC, one should apply the CC correction to a (typically variationally optimized) linear combination of Slater determinants:

$$|\Phi\rangle = \sum_{K\in\mu} c_K |\Phi_K\rangle \ . \tag{2.3.20}$$

The determinants in the linear combination span the so-called model space, their indices being collected in set μ. For practical reasons, one should try to keep the size of the model space small. The model space of the APSG wave function (discussed in Sec. 2.2) contains e.g.

$$|\mu_{APSG}| = \prod_{P=1}^{N/2} |\mathbb{G}_P|$$

determinants; for GVB this means $|\mu_{GVB}| \leq 2^{N/2}$. The dynamic correlation can then be introduced through an exponential wave operator resembling Eq. (2.3.2):

$$|\Psi\rangle = \hat{\Omega}|\Phi\rangle \ .$$

However, this is much more complicated than in the single determinantal case, since many different forms of $\hat{\Omega}$ can be chosen, each of them defining a possible MRCC model. While the different MRCC approaches often prove to be very efficient at combining strong and weak correlations effects, they also raise many new problems, both conceptual (treatment of active orbitals, possible violation of size consistency and unitary invariance) and practical (redundancy of the parametrization, numerical instabilities).
Here we attempt to highlight some of the more common MRCC methods. Since the literature of MRCC is quite extensive, only a brief and sketchy overview can be given here; see Refs. [106–108] for a much more detailed discussion.

Probably the first MRCC Ansatz that saw routine applications is originating from Jeziorski and Monkhorst [109], and is defined by the wave operator

$$\hat{\Omega}_{JM} = \sum_{K \in \mu} \exp(T_K) |\Phi_K\rangle\langle\Phi_K| \ ,$$

which assigns different exponential operators to each of the model space determinants, strictly mapping them unto the complement of the model space. On one hand, this is nice, since many useful (e.g. diagrammatic) techniques of single determinantal CC can be directly imported to facilitate the derivations (see e.g. Ref. [110]). On the other hand, such a parametrization is inherently redundant: it contains roughly $|\mu|$ times more parameters than the dimension of the CI space the exact wave function is expanded on. The excitation operators can map different model space determinants unto the very same excited determinant, meaning that the corresponding amplitudes are linearly dependent. This renders the Schrödinger equation by itself insufficient in uniquely determining the JM solution, and may lead to severe convergence problems. One can fix the amplitudes either by using the JM Ansatz to calculate multiple electronic states simultaneously (as done in the state-universal approach), or by concentrating on a single state and making the amplitudes well-defined through additional constraints (state-selective or state-specific approach). The Mk-MRCC method of Mukherjee et al. [111–113] or the MRexpT method of Hanrath [114, 115] are examples of the latter, as well as the variational MR-AQCC of Szalay and Bartlett [116]. Although in many cases highly accurate potential surfaces can be obtained from methods employing the JM Ansatz, the occurence of numerical instabilities and the deterioration of results with the increasing number of active orbitals necessitated the development of alternative MRCC parametrizations.

The so-called internally contracted (ic) state-specific MRCC methods use a (seemingly) simpler wave operator with a single exponential operator acting on the whole model space:

$$\hat{\Omega}_{ic} = \exp(T) \ . \tag{2.3.21}$$

This wave operator may lead to more compact equations than Ω_{JM}. While it may not be as explicit as for the JM Ansätze, ic-MRCC wave functions still suffer from linear dependence of excitations; see Chapter 5.

Although the original idea of ic-MRCC can be traced back to the seminal paper of Čížek [64], development of models has gained momentum only later, the works of Evangelista and Gauss [117] and Hanauer and Köhn [118,119] being important examples. An efficient unitary ic-MRCC model (with $T^\dagger = -T$) called canonical transformation theory has been worked out by Yanai and Chan [120, 121], while an equation-of-motion ic-MRCC has been proposed by Nooijen et al. for both ground and excited state calculations [122–125]. The otherwise rather complicated formalism of ic-MRCC can often be simplified by exploiting the structure of the underlying reference function (typically some geminal wave function); examples include the GVB-based block correlated CC of Li et al. [59] and the APSG-based linearized CC of Zoboki et al. [54].

There is an important alternative form of ic-MRCC. As $\exp(T)$ acts not on single determinants but on the complete MR function, the concept of particles and holes must be extended: occupied and active orbitals span the set of hole states, while active and

virtual orbitals span the set of particle states (see Sec. 2.1.2). Due to the appearance of active orbitals these two sets are no longer disjoint, so normal ordering of the wave operator now does make a difference. As an alternative to Eq. (2.3.21), one may thus choose

$$\hat{\Omega}_{ic,N} =:\exp(T): \ . \tag{2.3.22}$$

Such an operator was considered first probably by Lindgren in the context of open-shell CC [126]. An ic-MRCC theory using Eq. (2.3.22) with the corresponding MRGNO and MRGWT (see Secs. 2.1.5 and 2.1.6) was first outlined by Mukherjee et al. [30, 34, 127]. Despite its many appealing properties (no contractions between T operators, the possibility of a manifestly connected formalism) the severe difficulties that arise during derivations seriously limited the applicability of Eq. (2.3.22) Investigation of this ic-MRCC theory will be the main subject of the forthcoming chapters.

The so-called valence-universal formalism [128–132] presents a route to MRCC rather different from those above. Valence-universal models attempt to treat the different sectors of the Fock-space (with different number of electrons) on the same footing, and aim at a unified description of excited, ionized and electron attached states. For this reason, they are also called Fock-space methods; to distinguish, previously discussed MRCC theories (using only the N-electron sector of the Fock-space) are often referred to as Hilbert-space methods.

Besides all of the "genuine" MRCC models listed above, much work has been put into the development of MRCC methods resembling the traditional single determinantal CC. These methods are mostly based on the observation that the reference function can be written as

$$|\Phi\rangle = (1 + C_{int})|\Phi_0\rangle = \exp(T_{int})|\Phi_0\rangle \ ,$$

where $|\Phi_0\rangle$ is a determinant (practically the one with the largest weight in the model space), and action of the "internal wave operator" $(1 + C_{int}) = \exp(T_{int})$ on $|\Phi_0\rangle$ generates all other $|\Phi_K\rangle$ (for $K \in \mu$). Combining this reference function with an "external wave operator" generating excitations outside of the model space leads to an almost single determinantal CC parametrization: this is the starting point of the SRMRCC and CASCC developed by Adamowicz et al. [133, 134]. A clever alternative approach based on quasi-particle transformations was considered by Rolik and Kállay [135]. While often impressive accuracy can be reached this way, the biggest drawback of almost single determinantal models is the strong dependence of the results on the choice of $|\Phi_0\rangle$. Picking the determinant with the largest weight is ambiguous in quasi-degenerate situations (such as covalent bond dissociations). Also, unphysical discontinuities can appear on potential surfaces wherever the largest weight determinant changes, although the use of natural orbitals seems to dampen this effect [136].

As the above examples show, MRCC methods are capable of describing a wide set of strongly and weakly correlated phenomena, yet, there is still much to improve on. Hopefully, the next chapters help to gain some further understanding of ic-MRCC.

Chapter 3

A new look at internally contracted Multi-Reference Coupled Cluster

A novel approach to ic-MRCC is presented in this chapter, using the normal ordered parametrization of Mukherjee et al. [30, 34, 127] as a starting point. After setting up the general formulae and demonstrating their connectedness, diagrammatic methods are found and used to construct the building blocks of the theory, dressed interaction vertices; their interpretation and some important properties are reviewed. This is followed by the (also diagrammatic) derivation of the amplitude equation, after which the general ic-MRCC formalism can be presented. Finally, a formally MBPT-like analysis of the equations, spin-adaptation to singlet systems, as well as various approximation schemes are discussed.

The content of this chapter is based on Refs. [137, 138]. A precursor of this research was presented earlier in Ref. [139].

3.1 Parametrization of the wave function, setting up the MRCC equations

As discussed in Sec. 2.3.4, MRCC calculations require some multi-determinantal wave function $|\Phi\rangle$ (c.f. Eq. (2.3.20)). At this point, $|\Phi\rangle$ can be considered completely arbitrary, as no explicit reference will be made to its exact structure throughout the derivations; when it comes to actual calculations (see Chapter 6), a GVB reference will be used.

Our ic-MRCC parametrization uses the normal ordered wave operator of Eq. (2.3.22):

$$|\Psi\rangle = :\exp(T): |\Phi\rangle \ ,$$

where :(...): denotes the MRGNO as introduced in Sec. 2.1.5. The cluster operator T is decomposed to various K-body terms:

$$T = \sum_K T_K = \sum_K \frac{1}{(K!)^2} \sum_{\substack{i_1,\ldots,i_K \\ a_1,\ldots,a_K}} t^{i_1\ldots i_K}_{a_1\ldots a_K} a^{a_1\ldots a_K}_{i_1\ldots i_K} = \sum_{\substack{i \\ a}} t^i_a a^a_i + \frac{1}{4} \sum_{\substack{i,j \\ a,b}} t^{ij}_{ab} a^{ab}_{ij} + \ldots \ . \tag{3.1.1}$$

Note that the intermediate normalization (i.e. $\langle\Phi|\Psi\rangle = \langle{:}\exp(T){:}\rangle = 1$) still holds due to Eq. (2.1.37) and the fact that $:1:= 1$.
Despite the similar looking formulae, there are major differences between the single determinantal and the MR theory. The first one is the appearance of active orbitals, giving the sets of holes and particles (meant in the more general sense of Sec. 2.3.4) a non-empty intersection. In Eq. (3.1.1), indices $i, j, k, l, ...$ refer to occupied and active natural orbitals, while indices $a, b, c, d, ...$ denote active and virtual natural orbitals. One can think of this notation as simple bookkeeping to distinguish between annihilated and created orbitals. For the time being, no other restrictions are imposed on the excitations entering Eq. (3.1.1). Possible choices for the excitation manifold will be considered in Sec. 3.3.3. Also, in practice, certain excitations will need to be removed from the Ansatz in order to avoid numerical instabilities; this will be discussed in Sec. 6.1.1.
Another important difference concerns the truncation of the wave operator when acting on $|\Phi\rangle$. In the single determinantal treatment of an N-electron system, the Taylor expansion of $:\exp(T):$ naturally ends after the N-th order. In general, however, this is not the case for the MR theory, as normal ordered excitation operators with active indices contain various non-vanishing lower rank operators and even c-number valued terms (this can be seen by e.g. inverting Eq. (2.1.41)). This makes the expansion of $|\Psi\rangle$ non-terminating; nevertheless, as long as CC amplitudes are assumed to be small (which holds if $|\Phi\rangle$ provides a sufficiently good zeroth-order description), neglecting high order contributions is justified.

Since $|\Psi\rangle$ is an exact parametrization of the ground state, we can substitute it in the Schrödinger equation:

$$\mathcal{H}|\Psi\rangle = \Delta E|\Psi\rangle \ , \tag{3.1.2}$$

again with $\Delta E = E_{exact} - \langle H\rangle$, $\langle(...)\rangle$ referring to the expectation value calculated with $|\Phi\rangle$. Formally, the derivation of the ic-MRCC equations starts just like in the single determinantal case, by projecting Eq. (3.1.2) with $\langle\Phi|$ and $\langle\Phi|\ :a^{i_1\dots i_K}_{a_1\dots a_K}:$ (or, using the condensed notation, $\langle\Phi|\ :a^\dagger_K:$) for $K = 1, 2, ...$ Using the intermediate normalization, one finds the energy formula

$$\Delta E = \langle\mathcal{H}\ :\exp(T):\rangle \ , \tag{3.1.3}$$

while the amplitude equation reads

$$\langle :a^\dagger_K:\ \mathcal{H}\ :\exp(T):\rangle = \Delta E\langle :a^\dagger_K::\exp(T):\rangle \ . \tag{3.1.4}$$

These operator products are to be evaluated using the MRGWT giving rise to density cumulants. The above form of the amplitude equation suggests several factorizable terms: the right hand side contains so-called unlinked terms with a scalar multiplier in the form of ΔE, and the MRGWT would produce similar terms on the left as well. Also, the MRGWT generates other disconnected contributions with a product-like structure (e.g. $\Omega^i_a\tau^j_b$ as a formal example). It was claimed in Sec. 2.3.1 that these terms vanish, and only connected contributions need to be considered, where amplitudes and Hamiltonian parameters are all bound to each other by Wick-contractions.

In the first step, we show the cancellation of unlinked terms. Let us define the complete contraction between $\mathcal{H}$ and $:T^k:$ (for some power k) as the sum of all possible MRGWT

contractions between every creation-annihilation operator of $\mathcal{H}$ (both the one- and two-body part) and $:T^k:$ (all K-body terms); no operator is left uncontracted. This complete contraction is just

$$\overbrace{\mathcal{H}\ :T^k:} = \langle \mathcal{H}\ :T^k:\rangle\ .$$

The $k=0$ case simply gives $\langle\mathcal{H}\rangle = 0$. In the single determinantal case the number of non-zero complete contractions would be limited by the rank diference between the particular terms of $\mathcal{H}$ and $:T^k:$, but in the MR case $\mathcal{H}$ can be contracted to an arbitrarily high power of T (with sufficiently high rank cumulants). Of course, $\mathcal{H}$ does not need to be contracted to all amplitudes within the normal product:

$$\overbrace{\mathcal{H}\ :T^l}T^{k-l}\ : = \langle \mathcal{H}\ :T^l:\rangle\ :T^{k-l}:\ ,$$

for $l \leq k$. Since all cluster operators are equivalent within the normal ordering, the above contraction can be done in $\binom{k}{l}$ equivalent ways. The only scalar quantities one can construct are of the form $\langle \mathcal{H}\ :T^k:\rangle$, hence we see that complete contractions produce all the unlinked terms on the left hand side of Eq. (3.1.4). When considering only the completely contracted part of the left hand side, we find

$$\begin{aligned}
\langle :a_K^\dagger:\ \overbrace{\mathcal{H}\ :\exp(T):}\rangle &= \sum_{k=0}^{\infty}\sum_{l=0}^{k}\frac{1}{k!}\binom{k}{l}\langle :a_K^\dagger:\ \overbrace{\mathcal{H}\ :T^l}T^{k-l}\ :\rangle \\
&= \sum_{k=0}^{\infty}\sum_{l=0}^{k}\frac{1}{l!}\langle \mathcal{H}\ :T^l:\rangle\frac{1}{(k-l)!}\langle :a_K^\dagger::T^{k-l}:\rangle \\
&= \sum_{m=0}^{\infty}\frac{1}{m!}\langle \mathcal{H}\ :T^m:\rangle\sum_{n=0}^{\infty}\frac{1}{n!}\langle :a_K^\dagger::T^n:\rangle \\
&= \Delta E\langle :a_K^\dagger::\exp(T):\rangle\ .
\end{aligned}$$

The unlinked part of the left hand side cancels with the right hand side in Eq. (3.1.4), and we are left with

$$\langle :a_K^\dagger:\ \mathcal{H}\ :\exp(T):\rangle_L = 0\ , \tag{3.1.5}$$

L in the subscript standing for "linked".

Cancellation of disconnected terms in Eq. (3.1.5) still must be proved. To make things easier, let us introduce the following notations:

$$\begin{aligned}
\xi_K &= \langle :a_K^\dagger:\ \mathcal{H}\ :\exp(T):\rangle_L\ , \\
\Omega_K &= \langle\ \overbrace{:a_K^\dagger:\ \mathcal{H}\ :\exp(T):}\rangle_L = \langle :a_K^\dagger:\ \mathcal{H}\ :\exp(T):\rangle_C\ ,
\end{aligned} \tag{3.1.6}$$

and

$$\tau_K = \langle :a_K^\dagger::\exp(T):\rangle\ .$$

The complete contraction was defined above for the product of three operators, leading to connected expressions (hence the subscript C). Note that for $K=1$ the linkedness implies connectedness:

$$\xi_1 = \Omega_1\ .$$

For $K = 2$, we can partition ξ_2 to a connected and a disconnected part, the disconnected part being (since $:a_2:=:a_1 a_1:$)

$$\begin{aligned}
\langle :\overbrace{a_1^\dagger \overbrace{a_1^\dagger : \mathcal{H} :}\, \exp(T)}: \rangle_L &= \sum_{k=0}^{\infty}\sum_{l=0}^{k} \frac{1}{k!}\binom{k}{l} \langle :\overbrace{a_1^\dagger a_1^\dagger : \mathcal{H} : T^l}\, T^{k-l}: \rangle_L \\
&= \sum_{k=0}^{\infty}\sum_{l=0}^{k} \frac{1}{l!} \langle :\overbrace{a_1^\dagger : \mathcal{H} : T^l}: \rangle_L \wedge \frac{1}{(k-l)!} \langle :\overbrace{a_1^\dagger :: T^{k-l}}: \rangle_L \\
&= \sum_{m=0}^{\infty} \frac{1}{m!} \langle :\overbrace{a_1^\dagger : \mathcal{H} : T^m}: \rangle_L \wedge \sum_{n=0}^{\infty} \frac{1}{n!} \langle :\overbrace{a_1^\dagger :: T^n}: \rangle_L \\
&= \Omega_1 \wedge \tau_1 \ .
\end{aligned}$$

The wedge product appears because ξ_2 is antisymmetric under individual permutations of its upper or lower indices. We have found

$$\xi_2 = \Omega_2 + \Omega_1 \wedge \tau_1 \ .$$

For $K = 3$, one could similarly arrive at

$$\xi_3 = \Omega_3 + \Omega_2 \wedge \tau_1 + \Omega_1 \wedge \left(\tau_2 + \frac{1}{2}\tau_1 \wedge \tau_1\right) \ ,$$

by considering all possible disconnected partitions, and so on for any larger values of K. Similarly to the cumulant expansion (cf. Eq. (2.1.27)), we can introduce

$$\chi_K = \sum_{\nu=1}^{K} \sum_{K_1,\ldots,K_\nu \in \mathbb{P}_{K,\nu}} \frac{1}{w_{K_1 \ldots K_\nu}} \bigwedge_{\mu=1}^{\nu} \tau_{K_\mu} \ ,$$

with $\chi_0 = \tau_0 = 1$ (which is just the intermediate normalization, since $:a_0:= 1$). Then we can compactly write

$$\xi_K = \sum_{K'=0}^{K-1} \chi_{K'} \wedge \Omega_{K-K'} \ .$$

This means that $\xi_1 = 0$ implies $\Omega_1 = 0$, which leads to $\xi_2 = 0$ implying $\Omega_2 = 0$, and so on. One can conclude that the $\xi_K = 0$ amplitude equation is equivalent to $\Omega_K = 0$, that is:

$$\langle :a_K^\dagger: \, \mathcal{H} \, :\exp(T):\rangle_C = 0 \ . \tag{3.1.7}$$

Only connected terms need to be considered, as hinted before.

Let us briefly remark that in single determinantal CC, the usual proof of connectedness is quite different from the one given above, and uses the concept of a similarity transformed Hamiltonian $\bar{\mathcal{H}} = :\exp(T):^{-1} \, \mathcal{H} \, :\exp(T):$. In the case of a single determinant, the inverse of the wave operator is simply given by $:\exp(T):^{-1} = :\exp(-T):$, and one can use the Baker-Campbell-Hausdorff expansion of $\bar{\mathcal{H}}$ to argue connectedness [26]. This, however, would not have been applicable in our case, since in a general MR setting no closed form expression is known for $:\exp(T):^{-1}$. We have simply avoided the use of this inverse, but there other approaches actually tackling this problem [130].

3.2 Dressing of interaction vertices

The evaluation of Eqs. (3.1.3) and (3.1.7) based on the MRGWT is much more challenging than that of their single determinantal counterparts. These equations are non-terminating even for a truncated T-expansion, since any product of operators can be completely contracted with sufficiently high rank cumulants; later, this will make further approximations necessary. Any operator product will lead to much more complicated results in the MR case than in the single determinantal one due to the abundance of cumulants. It is too easy to get lost in either the algebraic or the (naive) diagrammatic derivation of the MRCC equations. These derivations can be largely facilitated by the introduction of so-called dressed interaction vertices.

Let us start with the simple case of the energy formula. Expanding the wave operator of Eq. (3.1.3) in terms of increasing operator rank gives

$$\begin{aligned}
\Delta E =& \langle \mathcal{H} : \exp(T) : \rangle \\
=& \langle \mathcal{H} : T_1 : \rangle + \langle \mathcal{H} : T_2 + \tfrac{1}{2} T_1^2 : \rangle + \ldots \\
=& \sum_{\substack{i \\ a}} \langle \mathcal{H} : a_i^a : \rangle t_a^i + \frac{1}{4} \sum_{\substack{i,j \\ a,b}} \langle \mathcal{H} : a_{ij}^{ab} : \rangle \left(t_{ab}^{ij} + t_a^i t_b^j - t_b^i t_a^j \right) + \ldots \\
=& \sum_{\substack{i \\ a}} n_i \bar{n}^a F_i^a t_a^i + \frac{1}{4} \sum_{\substack{i,j \\ a,b}} n_i n_j \bar{n}^a \bar{n}^b \tilde{V}_{ij}^{ab} \left(t_{ab}^{ij} + t_a^i t_b^j - t_b^i t_a^j \right) + \ldots \, .
\end{aligned} \tag{3.2.1}$$

In this expansion each rank K product of amplitudes is contracted to a so-called dressed particle-hole vertex. These particle-hole vertices $\tilde{W}_{i_1 \ldots i_K}^{a_1 \ldots a_K}$ are defined in general as

$$n_{i_1} \bar{n}^{a_1} \ldots n_{i_K} \bar{n}^{a_K} \tilde{W}_{i_1 \ldots i_K}^{a_1 \ldots a_K} = \langle \mathcal{H} : a_{i_1 \ldots i_K}^{a_1 \ldots a_K} : \rangle \, .$$

The one- and two-body vertices shown in Eq. (3.2.1) are especially important, and a unique notation is reserved for them:

$$n_i \bar{n}^a F_i^a = \langle \mathcal{H} : a_i^a : \rangle \, , \tag{3.2.2}$$

$$n_i n_j \bar{n}^a \bar{n}^b \tilde{V}_{ij}^{ab} = \langle \mathcal{H} : a_{ij}^{ab} : \rangle \, . \tag{3.2.3}$$

Their algebraic form can be found with the help of Eqs. (2.1.53) and (2.1.54). After simplifying the results, we are left with

$$n_i \bar{n}^a F_i^a = n_i \bar{n}^a f_i^a + \sum_{p,q} f_p^q \, \lambda_{qi}^{pa} + \frac{1}{2} \sum_{p,q,r} \left(\bar{n}^a \tilde{v}_{rp}^{aq} \, \lambda_{iq}^{rp} - n_i \tilde{v}_{iq}^{rp} \, \lambda_{rp}^{aq} \right) + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \, \lambda_{qsi}^{pra} \, , \tag{3.2.4}$$

and

$$
\begin{aligned}
n_i n_j \bar{n}^a \bar{n}^b \tilde{V}_{ij}^{ab} = & \sum_p \left[\mathcal{A}^{ab} \left(\bar{n}^a f_p^a \lambda_{ij}^{pb} \right) - \mathcal{A}_{ij} \left(n_i f_i^p \lambda_{pj}^{ab} \right) \right] + \sum_{p,q} f_p^q \lambda_{ijq}^{abp} + n_i n_j \bar{n}^a \bar{n}^b \tilde{v}_{ij}^{ab} \\
& + \frac{1}{2} \sum_{p,q} \left(\bar{n}^a \bar{n}^b \tilde{v}_{pq}^{ab} \lambda_{ij}^{pq} + n_i n_j \tilde{v}_{ij}^{pq} \lambda_{pq}^{ab} \right) + \sum_{p,q} \mathcal{A}_{ij} \mathcal{A}^{ab} \left(n_i \bar{n}^a \tilde{v}_{ip}^{aq} \lambda_{jq}^{bp} \right) \\
& + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \lambda_{ij}^{pr} \lambda_{qs}^{ab} + \frac{1}{2} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \mathcal{A}^{ab} \mathcal{A}_{ij} \left(\lambda_{qi}^{pa} \lambda_{sj}^{rb} \right) \\
& - \frac{1}{2} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \left[\mathcal{A}^{ab} \left(\lambda_{qs}^{pa} \lambda_{ij}^{rb} \right) + \mathcal{A}_{ij} \left(\lambda_{qi}^{pr} \lambda_{sj}^{ab} \right) \right] \\
& + \frac{1}{2} \sum_{p,q,r} \left[\mathcal{A}^{ab} \left(\bar{n}^a \tilde{v}_{pr}^{qa} \lambda_{ijq}^{rbp} \right) - \mathcal{A}_{ij} \left(n_i \tilde{v}_{ip}^{rq} \lambda_{rjq}^{abp} \right) \right] + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \lambda_{ijqs}^{abpr} \, . \qquad (3.2.5)
\end{aligned}
$$

Both F_i^a and $\tilde{V}_{ij}^{ab}$ contain "bare" contributions (in the form of f_i^a and $\tilde{v}_{ij}^{ab}$, respectively) and additional cumulant contributions, owing to the MR nature of $|\Phi\rangle$. These cumulants are interpreted as a dressing to the bare vertex part. Higher rank vertices do not have any bare parts, only cumulant-involving terms.

Note that these dressed one- and two-body quantities play formally the same role in the MR energy formula of Eq. (3.2.1) as the Fockian and the electron-electron vertex in the single determinantal case (cf. Eq. (2.3.5)). Indeed, if $|\Phi\rangle$ were a single Slater determinant, than all cumulant contributions of the MRGWT would vanish, and the dressed vertices would fall back to their bare counterparts: $F_i^a = f_i^a$, $\tilde{V}_{ij}^{ab} = \tilde{v}_{ij}^{ab}$, and $\tilde{W}_K = 0$ for $K \geq 3$.

Above we only introduced a special particle-hole case of dressed vertices. For dressed vertices to be useful in building the MRCC formalism, we need to further generalize their concept. This will be done in Sec. 3.2.2; before that, however, diagrammatic construction of particle-hole vertices must be discussed.

3.2.1 Diagram rules for dressed particle-hole vertices

The quantities we interpreted as dressed particle-hole vertices in the previous section have actually appeared in many-body theory before, but in a rather different context. Equations

$$
n_{i_1} \bar{n}^{a_1} \ldots n_{i_K} \bar{n}^{a_K} \tilde{W}_{i_1 \ldots i_K}^{a_1 \ldots a_K} = 0
$$

were investigated by Kutzelnigg and Mukherjee [140, 141] and by Herbert and Harriman [142, 143], as they are necessary conditions for the stationarity of energy. These equations are generally referred to as K-body irreducible contracted Schrödinger equations (ICSE$_K$). The above authors facilitated the construction of the ICSE$_K$ with the help of diagrams. This section can be thought of as a slight extension of their work, applied to the dressed particle-hole vertices.

In addition to the dashed lines representing Hamiltonian vertices (as in Sec. 2.3.2), wavy lines are introduced for cumulants:

$\lambda_2 \quad \Leftrightarrow$, $\lambda_3 \quad \Leftrightarrow$, ...

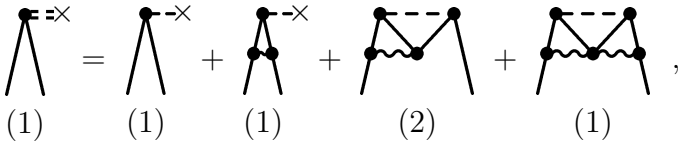
All unique terms of $n_{i_1}\bar{n}^{a_1}...n_{i_K}\bar{n}^{a_K}\tilde{W}^{a_1...a_K}_{i_1...i_K}$ can be taken into account by drawing all possible connected diagrams with K up- and incoming and K down- and outgoing external fermion lines ("legs"). Stemming from the normal ordered nature of operators, contracting a vertex to itself is forbidden. Connecting cumulants to each other is also forbidden as an immediate consequence of the MRGWT.
As a further notation, let us represent dressed particle-hole vertices with double dashed lines:

$F \Leftrightarrow$  , $\tilde{V} \Leftrightarrow$, $\tilde{W}_3 \Leftrightarrow$, ...

Note that the legs (that is, external lines) are not part of the definition. Equipped with the appropriate external lines (i.e. factors of n_i and $\bar{n}^a$), we have

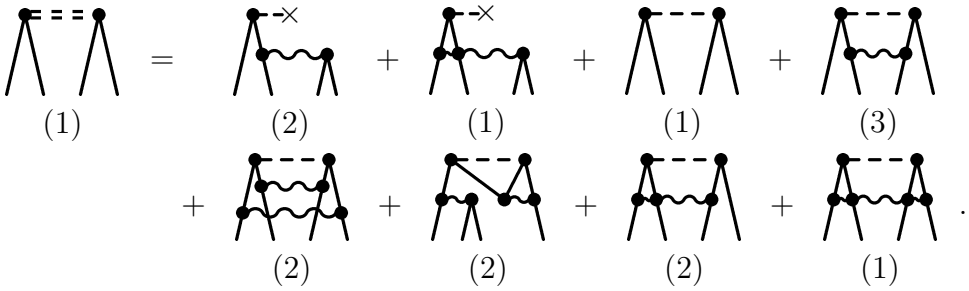
$n_i\bar{n}^a F^a_i =$  , $n_i n_j \bar{n}^a\bar{n}^b \tilde{V}^{ab}_{ij} =$  , $n_i n_j n_k \bar{n}^a\bar{n}^b\bar{n}^c \tilde{W}^{abc}_{ijk} =$  , ...

The following skeleton diagrams can be drawn for the one- and two-body dressed vertices (with legs):



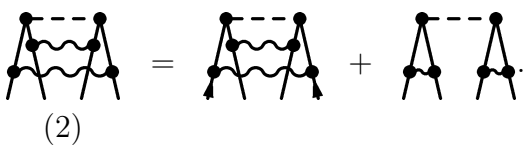
(3.2.6)

and



(3.2.7)

Again, the antisymmetry of Hamiltonian vertices and cumulants allows multiple equivalent ways to draw the diagrams. For example, the fifth skeleton of Eq. (3.2.7) can be expanded as

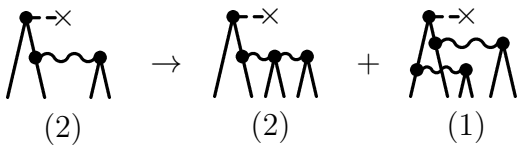


(3.2.8)

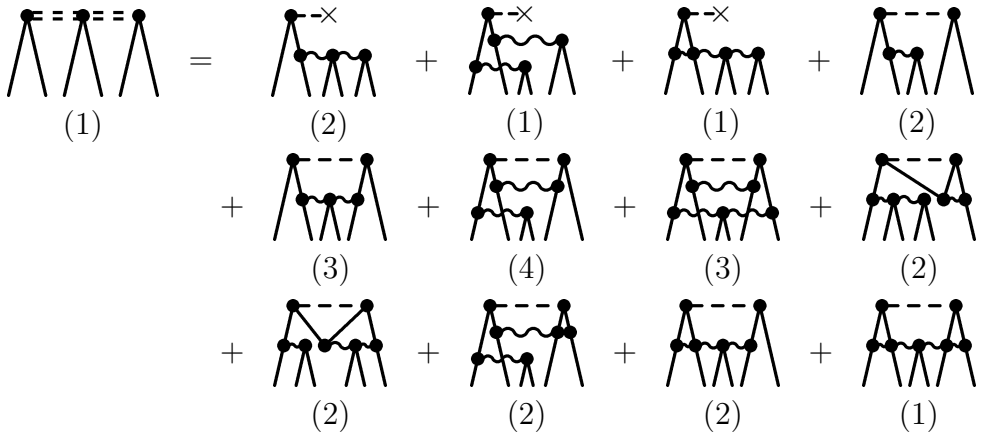
Furthermore, an extra flexibility in drawing diagrams is gained from the fact that the

orientation of internal cumulant lines is immaterial (that is, they do not carry any particle or hole character). One could thus position cumulants anywhere in the diagram relative to the Hamiltonian vertex or each other as long as the orientation of external lines remains the same.

For $K \geq 3$ one can systematically generate the K-particle diagrams from the $(K-1)$-particle diagrams by "promoting" the cumulants, that is, by inserting a new pair of cumulant legs in every allowed and unique way. The reason why this works is that the Hamiltonian contains at most 2-particle interactions, thus no new bare vertices appear for $K \geq 3$. A simple example of this promotion is given as follows:



Using this technique, it is easy to draw all skeletons for the 3-particle vertex starting from Eq. (3.2.7):



(3.2.9)

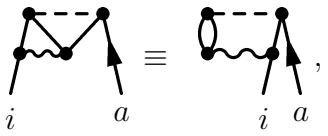
The interpretation rules for $n_{i_1}...n_{i_K}\bar{n}^{a_1}...\bar{n}^{a_K}\tilde{W}^{a_1...a_K}_{i_1...i_K}$ can be stated as follows:

- V1: Write the product of algebraic expressions corresponding to bare interaction and cumulant vertices.
- V2: Assign a factor of n_i or $\bar{n}^a$ to each down- or upgoing non-cumulant line.
- V3: Sum over orbitals at each internal line.
- V4: Assign $(-1)^{H+C-L}$, with H, L and C denoting the number of downgoing non-cumulant lines, loops (including quasi-loops) and the sum of cumulant ranks, respectively.
- V5: Assign a symmetry factor of $1/l!$ for each l-tuple of equivalent lines and $1/v!$ for each v-tuple of equivalent diagram parts.

- V6: Antisymmetrize all external upper/lower indices that are not manifestly antisymmetric.

These rules (in the context of ICSE_K) were stated without proof. Here a short rationalization is given for rules V4, V5 and V6, the only ones which do not follow immediately from the MRGWT.

In order to see why the phase rule V4 is valid, note that outer indices and Hamiltonian vertices imply "pairings" in the product: $\tilde{v}^{qs}_{pr}$ sets pairs pq and rs, while outer indices are paired like ia, jb, and so on. Of course, interchanging indices (using antisymmetry) would lead to different pairings.
Each diagram corresponds to a specific contraction pattern of operators. This pattern can always be brought to "maximal loop" form, in which upper-lower indices in the contraction are moved near each other to resemble the aforementioned pairing as much as possible. A diagrammatic example of this rearrangement (taken from the third skeleton of Eq. (3.2.6)) is



or, algebraically (currently not exploiting the diagonality of η_1):

$$\tilde{v}^{qs}_{pr}\langle : \overbrace{a^p a^r a_s a_q a^a a_i} :\rangle \equiv \tilde{v}^{qs}_{pr}\langle : \underbrace{a^p a_q}\,\underbrace{a^r a_i}\,\underbrace{a_s a^a} :\rangle = \tilde{v}^{qs}_{pr}\lambda^{pr}_{qi}\eta^a_s \ . \tag{3.2.10}$$

The braces represent the loops contractions form with $\tilde{v}^{qs}_{pr}$: one closed loop and one quasi-loop involving external indices i and a:

$$p \xrightarrow{\lambda} q \xrightarrow{\tilde{v}} p \quad , \quad a \xrightarrow{\eta} s \xrightarrow{\tilde{v}} r \xrightarrow{\lambda} i \xrightarrow{\text{outer}} a \ .$$

One can always separate the different loops with an even number of transpositions. For this reason, phase factors can be computed for each individual loop, and the overall phase will be given by their product. From now on we assume the maximal loop form of diagrams.
Let us say a loop contains h external downgoing lines (hole-contractions) and c "cumulant dots" (that is, $2c$ cumulant lines; $c = 0, 1, 2$ since cumulants cannot be connected to each other).
Let us first consider $c = 0$ (i.e. a loop formed by pair contractions only). Such a loop must contain at least one hole-contraction. For $h = 1$, the pattern looks like $: \overbrace{a^p \ldots a_q} :$, all other operators in-between being in particle contractions. This has a sign of $+1$. A second hole-contraction ($h = 2$) would look like e.g. $: \overbrace{a^p \ldots \overbrace{a^r \ldots a^t} \ldots a_s \ldots a_q} :$, changing the sign to -1 for permutational reasons. One could keep adding hole-contractions to the loops (each changing the sign), and conclude that the phase factor must be $(-1)^{h-1}$. See Ref. [26] for a more detailed discussion of this rule in the context of single determinantal theories.
Another special case is $h = 0$, like the loops in Eq. (3.2.10) (with $c = 1$ in both cases). Note that cumulants can always be written as

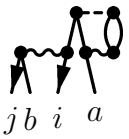
$$\lambda^{p_1 \ldots p_K}_{q_1 \ldots q_K} = : \overbrace{a^{p_1} a_{q_1} \ldots a^{p_K} a_{q_K}} : = (-1)^K : \overbrace{a_{q_1} a^{p_1} \ldots a_{q_K} a^{p_K}} : \ .$$

This suggests that cumulant dots play a role analogous to hole-contractions in loops with a phase factor of $(-1)^{c-1}$. Indeed, this correctly gives the factor of Eq. (3.2.10), $(-1)^{1-1}(-1)^{1-1} = 1$. Two additional examples from the second and sixth skeletons of Eq. (3.2.7) are



$$\rightarrow f^q_p \langle : \underbrace{a^p a_q}\, \underbrace{a^a a_i}\, \underbrace{a^b a_j} : \rangle = (-1)^{1-1}(-1)^{1-1}(-1)^{1-1} f^q_p \lambda^{pab}_{qij} \ ,$$

and



$$\rightarrow \tilde{v}^{qs}_{pr} \langle : \underbrace{a^p a_q}\, \underbrace{a^r a_s a^a a_i}\, \underbrace{a^b a_j} : \rangle = (-1)^{1-1}(-1)^{2-1}(-1)^{1-1} \tilde{v}^{qs}_{pr} \lambda^{pa}_{qs} \lambda^{rb}_{ij} \ .$$

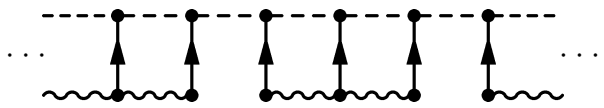
Now, h and c must be independent of each other in determining the sign: this means that the general phase factor (conforming to both special cases) must be $(-1)^{h+c-1}$. Then the overall phase factor for L loops is given by

$$\prod_{m=1}^{L} (-1)^{h_m + c_m - 1} = (-1)^{H+C-L} \ ,$$

where $H = h_1 + ... + h_L$ and $C = c_1 + ... + c_L$ are the number of hole-type external lines and the sum of cumulant ranks, respectively.

Symmetry factors of rule V5 follow from the antisymmetry of interaction vertices and cumulants. As seen in Eq. (2.1.4), a K'-body operator $X_{K'}$ carries a factor of $1/(K'!)^2$ to compensate for the fact that the summation runs over $(K'!)^2$ equivalent copies of the same term ($K'!$ for upper or lower indices). Symmetry factor $1/l!$ associated with equivalent lines simply counts the "residual symmetry" of $X_{K'}$ that remains after contractions. In our current case $l = 2$ is the only possibility, since the Hamiltonian is a two-body operator at most. Nevertheless, the result is presented for the general K'-body case as it will be useful later.

Let us concentrate on the lower operator indices of $X_{K'}$; the argument would be the same for upper indices. Let all K' indices be part of one of m pair- or cumulant contractions. The contractions use up $l_1, l_2, ..., l_m$ lower indices of $X_{K'}$ (so that $l_1 + ... + l_m = K'$); note that $l_1, l_2, ..., l_m$ do not necessarily equal the ranks of cumulants. This means $l_1, l_2, ..., l_m$ equivalent lines in each contraction. Showing only the relevant contractions, we illustrate this by



Unfinished cumulant lines symbolize contractions unrelated to $X_{K'}$.

In a specific term, the first l_1-tuple of indices can be contracted in $\binom{K'}{l_1}$ ways, the next one in $\binom{K'-l_1}{l_2}$ ways, and so on. Then the global symmetry factor becomes

$$\frac{1}{K'!}\binom{K'}{l_1}\binom{K'-l_1}{l_2}...\binom{K'-l_1-...-l_{m-1}}{l_m} = \frac{1}{l_1!...l_m!} ,$$

each l-tuple of equivalent lines bringing a factor of $1/l!$ to the diagram.

Equivalent diagram parts arise from contractions of similar topology involving the same number of external upper-lower indices. A simple example is given by the second diagram of Eq. (3.2.8), with underlying contraction pattern

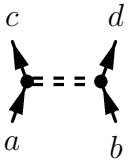
$$\tilde{v}^{qs}_{pr}\langle : \overbrace{a^p a_q a^r a_s a^a a_i a^b a_j} : \rangle = \tilde{v}^{qs}_{pr}\lambda^{pa}_{qi}\lambda^{rb}_{sj} .$$

Summation over p, q, r, s makes the two cumulant contractions equivalent (indistinguishable), as they can be transformed into each other by interchanging dummy indices. Due to this equivalence, antisymmetrization of indices ij and ab (enforced by rule V6) counts each unique term twice, and one must compensate by introducing a factor of $1/2$ for the two equivalent cumulants (see the second term in the third line of Eq. (3.2.5)).
More generally, let us assume a diagram having v equivalent parts with (say) m lower indices assigned to each of them. Let us further assume that upper indices are completely antisymmetric (either by default, or by a previous antisymmetrization). Lower indices must be antisymmetrized with $\mathcal{A}_{(i_1...i_m)...(i_{vm-m+1}...i_{vm})}$, the parantheses serving as a reminder that only indices between separate quantities are permuted (cf. the definition of $\mathcal{A}$ in Appendix A). This generates $(vm)!/(m!)^v$ terms with all unique terms counted $v!$ times. This overcounting stems from interchanging $(i_{k_1}...i_{k_m})$ and $(i_{l_1}...i_{l_m})$ between equivalent diagram parts. The role of symmetry factor $1/v!$ in rule V5 is simply to correct this redundancy.
Again, in the current case, the two-body nature of the Hamiltonian limits the possible number of equivalent diagram parts to $v = 2$.

3.2.2 The general form of dressed interaction vertices

The previous sections were about a special particle-hole case of dressed vertices with K up- and incoming legs and K down- and outgoing legs. We need to generalize the notion of dressed vertices so that they can be useful in the diagrammatic construction of ic-MRCC. Thinking of single determinantal CC, one can imagine that we would need vertices with alternative leg orientations, such as

$n_i n^k F^k_i =$ 

, $\bar{n}^a\bar{n}^b\bar{n}_c\bar{n}_d\tilde{V}^{ab}_{cd} =$ 

,

and so on. Examining the matrix element of $\tilde{W}^{a_1...a_K}_{i_1...i_K}$, we can see that the reason why only

"in-particle" or "out-hole" type legs are present is that all operators are standing right to $\mathcal{H}$. Having some of the $2K$ operators (like a^k, or a_c) on the left hand side of $\mathcal{H}$ would yield "in-hole" or "out-particle" type legs. For this reason, general K-body dressed vertices are defined as

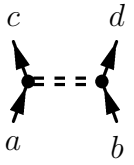
$$\pi^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu}\tilde{W}^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu} \;=\; (-1)^{\mu(\gamma+1)}\langle :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}:\,\mathcal{H}\,:a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C\;, \tag{3.2.11}$$

having μ creation and γ annihilation operators on the left, and δ creation and ν annihilation operators on the right hand side of $\mathcal{H}$ (with $\mu+\delta=\nu+\gamma=K$). The factor

$$\pi^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu} = n^{i_1}\ldots n^{i_\mu}n_{j_1}\ldots n_{j_\nu}\bar{n}_{a_1}\ldots\bar{n}_{a_\gamma}\bar{n}^{b_1}\ldots\bar{n}^{b_\delta}$$

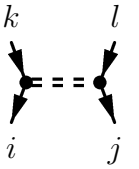
is introduced simply for notational convenience. A few examples are

$$\bar{n}^a\bar{n}^b\bar{n}_c\bar{n}_d\,\tilde{V}^{ab}_{cd} = \langle :a_{cd}:\,\mathcal{H}\,:a^{ab}:\rangle_C =$$



,

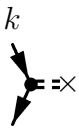
$$n_in_jn^kn^l\,\tilde{V}^{kl}_{ij} = \langle :a^{kl}:\,\mathcal{H}\,:a_{ij}:\rangle_C =$$



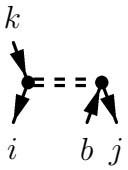
,

and

$$n_in^kF^k_i = -\langle :a^k:\,\mathcal{H}\,:a_i:\rangle_C =$$



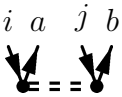
, $\quad n_in_jn^k\bar{n}^b\,\tilde{V}^{kb}_{ij} = -\langle :a^k:\,\mathcal{H}\,:a^b_{ij}:\rangle_C =$



.

Note that

$$n^in^j\bar{n}_a\bar{n}_b\tilde{V}^{ij}_{ab} = \langle :a^{ij}_{ab}:\,\mathcal{H}\rangle =$$



.

is simply the complex conjugate of $n_in_j\bar{n}^a\bar{n}^b\tilde{V}^{ab}_{ij}$.
The awkward looking phase factor $(-1)^{\mu(\gamma+1)}$ is chosen in a way that the sign of dressed vertices always matches that of their bare counterparts. For example, $\tilde{V}^{kb}_{ij} = \tilde{v}^{kb}_{ij} + \ldots$ for $\mu=1$, $\gamma=0$. We will elaborate on this phase factor shortly.

It is important to keep in mind that the sets of holes and particles are not disjoint in an MR theory. Indices $i,j,k,l,\ldots$ and $a,b,c,d,\ldots$ only serve as bookkeeping for the type of pair-contraction their respective operators can take part in, and as labels to classify the different leg orientations of a dressed vertex.
In the general K-body case, there exist $(K+1)^2$ different, unique leg orientations not related through antisymmetry (but possibly related through complex conjugation).

At first, working out Eq. (3.2.11) for all possible left-right partitioning of $2K$ operators seems challenging. Fortunately, evaluation of a general K-body dressed vertex can always be traced back to that of the original particle-hole vertex that was solved in the previous sections. To see this, let us compare the two matrix elements below:

$$\langle :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}: \mathcal{H} :a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C \quad \leftrightarrow \quad \langle \mathcal{H} :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}::a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C \ . \tag{3.2.12}$$

The key observation is that the MRGWT expansions of these matrix elements are very similar. As the Hamiltonian contains an even number of creation-annihilation operators, the parity of operator products $:a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}: \mathcal{H}$ and $\mathcal{H} :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}:$ is the same. For this reason, both products generate the same cumulant contractions. However, while $:a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}: \mathcal{H}$ yields pair-contractions like $\gamma^i_q = \delta^i_q n^i$ or $\eta^p_a = \delta^p_a \bar{n}_a$, in $:a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}: \mathcal{H}$ the particle-hole character is switched, resulting in $-\eta^i_q = \delta^i_q(-\bar{n}^i)$ or $-\gamma^p_a = \delta^p_a(-n_a)$. The additional signs stem from the parity change associated with hole-, or particle-contractions.
There are no other differences between the two matrix elements. This means that the MRGWT expansion of $\langle :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}: \mathcal{H} :a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C$ equals the expansion of $\langle \mathcal{H} :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}::a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C$ after the replacements $n_a \to -\bar{n}_a$, $\bar{n}^i \to -n^i$. The relation of the two sides of Eq. (3.2.12) can be written more formally as

$$\langle :a^{i_1\ldots i_\mu}_{a_1\ldots a_\gamma}: \mathcal{H} :a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C = \langle \mathcal{H} :a^{\bar{i}_1\ldots\bar{i}_\mu}_{\bar{a}_1\ldots\bar{a}_\gamma}::a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C \ ,$$

where "irregular" occupation numbers $n_{\bar{a}} = -\bar{n}_a$ and $\bar{n}^{\bar{i}} = -n^i$ are assigned to indices $\bar{a}$ and $\bar{i}$ with an overline. These formal occupation numbers are only used in pair-contractions. Cumulants are insensitive to the overline of indices.
As a next step, note that operators $:a^{\bar{i}_1\ldots\bar{i}_\mu}_{\bar{a}_1\ldots\bar{a}_\gamma}:$ and $:a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:$ can be merged into one, as connectedness forbids contractions between them:

$$\langle \mathcal{H} :a^{\bar{i}_1\ldots\bar{i}_\mu}_{\bar{a}_1\ldots\bar{a}_\gamma}::a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle_C = \langle \mathcal{H} :a^{\bar{i}_1\ldots\bar{i}_\mu}_{\bar{a}_1\ldots\bar{a}_\gamma} a^{b_1\ldots b_\delta}_{j_1\ldots j_\nu}:\rangle = (-1)^{\gamma(\delta+\nu)}\langle \mathcal{H} :a^{\bar{i}_1\ldots\bar{i}_\mu b_1\ldots b_\delta}_{\bar{a}_1\ldots\bar{a}_\gamma j_1\ldots j_\nu}:\rangle \ . \tag{3.2.13}$$

In the second step above, we brought the merged excitation operator into a canonical form, which necessitates $(\delta+\nu)$ transpositions for each of the $a_{\bar{a}}$ operators.
Comparison of Eqs. (3.2.11) and (3.2.13) reveals the phase factor $(-1)^{\mu(\gamma+1)+\gamma(\delta+\nu)}$. Since only the parity of the exponent matters, we can simplify it to

$$\mu(\gamma+1)+\gamma(\delta+\nu) = \mu+\gamma(2K-\gamma) \equiv \mu+\gamma \pmod 2 \ .$$

We then have

$$\pi^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu}\tilde{W}^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu} = (-1)^{\mu+\gamma}\langle \mathcal{H} :a^{\bar{i}_1\ldots\bar{i}_\mu b_1\ldots b_\delta}_{\bar{a}_1\ldots\bar{a}_\gamma j_1\ldots j_\nu}:\rangle \ .$$

After introducing

$$\bar{\pi}^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu} = (-n^{i_1})\ldots(-n^{i_\mu})n_{j_1}\ldots n_{j_\nu}(-\bar{n}_{a_1})\ldots(-\bar{n}_{a_\gamma})\bar{n}^{b_1}\ldots\bar{n}^{b_\delta} \ ,$$

the formula for general dressed vertices reads

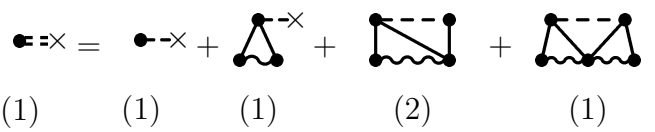
$$\bar{\pi}^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu}\tilde{W}^{i_1\ldots i_\mu b_1\ldots b_\delta}_{a_1\ldots a_\gamma j_1\ldots j_\nu} = \langle \mathcal{H} :a^{\bar{i}_1\ldots\bar{i}_\mu b_1\ldots b_\delta}_{\bar{a}_1\ldots\bar{a}_\gamma j_1\ldots j_\nu}:\rangle \ . \tag{3.2.14}$$

Dressed vertices with arbitrary orientation of legs can thus always be evaluated using the original particle-hole vertex with appropriate barred indices and substitutions $n_{\bar{a}} = -\bar{n}_a$, $\bar{n}^{\bar{i}} = -n^i$. The form of Eq. (3.2.14) ensures that dressed and bare vertices appear with the same sign, as claimed before.

The main conclusion is that all variants of a dressed vertex are represented by essentially the same diagrams, with only the leg orientations being different. Interpretation rules of the previous section have to be only marginally modified to account for the different factors associated with legs.
It is somewhat cumbersome to evaluate all other variants of a vertex starting from the particle-hole one through substitutions. Fortunately, this dependence on the particle-hole vertex can be eliminated, and all variants can be treated on the same footing after introducing "neutral" factors

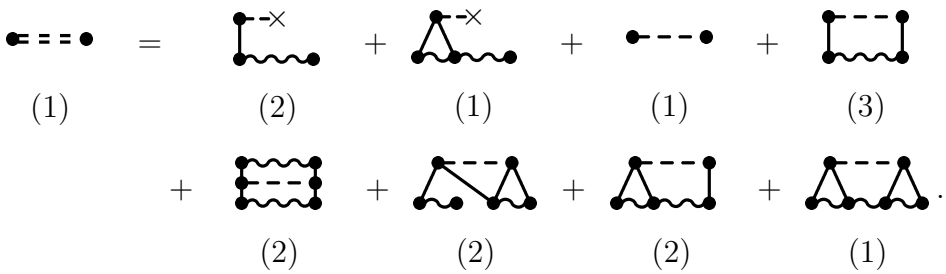
$$\zeta_u = \begin{cases} -n_u & \text{if } u \in \{i,j,k,l,...\} \\ \bar{n}_u & \text{if } u \in \{a,b,c,d,...\} \end{cases},$$

and so-called amputated diagrams (terminology borrowed from quantum field theory [144]). Amputated diagrams are simply diagrams without legs, thus being the same for all variants. For example, amputated diagrams representing one- and two-body dressed vertices are



(3.2.15)

and



(3.2.16)

The general form of dressed vertices equipped with arbitrary legs reads $\zeta_{u_1}...\zeta_{u_K}\zeta^{x_1}...\zeta^{x_K}\tilde{W}^{x_1...x_K}_{u_1...u_K}$, where $u, w, ...$ and $x, y, ...$ can stand for any label. The main advantage of using ζ_u is that phase factors associated with downgoing non-cumulant lines and quasi-loops can be absorbed into them, making interpretation rules formally independent of leg orientation:

- G1: Write the product of algebraic expressions corresponding to bare interaction and cumulant vertices.
- G2: Assign a factor of ζ_u or ζ^x to any outgoing or incoming leg
- G3: Sum over orbitals at each internal line.
- G4: Assign $(-1)^{C-L'}$, with L' and C denoting the number of loops (excluding quasi-loops) and the sum of cumulant ranks, respectively.

- G5: Assign a symmetry factor of $1/l!$ for each l-tuple of equivalent lines and $1/v!$ for each v-tuple of equivalent diagram parts.
- G6: Antisymmetrize all external upper/lower indices that are not manifestly antisymmetric.

Algebraic form of the general one- and two-body dressed vertex reads

$$\zeta_u \zeta^x F_u^x = \zeta_u \zeta^x f_u^x - \sum_{p,q} f_p^q \; \lambda_{qu}^{px} - \frac{1}{2} \sum_{p,q,r} \left(\zeta^x \tilde{v}_{rp}^{xq} \; \lambda_{uq}^{rp} + \zeta_u \tilde{v}_{uq}^{rp} \; \lambda_{rp}^{xq} \right) - \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \; \lambda_{qsu}^{prx} \; , \quad (3.2.17)$$

and

$$\begin{aligned} \zeta_u \zeta^x \zeta_w \zeta^y \tilde{V}_{uw}^{xy} = & \sum_p \left[\mathcal{A}^{xy} \left(\zeta^x f_p^x \lambda_{uw}^{py} \right) + \mathcal{A}_{uw} \left(\zeta_u f_u^p \lambda_{pw}^{xy} \right) \right] + \sum_{p,q} f_p^q \lambda_{uwq}^{xyp} + \zeta_u \zeta^x \zeta_w \zeta^y \tilde{v}_{uw}^{xy} \\ & + \frac{1}{2} \sum_{p,q} \left(\zeta^x \zeta^y \tilde{v}_{pq}^{xy} \lambda_{uw}^{pq} + \zeta_u \zeta_w \tilde{v}_{uw}^{pq} \lambda_{pq}^{xy} \right) - \sum_{p,q} \mathcal{A}_{uw} \mathcal{A}^{xy} \left(\zeta_u \zeta^x \tilde{v}_{up}^{xq} \lambda_{wq}^{yp} \right) \\ & + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \lambda_{uw}^{pr} \lambda_{qs}^{xy} + \frac{1}{2} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \mathcal{A}^{xy} \mathcal{A}_{uw} \left(\lambda_{qu}^{px} \lambda_{sw}^{ry} \right) \\ & - \frac{1}{2} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \left[\mathcal{A}^{xy} \left(\lambda_{qs}^{px} \lambda_{uw}^{ry} \right) + \mathcal{A}_{uw} \left(\lambda_{qu}^{pr} \lambda_{sw}^{xy} \right) \right] \\ & + \frac{1}{2} \sum_{p,q,r} \left[\mathcal{A}^{xy} \left(\zeta^x \tilde{v}_{pr}^{qx} \lambda_{uwq}^{ryp} \right) + \mathcal{A}_{uw} \left(\zeta_u \tilde{v}_{up}^{rq} \lambda_{rwq}^{xyp} \right) \right] + \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \lambda_{uwqs}^{xypr} \; . \end{aligned} \quad (3.2.18)$$

One can find the algebraic form of any variant with these formulae by simple substitution. For example, the hole-hole variant of the one-body vertex reads ($\zeta_u \to \zeta_i = -n_i$, $\zeta^x \to \zeta^k = -n^k$)

$$n_i n^k F_i^k = n_i n^k f_i^k - \sum_{p,q} f_p^q \; \lambda_{qi}^{pk} + \frac{1}{2} \sum_{p,q,r} \left(n^k \tilde{v}_{rp}^{kq} \; \lambda_{iq}^{rp} + n_i \tilde{v}_{iq}^{rp} \; \lambda_{rp}^{kq} \right) - \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \; \lambda_{qsi}^{prk} \; , \quad (3.2.19)$$

while for the particle-particle variant we find ($\zeta_u \to \zeta_c = \bar{n}_c$, $\zeta^x \to \zeta^a = \bar{n}^a$)

$$\bar{n}^a \bar{n}_c F_c^a = \bar{n}^a \bar{n}_c f_c^a - \sum_{p,q} f_p^q \; \lambda_{qc}^{pa} - \frac{1}{2} \sum_{p,q,r} \left(\bar{n}^a \tilde{v}_{rp}^{aq} \; \lambda_{cq}^{rp} + \bar{n}_c \tilde{v}_{cq}^{rp} \; \lambda_{rp}^{aq} \right) - \frac{1}{4} \sum_{p,q,r,s} \tilde{v}_{pr}^{qs} \; \lambda_{qsc}^{pra} \; . \quad (3.2.20)$$

This concludes the general construction of dressed interaction vertices. We will see in the next section, that these quantities arise naturally in the ic-MRCC framework.
Interpretation of dressed vertices is probably the most transparent in real space coordinate representation. This is outlined in Appendix D.

A few simple, yet important properties of dressed vertices are to be discussed.
Dressed vertices are antisymmetric under permutation of upper or lower indices; for example $\tilde{V}_{ic}^{ak} = -\tilde{V}_{ic}^{ka} = -\tilde{V}_{ci}^{ak} = \tilde{V}_{ci}^{ka}$.
Particle-hole dressed vertices vanish identically whenever reference function $|\Phi\rangle$ happens to be an exact eigenstate of H, since in such cases $\mathcal{H}|\Phi\rangle = |\emptyset\rangle$. This is a desirable property, as it guarantees that many-body methods employing these vertices do not add any spurious

correction to the already exact solution via Eq. (3.2.1). Note however, that this vanishing can only happen if the vertex is evaluated exactly, and no cumulants are neglected based on e.g. rank. This highlights the danger of neglecting EPV cumulants on an ad-hoc basis (see Sec. 2.1.4). Consider a two-electron system with exact ground state $|\Phi\rangle$ (provided by e.g. APSG if all orbitals are assigned to the geminal): in this case, three- and four-body cumulants in Eqs. (3.2.6) and (3.2.7) are EPV, but dropping them would violate the vanishing of vertices. Besides the plausible reasoning of Hanauer and Köhn [13], this is the main argument why EPV cumulants are always to be kept.
Dressed interaction vertices transform as tensor components under unitary rotations in the subspace of core or virtual orbitals. This simply follows from their structure, and the fact that working in the natural basis does not affect core and virtual orbitals. For example, let us consider the particle-hole type one-body dressed vertex with ϕ^i being a core orbital (with $n_i = 1$):

$$\bar{n}^a F_i^a = \bar{n}^a f_i^a - \frac{1}{2} \sum_{p,q,r} \tilde{v}_{iq}^{rp} \ ,$$

all other terms of Eq. (3.2.4) vanishing in accordance with Sec. 2.1.4; correct transformation properties in the core or virtual subspace then follow from the transformation properties of bare vertices. This means that the logic of Sec. 2.3.3 applies, and the ic-MRCC energy calculated with Eq. (3.2.1) is invariant under rotations in the core or virtual subspace. Note that in general, no such rotational freedom is present for active orbitals, as they are fixed by the requirement of being natural.
Intersystem cumulants vanish for infinitely separated singlet subsystems, as it was discussed in Sec. 2.1.4. One can assume that all orbitals are localized on subsystems in such a case, hence intersystem bare vertices also vanish. These two properties combined with the manifest connectedness of the equations make all intersystem dressed vertices disappear. This feature ensures size consistency over singlet subsystems for any correction method built with dressed vertices. The harder question of size consistency over non-trivially spin coupled subsystems will not be addressed here.
Finally note that dressed vertices for singlet systems are invariant under the action of spin rotation group $SU(2)$ (since both bare vertices and cumulants are invariant), and so the results of Appendix C are applicable. We will make use of this during the singlet spin-adaptation of ic-MRCC (see Sec. 3.3.2).

3.3 The general ic-MRCC formalism

Having established the concept of dressed interaction vertices in the previous sections, the ic-MRCC equations can finally be derived. In order not to get lost in details, we present the derivations in the CCSD approximation (i.e. $T = T_1 + T_2$); generalizing the results to higher excitation levels is straightforward.

In the CCSD scheme, Eq. (3.1.7) reads

$$\langle :a_a^i: \mathcal{H} :\exp(T):\rangle_C = 0 \quad , \quad \langle :a_{ab}^{ij}: \mathcal{H} :\exp(T):\rangle_C = 0 \ .$$

Upon expanding the wave operators, the inhomogeneous terms turn out to be the complex conjugates of Eqs. (3.2.2) and (3.2.3):

$$\langle :a_a^i: \mathcal{H}\rangle = n^i\bar{n}_a F_a^i \quad , \quad \langle :a_{ab}^{ij}: \mathcal{H}\rangle = n^i n^j \bar{n}_a \bar{n}_b \tilde{V}_{ab}^{ij} \ . \tag{3.3.1}$$

In the terms involving T, dressed interaction vertices can be recognized again by systematically evaluating contractions between the excitation operator used for projection and the excitation involved in $:\exp(T):$ As a simple example, let us consider the linear homogeneous part of the T_1 equation:

$$L_a^i = \langle :a_a^i: \mathcal{H} :T_1:\rangle + \langle :a_a^i: \mathcal{H} :T_2:\rangle \ . \tag{3.3.2}$$

The first term contains matrix elements of the form $\langle :a_a^i: \mathcal{H} :a_k^c:\rangle$. Let us group its MRGWT expansion into the sum of three terms based on the three possible contractions purely between operators $:a_a^i:$ and $:a_k^c:$: hole-type pair contraction, particle-type pair contraction and no contraction. The matrix element then reads

$$\begin{aligned}\langle :a_a^i: \mathcal{H} :a_k^c:\rangle &= \delta_k^i n^i \langle :a_a: \mathcal{H} :a^c:\rangle_X + \delta_a^c \bar{n}_a \langle :a^i: \mathcal{H} :a_k:\rangle_X + \langle :a_a^i: \mathcal{H} :a_k^c:\rangle_X \\ &= \delta_k^i n^i \bar{n}_a \bar{n}^c F_a^c - \delta_a^c \bar{n}_a n^i n_k F_k^i + n^i \bar{n}^c \bar{n}_a n_k \tilde{V}_{ak}^{ic} \ .\end{aligned} \tag{3.3.3}$$

In the first equality, X in the subscript refers to a reduced MRGWT expansion forbidding any further contraction that would connect solely the left and right operators without touching $\mathcal{H}$. Of course, cumulant contractions connecting all three operators are allowed. This way, a unique partitioning of all contributions to the matrix element is achieved, all of them being counted only once. Dressed vertices were recognized in the second equality of Eq. (3.3.3) using Eq. (3.2.11) (note that X implies connectedness of the remainder of the matrix element).

One can proceed similarly with the second term of Eq. (3.3.2). Contractions of matrix element $\langle :a_a^i: \mathcal{H} :a_{kl}^{cd}:\rangle$ can be uniquely classified by listing contractions between $:a_a^i:$ and $:a_{kl}^{cd}:$. The major difference from the previous case is the appearance of cumulant contractions between the two operators.

Terms in the MRGWT expansion of $\langle :a_a^i: \mathcal{H} :a_{kl}^{cd}:\rangle$ can be grouped as

$$\begin{aligned}\langle :a_a^i: \mathcal{H} :a_{kl}^{cd}:\rangle =& \mathcal{A}^{cd}\mathcal{A}_{kl}\Big[\left(\delta_a^c\delta_k^i\bar{n}_a n^i + \lambda_{ak}^{ic}\right)\langle \mathcal{H} :a_l^d:\rangle_X\Big] \\ &+ \mathcal{A}^{cd}\Big[\lambda_{kl}^{id}\langle :a_a: \mathcal{H} :a^c:\rangle_X\Big] - \mathcal{A}_{kl}\Big[\lambda_{al}^{cd}\langle :a^i: \mathcal{H} :a_k:\rangle_X\Big] \\ &+ \mathcal{A}^{cd}\Big[\delta_a^c\bar{n}_a\langle :a^i: \mathcal{H} :a_{kl}^d:\rangle_X\Big] + \mathcal{A}_{kl}\Big[\delta_k^i n^i\langle :a_a: \mathcal{H} :a_l^{cd}:\rangle_X\Big] \\ &+ \langle :a_a^i: \mathcal{H} :a_{kl}^{cd}:\rangle_X \\ =& \mathcal{A}^{cd}\mathcal{A}_{kl}\Big[\bar{n}^d n_l\left(\delta_a^c\delta_k^i\bar{n}_a n^i + \lambda_{ak}^{ic}\right)F_l^d\Big] \\ &+ \mathcal{A}^{cd}\Big[\bar{n}_a\bar{n}^c\lambda_{kl}^{id}F_a^c\Big] + \mathcal{A}_{kl}\Big[n^i n_k\lambda_{al}^{cd}F_k^i\Big] \\ &- \mathcal{A}^{cd}\Big[\delta_a^c\bar{n}_a n^i\bar{n}^d n_k n_l\tilde{V}_{kl}^{id}\Big] + \mathcal{A}_{kl}\Big[\delta_k^i n^i\bar{n}_a\bar{n}^c\bar{n}^d n_l\tilde{V}_{al}^{cd}\Big] \\ &+ n^i\bar{n}^c\bar{n}^d\bar{n}_a n_k n_l\tilde{W}_{akl}^{icd} \ .\end{aligned} \tag{3.3.4}$$

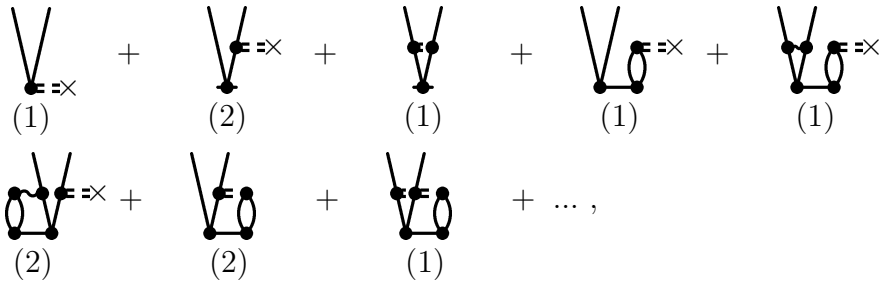
The "explicit" cumulant terms appearing above (the ones with the product of F and λ_2) signal the fact that not all cumulants can be folded into the dressing of interaction vertices. Contracting Eqs. (3.3.3) and (3.3.4) with t_c^k and $1/4\, t_{cd}^{kl}$ gives the linear part of the T_1 equation.

After the above examples, the general relation between matrix elements and dressed vertices can be formulated as follows. In a general term of the T_K equation, one must process matrix elements of the type $\langle :a_K^\dagger: \mathcal{H} :a_L:\rangle$, $:a_L:$ being some excitation operator from the expansion of $:\exp(T):$. All contractions are omitted which would make the overall term (contracted with CC amplitudes) disconnected. This matrix element is symbolically evaluated as

$$\langle :a_K^\dagger: \mathcal{H} :a_L:\rangle = \sum_{K'=0}^{K+L-1} \sum_{\tau} C_{K'}(\tau) \tilde{W}_{K+L-K'}(\tau) \ , \tag{3.3.5}$$

$C_{K'}(\tau)$ standing for contractions between $:a_K:$ and $:a_L:$ altogether of rank K', and index τ keeping track of all possible ways of writing such contractions ("topologies"). For either of the contractions collected in $C_{K'}(\tau)$, the remainder of the MRGWT expansion is a dressed interaction vertex $\tilde{W}_{K+L-K'}(\tau)$. Obviously, $K+L$ is the highest rank dressed vertex contributing to Eq.(3.3.5), for which $C_0 = 1$. The possible number of topologies varies with K, L and K' in general. In the second equality of Eq. (3.3.4) ($K=1$ and $L=2$), the first two, third and fourth lines correspond to $K' = 2, 1, 0$, respectively. The number of topologies (skeleton diagrams) associated with them is three, one and one, in the same order.

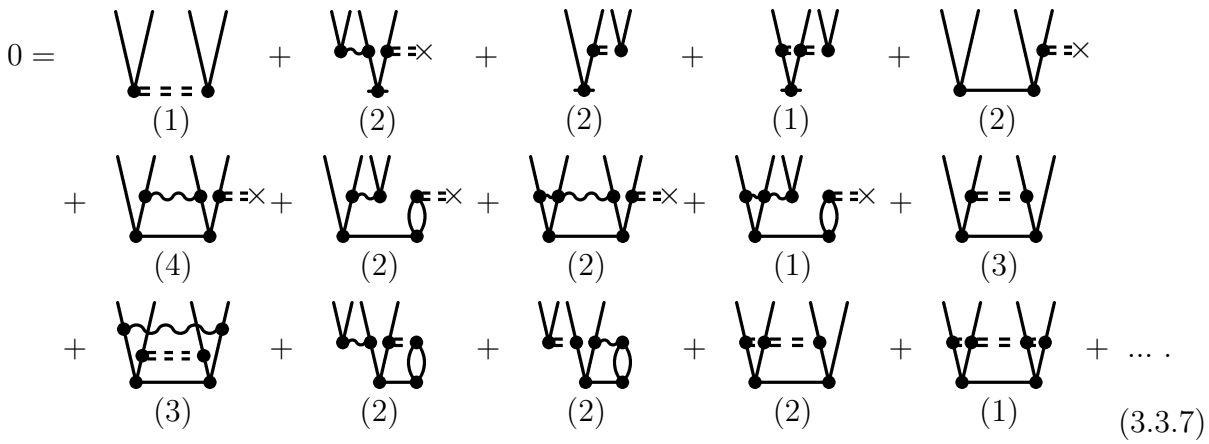
Based on the above, construction rules of diagrams contributing to the amplitude equation are relatively simple to formulate with dressed interaction vertices. Essentially the same rules and notations are to be used as in Secs. 2.3.2 and 3.2.1. One must draw all topologically distinct connected diagrams with $2K$ external legs, using one dressed interaction vertex and any number of amplitudes and cumulants. Self-contraction of any vertex is to be avoided, just like any direct cumulant-cumulant or cumulant-dressed interaction vertex contraction. The linear part of the T_1 equation takes the form (cf. Eqs. (3.3.1)-(3.3.4))

$0 =$  (3.3.6)

dots symbolizing all further contributions coming from the non-linear part of $:\exp(T):$. Note that − unlike in the case of dressed vertex diagrams − internal cumulant lines formally acquire a particle-hole character by being connected to amplitudes. This is somewhat deceiving though, as cumulants are non-vanishing for active orbitals only, so their indices run through the same domain regardless of the label of the internal lines.
Similarly to the T_1 case, one can examine to linear homogeneous part of the T_2 equation:

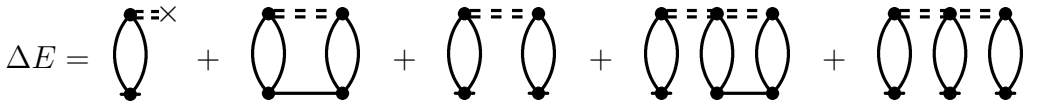
$$L_{ab}^{ij} = \langle :a_{ab}^{ij}: \mathcal{H} :T_1:\rangle + \langle :a_{ab}^{ij}: \mathcal{H} :T_2:\rangle .$$

The linear part of the T_2 equation is represented by



(3.3.7)

For the sake of completeness, it is appropriate to present here the diagrams of the energy formula as well (cf. Eq. (3.2.1)):



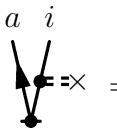
(3.3.8)

Energy diagrams have no external lines. Also, there are no explicit cumulants appearing in them, as cumulants and dressed vertices cannot be connected to each other.

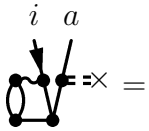
The ic-MRCC interpretation rules are essentially identical to the ones derived in Secs. 3.2.1 and 3.2.2. They are summarized here:

- A1: Write the product of algebraic expressions corresponding to dressed interaction, cumulant and amplitude vertices.
- A2: Assign a factor of n^i or $\bar{n}_a$ to each down- or upgoing non-cumulant line.
- A3: Sum over orbitals at each internal line.
- A4: Assign $(-1)^{H+C-L}$, with H, L and C denoting the number of downgoing non-cumulant lines, loops (including quasi-loops) and the sum of cumulant ranks, respectively.
- A5: Assign a symmetry factor of $1/l!$ for each l-tuple of equivalent lines and $1/v!$ for each v-tuple of equivalent diagram parts.
- A6: Antisymmetrize all external upper/lower indices that are not manifestly antisymmetric.

A few examples of linear terms in the T_1, T_2 equations include

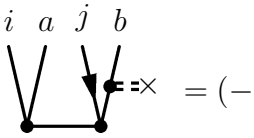


$$= (-1)^{2+0-1} n^i \bar{n}_a \sum_k t_a^k F_k^i n^k \ , \tag{3.3.9}$$

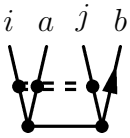


$$= \frac{(-1)^{0+2-2}}{2!} n^i \bar{n}_a \sum_{\substack{k,l \\ c,d}} \lambda_{kl}^{id} t_{cd}^{kl} F_a^c \bar{n}_c \ ,$$

and

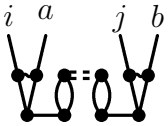


$$= (-1)^{2+0-2} \mathcal{A}_{ab} \Big(n^i n^j \bar{n}_a \bar{n}_b \sum_d t_{ad}^{ij} F_b^d \bar{n}_d \Big) \ , \tag{3.3.10}$$

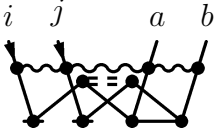


$$= \frac{(-1)^{4+0-2}}{2!} \mathcal{A}_{ab} \Big(n^i n^j \bar{n}_a \bar{n}_b \sum_{\substack{k,l \\ c}} t_{cb}^{kl} \tilde{W}_{kal}^{icj} n^k n^l \bar{n}_c \Big) \ .$$

Some more complicated examples:



$$= \frac{(-1)^{2+4-4}}{2!} \mathcal{A}^{ij} \mathcal{A}_{ab} \Big(\sum_{\substack{k,l,m,n \\ c,d,e,f}} \lambda_{ma}^{ie} \lambda_{nb}^{jf} t_{ce}^{km} t_{df}^{ln} \tilde{V}_{kl}^{cd} n^k n^l \bar{n}_c \bar{n}_d \Big) \ ,$$



$$= \frac{(-1)^{2+4-2}}{(2!)^3} \sum_{\substack{k,l,m,n \\ c,d,e,f}} \lambda_{klab}^{ijcd} t_e^k t_f^l t_{cd}^{mn} V_{mn}^{ef} n^m n^n \bar{n}_e \bar{n}_f \ .$$

Both of the above diagrams contain a pair of equivalent diagram parts; furthermore, the second diagram contains two pairs of equivalent lines.

With the diagram rules stated, the general formulation of ic-MRCC is completed. In order to gain some further insight into the structure of the equations, a diagrammatic expansion of the amplitudes and ΔE is investigated in the next section.

3.3.1 Intra- and intervertex electron correlation

The ic-MRCC equations are non-linear and must be solved in an iterative way. Just like in the single determinantal case (cf. Eq. (2.3.13)), we can set up the iterative form of the equations as

$$(t^{(n)})^{i_1\ldots i_K}_{a_1\ldots a_K} = (t^{(n-1)})^{i_1\ldots i_K}_{a_1\ldots a_K} + \frac{\Omega^{i_1\ldots i_K}_{a_1\ldots a_K}(t^{(n-1)})}{D^{i_1\ldots i_K}_{a_1\ldots a_K}} \,, \tag{3.3.11}$$

where $\Omega^{i_1\ldots i_K}_{a_1\ldots a_K}$ represents the sum of all connected amplitude diagrams (cf. Eq. (3.1.6)), and $D^{i_1\ldots i_K}_{a_1\ldots a_K}$ are control parameters of the iteration which can be freely chosen (within reasonable limits). In a single determinantal theory, these parameters (energy denominators) are typically built from one-body excitation energies (see Eq. (2.3.14)). To pursue a formal analogy, let us choose the energy denominators in the following way:

$$D^{i_1\ldots i_K}_{a_1\ldots a_K} = n^{i_1}F^{i_1}_{i_1} - \bar{n}_{a_1}F^{a_1}_{a_1} + \ldots + n^{i_K}F^{i_K}_{i_K} - \bar{n}_{a_K}F^{a_K}_{a_K} \,. \tag{3.3.12}$$

The initial guess $t^{(0)}$ of the amplitudes is obtained by neglecting all terms of the amplitude equations, except for the inhomogeneous term, and the diagonal part of the linear term containing F^i_k or F^c_a (as in Eqs. (3.3.9) or (3.3.10)). The zeroth-order T_1 and T_2 equations are immediately solved by

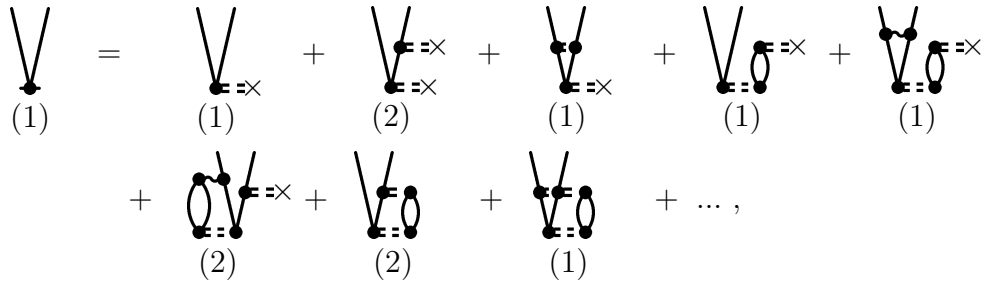
$$(t^{(0)})^i_a = \frac{F^i_a}{n^iF^i_i - \bar{n}_aF^a_a} \quad , \quad (t^{(0)})^{ij}_{ab} = \frac{\tilde{V}^{ij}_{ab}}{n^iF^i_i + n^jF^j_j - \bar{n}_aF^a_a - \bar{n}_bF^b_b} \,, \tag{3.3.13}$$

and in the general case by

$$(t^{(0)})^{i_1\ldots i_K}_{a_1\ldots a_K} = \frac{\tilde{W}^{i_1\ldots i_K}_{a_1\ldots a_K}}{D^{i_1\ldots i_K}_{a_1\ldots a_K}} \,.$$

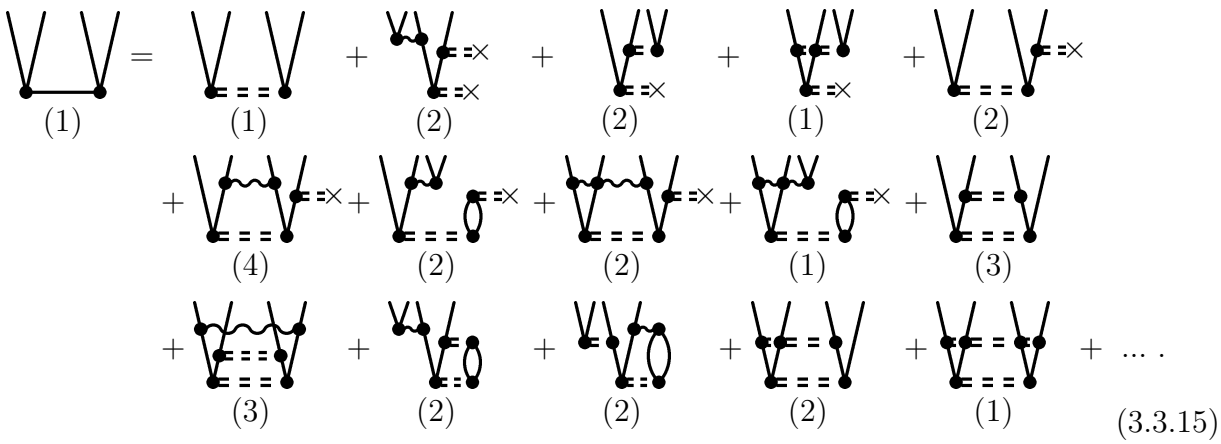
Note that, unlike in the single determinantal case, these solutions are not zero for $K \geq 3$, due to the presence of $\tilde{W}_K$.
Starting from $t^{(0)}$ and plugging the higher and higher iterations of t into Eq. (3.3.11), the sequence $t^{(n)}$ converges to the solution of the amplitude equation, at which point $\Omega^{i_1\ldots i_K}_{a_1\ldots a_K}$ vanishes.

The iteration process can be represented by diagrams. In the CCSD scheme, we find the following series for t_1 (cf. Eq (3.3.6)):



(3.3.14)

while t_2 is expanded as

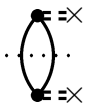


(3.3.15)

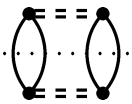
Note that cumulants can be connected to dressed interaction vertices and even to each other during the iteration process. In order to avoid ambiguities, the vertical position of cumulant vertices must be fixed: in each iteration, they must be placed as high in the diagram as possible without breaking the ordering of interaction vertices. Substituting the above forms of t_1 and t_2 in Eq. (3.3.8) leads to a similar expansion of the energy correction. The diagrams are interpreted similarly to the ones of the ic-MRCC equations, with three further rules:

- P1: assign a factor of $1/D^{i_1 \ldots i_K}_{a_1 \ldots a_K}$ to all fictitious (dotted) horizontal lines between interaction vertices (above the respective cumulant vertices) crossing K pairs of fermion lines.
- P2: restrict the summation over internal lines, so that $i \neq k$ for F^i_k and $a \neq c$ for F^c_a in the numerator.
- P3: in the case of $2K$ external lines with labels $i_1, ..., i_K, a_1, ..., a_K$, divide the term by $n^{i_1} ... n^{i_K} \bar{n}_{a_1} \bar{n}_{a_K}$.

Examples of energy diagrams include (dotted horizontal lines symbolizing energy denominators):

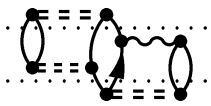


$$= (-1)^{1+0-1} \sum_{\substack{i \\ a}} \frac{F^i_a F^a_i}{D^i_a} n^i \bar{n}_a \quad , \quad$$



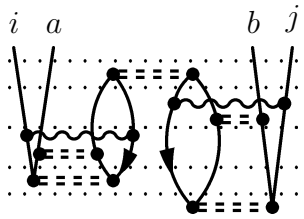
$$= \frac{(-1)^{2+0-2}}{(2!)^2} \sum_{\substack{i \\ a}} \frac{\tilde{V}^{ij}_{ab} \tilde{V}^{ab}_{ij}}{D^{ij}_{ab}} n^i n^j \bar{n}_a \bar{n}_b \ , \quad (3.3.16)$$

and



$$= \frac{(-1)^{3+2-3}}{(2!)^2} \sum_{\substack{i,j,k,l \\ a,b,c,d}} \frac{\lambda^{ab}_{cj} \tilde{V}^{ij}_{ab} \tilde{V}^{cd}_{kl} \tilde{V}^{kl}_{id}}{D^{ij}_{ab} D^{kl}_{cd}} n^i n^k n^l \bar{n}_d \ .$$

A more involved diagram contributing to t_2 reads



$$= \frac{(-1)^{0+4-4}}{(2!)^4} \sum_{\substack{k,l,m,n,\mu,\nu \\ c,d,e,f,\gamma,\delta}} \mathcal{A}^{ij}\mathcal{A}_{ab} \frac{\lambda^{ik}_{mn}\tilde{V}^{mn}_{ef}\tilde{V}^{ef}_{ac}\tilde{V}^{cd}_{kl}\tilde{V}^{\mu\nu}_{\gamma\delta}\tilde{V}^{\gamma\delta}_{db}\lambda^{lj}_{\mu\nu}}{D^{ij}_{ab}D^{ijkl}_{abcd}D^{ik\mu\nu}_{ac\gamma\delta}D^{mn\mu\nu}_{ef\gamma\delta}D^{\mu\nu}_{\gamma\delta}} \frac{\bar{n}_c\bar{n}_d\bar{n}_e\bar{n}_f\bar{n}_\gamma\bar{n}_\delta}{n^i n^j} .$$

The above expressions are reminiscent of the MBPT expansion of single determinantal CC amplitudes (hence of the wave function) and of the energy (cf. Eqs. (2.3.17), (2.3.18) and (2.3.19)). For example, Eq. (3.3.16) is just like the MBPT(2) energy, with dressed vertices instead of bare ones. While the analogy is clearly formal − none of these expansions are backed up by any partitioning of H −, it is still instructive to talk about the series in a "pseudo-perturbative" sense. Diagrams containing n dressed interaction vertices are said to be of order n.

One of the new features of this expansion compared with MBPT is the appearance of arbitrary K-body vertices. In general, all t_K contain a first-order diagram due to $\tilde{W}_K$, hence, they can all contribute to the second-order correction of ΔE. In a CCSD scheme, Eqs. (3.3.14) and (3.3.15) show all diagrams contributing to t_1 and t_2 up to second order. At variance with the single determinantal theory, inclusion of higher excitations would bring further terms.

The other unusual feature of the expansion is the explicit appearance of cumulants connecting interaction vertices. We can see that strong electron correlation encoded in cumulants manifests itself in two ways: on one hand it provides dressing for bare one- and two-body vertices and creates new K-body vertices for $K \geq 3$, while on the other hand it also provides couplings between these vertices beyond mean-field level pair-contractions. This observation leads us to partition strong correlation into intravertex (implicit) and intervertex (explicit) parts. This interpretation is also in agreement with the scattering picture assigned to dressed vertices and cumulants in Appendix D. From a more practical point of view, thinking with intra- and intervertex correlation may help in developing approximation schemes for the general ic-MRCC theory (see Sec. 3.3.3).

3.3.2 Singlet spin-adaptation

For computational purposes, the ic-MRCC formulae need to be spin-adapted. Here we show how to obtain the spin-free form of the equations for a singlet system. Spatial and spin indices are written separately in this section, latin indices denote spatial orbitals.

A direct algebraic route to spin-free expressions is provided by the spin-summed operators of the UGA framework (see Sec. 2.1.7). Using the results of Appendix C, we can write the spin orbital amplitudes as

$$t^{i\sigma}_{a\sigma'} = \delta^{\sigma}_{\sigma'}t^i_a \quad , \quad t^{i\sigma_1 j\sigma_2}_{a\sigma'_1 b\sigma'_2} = \delta^{\sigma_1}_{\sigma'_1}\delta^{\sigma_2}_{\sigma'_2}t^{ij}_{ab} - \delta^{\sigma_1}_{\sigma'_2}\delta^{\sigma_2}_{\sigma'_1}t^{ij}_{ba} \quad , \ \ldots \ ,$$

where amplitudes $t_a^i, t_{ab}^{ij}, ...$ only depend on spatial degrees of freedom. The spin indices of the cluster operator in Eq. (3.1.1) can now be easily summed up to yield

$$T = \sum_K T_K = \sum_K \frac{1}{K!} \sum_{\substack{i_1,...,i_K \\ a_1,...,a_K}} t_{a_1...a_K}^{i_1...i_K} E_{i_1...i_K}^{a_1...a_K} = \sum_{\substack{i \\ a}} t_a^i E_i^a + \frac{1}{2} \sum_{\substack{i,j \\ a,b}} t_{ab}^{ij} E_{ij}^{ab} + ... \ .$$

The spin-summed form of the amplitude equation can be obtained either by projecting the Schrödinger equation with $\langle\Phi| :E_{a_1...a_K}^{i_1...i_K}:$, or by contracting the spin indices of the respective spin orbital formula in Eq. (3.1.7):

$$\bar{\Omega}_{a_1...a_K}^{i_1...i_K} = \sum_{\sigma_1,...,\sigma_K} \langle :a_{a_1\sigma_1...a_K\sigma_K}^{i_1\sigma_1...i_K\sigma_K}: \mathcal{H} :\exp(T):\rangle_C = \langle :E_{a_1...a_K}^{i_1...i_K}: \mathcal{H} :\exp(T):\rangle_C = 0 \ . \tag{3.3.17}$$

With the results of Appendix C, it is in principle straightforward to convert the spin orbital ic-MRCC equations to spin-summed ones. The advantage of this approach is its immediate applicability to any excitation level of CC. Its major drawback is that it makes the derivations rather tedious, and may result in very complicated formulae, as performing the spin-summation for a product of cumulants, or dressed vertices often multiplies the number of terms (cf. Eq. (2.1.62)).

We proceed to discuss a somewhat different but equivalent approach for CCSD. An antisymmetric, spin rotational invariant K-body tensor χ can be written as

$$\chi_{q_1\sigma_1p_2\sigma_2...q_K\sigma_K}^{p_1\sigma_1p_2\sigma_2...p_K\sigma_K} = \delta_{\sigma_1'}^{\sigma_1}\delta_{\sigma_2'}^{\sigma_2}...\delta_{\sigma_K'}^{\sigma_K}\chi_{q_1q_2...q_K}^{p_1p_2...p_K} - \delta_{\sigma_2'}^{\sigma_1}\delta_{\sigma_1'}^{\sigma_2}...\delta_{\sigma_K'}^{\sigma_K}\chi_{q_2q_1...q_K}^{p_1p_2...p_K} + ... - \delta_{\sigma_K'}^{\sigma_1}\delta_{\sigma_2'}^{\sigma_2}...\delta_{\sigma_1'}^{\sigma_K}\chi_{q_Kq_2...q_1}^{p_1p_2...p_K}$$

that is, in a factorized form with spatial part $\chi_{q_1...q_K}^{p_1...p_K}$ and diagonal spin part. The spatial part is symmetric under simultaneous permutations of upper and lower indices.
For one- and two-body quantities particularly (like t_1, F, or $t_2, \tilde{V}, \lambda_2$), we have (cf. Eq. (C.0.14))

$$\chi_{q\sigma'}^{p\sigma} = \delta_{\sigma'}^{\sigma}\chi_q^p \quad , \quad \tilde{\chi}_{q\sigma_1's\sigma_2'}^{p\sigma_1r\sigma_2} = \delta_{\sigma_1'}^{\sigma_1}\delta_{\sigma_2'}^{\sigma_2}\chi_{qs}^{pr} - \delta_{\sigma_2'}^{\sigma_1}\delta_{\sigma_1'}^{\sigma_2}\chi_{sq}^{pr} \ .$$

A simple inversion immediately reveals $\chi_q^p = \chi_{q\alpha}^{p\alpha}$ and $\chi_{qs}^{pr} = \chi_{q\alpha s\beta}^{p\alpha r\beta}$.
Just like amplitudes, cumulants or dressed vertices, the sum of connected amplitude diagrams, $\Omega_{a_1\sigma_1'...a_K\sigma_K'}^{i_1\sigma_1...i_K\sigma_K}$ is invariant under spin rotations, hence in the case of CCSD we can write

$$\Omega_{a\sigma'}^{i\sigma} = \delta_{\sigma'}^{\sigma}\Omega_a^i \quad , \quad \Omega_{a\sigma_1'b\sigma_2'}^{i\sigma_1j\sigma_2} = \delta_{\sigma_1'}^{\sigma_1}\delta_{\sigma_2'}^{\sigma_2}\Omega_{ab}^{ij} - \delta_{\sigma_2'}^{\sigma_1}\delta_{\sigma_1'}^{\sigma_2}\Omega_{ba}^{ij} \ ,$$

with $\Omega_a^i = \Omega_{a\alpha}^{i\alpha}$ and $\Omega_{ab}^{ij} = \Omega_{a\alpha b\beta}^{i\alpha j\beta}$. This suggests that all information of the T_1 and T_2 amplitude equations is contained in equations

$$\Omega_a^i = 0 \quad , \quad \Omega_{ab}^{ij} = 0 \ , \tag{3.3.18}$$

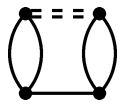
corresponding to projection of the Schrödinger equation by $\langle\Phi| :a_{a\alpha}^{i\alpha}:$ and $\langle\Phi| :a_{a\alpha b\beta}^{i\alpha j\beta}:$. The fact that only one particular value of spin or spin pairs (α or α, β) is required in each projection, makes these amplitude equations spin-free. This alternative approach is equivalent to the spin-summed method described above; they are simply related by

$$\bar{\Omega}_a^i = 2\Omega_a^i \quad , \quad \bar{\Omega}_{ab}^{ij} = 4\Omega_{ab}^{ij} - 2\Omega_{ba}^{ij} \ . \tag{3.3.19}$$

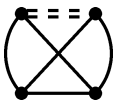
Let us suppose first that − as a consequence of some approximation, as described in the next section − at most two-body quantities appear in the CCSD equations. In this case expressing Ω_a^i and Ω_{ab}^{ij} can be easily achieved by diagrammatic methods, vertices representing spin-free quantities (collectively denoted by χ_q^p, χ_{qs}^{pr}).
Explicit treatment of spin means that direct and exchange parts of an expression cannot be bundled in the same antisymmetric diagram anymore: one must separately draw all Goldstone diagrams for each possible exchange process. Spin-free terms can be then directly obtained: the only difference in the interpretation is that numerical factors are no longer determined by equivalent lines and diagram parts. In the spin-free formalism, rules A5 and A6 of Sec. 3.3 are replaced by rules S5 and S6:

- S5: Assign a factor of $2^{L'}$ for diagrams with L' loops (quasi-loops excluded) and a factor of $1/K!$ for all diagram parts with a K-fold vertical symmetry.
- S6: Symmetrize upper-lower indices ia and jb.

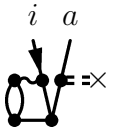
A few examples include



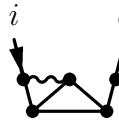
$$= 2\sum_{\substack{k,l\\c,d}} t_{cd}^{kl} V_{kl}^{cd} n^k n^l \bar{n}_c \bar{n}_d \quad ,$$



$$= -\sum_{\substack{k,l\\c,d}} t_{dc}^{kl} V_{kl}^{cd} n^k n^l \bar{n}_c \bar{n}_d \ ,$$



$$= 2n^i \bar{n}_a \sum_{\substack{k,l\\c,d}} \lambda_{kl}^{id} t_{cd}^{kl} F_a^c \bar{n}_c \quad ,$$



$$= -n^i \bar{n}_a \sum_{\substack{k,l\\c,d}} \lambda_{kl}^{id} t_{dc}^{kl} F_a^c \bar{n}_c \ .$$

This reduction of spin degrees of freedom is rather straightforward as long as $K < 3$ for the ranks of all many-body quantities. For higher ranks, diagrammatic representation of spin-free quantities becomes less trivial. Let us take a look at the $K = 3$ case:

$$\begin{aligned}\chi_{q\sigma_1' s\sigma_2' u\sigma_3'}^{p\sigma_1 r\sigma_2 t\sigma_3} = &+ \delta_{\sigma_1'}^{\sigma_1}\delta_{\sigma_2'}^{\sigma_2}\delta_{\sigma_3'}^{\sigma_3}\chi_{qsu}^{prt} + \delta_{\sigma_3'}^{\sigma_1}\delta_{\sigma_1'}^{\sigma_2}\delta_{\sigma_2'}^{\sigma_3}\chi_{uqs}^{prt} + \delta_{\sigma_2'}^{\sigma_1}\delta_{\sigma_3'}^{\sigma_2}\delta_{\sigma_1'}^{\sigma_3}\chi_{suq}^{prt} \\ &- \delta_{\sigma_2'}^{\sigma_1}\delta_{\sigma_1'}^{\sigma_2}\delta_{\sigma_3'}^{\sigma_3}\chi_{squ}^{prt} - \delta_{\sigma_1'}^{\sigma_1}\delta_{\sigma_3'}^{\sigma_2}\delta_{\sigma_2'}^{\sigma_3}\chi_{qus}^{prt} - \delta_{\sigma_3'}^{\sigma_1}\delta_{\sigma_2'}^{\sigma_2}\delta_{\sigma_1'}^{\sigma_3}\chi_{usq}^{prt}\end{aligned} \tag{3.3.20}$$

Unlike in the $K = 1, 2$ case, this formula cannot be inverted to yield a unique expression for χ_{qsu}^{prt}. To circumvent this problem, let us introduce the spin-summed tensor

$$\bar{\chi}_{qsu}^{prt} = \sum_{\sigma_1,\sigma_2,\sigma_3} \chi_{q\sigma_1 s\sigma_2 u\sigma_3}^{p\sigma_1 r\sigma_2 t\sigma_3} \ .$$

Upon symmetrizing this tensor with respect to independent permutations of upper-lower indices, we get zero (just like in Eq. (C.0.8)):

$$\bar{\chi}_{qsu}^{prt} + \bar{\chi}_{uqs}^{prt} + \bar{\chi}_{suq}^{prt} + \bar{\chi}_{squ}^{prt} + \bar{\chi}_{qus}^{prt} + \bar{\chi}_{usq}^{prt} = 0 \ . \tag{3.3.21}$$

Hence, the six components of $\bar{\chi}_3$ are linearly dependent for any given p, q, r, s, t, u. Performing the spin-summation of Eq. (3.3.20) for all six permutations gives a redundant system of equations:

$$\begin{bmatrix} A_+ & A_- \\ A_- & A_+ \end{bmatrix} \begin{bmatrix} \chi_+ \\ \chi_- \end{bmatrix} = \begin{bmatrix} \bar{\chi}_+ \\ \bar{\chi}_- \end{bmatrix} ,$$

with

$$\chi_+ = \begin{bmatrix} \chi^{prt}_{qsu} \\ \chi^{prt}_{uqs} \\ \chi^{prt}_{suq} \end{bmatrix} , \quad \bar{\chi}_+ = \begin{bmatrix} \bar{\chi}^{prt}_{qsu} \\ \bar{\chi}^{prt}_{uqs} \\ \bar{\chi}^{prt}_{suq} \end{bmatrix} , \quad \chi_- = \begin{bmatrix} \chi^{prt}_{squ} \\ \chi^{prt}_{qus} \\ \chi^{prt}_{usq} \end{bmatrix} , \quad \bar{\chi}_- = \begin{bmatrix} \bar{\chi}^{prt}_{squ} \\ \bar{\chi}^{prt}_{qus} \\ \bar{\chi}^{prt}_{usq} \end{bmatrix} ,$$

and

$$A_+ = \begin{bmatrix} 8 & 2 & 2 \\ 2 & 8 & 2 \\ 2 & 2 & 8 \end{bmatrix} , \quad A_- = \begin{bmatrix} -4 & -4 & -4 \\ -4 & -4 & -4 \\ -4 & -4 & -4 \end{bmatrix} .$$

Solving this equation by pseudo-inversion and using Eq. (3.3.21), a simple formula can be found for χ_+, χ_-:

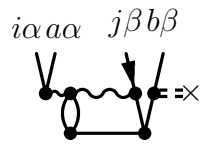
$$\begin{bmatrix} \chi_+ \\ \chi_- \end{bmatrix} = \begin{bmatrix} C & \emptyset \\ \emptyset & C \end{bmatrix} \begin{bmatrix} \bar{\chi}_+ \\ \bar{\chi}_- \end{bmatrix} , \quad C = \frac{1}{24} \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix} .$$

All permutations have the same form, so we can generally write

$$\chi^{prt}_{qsu} = \frac{1}{24} \left(3\bar{\chi}^{prt}_{qsu} - \bar{\chi}^{prt}_{uqs} - \bar{\chi}^{prt}_{suq} \right) . \tag{3.3.22}$$

This relationship is important, as χ^{prt}_{qsu} is the quantity that naturally arises from diagram rules, while $\bar{\chi}^{prt}_{qsu}$ is the one that has a direct algebraic form. We can use the diagram rules discussed above for three-body quantities as well, with χ^{prt}_{qsu} appearing during interpretation of diagrams.

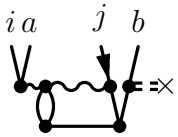
As a simple demonstration for three-body cumulants, let us evaluate the following spin orbital diagram and do the spin-summation manually:



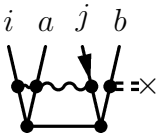
$$\begin{aligned} &= \frac{(-1)^{0+3-3}}{2!} \mathcal{A}_{a\alpha b\beta} \left(\bar{n}_b \sum_{\substack{k,l \\ c,d}} \sum_{\sigma_1,\ldots,\sigma_4} \lambda^{i\alpha c\sigma_2 j\beta}_{a\alpha k\sigma_1 l\sigma_3} t^{k\sigma_1 l\sigma_3}_{c\sigma_2 d\sigma_4} F^{d\sigma_4}_{b\beta} \bar{n}_d \right) \\ &= \mathcal{S}^{ij}_{ab} \left(\bar{n}_b \sum_{\substack{k,l \\ c,d}} \sum_{\sigma} \lambda^{i\alpha j\beta c\sigma}_{a\alpha l\beta k\sigma} t^{kl}_{cd} F^{d}_{b} \bar{n}_d \right) \\ &= \mathcal{S}^{ij}_{ab} \left(\bar{n}_b \sum_{\substack{k,l \\ c,d}} \frac{1}{6} \left[2\Lambda^{ijc}_{alk} + \Lambda^{ijc}_{lak} \right] t^{kl}_{cd} F^{d}_{b} \bar{n}_d \right) . \end{aligned}$$

The last equality follows from Eq. (2.1.59) and the spin partial trace property of cumulants. Symbol $\mathcal{S}^{ij}_{ab}$ symmetrizes the expression with respect to simultaneous permutation of ia and

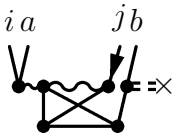
jb.
Let us now obtain the same term using spin-free diagram rules:



$$= 2\mathcal{S}_{ab}^{ij}\left(\bar{n}_b \sum_{\substack{k,l\\c,d}} \lambda_{akl}^{icj} t_{cd}^{kl} F_b^d \bar{n}_d\right) ,$$



$$= -\mathcal{S}_{ab}^{ij}\left(\bar{n}_b \sum_{\substack{k,l\\c,d}} \lambda_{kal}^{icj} t_{cd}^{kl} F_b^d \bar{n}_d\right) ,$$



$$= -\mathcal{S}_{ab}^{ij}\left(\bar{n}_b \sum_{\substack{k,l\\c,d}} \lambda_{alk}^{icj} t_{cd}^{kl} F_b^d \bar{n}_d\right) .$$

The two approaches indeed lead to the same result, according to Eqs. (3.3.21) and (3.3.22):

$$2\lambda_{akl}^{icj} - \lambda_{kal}^{icj} - \lambda_{alk}^{icj} = \frac{1}{6}\left[2\Lambda_{akl}^{icj} + \Lambda_{lka}^{icj}\right] .$$

Although the calculations are getting more and more tedious, one could follow the same recipe to find the diagrammatic representation and interpretation of higher rank spin-free quantities.

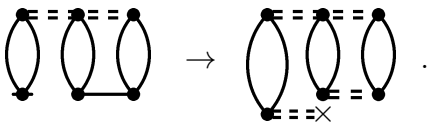
In principle, the two routes to singlet spin-adaptation outlined above are equivalent, and one is free to choose between Eqs. (3.3.17) and (3.3.18). Transparent diagrammatic formulation is an obvious advantage of the latter. However, in practice, solving the amplitude equations is not that simple, since linear dependence of the projecting excited states must be treated as well. The handling of this redundancy turns out to be more straightforward in the UGA-like scheme, where the overlap of spin-summed states $:E_{i_1\ldots i_K}^{a_1\ldots a_K}: |\Phi\rangle$ is to be examined. For this reason, we are adopting a "hybrid" approach: in the next section, we will derive spin-free approximations to ic-MRCC using diagrams, and with Eq. (3.3.18) in mind. These equations are then converted to UGA formulae with the help of Eq. (3.3.19). See Chapter 5 for details on the treatment of redundancy.

3.3.3 Approximations to the general formalism

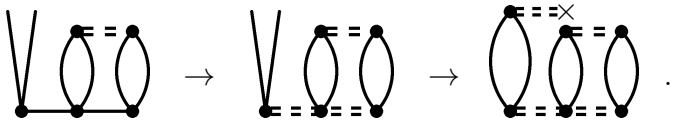
The usefulness of the general theory outlined in previous sections is seriously hindered by the extreme number of diagrams even in the CCSD scheme: none of the above ic-MRCC equations are truncated at any finite order of T. Diagrammatically speaking, this is

because an arbitrary number of amplitude vertices can be connected to sufficiently high rank cumulants and dressed vertices, while still leaving $2K$ legs free. Thinking of the discussion of Sec. 2.1.4, in general, we cannot say that cumulants become negligible at a rank that would allow for substantial simplification of the equations. It is the assumption on the small value of the amplitudes that provides a basis for arguing the neglect of higher orders in the wave operator.
In this section, a couple of approximation schemes are proposed for ic-MRCCSD. Latin indices again denote spatial orbitals.

Based on Sec. 3.3.1, our approximations can go along basically two paths: neglecting intravertex or intervertex correlation effects. The first option can be reached by leaving out diagrams containing high rank dressed interaction vertices. Since the Hamiltonian contains only one- and two-body parts, it seems reasonable to keep vertices F and $\tilde{V}$ only, while discarding all $\tilde{W}_K$ for $K \geq 3$. Examples of omitted diagrams include the fourth and the last two of diagrams of Eq. (3.3.7), and all but the first three diagrams of Eq. (3.3.8).
While this approximation based on a Hamiltonian analogy is plausible, it is nevertheless an assumption that we cannot prove in a rigorous way. We can argue that the direct effect of odd-rank dressed vertices on the energy is small. This is simply due to the fact that each of the odd rank dressed vertices must be connected to at least one t_1 in Eq. (3.3.8), whose effect can be safely considered small (given that orbitals of $|\Phi\rangle$ are optimized, which is the typical case). Some further rationalization for the GVB wave function is given in Sec. 6.1. Note that indirect effect of high rank vertices on the energy (coming from their contributions to amplitudes) cannot be estimated this way.
From a certain point of view, neglecting some diagrams with high rank vertices is actually recommended to avoid unphysical results. Let us show on a simple example why one must be cautious with inconsistent approximations regarding dressed vertices. In the general ic-MRCCSD theory, we could obtain a third-order correction to the energy by substituting the leading order contributions of t_1 and t_2 to the third diagram of Eq. (3.3.8):



Note that in a general setting using complex spin orbitals, this energy diagram is complex valued. Its unphysical imaginary part is cancelled by its conjugate diagram, which, however, can only be introduced via triple excitations:



In a CCSD scheme, keeping the three-body vertex in the energy formula could thus generate a spurious complex valued energy correction. Even when using real orbitals, keeping this vertex would correspond to keeping only one half of the respective energy term, which seems

rather ad-hoc. It might be better to drop the three-body vertex from the energy formula. To be fair, this is not just an MR issue: one could obtain such strange results even in single determinantal CC if unbalanced approximations were introduced (non-variational CCS being a simple example).

Unfortunately, the general amplitude equations remain quite untractable even after excluding high rank dressed vertices. In the next step of approximations, some of the intervertex correlation must be neglected. The simplest possible choice is to drop this part of correlation completely (thus erasing cumulant couplings between dressed vertices), and treat only intravertex correlation hidden within F and $\tilde{V}$. While this approximation (to be referred to as idCCSD after interaction dressing) is rather crude, it is quite appealing from a formal point of view. It provides a natural truncation for the amplitude equations.
With explicitly cumulant-dependent diagrams removed, all remaining spin orbital diagrams formally resemble those of single determinantal CC (see Eqs. (2.3.7) and (2.3.8)), the only difference being the replacement of bare vertices by their dressed counterparts. Algebraic form of the equations can then be easily found; spin free form of Ω_a^i and Ω_{ab}^{ij} can also be directly obtained with the diagrammatic techniques of the previous section. The UGA form of the spin-free idCCSD equations read

$$\Delta E = 2\sum_{\substack{k\\c}} F_k^c t_c^k n^k \bar{n}_c + \sum_{\substack{k,l\\c,d}} (2V_{kl}^{cd} - V_{lk}^{cd}) \tau_{cd}^{kl} n^k n^l \bar{n}_c \bar{n}_d \ , \tag{3.3.23}$$

and

$$n^i \bar{n}_a \omega_a^i = 0 \quad , \quad n^i n^j \bar{n}_a \bar{n}_b (2\omega_{ab}^{ij} - \omega_{ba}^{ij}) = 0 \ ,$$

where occupation factors belonging to external lines were separated for notational convenience. Most of the terms can be collected into intermediate quantities similar to the ones introduced by Scuseria et al. [66] in the context of single determinantal CCSD:

$$\begin{aligned}\omega_a^i =& F_a^i - 2\sum_{\substack{k\\c}} F_k^c t_a^k t_c^i n^k \bar{n}_c + \sum_c H_a^c t_c^i \bar{n}_c - \sum_k H_k^i t_a^k n^k + \sum_{\substack{k\\c}} H_k^c \left(2t_{ca}^{ki} - t_{ac}^{ki} + t_c^i t_a^k\right) n^k \bar{n}_c \\ &+ \sum_{\substack{k\\c}} \left(2V_{ka}^{ci} - V_{ak}^{ci}\right) t_c^k n^k \bar{n}_c + \sum_{\substack{k\\c,d}} \left(2V_{ka}^{cd} - V_{ak}^{cd}\right) \tau_{cd}^{ki} n^k \bar{n}_c \bar{n}_d - \sum_{\substack{k,l\\c}} \left(2V_{kl}^{ci} - V_{lk}^{ci}\right) \tau_{ca}^{kl} n^k n^l \bar{n}_c \ ,\end{aligned} \tag{3.3.24}$$

$$\begin{aligned}\omega_{ab}^{ij} =& \mathcal{S}_{ab}^{ij} \Bigg[\frac{1}{2} V_{ab}^{ij} + \sum_c G_a^c t_{cb}^{ij} \bar{n}_c - \sum_k G_k^i t_{ab}^{kj} n^k + \frac{1}{2} \sum_{k,l} A_{kl}^{ij} \tau_{ab}^{kl} n^k n^l + \frac{1}{2} \sum_{c,d} B_{ab}^{cd} \tau_{cd}^{ij} \bar{n}_c \bar{n}_d \\ &+ \sum_c \left(V_{ab}^{ic} - \sum_k V_{kb}^{ic} t_a^k n^k \right) t_c^j \bar{n}_c - \sum_k \left(V_{ak}^{ij} + \sum_c V_{ak}^{ic} t_c^j \bar{n}_c \right) t_b^k n^k \\ &+ \sum_{\substack{k\\c}} J_{ak}^{ic} \left(2t_{cb}^{kj} - t_{bc}^{kj}\right) n^k \bar{n}_c - \sum_{\substack{k\\c}} K_{ka}^{ic} t_{cb}^{kj} n^k \bar{n}_c - \sum_{\substack{k\\c}} K_{kb}^{ic} t_{ac}^{kj} n^k \bar{n}_c \Bigg] \ .\end{aligned} \tag{3.3.25}$$

The following notations and intermediate quantities were introduced:

$$\tau_{ab}^{ij} = t_{ab}^{ij} + t_a^i t_b^j \ ,$$

$$T_{ab}^{ij} = \frac{1}{2} t_{ab}^{ij} + t_a^i t_b^j \ ,$$

$$H_k^i = F_k^i + \sum_{\substack{l \\ c,d}} \left(2V_{kl}^{cd} - V_{lk}^{cd} \right) \tau_{cd}^{il} n^l \bar{n}_c \bar{n}_d \ ,$$

$$H_a^c = F_a^c - \sum_{\substack{k,l \\ d}} \left(2V_{kl}^{cd} - V_{lk}^{cd} \right) \tau_{ad}^{kl} n^k n^l \bar{n}_d \ ,$$

$$H_k^c = F_k^c + \sum_{\substack{l \\ d}} \left(2V_{kl}^{cd} - V_{lk}^{cd} \right) t_d^l n^l \bar{n}_d \ ,$$

$$G_k^i = H_k^i + \sum_c F_k^c t_c^i \bar{n}_c + \sum_{\substack{l \\ d}} \left(2V_{kl}^{id} - V_{lk}^{id} \right) t_d^l n^l \bar{n}_d \ ,$$

$$G_a^c = H_a^c - \sum_k F_k^c t_a^k n^k + \sum_{\substack{l \\ d}} \left(2V_{al}^{cd} - V_{la}^{cd} \right) t_d^l n^l \bar{n}_d \ ,$$

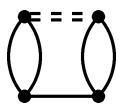
$$A_{kl}^{ij} = V_{kl}^{ij} + \sum_d V_{kl}^{id} t_d^j \bar{n}_d + \sum_c V_{kl}^{cj} t_c^i \bar{n}_c + \sum_{c,d} V_{kl}^{cd} \tau_{cd}^{ij} \bar{n}_c \bar{n}_d \ ,$$

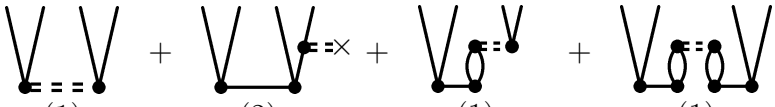
$$B_{ab}^{cd} = V_{ab}^{cd} - \sum_l V_{al}^{cd} t_b^l n^l - \sum_k V_{kb}^{cd} t_a^k n^k \ ,$$

$$J_{ak}^{ic} = V_{ak}^{ic} - \sum_l V_{lk}^{ic} t_a^l n^l + \sum_d V_{ak}^{dc} t_d^i \bar{n}_d - \sum_{\substack{l \\ d}} V_{kl}^{cd} T_{da}^{il} n^l \bar{n}_d + \frac{1}{2} \sum_{\substack{l \\ d}} \left(2V_{kl}^{cd} - V_{lk}^{cd} \right) t_{ad}^{il} n^l \bar{n}_d \ ,$$

$$K_{ka}^{ic} = V_{ka}^{ic} - \sum_l V_{kl}^{ic} t_a^l n^l + \sum_d V_{ka}^{dc} t_d^i \bar{n}_d - \sum_{\substack{l \\ d}} V_{lk}^{cd} T_{da}^{il} n^l \bar{n}_d \ .$$

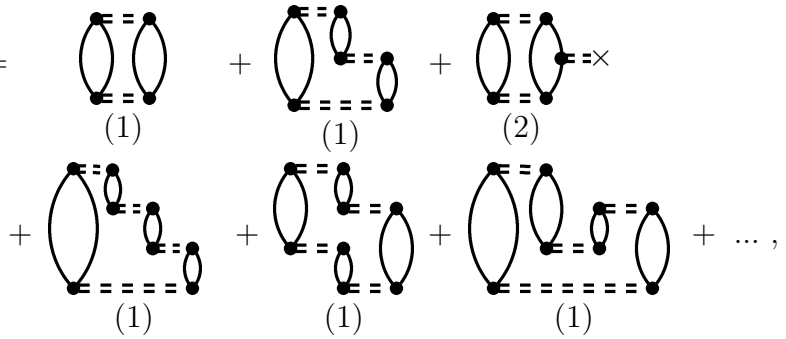
It is important to note that dressed vertices naturally generalize the concept of certain diagram classes from single determinantal CC. Therefore, many well-known approximations of CC can be immediately extended to MR in the idCCSD framework.
Among the classes of ring, ladder, and mosaic diagrams, ring diagrams are probably the most important. Ground state energy in the Random Phase Approximation (RPA) [29] is given by the all-order inclusion of direct ring diagrams (exchange processes completely neglected). In the idCCSD framework this is generalized as

$\Delta E_{RPA} =$  ,

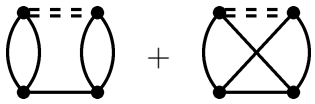
$0 =$ 

all of the above diagrams being spin-free. The only non-ring diagrams in the above equations

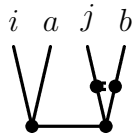
are the ones with $h-h$ and $p-p$ type one-body vertices (represented by the second skeleton). The all-order summation of diagrams during the iteration takes the form

$\Delta E_{RPA} =$  $+ \ \dots \ ,$

particle-hole pairs being created and destroyed at each intermediate state of dressed ring diagrams. Note that keeping only direct ring terms (while neglecting exchange) breaks the antisymmetry of the wave function. Some exchange effects can be added to the RPA energy with the complete CCD energy formula

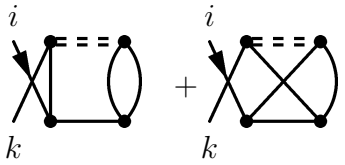
$\Delta E_{CCD} =$  , (3.3.26)

which can be viewed as an MR generalization of RPA with second-order screened exchange (SOSEX) [145, 146].
This model can be further upgraded either by adding exchange ring contributions to the amplitude equation, such as



$= -\mathcal{S}_{ab}^{ij}\left(n^i n^j \bar{n}_a \bar{n}_b \sum_{\substack{k \\ c}} t_{ac}^{ik} V_{kb}^{jc} n^k \bar{n}_c\right) ,$

or by mosaic diagram parts providing a screening to the $h-h$ one-body vertex (and similarly to the $p-p$ one):



$= \sum_{\substack{l \\ c,d}} \left(2V_{kl}^{cd} - V_{lk}^{cd}\right) t_{cd}^{il} n^l \bar{n}_c \bar{n}_d \ .$

Occupation factors of external lines were omitted. Adding these terms results in amplitude equation $0 = 2\breve{\omega}_{ab}^{ij} - \breve{\omega}_{ba}^{ij}$ with

$$\breve{\omega}_{ab}^{ij} = \mathcal{S}_{ab}^{ij}\left[\frac{1}{2}V_{ab}^{ij} + \sum_c G_a^c t_{cb}^{ij} \bar{n}_c - \sum_k G_k^i t_{ab}^{kj} n^k + \sum_{\substack{k \\ c}} J_{ak}^{ic}\left(2t_{cb}^{kj} - t_{bc}^{kj}\right) n^k \bar{n}_c - \sum_{\substack{k \\ c}} K_{ka}^{ic} t_{cb}^{kj} n^k \bar{n}_c\right] , \tag{3.3.27}$$

where intermediate quantities are the same as in idCCSD, only without t_1 contributions. In a single determinantal setting, this model is known as rmCCD (rm standing for ring and mosaic). Inclusion of mosaic terms has been shown to often improve the quality of ring-based approximations in simple systems [70]. Note that the so-called crossed ring term (the last term of Eq. (3.3.25)) is traditionally left out in rmCCD.
With dressed vertices, one could analogously define the MR generalization of other CC approximations, such as ladder diagram approximations; however, these will not be considered here.

It has to be said that development of approximations to this theory took quite a different route from how it is presented now. Our first model (now referred to as rCCD(UGA)) was a purely algebraic ring approximation to the UGA ic-MRCCD amplitude equation without considering diagrams at all [138]. In the algebraic ring approximation all terms are discarded where the original "composite indices" ia and jb of the projecting excitations appear "broken" on the amplitudes: that is, terms like $A^{ic}_{ak}t^{jk}_{bc}$ or $A^{ic}_{ka}t^{jk}_{bc}$ are kept, while terms like $A^{ic}_{ak}t^{jk}_{cb}$ are not. The only exceptions are the $p-p$ and $h-h$ one-body vertices. The energy formula is taken from idCCD (cf. Eq. (3.3.26)):

$$\Delta E_{CCD} = \sum_{\substack{k,l\\c,d}} (2V^{cd}_{kl} - V^{cd}_{lk}) t^{kl}_{cd} n^k n^l \bar{n}_c \bar{n}_d \ ,$$

while the amplitude equation reads

$$0 = 2\mathcal{S}^{ij}_{ab}\left[\frac{1}{2}V^{ij}_{ab} + \sum_c F^c_a t^{ij}_{cb}\bar{n}_c - \sum_k F^i_k t^{kj}_{ab} n^k + \sum_{\substack{k\\c}} \left(2\hat{J}^{ic}_{ak} - V^{ic}_{ka}\right) t^{kj}_{cb} n^k \bar{n}_c\right] - V^{ij}_{ba} \ ,$$

with

$$\hat{J}^{ic}_{ak} = V^{ic}_{ak} + \frac{1}{2}\sum_{\substack{l\\d}} \left(2V^{cd}_{kl} - V^{cd}_{lk}\right) t^{il}_{ad} n^l \bar{n}_d \ .$$

Up to this point, intervertex correlation has been neglected in all models discussed. As a very simple model exhibiting cumulant coupling between vertices, we consider the effect of the first and third skeletons in the second row of Eq. (3.3.7). These diagrams provide corrections to the $p-p$ and $h-h$ one-body vertex diagrams generating the energy dominator during iteration, hence, their role can be argued on the pseudo-perturbative grounds of Sec.

3.3.1. This model is called idCCSD+λF; the diagrams are easily evaluated to give

$$\begin{aligned}\hat{\omega}_{ab}^{ij} = \omega_{ab}^{ij} + \mathcal{S}_{ab}^{ij}\Bigg[& \sum_{k,l} \frac{\lambda_{kl}^{ij}}{n^i n^j} t_{ad}^{kl} F_b^d \bar{n}_d - \sum_{c,d} \frac{\lambda_{ab}^{cd}}{\bar{n}_a \bar{n}_b} t_{cd}^{il} F_l^j n^l + \\ & + \sum_{\substack{k \\ c,d}} \left(\frac{2\lambda_{ak}^{ic} - \lambda_{ka}^{ic}}{n^i \bar{n}_a} t_{cd}^{kj} F_b^d \bar{n}_d - \frac{\lambda_{ak}^{ic}}{n^i \bar{n}_a} t_{dc}^{kj} F_b^d \bar{n}_d - \frac{\lambda_{kb}^{ic}}{n^i \bar{n}_b} t_{dc}^{kj} F_a^d \bar{n}_d \right) - \\ & - \sum_{\substack{k,l \\ c}} \left(\frac{2\lambda_{ak}^{ic} - \lambda_{ka}^{ic}}{n^i \bar{n}_a} t_{cb}^{kl} F_l^j n^l - \frac{\lambda_{ak}^{ic}}{n^i \bar{n}_a} t_{bc}^{kl} F_l^j n^l - \frac{\lambda_{kb}^{ic}}{n^i \bar{n}_b} t_{ac}^{kl} F_l^j n^l \right) + \\ & + \sum_{\substack{k,l \\ c,d}} \frac{2\lambda_{akl}^{icj} - \lambda_{kal}^{icj} - \lambda_{alk}^{icj}}{n^i n^j \bar{n}_a} t_{cd}^{kl} F_b^d \bar{n}_d + \sum_{\substack{k,l \\ c,d}} \frac{2\lambda_{akb}^{icd} - \lambda_{kab}^{icd} - \lambda_{abk}^{icd}}{n^i \bar{n}_a \bar{n}_b} t_{cd}^{kl} F_l^j n^l \Bigg] ,\end{aligned}$$

ω_{ab}^{ij} denoting the diagrams of idCCSD. The diagrams generating the first term on the fourth line were evaluated in the previous section; the same notations are used.

Above we have defined four approximate ic-MRCC models: id-CCSD, idCCSD+λF, rmCCD and rCCD(UGA). Performance of these methods with a GVB reference function will be assessed in Chapter 6.1.

Chapter 4

Calculation of dispersion interactions

In this chapter, certain approximation schemes of ic-MRCC are examined in the context of dispersion interactions. Investigations are carried out for closed-shell electrically neutral subsystems separated by a large distance R. Following a brief overview of the literature on standard calculation techniques, the emergence of the London dispersion formula

$$E_{disp.} \approx -\frac{C_6}{R^6} \tag{4.0.1}$$

is demonstrated for the second-order idCCSD energy. In this approximation, we also show that dispersion coefficient C_6 can be expressed in terms of dressed quantities formally resembling the frequency-dependent polarizabilites (FDP) known from the analogous single determinantal theory. Finally, the general $E_{disp} \sim -1/R^6$ decay is argued to be valid for the complete idCCSD, as well as for the general ic-MRCC formalism.

Latin indices are reserved for spatial orbitals in this chapter; spin indices are shown explicitly whenever they are needed.

This chapter is based on Ref. [147].

4.1 Overview

Dispersion interaction arises from the charge density fluctuations of the subsystems, leading to weakly bound states between e.g. closed-shell partners even if they lack permanent electric moments [12, 148–150]. The importance of dispersion interaction in both chemical and biological systems cannot be overemphasized.
For two subsystems at a large separation distance R, the asymptotic dispersion energy in leading order is given by Eq. (4.0.1) provided that R is not large enough for electromagnetic retardation effects to become noticeable. The dispersion coefficient C_6 can be expressed in terms of the FDP of each subsystem. Providing an equally accurate description of dispersion both in the equilibrium and in the asymptotic regime has remained a challenge for electronic structure methods to this day.

Mean-field methods cannot even qualitatively account for dispersion; such phenomena are usually introduced via correlation corrections, obtained, for example by MBPT. Szabo and Ostlund showed [151] that a supermolecular MBPT(2) calculation gives rise to dispersion energy exhibiting the correct asymptotic behaviour (i.e. conforming Eq. (4.0.1)), but with C_6 calculated only at the level of so-called uncoupled HF [152]. An alternative of the supermolecule approach is the Symmetry-Adapted Perturbation Theory (SAPT) of Jeziorski et al. (see Ref. [153] and references cited therein), in which the intersystem interaction is treated as perturbation. The relationship between supermolecular MBPT and SAPT has been investigated in detail [154].
The quality of the supermolecular approach can be improved by applying corrections more involved than MBPT(2), e.g. the all-order inclusion of direct ring diagrams (and possibly certain kinds of exchange ring diagrams) [71]. The effect of these diagrams can be easily taken into account in RPA. Different variants of RPA (distinguished by the different degree of including exchange diagrams) improve quantitatively on the MBPT(2) results, but not all of them lead to a correct expression for the dispersion coefficient [151, 155]. Another way to include higher-order diagrams in the energy is through CC, of course. CCSD(T) yields essentially exact results for intermolecular interactions and is often used as benchmark in such cases. The CC model also provides a way to sum up only ring diagrams (as was shown in Sec. 3.3.3 for the MR case), leading to e.g. the rCCD approach [72], a computationally less demanding alternative to CCSD(T). The formal equivalence of certain kinds of RPA and rCCD methods has been demonstrated on many occasions [63,71,72,156]. Most RPA/rCCD-like methods do not give a satisfactory description of long range dispersion interactions [151, 157]: while they produce the correct algebraic decay, the dispersion coefficient C_6 can rarely be expressed in terms of FDP-s.

Even in MR methods, dispersion effects are typically introduced in a second step via correction methods. For example, strongly orthogonal geminal approaches alone (geminals being localized on subsystems) are no better at describing such phenomena than mean-field methods; however, quite accurate results can be obtained with linearized CC [54] or RPA-type [57, 58] corrections. Therefore, it is still interesting to investigate MR methods in the context of dispersion energy calculations, as will be done in the next section.

4.2 Derivation of the London formula in the leading order of idCCSD

Before turning to the calculation of dispersion effects, let us investigate the long range behaviour of dressed interaction vertices. Consider a supersystem composed of two closed shell, electrically neutral atoms $\mathbb{A}$ and $\mathbb{B}$, separated at a large distance R (large enough to neglect overlap of charge clouds). Orbitals are assumed to be localized on either $\mathbb{A}$ or $\mathbb{B}$. Far away from the subsystem they are localized on, orbitals fall off exponentially fast, which renders most of the intersystem bare vertices negligible. We shall make use of the multipole

expansion of the Coulomb potential:

$$\frac{1}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}=\frac{1}{|\boldsymbol{r}-\boldsymbol{R}|}=\frac{1}{R}+\frac{\boldsymbol{n}\cdot\boldsymbol{r}}{R^2}+\frac{3(\boldsymbol{n}\cdot\boldsymbol{r})^2-r^2}{2R^3}+\mathcal{O}(R^{-4})\ , \tag{4.2.1}$$

$\boldsymbol{n}$ being a unit vector parallel to the center distance $\boldsymbol{R}$, and $\boldsymbol{r}=\boldsymbol{r}_{\mathbb{A}1}-\boldsymbol{r}_{\mathbb{B}2}$ being the difference of electronic coordinates measured from the centers. Neither of the subsystems have any permanent electric moments (e.g. $\langle\phi_p|\boldsymbol{r}|\phi^p\rangle=0$). Using this fact and Eq. (4.2.1), one can show that all intersystem components of the Fockian f_p^q (say $p\in\mathbb{A}$, $q\in\mathbb{B}$) are essentially zero, and that intrasystem components (like $p,q\in\mathbb{A}$) do not contain any electrostatic contribution from the other subsystem. The only non-negligible components of intersystem electron-electron interaction v_{pr}^{qs} are those with $p,q\in\mathbb{A}$ and $r,s\in\mathbb{B}$.
In Sec. 2.1.4 we showed that all intersystem cumulants can be safely neglected for well separated singlet subsystems. This seriously reduces the number of important components for dressed vertices. Inspection of Eqs. (3.2.17) and (3.2.18) reveals that all intersystem components of $F_u^x=F_{u\alpha}^{x\alpha}$ are negligible, and that only case $u,x\in\mathbb{A}$, $w,y\in\mathbb{B}$ has to be considered for $V_{uw}^{xy}=\tilde{V}_{u\alpha w\beta}^{x\alpha y\beta}$. For the latter we find

$$\begin{aligned}\zeta_u\zeta_w\zeta^x\zeta^y V_{uw}^{xy}\approx&\zeta_u\zeta_w\zeta^x\zeta^y v_{uw}^{xy}-\zeta_w\zeta^y\sum_{p,q}v_{wp}^{yq}(2\lambda_{uq}^{xp}-\lambda_{qu}^{xp})-\zeta_u\zeta^x\sum_{p,q}v_{up}^{xq}(2\lambda_{wq}^{yp}-\lambda_{qw}^{yp})\\&+\sum_{p,q,r,s}v_{pr}^{qs}\ (2\lambda_{uq}^{xp}-\lambda_{qu}^{xp})(2\lambda_{ws}^{yr}-\lambda_{sw}^{yr})\\=&\zeta_u\zeta_w\zeta^x\zeta^y v_{uw}^{xy}-\frac{1}{2}\zeta_w\zeta^y\sum_{p,q}v_{wp}^{yq}\Lambda_{uq}^{xp}-\frac{1}{2}\zeta_u\zeta^x\sum_{p,q}v_{up}^{xq}\Lambda_{wq}^{yp}\\&+\frac{1}{4}\sum_{p,q,r,s}v_{pr}^{qs}\ \Lambda_{uq}^{xp}\Lambda_{ws}^{yr}\\=&\int\mathrm{d}^3r_1\int\mathrm{d}^3r_2\frac{\rho_u^x(\boldsymbol{r}_1)\rho_w^y(\boldsymbol{r}_2)}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}\ ,\end{aligned} \tag{4.2.2}$$

where in the last equality we used Eq. (2.1.55), and realized that the integrand can be factorized with

$$\rho_u^x(\boldsymbol{r})=\zeta_u\zeta^x\phi_u(\boldsymbol{r})\phi^x(\boldsymbol{r})-\frac{1}{2}\sum_{p,q}\Lambda_{uq}^{xp}\phi_p(\boldsymbol{r})\phi^q q(\boldsymbol{r})\ .$$

Case $u=i$, $x=a$ of $\rho_u^x(\boldsymbol{r})$ can be interpreted as a transition density between states $|\Phi\rangle$ and $:a_{i\alpha}^{a\alpha}:|\Phi\rangle$ (or $:E_i^a:|\Phi\rangle$):

$$\rho_i^a(\boldsymbol{r})=-\langle\hat{\rho}(\boldsymbol{r})\ :a_{i\alpha}^{a\alpha}:\rangle=-\frac{1}{2}\langle\hat{\rho}(\boldsymbol{r})\ :E_i^a:\rangle\ , \tag{4.2.3}$$

the spin-summed density operator being (cf. Appendix A)

$$\hat{\rho}(\boldsymbol{r})=\psi^\dagger(\boldsymbol{r},\alpha)\psi(\boldsymbol{r},\alpha)+\psi^\dagger(\boldsymbol{r},\beta)\psi(\boldsymbol{r},\beta)=\sum_{p,q}\phi^q(\boldsymbol{r})\phi_p(\boldsymbol{r})E_q^p\ .$$

Using the spin-summed form of the partial trace relation in Eq. (2.1.31)

$$\sum_p\Lambda_{up}^{xp}=-2\delta_u^x n_u\bar{n}^x\ ,$$

it is easy to see that $\rho_u^x(\boldsymbol{r})$ integrates to

$$\int \mathrm{d}^3 r \rho_u^x(\boldsymbol{r}) = \zeta_u \zeta^x \delta_u^x - \frac{1}{2}\sum_p \Lambda_{up}^{xp} = \delta_u^x \left(\zeta_u \zeta^x + n_u \bar{n}^x\right) \ . \tag{4.2.4}$$

For different u, x, we have

$$\int \mathrm{d}^3 r \rho_i^k(\boldsymbol{r}) = \delta_i^k n_i = \gamma_i^k \quad , \quad \int \mathrm{d}^3 r \rho_c^a(\boldsymbol{r}) = \delta_c^a \bar{n}^a = \eta_c^a \quad , \quad \int \mathrm{d}^3 r \rho_i^a(\boldsymbol{r}) = 0 \ .$$

The last equality can also be obtained from Eq. (4.2.3):

$$\int \mathrm{d}^3 r \rho_i^a(\boldsymbol{r}) = -\langle \hat{N} : a_{i\alpha}^{a\alpha} : \rangle = -N \langle : a_{i\alpha}^{a\alpha} : \rangle = 0 \ . \tag{4.2.5}$$

This equality ensures the cancellation of terms proportional to $1/R$ or $1/R^2$ in the multipole expansion of intersystem V_{ij}^{ab} (cf. Eqs. (4.2.1) and (4.2.2)):

$$V_{ij}^{ab} \approx \frac{1}{R^3} \boldsymbol{\mu}_i^a \boldsymbol{\tau} \boldsymbol{\mu}_j^b \quad (i, a \in \mathbb{A} \ , \ j, b \in \mathbb{B}) \ . \tag{4.2.6}$$

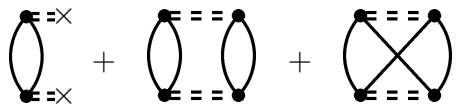
Here $\boldsymbol{\tau} = \boldsymbol{I} - 3\boldsymbol{n} \otimes \boldsymbol{n}$ is a symmetric matrix ($\otimes$ being the dyadic product of the 3-vectors), and $\boldsymbol{\mu}_i^a$ can be interpreted − in analogy with $\rho_i^a(\boldsymbol{r})$ − as the transition dipole moment with a cumulant correction:

$$n_i \bar{n}^a \boldsymbol{\mu}_i^a = \int \mathrm{d}^3 r \rho_i^a(\boldsymbol{r}) \boldsymbol{r} = \langle \hat{\boldsymbol{\mu}} : a_{i\alpha}^{a\alpha} : \rangle \ .$$

The dipole moment operator is

$$\hat{\boldsymbol{\mu}} = \sum_{p,q} (-\boldsymbol{r})_p^q E_q^p \quad , \quad (-\boldsymbol{r})_p^q = \int \mathrm{d}^3 r \phi_p(\boldsymbol{r}) (-\boldsymbol{r}) \phi^q(\boldsymbol{r}) \ .$$

Let us now calculate the second-order dispersion energy in the idCCSD framework. The derivation follows closely the MBPT(2) analysis of Szabo and Ostlund [151, 155], and can be thought of as an MR extension. Using the results of Secs. 3.3.1 and 3.3.2, we find

$E^{(2)} =$  $+$ $+$

$$= -2 \sum_{\substack{i \\ a}} \frac{F_a^i F_i^a}{\epsilon_a^i} n^i \bar{n}_a - \sum_{\substack{i,j \\ a,b}} \frac{V_{ab}^{ij} \left(2V_{ij}^{ab} - V_{ji}^{ab}\right)}{\epsilon_a^i + \epsilon_b^j} n^i n^j \bar{n}_a \bar{n}_b \ , \tag{4.2.7}$$

where $\epsilon_a^i = \bar{n}_a F_a^a - n^i F_i^i$. For large enough distances between the subsystems, $E^{(2)}$ can be written as

$$E^{(2)} \approx E_{\mathbb{A}}^{(2)} + E_{\mathbb{B}}^{(2)} + E_{disp}^{(2)} \ ,$$

where $E_{\mathbb{A}}^{(2)}$ and $E_{\mathbb{B}}^{(2)}$ are the second-order energies of the separate subsystems (all summation indices belonging to either $\mathbb{A}$ or $\mathbb{B}$), and $E_{disp}^{(2)}$ represents the dispersion interaction:

$$E_{disp}^{(2)} = \quad = -4 \sum_{\substack{i,a\in\mathbb{A}\\ j,b\in\mathbb{B}}} \frac{V_{ij}^{ab} V_{ab}^{ij}}{\epsilon_i^a + \epsilon_j^b} n_i \bar{n}^a n_j \bar{n}^b \ . \tag{4.2.8}$$

Vanishing of most of the intersystem dressed vertices was exploited. Note that the one-body and exchange parts of $E^{(2)}$ (the first and third diagrams of Eq. (4.2.7)) vanish for large distances and only the direct part contributes to dispersion.
Let us now assume the positivity of ϵ_i^a. This might not be as obvious as in the single determinantal theory, as ϵ_i^a is not a true energy denominator of perturbation theory. Nevertheless, the assumption is quite plausible on physical grounds; a further rationalization for the GVB wave function can be found in Appendix E. With $\epsilon_i^a > 0$, we can use the identity

$$\frac{1}{\epsilon + \epsilon'} = \frac{2}{\pi} \int_0^\infty \mathrm{d}\omega \frac{\epsilon\epsilon'}{(\epsilon^2 + \omega^2)(\epsilon'^2 + \omega^2)} \qquad (\epsilon, \epsilon' > 0) \ ,$$

and Eq. (4.2.2) to recast the dispersion formula into

$$E_{disp}^{(2)} = -\int \mathrm{d}^3 r_1 \int \mathrm{d}^3 r_2 \int \mathrm{d}^3 r_1' \int \mathrm{d}^3 r_2' \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \frac{1}{|\boldsymbol{r}_1' - \boldsymbol{r}_2'|} \int_0^\infty \frac{\mathrm{d}\omega}{2\pi} \chi_{\mathbb{A}}(\boldsymbol{r}_1, \boldsymbol{r}_1'; i\omega) \chi_{\mathbb{B}}(\boldsymbol{r}_2, \boldsymbol{r}_2'; i\omega) \ , \tag{4.2.9}$$

$\chi(\boldsymbol{r}, \boldsymbol{r}'; i\omega)$ formally standing for a density-density response function of imaginary frequencies:

$$\chi(\boldsymbol{r}, \boldsymbol{r}'; i\omega) = -4 \sum_{i,a} \frac{\epsilon_i^a}{n_i \bar{n}^a} \frac{\rho_i^a(\boldsymbol{r}) \rho_a^i(\boldsymbol{r}')}{(\epsilon_i^a)^2 - (i\omega)^2} = -\sum_{i,a} \frac{\epsilon_i^a}{n_i \bar{n}^a} \frac{\langle \hat{\rho}(\boldsymbol{r}) :E_i^a:\rangle \langle :E_a^i: \hat{\rho}(\boldsymbol{r}')\rangle}{(\epsilon_i^a)^2 - (i\omega)^2} \ . \tag{4.2.10}$$

Subscript $\mathbb{A}$ or $\mathbb{B}$ in Eq. (4.2.9) means restriction of summation indices to the respective subsystem. Also, orbitals were taken to be real for simplicity. An immediate consequence of Eq. (4.2.5) is

$$\int \mathrm{d}^3 r \chi(\boldsymbol{r}, \boldsymbol{r}'; i\omega) = \int \mathrm{d}^3 r' \chi(\boldsymbol{r}, \boldsymbol{r}'; i\omega) = 0 \ .$$

Therefore, the leading order (dipole-dipole) term of $E_{disp}^{(2)}$ in the multipole expansion just gives the London formula of Eq. (4.0.1) with dispersion coefficient

$$C_6 = \int_0^\infty \frac{\mathrm{d}\omega}{2\pi} \mathrm{Tr}\left[\boldsymbol{\alpha}_{\mathbb{A}}(i\omega) \boldsymbol{\tau} \boldsymbol{\alpha}_{\mathbb{B}}(i\omega) \boldsymbol{\tau}\right] \ . \tag{4.2.11}$$

Tensorial quantities $\boldsymbol{\alpha}_{\mathbb{A}}(i\omega)$ and $\boldsymbol{\alpha}_{\mathbb{B}}(i\omega)$ formally resemble the FDP-s of the separate subsystems:

$$\boldsymbol{\alpha}_{\mathbb{A}}(i\omega) = -\int \mathrm{d}^3 r \int \mathrm{d}^3 r' \chi(\boldsymbol{r}, \boldsymbol{r}'; i\omega) \boldsymbol{r} \otimes \boldsymbol{r}' = 4 \sum_{i,a\in\mathbb{A}} \frac{\epsilon_i^a n_i \bar{n}^a}{(\epsilon_i^a)^2 - (i\omega)^2} \boldsymbol{\mu}_i^a \otimes \boldsymbol{\mu}_a^i \ .$$

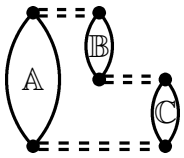
For isotropic subsystems $\boldsymbol{\alpha}_{\mathbb{A}}(i\omega)$ must be proportional to the unit tensor: $\boldsymbol{\alpha}(i\omega) = \alpha_{\mathbb{A}}(i\omega)\boldsymbol{I}$. Using this and $\mathrm{Tr}[\boldsymbol{\tau}^2] = 6$ allows us to write Eq. (4.2.11) as

$$C_6 = \frac{3}{\pi}\int_0^\infty \mathrm{d}\omega \alpha_{\mathbb{A}}(i\omega)\alpha_{\mathbb{B}}(i\omega) \ .$$

The above expressions are formally similar to the well known formulae of single determinantal theory [151], the main difference being that quantities like $\boldsymbol{\mu}_i^a$, $\chi(\boldsymbol{r}, \boldsymbol{r}', i\omega)$ or $\boldsymbol{\alpha}(i\omega)$ are all cumulant-dependent. Although the derivations were carried out for atoms, similar results could be found for non-spherical subsystems (after suitable rotational averaging). The cumulant corrections to the otherwise mean-field-like quantities possibly make the model more suitable for describing dispersion interaction between strongly correlated systems.

Following the same arguments, one could show that intersystem components of higher rank particle-hole dressed vertices $W^{i_1\ldots i_K}_{a_1\ldots a_K}$ fall off at least as $1/R^3$ too, similarly to Eq. (4.2.6). Since these vertices provide the leading-order contributions to amplitudes through the CC iteration, one can conclude that intersystem amplitudes also fall off at least as $1/R^3$, leading to the $E_{disp} \sim -1/R^6$ decay of dispersion energy in the general ic-MRCC framework as well. Whether one can extract a closed form expression for C_6 strongly depends on the underlying approximation schemes.

We mention that when dealing with more than two subsystems, higher-order diagrams include pair-wise non-additive dispersion effects as well. The simplest example is given by the third-order direct ring diagram for three well-separated subsystems $\mathbb{A}$, $\mathbb{B}$ and $\mathbb{C}$ (cf. Eq. (4.2.6)):



$$\sim \frac{1}{R^3_{\mathbb{AB}} R^3_{\mathbb{BC}} R^3_{\mathbb{CA}}} \ ,$$

describing a genuine three-atom interaction that cannot be decomposed as a sum of pair contributions. This energy diagram can be thought of as a simple MR example of the Axilrod-Teller interaction [150].

Finally, let us comment on the interpretation (or interpretability) of the obtained results. Cumulant dressing of quantities like $\boldsymbol{\mu}_i^a$, or $\boldsymbol{\alpha}(i\omega)$ goes well with the standard interpretation of dispersion interaction. The usual explanation is as follows [12]: Eq. (4.2.9) describes Coulomb interactions between points $\boldsymbol{r}_1, \boldsymbol{r}_1'$ and $\boldsymbol{r}_2, \boldsymbol{r}_2'$ of atoms $\mathbb{A}$ and $\mathbb{B}$, respectively, propagation of density fluctuations on the separate atoms being described by response functions $\chi_{\mathbb{A}}(\boldsymbol{r}_1, \boldsymbol{r}_1', i\omega)$ and $\chi_{\mathbb{B}}(\boldsymbol{r}_2, \boldsymbol{r}_2', i\omega)$. Taking Eq. (4.2.1) into account, we see that dispersion arises from the interaction of induced electric moments generated by these density disturbances. The very role of cumulant dressing is to provide corrections to the otherwise mean-field level propagation of density fluctuations.
On a formal level, identifying Eq. (4.2.10) as a response function checks out: it describes density disturbances mediated by single excitations manifesting themselves not just via pair-contractions but cumulants as well. On the other hand, this analogy has limitations, as shown

below.
The general definition of the density-density response function (using real functions) reads

$$\chi(\boldsymbol{r},\boldsymbol{r}',i\omega) = -2\sum_{n\neq 0}\omega_{n0}\frac{\langle\Psi_0|\hat{\rho}(\boldsymbol{r})|\Psi_n\rangle\langle\Psi_n|\hat{\rho}(\boldsymbol{r}')|\Psi_0\rangle}{\omega_{n0}^2-(i\omega)^2} \ ,$$

$|\Psi_0\rangle$ and $|\Psi_n\rangle$ being ground and excited states with excitation energy $\omega_{n0} = E_n - E_0$. If a single determinant is used for $|\Psi_0\rangle$ (and its various substitutions as excited states), then only single excitations can contribute to $\langle\Psi_0|\hat{\rho}(\boldsymbol{r})|\Psi_n\rangle$ with excitation energies $\epsilon_i^a = f_a^a - f_i^i$, and we are left with

$$\chi(\boldsymbol{r},\boldsymbol{r}';i\omega) = -\sum_{i,a}\epsilon_i^a\frac{\langle\hat{\rho}(\boldsymbol{r})\ {:}E_i^a{:}\rangle\langle{:}E_a^i{:}\ \hat{\rho}(\boldsymbol{r}')\rangle}{(\epsilon_i^a)^2-(i\omega)^2} \ . \tag{4.2.12}$$

Eq. (4.2.10) using MR function $|\Phi\rangle$ and its single substitutions does look very similar to Eq. (4.2.12) and exactly reduces to it in the single determinantal limit. However, "excited" states $\{{:}E_K{:}\ |\Phi\rangle\}$ are not orthogonal to each other, meaning that all of them could contribute to $\langle\hat{\rho}(\boldsymbol{r})\ {:}E_K{:}\rangle$. This would generate further terms not included in Eq. (4.2.10) with transition densities of higher excitations. These terms would correspond to the contribution of higher rank dressed vertices to second-order dispersion energy which is neglected in the idCCSD approximation.
The other problem is identifying ϵ_K with the excitation energy associated to ${:}E_K{:}\ |\Phi\rangle$. While this is natural in a single determinantal theory (cf. Eq. (2.3.15)), it is highly non-trivial in the MR case due to the overlap of states. It might be tempting to define some zeroth-order model Hamiltonian in spectral form, like

$$\begin{aligned} H^{(0)} &= \sum_K \frac{1}{K!}(E_{ref}+\epsilon_K)\ {:}E_K{:}\ |\Phi\rangle\langle\tilde{\Phi}_K| \\ &= E_{ref}|\Phi\rangle\langle\Phi| + \sum_{i,a}(E_{ref}+\epsilon_i^a)\ {:}E_i^a{:}\ |\Phi\rangle\langle\tilde{\Phi}_a^i| + \frac{1}{2}\sum_{\substack{i,a\\j,b}}(E_{ref}+\epsilon_i^a+\epsilon_j^b)\ {:}E_{ij}^{ab}{:}\ |\Phi\rangle\langle\tilde{\Phi}_{ab}^{ij}| + \dots \ , \end{aligned}$$

with $E_{ref} = \langle H\rangle$, and reciprocal bra vectors $\{\langle\tilde{\Phi}_K|\}$ being orthogonal to the substituted ket vectors. By construction, this model Hamiltonian would satisfy the eigenvalue equations

$$H^{(0)}\ {:}E_K{:}\ |\Phi\rangle = (E_{ref}+\epsilon_K)\ {:}E_K{:}\ |\Phi\rangle \quad , \quad \langle\tilde{\Phi}_K|H^{(0)} = (E_{ref}+\epsilon_K)\langle\tilde{\Phi}_K| \ ,$$

so ϵ_K could be interpreted as excitation energy. However, this choice would lead to transition densities like $\langle\tilde{\Phi}_a^i|\hat{\rho}(\boldsymbol{r})|\Phi\rangle$, making it impossible to exactly recover Eq. (4.2.10). The conclusion is that, although the analogy between the dressed formula of Eq. (4.2.10) and a density-density response function intuitively makes sense, their relationship is not exact.

In the next chapter, we address further issues related to the overlap of states $\{{:}E_K{:}\ |\Phi\rangle\}$.

Chapter 5

Treating the linear dependence of the excitation manifold

During the derivation of the ic-MRCC formalism in Chapter 3, we included all kinds of excitations in the parametrization of T and in the projection equations without considering the possible linear dependence of these excited states. Unlike in a single determinantal theory, automatic inclusion of all possible excitations involving active orbitals leads to a redundant set of vectors $\{: E_K : |\Phi\rangle\}$. This is a general problem of ic-MRCC, as one may encounter serious numerical instabilities when trying to solve an ill-posed system of equations.
Redundancy is usually handled by elimination of certain amplitudes or linear combinations thereof. An unfortunate consequence of amplitude elimination is the possible appearance of discontinuities on the Potential Energy Surface (PES). Interference with orbital invariance and size consistency has been reported for some of the redundancy treatments investigated in Ref. [118], with size extensivity restored in a follow-up work [119]. To avoid such pitfalls, the use of so-called many-body residual equations is often advocated instead of projection equations [130].
In the present chapter, a hitherto unexplored scheme is introduced for handling redundancy when CC amplitude equations are obtained as ordinary projections (as was done in Sec. 3.1). We rely on the concept of so-called frames which facilitates keeping all the amplitudes corresponding to the redundant set of excited functions. Frames are generalizations of basis sets, relaxing the requirement of linear independence on the constituting vectors. They were originally introduced in connection with signal analysis, and have found ever increasing applications in physics, engineering and computer science [158, 159].

In this chapter we first show the emergence of linear dependence on a simple example. This is followed by the general procedure to solve the redundant system of amplitude equations without removing excitations. Finally, this procedure is also demonstrated using a simple example.
Spatial orbitals are used throughout this chapter, denoted by latin indices.

This chapter is based on Ref. [138].

5.1 Linear dependence of UGA excitations

In the ic-MRCCSD framework, the Schrödinger equation is projected with singly and doubly excited states of $|\Phi\rangle$. In the spin-free formalism (as discussed in Sec. 3.3.2), we shall use the UGA parametrization of these states to investigate redundancy: $\{:E_i^a:|\Phi\rangle\}$ and $\{:E_{ij}^{ab}:|\Phi\rangle\}$, with $i,j,k,l,...$ and $a,b,c,d,...$ standing for core/active and active/virtual spatial orbitals, respectively.
The overlap of such states can be found in a straightforward although somewhat tedious manner by writing out explicitly and contracting the spin indices in the expectation values of Eqs. (2.1.52), (2.1.53) and (2.1.54) with the help of Eqs. (2.1.58) and (2.1.59). We then have

$$S_{ci}^{ka} = \langle : E_c^k :: E_i^a : \rangle = 2\delta_i^k \delta_c^a n_i \bar{n}^a + \Lambda_{ic}^{ak} \ ,$$

$$S_{cij}^{kab} = \langle : E_c^k :: E_{ij}^{ab} : \rangle = \delta_c^a \bar{n}^a \Lambda_{ij}^{kb} + \delta_c^b \bar{n}^b \Lambda_{ij}^{ak} - \delta_i^k n_i \Lambda_{cj}^{ab} - \delta_j^k n_j \Lambda_{ic}^{ab} + \Lambda_{ijc}^{abk} \ .$$

and

$$\begin{aligned}
S_{cdij}^{klab} =& \langle : E_{cd}^{kl} :: E_{ij}^{ab} : \rangle \\
=& 4\delta_i^k \delta_j^l \delta_c^a \delta_d^b n_i n_j \bar{n}^a \bar{n}^b - 2\delta_j^k \delta_i^l \delta_c^a \delta_d^b n_i n_j \bar{n}^a \bar{n}^b - 2\delta_i^k \delta_j^l \delta_c^b \delta_d^a n_i n_j \bar{n}^a \bar{n}^b + 4\delta_j^k \delta_i^l \delta_c^b \delta_d^a n_i n_j \bar{n}^a \bar{n}^b \\
&+ \delta_i^k \delta_j^l n_i n_j \Lambda_{cd}^{ab} + \delta_j^k \delta_i^l n_i n_j \Lambda_{dc}^{ab} + \delta_c^a \delta_d^b \bar{n}^a \bar{n}^b \Lambda_{ij}^{kl} + \delta_c^b \delta_d^a \bar{n}^a \bar{n}^b \Lambda_{ij}^{lk} \\
&+ 2\delta_i^k \delta_c^a n_i \bar{n}^a \Lambda_{jd}^{bl} - \delta_i^l \delta_c^a n_i \bar{n}^a \Lambda_{jd}^{bk} - \delta_i^k \delta_d^a n_i \bar{n}^a \Lambda_{jc}^{bl} + 2\delta_i^l \delta_d^a n_i \bar{n}^a \Lambda_{jc}^{bk} \\
&+ 2\delta_j^k \delta_c^b n_j \bar{n}^b \Lambda_{id}^{al} - \delta_j^l \delta_c^b n_j \bar{n}^b \Lambda_{id}^{ak} - \delta_j^k \delta_d^b n_j \bar{n}^b \Lambda_{ic}^{al} + 2\delta_j^l \delta_d^b n_j \bar{n}^b \Lambda_{ic}^{ak} \\
&- \delta_j^k \delta_c^a n_j \bar{n}^a \Lambda_{id}^{bl} - \delta_j^k \delta_d^a n_j \bar{n}^a \Lambda_{ic}^{lb} - \delta_j^l \delta_d^a n_j \bar{n}^a \Lambda_{ic}^{bk} - \delta_j^l \delta_c^a n_j \bar{n}^a \Lambda_{id}^{kb} \\
&- \delta_i^k \delta_c^b n_i \bar{n}^b \Lambda_{jd}^{al} - \delta_i^k \delta_d^b n_i \bar{n}^b \Lambda_{jc}^{la} - \delta_i^l \delta_d^b n_i \bar{n}^b \Lambda_{jc}^{ak} - \delta_i^l \delta_c^b n_i \bar{n}^b \Lambda_{jd}^{ka} \\
&+ \frac{1}{6}\left(2\Lambda_{ij}^{kl}\Lambda_{cd}^{ab} + \Lambda_{ji}^{kl}\Lambda_{cd}^{ab} + \Lambda_{ij}^{kl}\Lambda_{dc}^{ab} + 2\Lambda_{ji}^{kl}\Lambda_{dc}^{ab}\right) \\
&+ \Lambda_{ic}^{ak}\Lambda_{jd}^{bl} - \frac{1}{2}\Lambda_{id}^{ak}\Lambda_{jc}^{bl} - \frac{1}{2}\Lambda_{ic}^{al}\Lambda_{jd}^{bk} + \Lambda_{id}^{al}\Lambda_{jc}^{bk} - \frac{1}{2}\Lambda_{ic}^{bk}\Lambda_{jd}^{al} - \frac{1}{2}\Lambda_{id}^{bl}\Lambda_{jc}^{ak} \\
&+ \frac{1}{6}\left(2\Lambda_{ic}^{lb}\Lambda_{jd}^{ka} + \Lambda_{ci}^{lb}\Lambda_{jd}^{ka} + \Lambda_{ic}^{lb}\Lambda_{dj}^{ka} + 2\Lambda_{ci}^{lb}\Lambda_{dj}^{ka}\right) + \frac{1}{6}\left(2\Lambda_{id}^{kb}\Lambda_{jc}^{la} + \Lambda_{di}^{kb}\Lambda_{jc}^{la} + \Lambda_{id}^{kb}\Lambda_{cj}^{la} + 2\Lambda_{di}^{kb}\Lambda_{cj}^{la}\right) \\
&- \frac{1}{2}\Lambda_{ij}^{ak}\Lambda_{cd}^{bl} - \frac{1}{2}\Lambda_{ij}^{al}\Lambda_{cd}^{kb} - \frac{1}{2}\Lambda_{ij}^{kb}\Lambda_{cd}^{al} - \frac{1}{2}\Lambda_{ij}^{lb}\Lambda_{cd}^{ka} - \frac{1}{2}\Lambda_{id}^{ab}\Lambda_{cj}^{kl} - \frac{1}{2}\Lambda_{ic}^{ab}\Lambda_{jd}^{kl} - \frac{1}{2}\Lambda_{dj}^{ab}\Lambda_{ci}^{kl} - \frac{1}{2}\Lambda_{cj}^{ab}\Lambda_{id}^{kl} \\
&+ \delta_c^a \bar{n}^a \Lambda_{ijd}^{kbl} + \delta_d^a \bar{n}^a \Lambda_{ijc}^{lbk} + \delta_c^b \bar{n}^b \Lambda_{ijd}^{akl} + \delta_d^b \bar{n}^b \Lambda_{ijc}^{alk} - \delta_i^k n_i \Lambda_{cjd}^{abl} - \delta_i^l n_i \Lambda_{djc}^{abk} - \delta_j^k n_j \Lambda_{icd}^{abl} - \delta_j^l n_j \Lambda_{idc}^{abk} \\
&+ \Lambda_{ijcd}^{abkl} \ . \qquad (5.1.1)
\end{aligned}$$

Let us introduce the compact notation of $\{|\psi_\mu\rangle\}$ for the set of general excited states (index μ running over all singly and doubly excited states). Then the overlap matrix can be written as

$$S_{\mu\nu} = \langle \psi_\mu | \psi_\nu \rangle \ .$$

Zero eigenvalues of this matrix signal redundancy among the excited states: the amplitude equation contains less information than the number of parameters. Among such

circumstances the amplitude vector $\boldsymbol{t}$, collecting elements t_μ (assigned to excitation $|\psi_\mu\rangle$), is ill defined.
Indeed, it is not hard to find an example of such singular behaviour. Let us consider a GVB wave function with orbitals ϕ^p and ϕ^q assigned to the same geminal (so that $n_p + n_q = 1$). Furthermore, let ϕ^i be a core orbital, while ϕ^a is virtual. Due to the strongly orthogonal structure of GVB, the overlap matrix takes a simple block-diagonal form (this will be used in Chapter 6). It is not hard to see that so-called exchange spectator-type excitations $|\psi_1\rangle =: E^{ap}_{pi} : |\Phi\rangle$ and $|\psi_2\rangle =: E^{aq}_{qi} : |\Phi\rangle$ form a two-by-two block of $\boldsymbol{S}$ with a zero eigenvalue:

$$\boldsymbol{S}_{2\times 2} = 2n_p n_q \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} ,$$

hence, they are linearly dependent (actually, $|\psi_1\rangle = -|\psi_2\rangle$). The above matrix can be obtained either with the help of Eqs. (5.1.1) and (2.2.8), or by calculating the action of $:E^{ap}_{pi}:= E^p_i E^a_p + n_p E^a_i$ on $|\Phi\rangle$ (and similarly for $:E^{aq}_{qi}:$), and constructing the overlap manually.
In order to keep the argument simple, we do not consider direct spectator-type excitations $: E^{pa}_{pi} : |\Phi\rangle$ and $: E^{qa}_{qi} : |\Phi\rangle$. In general, these would couple to $|\psi_1\rangle$ and $|\psi_2\rangle$, generating a four-by-four block.
The zero eigenvalue of $\boldsymbol{S}_{2\times 2}$ means that − although looking different on the surface − vectors $|\psi_1\rangle$ and $|\psi_2\rangle$ (or an arbitrary linear combination of them) represent the very same state, and, in principle, no information would be lost by discarding either one of them. However, in general, this could lead to undesired side effects, as mentioned above. To prevent this, we proceed to find a way to solve the amplitude equation for the redundant vectors without the need to remove any of them.

5.2 Extracting unique amplitudes from the redundant amplitude equation

Before dealing with the amplitude equation, certain sets of vectors (and their associated dual vector sets) must be defined.
Let us write the spectral decomposition of $\boldsymbol{S}$ as

$$\boldsymbol{S} = \boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{V}^\dagger , \tag{5.2.1}$$

where diagonal matrix $\boldsymbol{\Sigma}$ contains the Σ_μ eigenvalues of $\boldsymbol{S}$. These are nonzero for $\mu = 1, \ldots, \mathcal{M}$ and zero for $\mu = \mathcal{M}+1, \ldots, \mathcal{N}$ ($\mathcal{M}$ and $\mathcal{N}$ being the rank of $\boldsymbol{S}$ and the overall number of excitations in T, respectively). For further convenience we define $\boldsymbol{\Sigma'} = \mathrm{diag}(\Sigma_\gamma)$ as an $\mathcal{M} \times \mathcal{M}$ matrix collecting nonzero Σ_γ, and $\boldsymbol{\sigma'} = \boldsymbol{\Sigma'}^{1/2}$ (that is, $\sigma'_\gamma = \sqrt{\Sigma'_\gamma}$). Unitary matrix $\boldsymbol{V}$ contains the eigenvectors of $\boldsymbol{S}$ as columns. To make things somewhat more transparent, from now on, we distinguish between indices based on their range: $\mu, \nu = 1, ..., \mathcal{N}$ and $\gamma, \delta = 1, .., \mathcal{M}$.
Let us also introduce an auxiliary orthonormal basis obtained by the canonical

orthogonalization scheme of Löwdin [14]:

$$|e_\gamma\rangle = \sum_\nu |\psi_\nu\rangle V_{\nu\gamma}\sigma_\gamma^{-1} \quad (\gamma = 1, \ldots, \mathcal{M}) \ , \tag{5.2.2}$$

so that $\langle e_\gamma | e_\delta\rangle = \delta_{\gamma\delta}$. Rectangular matrix $\boldsymbol{F}$ composed of elements $F_{\gamma\mu} = \langle e_\gamma|\psi_\mu\rangle$ provides the matrix of the so-called synthesis operator [159]: its Singular Value Decomposition (SVD) reads

$$\boldsymbol{F} = \boldsymbol{U}\boldsymbol{\sigma}\boldsymbol{V}^\dagger \ , \tag{5.2.3}$$

where

$$\boldsymbol{\sigma} \ = \ \begin{bmatrix}\boldsymbol{\sigma'} & \mathbf{0}\end{bmatrix}$$

is the $\mathcal{M}\times\mathcal{N}$ extension of $\boldsymbol{\sigma'}$, and $\boldsymbol{U}$ is just an $\mathcal{M}\times\mathcal{M}$ unit matrix. Matrix $\boldsymbol{F}$ obviously fulfills $\boldsymbol{F}^\dagger\boldsymbol{F} = \boldsymbol{S}$. The product in reverse order, $\boldsymbol{F}\boldsymbol{F}^\dagger = \boldsymbol{\Sigma'}$ obeys the so-called frame condition:

$$A\boldsymbol{P}_U \leq \boldsymbol{F}\boldsymbol{F}^\dagger \leq B\boldsymbol{P}_U \ ,$$

where $\boldsymbol{P}_U = \boldsymbol{U}\boldsymbol{U}^\dagger = \boldsymbol{U}$ is a trivial projector. Choosing e.g. $A = \min_\gamma \Sigma'_\gamma$, and $B = \max_\gamma \Sigma'_\gamma$ fulfills this condition, showing that overcomplete vectors $\{|\psi_\mu\rangle\}$ constitute a frame with A and B.
Projector $\boldsymbol{P}_U$ can be expressed with $\boldsymbol{F}$ in the form

$$\boldsymbol{F}\widetilde{\boldsymbol{F}}^\dagger \ = \ \boldsymbol{P}_U \ , \tag{5.2.4}$$

where

$$\widetilde{\boldsymbol{F}} \ = \ \boldsymbol{F}\boldsymbol{S}^- \ = \ \boldsymbol{U}\begin{bmatrix}\boldsymbol{\sigma'}^{-1} & \mathbf{0}\end{bmatrix}\boldsymbol{V}^\dagger \tag{5.2.5}$$

is the synthesis matrix of the so-called canonical dual frame. In Eq. (5.2.5), $\boldsymbol{S}^-$ is understood as a Moore–Penrose pseudoinverse:

$$\boldsymbol{S}^- = \boldsymbol{V}\begin{bmatrix}\boldsymbol{\Sigma'}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0}\end{bmatrix}\boldsymbol{V}^\dagger \ .$$

The dual frame vectors, $\langle\widetilde{\psi}_\mu|$ are associated with columns of $\widetilde{\boldsymbol{F}}$. It follows from Eq. (5.2.4) that $\langle\widetilde{\psi}_\mu|$ can be used as bra vectors and $|\psi_\mu\rangle$ are considered as ket vectors when writing the matrix vector form of the amplitude equations. Bra vectors $\langle\widetilde{\psi}_\mu|$ read

$$\langle\widetilde{\psi}_\mu| = \sum_\nu (\boldsymbol{S}^-)_{\mu\nu}\langle\psi_\nu| \ . \tag{5.2.6}$$

A justification of working with the overcomplete set of frame vectors and their dual vectors may be given by introducing a linearly independent set via extension of matrix $\boldsymbol{\sigma}$ to dimension $\mathcal{N}\times\mathcal{N}$, according to

$$\overline{\boldsymbol{\sigma}} = \begin{bmatrix}\boldsymbol{\sigma'} & \mathbf{0} \\ \mathbf{0} & \mathbf{1}\end{bmatrix} \ .$$

The SVD form of the extended synthesis matrix is constructed in complete analogy with Eq. (5.2.3) as

$$\overline{\boldsymbol{F}} = \overline{\boldsymbol{U}}\overline{\boldsymbol{\sigma}}\boldsymbol{V}^\dagger \ , \tag{5.2.7}$$

with $\overline{\boldsymbol{U}}$ denoting simply the unit $\mathcal{N} \times \mathcal{N}$ matrix. Columns of $\overline{\boldsymbol{F}}$ represent $\mathcal{N}$ linearly independent vectors $\{|\overline{\psi}_\mu\rangle\}$, which property follows from the overlap matrix $\overline{\boldsymbol{S}} = \overline{\boldsymbol{F}}^\dagger\overline{\boldsymbol{F}}$ (that is, $\overline{S}_{\mu\nu} = \langle\overline{\psi}_\mu|\overline{\psi}_\nu\rangle$) lacking zero eigenvalues:

$$\overline{\boldsymbol{S}} = \overline{\boldsymbol{F}}^\dagger\overline{\boldsymbol{F}} = \boldsymbol{V}\overline{\boldsymbol{\Sigma}}\boldsymbol{V}^\dagger \ ,$$

where

$$\overline{\boldsymbol{\Sigma}} = \begin{bmatrix} \boldsymbol{\Sigma}' & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} \end{bmatrix} \ .$$

Inverse of $\overline{\boldsymbol{S}}$ facilitates the construction of the dual set in the extended space as

$$\widetilde{\overline{\boldsymbol{F}}} = \overline{\boldsymbol{F}}\,\overline{\boldsymbol{S}}^{-1} \ = \ \overline{\boldsymbol{U}}\overline{\boldsymbol{\sigma}}^{-1}\boldsymbol{V}^\dagger \ .$$

Columns of $\overline{\boldsymbol{F}}$ and $\widetilde{\overline{\boldsymbol{F}}}$ ($\{|\overline{\psi}_\mu\rangle\}$ and $\{\langle\widetilde{\overline{\psi}}_\mu|\}$) are biorthonormal, characterized by $\widetilde{\overline{\boldsymbol{F}}}^\dagger\overline{\boldsymbol{F}} = \boldsymbol{V}\boldsymbol{V}^\dagger = \overline{\boldsymbol{U}}$ (that is, $\langle\widetilde{\overline{\psi}}_\mu|\overline{\psi}_\nu\rangle = \delta_{\mu\nu}$). Again, similarly to Eq. (5.2.6)

$$\langle\widetilde{\overline{\psi}}_\mu| = \sum_\nu (\overline{\boldsymbol{S}}^{-1})_{\mu\nu}\langle\overline{\psi}_\nu| \ .$$

In comparison, frame vectors and their dual vectors are not biorthonormal, their product yielding

$$\widetilde{\overline{\boldsymbol{F}}}^\dagger\boldsymbol{F} \ = \ \boldsymbol{S}^-\boldsymbol{S} \ = \ \boldsymbol{V}\begin{bmatrix} \boldsymbol{P}_U & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}\boldsymbol{V}^\dagger = \boldsymbol{R} \ . \tag{5.2.8}$$

Note that $(\widetilde{\overline{\boldsymbol{F}}}^\dagger\boldsymbol{F})_{\mu\nu} = \langle\widetilde{\psi}_\mu|\psi_\nu\rangle$ (cf. Eq. (5.2.6)). The last equality of Eq. (5.2.8) defines idempotent matrix $\boldsymbol{R}$. It is a key quantity, connecting synthesis matrices via

$$\begin{bmatrix} \boldsymbol{F} \\ \boldsymbol{0} \end{bmatrix} \ = \ \overline{\boldsymbol{F}}\boldsymbol{R} \ , \tag{5.2.9}$$

$$\begin{bmatrix} \widetilde{\overline{\boldsymbol{F}}} \\ \boldsymbol{0} \end{bmatrix} \ = \ \widetilde{\overline{\boldsymbol{F}}}\boldsymbol{R} \ , \tag{5.2.10}$$

with the bottom zero block on the left hand side of the above equations being of dimension $(\mathcal{N} - \mathcal{M}) \times \mathcal{N}$.
As $\{|\overline{\psi}_\nu\rangle\}$ are associated with the columns of $\overline{\boldsymbol{F}}$, the relation

$$|\psi_\mu\rangle = \sum_\nu |\overline{\psi}_\nu\rangle R_{\nu\mu} \tag{5.2.11}$$

holds based on Eq. (5.2.9).
Finally, we introduce the "extended" Löwdin vectors:

$$|\overline{e}_\mu\rangle = \sum_\nu |\overline{\psi}_\nu\rangle V_{\nu\mu}\overline{\sigma}_\mu^{-1} \quad (\mu = 1, \ldots, \mathcal{N}) \ . \tag{5.2.12}$$

These are orthonormal by construction: $\langle\overline{e}_\mu|\overline{e}_\nu\rangle = \delta_{\mu\nu}$.

To summarize, we have four sets of ket vectors:

- overcomplete frame vectors $\{|\psi_\mu\rangle\}$,
- orthonormal Löwdin vectors $\{|e_\gamma\rangle\}$,
- linearly independent extended vectors $\{|\overline{\psi}_\mu\rangle\}$,
- orthonormal extended Löwdin vectors $\{|\overline{e}_\mu\rangle\}$.

Bra sets $\{\langle\overline{\psi}_\mu|\}$ and $\{\langle\widetilde{\overline{\psi}}_\mu|\}$ are defined to be duals of the first and third sets, respectively. We can now proceed to the expansion of the ic-MRCC wavefunction and the amplitude equations. We will argue that amplitudes expanded in basis $\{|\overline{\psi}_\mu\rangle\}$ keep their unique character upon returning to overcomplete set $\{|\psi_\mu\rangle\}$. Actually, it will turn out that we do not need to explicitly construct $\{|\overline{\psi}_\mu\rangle\}$ and $\{\langle\widetilde{\overline{\psi}}_\mu|\}$, but are still able to assign unique values to amplitudes.

With the help of frame vectors one can write the $\overline{U}$-component of $|\Psi\rangle$ as

$$P_{\overline{U}}|\Psi\rangle = \sum_\mu |\psi_\mu\rangle t_\mu \;+\; \mathcal{O}(\boldsymbol{t}^2)\;, \tag{5.2.13}$$

while the extended frame vectors allow us to parametrize the same component as

$$P_{\overline{U}}|\Psi\rangle = \sum_\mu |\overline{\psi}_\mu\rangle \overline{t}_\mu \;+\; \mathcal{O}(\overline{\boldsymbol{t}}^2)\;. \tag{5.2.14}$$

$P_{\overline{U}}$ stands for the projector of the space spanned by $\{|\overline{\psi}_\mu\rangle\}$. Substituting Eq. (5.2.11) into Eq. (5.2.13) results in

$$\begin{aligned} P_{\overline{U}}|\Psi\rangle &= \sum_\mu \sum_\nu |\overline{\psi}_\nu\rangle R_{\nu\mu} t_\mu \;+\; \mathcal{O}(\boldsymbol{t}^2) \\ &= \sum_\mu |\overline{\psi}_\mu\rangle (\boldsymbol{R}\boldsymbol{t})_\mu \;+\; \mathcal{O}(\boldsymbol{t}^2)\;. \end{aligned} \tag{5.2.15}$$

Comparing Eqs. (5.2.15) and (5.2.14), one can deduce

$$\boldsymbol{R}\boldsymbol{t} = \overline{\boldsymbol{t}}\;, \tag{5.2.16}$$

omitting terms quadratic in the amplitude.

Elaborating further on amplitudes, let us relate $\boldsymbol{t}$ and $\overline{\boldsymbol{t}}$ to a third amplitude vector, corresponding to $\{|e_\gamma\rangle\}$ denoted by $\boldsymbol{t}^e$ (a column vector of length $\mathcal{M}$). Based on Eq. (5.2.2), and essentially repeating Eq. (5.2.15), the relation between $\boldsymbol{t}$ and $\boldsymbol{t}^e$ is found to be

$$\boldsymbol{t} = \boldsymbol{V}\begin{bmatrix}\boldsymbol{\sigma}'^{-1}\\ \boldsymbol{0}\end{bmatrix}\boldsymbol{t}^e = \boldsymbol{V}\begin{bmatrix}\boldsymbol{\sigma}'^{-1}\boldsymbol{t}^e\\ \boldsymbol{0}\end{bmatrix}\;. \tag{5.2.17}$$

At the same time, extended orthonormal vectors $\{|\overline{e}_\mu\rangle\}$ facilitate to relate $\overline{\boldsymbol{t}}$ and $\boldsymbol{t}^e$ as

$$\overline{\boldsymbol{t}} = \boldsymbol{V}\begin{bmatrix}\boldsymbol{\sigma}'^{-1} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{1}\end{bmatrix}\begin{bmatrix}\boldsymbol{t}^e\\ \boldsymbol{0}\end{bmatrix} = \boldsymbol{V}\begin{bmatrix}\boldsymbol{\sigma}'^{-1}\boldsymbol{t}^e\\ \boldsymbol{0}\end{bmatrix}\;. \tag{5.2.18}$$

Note that amplitudes in basis $\{|\overline{e}_\mu\rangle\}$ are necessarily equal to t^e_μ for $\mu \leq \mathcal{M}$ and zero otherwise. Comparison of Eqs. (5.2.17) and (5.2.18) confirms not only that Eq. (5.2.16) holds, but reveals that $\boldsymbol{t}$ and $\overline{\boldsymbol{t}}$ are actually the same. Note, however, that Eq. (5.2.2) is not the only way to express $\{|\overline{e}_\mu\rangle\}$ with $\{|\psi_\mu\rangle\}$ due to overcompleteness of the latter set. Consequently, Eq. (5.2.17) is just one of the possibilities of generating $\boldsymbol{t}$ from $\boldsymbol{t}^e$. On the other hand, expression of $\{|\overline{e}_\mu\rangle\}$ with $\{|\overline{\psi}_\mu\rangle\}$ is unique according to Eq. (5.2.12). The equivalence of vectors $\boldsymbol{t}$ and $\overline{\boldsymbol{t}}$ of Eqs. (5.2.17) and (5.2.18) can be interpreted as follows. Introduction of the linearly independent extended functions singles out one particular vector $\boldsymbol{t}$, that is related to $\boldsymbol{t}^e$ according to Eq. (5.2.17). This is the origin of removing the ill-defined nature of $\boldsymbol{t}$ while conserving an amplitude vector of length $\mathcal{N}$. Other vectors $\boldsymbol{t}$, fulfilling Eq. (5.2.13) and not matching $\overline{\boldsymbol{t}}$, can be related to it via Eq. (5.2.16).

It is also instructive to project Eq. (5.2.15) with functions $\langle\widetilde{\overline{\psi}}_\nu|$, as well as project Eq. (5.2.13) with the dual frame vectors $\langle\widetilde{\psi}_\nu|$. In both cases one obtains

$$(\boldsymbol{R}\boldsymbol{t})_\nu + \mathcal{O}(\boldsymbol{t}^2) = \langle\widetilde{\overline{\psi}}_\nu|\Psi\rangle = \langle\widetilde{\psi}_\nu|\Psi\rangle \ ,$$

based on Eqs. (5.2.16) and (5.2.8). This indicates that it is not necessary to explicitly construct $\widetilde{\overline{\boldsymbol{F}}}$ and $\overline{\boldsymbol{F}}$, since the same result is obtained when working with the set $\widetilde{\boldsymbol{F}}, \boldsymbol{F}$.

With this in hand, we can finally consider the amplitude equation of an arbitrary excitation written in the form

$$\langle\widetilde{\psi}_\mu|\mathcal{H}|\Phi\rangle + \sum_\nu \langle\widetilde{\psi}_\mu|\mathcal{H}|\psi_\nu\rangle t_\nu + \mathcal{O}(\boldsymbol{t}^2) = 0 \ . \tag{5.2.19}$$

As previously, terms nonlinear in t_μ are implicitly contained in $\mathcal{O}(\boldsymbol{t}^2)$. After introducing coefficient matrix

$$A_{\mu\nu} = \langle\widetilde{\psi}_\mu|\mathcal{H}|\psi_\nu\rangle \ ,$$

and vector

$$b_\mu = \langle\widetilde{\psi}_\mu|\mathcal{H}|\Phi\rangle + \mathcal{O}(\boldsymbol{t}^2) \ ,$$

the amplitude equation can be recast as a pseudo-linear system of equations, where nonlinear terms are "hidden" in the inhomogeneous term in each iteration:

$$\sum_\nu A_{\mu\nu} t_\nu = -b_\mu(\boldsymbol{t}) \ . \tag{5.2.20}$$

More compactly:

$$\boldsymbol{A}\boldsymbol{t} = -\boldsymbol{b}(\boldsymbol{t}) \ . \tag{5.2.21}$$

CC iteration usually proceeds via rearranging the equation so that

$$A_{\mu\mu} t_\mu = -b_\mu(\boldsymbol{t}) - \sum_{\nu\neq\mu} A_{\mu\nu} t_\nu \ .$$

Updating t_μ is then achieved via division by diagonal element $A_{\mu\mu}$. Note that often an even simpler rearrangement is chosen, motivated by MBPT, where only one-body parts of the diagonal term are used (see Secs. 2.3.2 and 3.3.1).
Instead of this route, let us consider updating $\boldsymbol{t}$ via inversion of the whole coefficient matrix

in Eq. (5.2.21). In the case where the set of $\{|\psi_\mu\rangle\}$ is overcomplete, the coefficient matrix is invertible only in the Moore-Penrose sense. To make this more transparent, let us rewrite $A_{\mu\nu}$ using Eqs. (5.2.3) and (5.2.5), to get

$$A_{\mu\nu} = \sum_{\gamma,\delta} V_{\mu\gamma} \sigma'^{-1}_{\gamma} \alpha_{\gamma\delta} \sigma'_{\delta} V^*_{\nu\delta} \tag{5.2.22}$$

with

$$\alpha_{\gamma\delta} = \langle e_\gamma | \mathcal{H} | e_\delta \rangle \qquad (\gamma, \delta = 1, ..., \mathcal{M}) \ . \tag{5.2.23}$$

Since the set $\{|e_\mu\rangle\}$ is linearly independent, matrix $\boldsymbol{\alpha}$ is invertible. Eq. (5.2.22) can be written more compactly as

$$\boldsymbol{A} = \boldsymbol{V} \begin{bmatrix} \boldsymbol{\sigma}'^{-1} \\ \mathbf{0} \end{bmatrix} \boldsymbol{\alpha} \begin{bmatrix} \boldsymbol{\sigma}' & \mathbf{0} \end{bmatrix} \boldsymbol{V}^\dagger \ .$$

The Moore-Penrose pseudoinverse is quite straightforward in this notation:

$$\boldsymbol{A}^- = \boldsymbol{V} \begin{bmatrix} \boldsymbol{\sigma}'^{-1} \\ \mathbf{0} \end{bmatrix} \boldsymbol{\alpha}^{-1} \begin{bmatrix} \boldsymbol{\sigma}' & \mathbf{0} \end{bmatrix} \boldsymbol{V}^\dagger \ . \tag{5.2.24}$$

We can similarly write

$$\boldsymbol{b} = \boldsymbol{V} \begin{bmatrix} \boldsymbol{\sigma}'^{-1} \\ \mathbf{0} \end{bmatrix} \boldsymbol{\beta} \ ,$$

with

$$\beta_\delta(\boldsymbol{t}) = \langle e_\delta | \mathcal{H} | \Phi \rangle + \mathcal{O}(\boldsymbol{t}^2) \qquad (\delta = 1, ..., \mathcal{M}) \ . \tag{5.2.25}$$

Then the update of the amplitude can be carried out to give

$$\boldsymbol{t} = -\boldsymbol{A}^- \boldsymbol{b}(\boldsymbol{t}) = -\boldsymbol{V} \begin{bmatrix} \boldsymbol{\sigma}'^{-1} \\ \mathbf{0} \end{bmatrix} \boldsymbol{\alpha}^{-1} \boldsymbol{\beta}(\boldsymbol{t}) \ . \tag{5.2.26}$$

Comparing Eqs. (5.2.26) and (5.2.17) one can deduce

$$\boldsymbol{t}^e = -\boldsymbol{\alpha}^{-1} \boldsymbol{\beta}(\boldsymbol{t}) \ . \tag{5.2.27}$$

Amplitude update based on the Moore-Penrose inverse of coefficient matrix $\boldsymbol{A}$ is therefore equivalent to updating in the Löwdin-basis according to Eq. (5.2.27) and transforming $\boldsymbol{t}^e$ with the help of Eq. (5.2.17). Note that we usually know terms of the amplitude equation expressed in the overcomplete set $\{|\psi_\mu\rangle\}$ (like $A^\psi_{\mu\nu} = \langle \psi_\mu | \mathcal{H} | \psi_\nu \rangle$, or $b^\psi_\mu = \langle \psi_\mu | \mathcal{H} | \Phi \rangle + \mathcal{O}(\boldsymbol{t}^2)$), since these are the ones that can be directly found using algebraic or diagrammatic techniques. Their Löwdin basis counterparts ($\alpha_{\gamma\delta}$ and $\beta_\delta(\boldsymbol{t})$) are easily constructed using Eq. (5.2.2):

$$\boldsymbol{\alpha} = \begin{bmatrix} \boldsymbol{\sigma}'^{-1} & \mathbf{0} \end{bmatrix} \boldsymbol{V}^\dagger \boldsymbol{A}^\psi \boldsymbol{V} \begin{bmatrix} \boldsymbol{\sigma}'^{-1} \\ \mathbf{0} \end{bmatrix} \quad , \quad \boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\sigma}'^{-1} & \mathbf{0} \end{bmatrix} \boldsymbol{V}^\dagger \boldsymbol{b}^\psi \ . \tag{5.2.28}$$

and then Eq. (5.2.27) can be applied immediately.

This concludes the construction of unique amplitudes. Note that the treatment is general in the sense that it is applicable to any excitation level or approximate form of the CC equations. Merely matrices $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}(\boldsymbol{t})$ need to be adapted to the particular CC scheme.

In general, diagonalization of $\boldsymbol{S}$ and inversion of $\boldsymbol{\alpha}$ could be quite demanding computation-wise. Note however, that in practice (for example, when using a strongly orthogonal geminal reference), the overlap matrix admits a block-diagonal form, and both the diagonalization and the redundancy handling can be done independently for each (typically small) block.

Let us revisit the redundant two-by-two block of the overlap matrix from the previous section (now simply called $\boldsymbol{S}$), to illustrate the above procedure. The two states considered are $|\psi_1\rangle =: E^{ap}_{pi} : |\Phi\rangle$ and $|\psi_2\rangle =: E^{aq}_{qi} : |\Phi\rangle$. Spectral decomposition of

$$\boldsymbol{S} = 2n_p n_q \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

reveals

$$\Sigma' = 4n_p n_q \quad , \quad \sigma' = 2\sqrt{n_p n_q} \quad , \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} .$$

The coefficient matrix and the inhomogeneous term read

$$\boldsymbol{A}^{\psi} = \begin{bmatrix} A^{\psi}_{11} & A^{\psi}_{12} \\ A^{\psi}_{21} & A^{\psi}_{22} \end{bmatrix} \quad , \quad \boldsymbol{b}^{\psi} = \begin{bmatrix} b^{\psi}_1 \\ b^{\psi}_2 \end{bmatrix} .$$

After carrying out the transformation in Eq. (5.2.28), we have

$$\alpha = \frac{A^{\psi}_{11} + A^{\psi}_{22} - A^{\psi}_{12} - A^{\psi}_{21}}{2\Sigma'} \quad , \quad \beta = \frac{b^{\psi}_1 - b^{\psi}_2}{\sqrt{2\Sigma'}} ,$$

which immediately leads to

$$t^e = -\frac{\beta}{\alpha} = -\sqrt{2\Sigma'} \frac{b^{\psi}_1 - b^{\psi}_2}{A^{\psi}_{11} + A^{\psi}_{22} - A^{\psi}_{12} - A^{\psi}_{21}} .$$

Transformation with Eq. (5.2.17) gives the two final amplitudes:

$$\boldsymbol{t} = -\frac{b^{\psi}_1 - b^{\psi}_2}{A^{\psi}_{11} + A^{\psi}_{22} - A^{\psi}_{12} - A^{\psi}_{21}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} .$$

We have thus found $t_1 = -t_2$, which is, of course, not surprising, as $|\psi_1\rangle = -|\psi_2\rangle$. Nevertheless, the procedure gave us unique amplitudes for both states, which was our goal. The redundancy treatment works just like that in larger blocks too, where linear dependence of vectors may take a less trivial form.

Other, more practical questions concerning the redundancy handling in GVB excitations will be addressed in the next chapter.

Chapter 6

Computation of Potential Energy Surfaces

This chapter deals with some practical aspects of the approximate ic-MRCC models idCCSD, idCCSD+λF, rmCCD and rCCD(UGA), all of them having been introduced in Sec. 3.3.3. We present the details of ic-MRCC calculations built on the GVB wave function. Computational benefits of choosing GVB as reference function are discussed in detail, just like the hardships associated with our ic-MRCC calculations (convergence issues, and the need to remove certain excitations from the Ansatz).

This is followed by numerical illustrations involving Potential Energy Surface (PES) calculations to test the capability of the ic-MRCC approximations in describing strongly and weakly correlated systems. The examples investigated include breaking of covalent bonds, conformational changes, deprotonation processes and dispersion interactions. Strengths and weaknesses of the approximation schemes are discussed.

In this chapter, latin indices represent spatial orbitals.

This chapter is based on Refs. [137, 138, 147].

6.1 Computational details

In practice, all of our ic-MRCC corrections are built on the GVB wave function (see Sec. 2.2.1). GVB calculations start from restricted HF orbitals localized to bonds or atoms using the Foster-Boys procedure [160]. These localized spatial orbitals are then assigned to geminals. Two spatial orbitals are assigned to all valence geminals corresponding to bonding and lone electron pairs. On the other hand, core geminals describing inner-shell electrons are kept on the HF level and get only one spatial orbital. All other orbitals not assigned to any of the geminals are virtual.
The GVB procedure optimizes orbitals and geminal coefficients simultaneously. The resulting active orbitals (belonging to valence geminals) are chosen to be natural orbitals diagonalizing the 1-RDM. Core and virtual orbitals remain to be the very same localized HF

orbitals we started with and are natural orbitals by construction (with eigenvalues $n_i = 1$ and $n_a = 0$, respectively).
Let the number of core, active and virtual spatial orbitals be denoted by m_c, m_a and m_v, respectively, with the total number of spatial orbitals given by $\bar{M} = m_c + m_a + m_v$.

Using the results of Sec. 2.2.2, cumulants of the GVB wave function can be constructed in a straightforward manner. At first, storing these cumulants appears to be an obstacle, since a K-body cumulant is composed of m_a^{2K} elements; when dealing with a large molecule having many active orbitals, this number could be extremely large even for relatively low-rank cumulants. This problem, however, is easy to overcome, as the strongly orthogonal nature of GVB makes cumulants very sparse: all of their $2K$ indices must belong to the same geminal, otherwise they vanish. This makes the maximal number of non-zero entries 2^{2K} per geminal, hence $2^{2K}N/2 = 2^{2K-1}N$ for the complete wave function. Furthermore, GVB cumulants are identically zero for $K > 4$ (see Sec. 2.2.2). Therefore, GVB cumulants are quite easy to store as long as the number of geminals does not become too high.

Having obtained the necessary cumulants, dressed interaction vertices are to be constructed in the next step. Similarly to cumulants, dressed vertices need to be calculated only once before starting the CC iteration. Each variant of a K-body dressed vertex (corresponding to the different leg orientations) is an array containing roughly $\bar{M}^{2K}$ elements. The actual number of elements is smaller when excluding identically zero entries (e.g. V_{ic}^{ak} has no more than $(m_c + m_a)^2(m_a + m_v)^2$ non-zero elements); exploiting permutational symmetry could lead to further reduction.
Calculation of dressed vertices could become quite costly with a general MR function; as an example, let us concentrate on the two-body dressed vertex. Looking at Eq. (3.2.18), we can see that terms involving the four-body cumulant or a product of two-body cumulants are the most demanding computation-wise, requiring roughly $\bar{M}^4 n_a^4$ steps to build. Sparsity of GVB cumulants greatly reduces this number, as all indices belonging to a cumulant must run over the same geminal. In the case of the two-body vertex, this renders the number of steps to build the cumulant dressing at most $2^7 N$, meaning that dressing terms can be added quite inexpensively to bare vertices.

In the next step, the overlap matrix $\boldsymbol{S}$ needs to be constructed and diagonalized for all spin-summed excitations of the GVB function that we plan to use for the ic-MRCC correction (see Sec. 6.1.1). Due to the strong orthogonality of GVB, $\boldsymbol{S}$ is block-diagonal, and one can investigate the small blocks separately. Excited states are to be classified based on the eigenvalues of these blocks; there are three possible cases.
The simplest case is when none of the block eigenvalues are zero: this means that all of the constituent excited states are unique, and no further redundancy treatment is needed. During the iterative solution of the amplitude equation, amplitudes assigned to these excitations can be updated with the usual technique of Sec. 3.3.1.
If some of the block eigenvalues are zero then there is redundancy in the states, and the corresponding amplitudes are to be updated with the technique of Sec. 5.2.
Finally, the special case of all block eigenvalues being zero signals that excited states of the block are actually zero vectors. These states can be discarded, as the procedure of Sec. 5.2 would assign a zero amplitude to all of them anyway. This is also the reason why the results

do not depend on the block structure of $\boldsymbol{S}$: one could in principle append any block with additional zero states, as the redundancy treatment would always make their amplitudes vanish. In practice, however, it is recommended to discard zero states, as division by very small numbers may introduce numerical instabilities.

Of course, in order to classifiy states the above way, a numerical threshold must be set up to distinguish between zero and non-zero eigenvalues. A threshold of $\Sigma_{min} = 10^{-10}$ is used uniformly for all rCCD(UGA) calculations, while calculations done with rmCCD, idCCSD and idCCSD+λF employ $\Sigma_{min} = 10^{-9}$. In the latter three cases, a second threshold of $\Sigma_{block} = 10^{-8}$ was set for the trace of overlap blocks: all states are discarded if the trace of their block falls below Σ_{block}.

Scaling as $\mathcal{O}((n_c + n_a)^3(n_a + n_v)^3)$, solution of the approximate ic-MRCCSD equations has roughly the same computational cost per iteration as single determinantal CCSD. However, in most cases, making the solution of amplitude equations converge is not straightforward: poor choice of control parameters may lead to very slow convergence or even divergence of the iteration. While the MBPT-like energy denominator of Eq. (3.3.12) seems to be the best from an interpretational point of view, in practice it is much better to use the diagonal part of the complete coefficient matrix (in the given approximation scheme) to update the non-redundant amplitudes. In redundant blocks, amplitudes are updated via inversion of the whole coefficient matrix, as discussed in Sec. 5.2. In the strongly correlated parts of the PES, shifting the control parameters by a small constant value often improves convergence. In order to achieve convergence with the rCCD(UGA) and rmCCD methods, an additional scaling of the control parameters is required.

6.1.1 Pruning the excitation manifold

Up to this point, we have not discussed the necessity to omit certain excitations from the parametrization of T; we have simply included all possible excitations from core/active orbitals to active/virtual orbitals. Sadly, this approach might run into trouble if we try to excite either from very weakly occupied orbitals (with $n_i \ll 1$) or to very strongly occupied orbitals (with $\bar{n}_a \ll 1$).
The reason can be understood by analyzing the leading order contributions to amplitudes (cf. Eq. (3.3.13)). To keep things simple, we present the investigation with Eq. (3.3.18) in mind; the same conclusions could be drawn from the UGA form of the amplitude equation (cf. Eqs. (3.3.17) and (3.3.19)). For the t_2, we find

$$(t^{(0)})^{ij}_{ab} = \frac{V^{ij}_{ab}}{D^{ij}_{ab}} \ , \tag{6.1.1}$$

where D^{ij}_{ab} are the control parameters of the iteration; these may come from e.g. the diagonal part of the linear amplitude equation terms, as discussed in the previous section. Without its "legs" (the factors of $n^i, n^j, \bar{n}_a, \bar{n}_b$), vertex V^{ij}_{ab} may grow quite large if orbitals ϕ^i, ϕ^j are weakly occupied, or if orbitals ϕ^a, ϕ^b are strongly occupied. For example, let us investigate

the two terms of V_{ab}^{ij} below (cf. the first two terms in the second row of Eq. (3.2.5) after division by $n^i n^j \bar{n}_a \bar{n}_b$):

$$\begin{aligned}\frac{1}{n^i n^j}\sum_{p,q} v_{ab}^{pq}\lambda_{pq}^{ij} &= \delta^{ij}\frac{c^i}{(n^i)^2}\sum_{p\in\mathbb{G}_I} v_{ab}^{pp}c_p - \delta^{IJ}v_{ab}^{ij} \\ &= \delta^{ij}\frac{1}{n^i}\left(v_{ab}^{ii} + \frac{c^i}{n^i}\sum_{\substack{p\in\mathbb{G}_I \\ p\neq i}} v_{ab}^{pp}c_p\right) - \delta^{IJ}v_{ab}^{ij} \ , \end{aligned} \tag{6.1.2}$$

and

$$\frac{1}{\bar{n}_a\bar{n}_b}\sum_{p,q} v_{pq}^{ij}\lambda_{ab}^{pq} = \delta_{ab}\frac{c_a}{(\bar{n}_a)^2}\sum_{p\in\mathbb{G}_A} v_{pp}^{ij}c^p - \delta_{AB}\frac{n_a n_b}{\bar{n}_a\bar{n}_b}v_{ab}^{ij} \ . \tag{6.1.3}$$

Note that we use the notations of Sec. 3.3.2 for spatial orbital quantities (e.g. $\lambda_{qs}^{pr} = \lambda_{q\alpha s\beta}^{p\alpha r\beta}$). The above expressions were obtained with (cf. Eqs (2.2.5) and (2.2.7))

$$\lambda_{qs}^{pr} = \delta_Q^P\delta^{pr}\delta_{qs}c^p c_q - \delta_{QS}\delta_q^p\delta_s^r n_q n_s \ ,$$

where geminal indices of the coefficients were omitted to avoid clutter.
The expression of Eq. (6.1.2) is perfectly well-defined for $i \neq j$: it describes an excitation from two different orbitals of the same geminal. On the other hand, case $i = j$ (corresponding to a double excitation from the same orbital) is problematic if ϕ^i is weakly occupied due to the small occupation numbers appearing in the denominator.
Eq. (6.1.3) has the same problem: while case $a \neq b$ is always bounded (note that $n_a n_b/(\bar{n}_a\bar{n}_b) \in [0,1]$ for a general APSG geminal, and is equal to one for GVB), the expression can grow arbitrarily large when $a = b$.
Other parts of V_{ab}^{ij} might show similarly pathological behaviour. For example, the last term in the second row of Eq. (3.2.5) gives rise to

$$-\frac{1}{n^i\bar{n}_b}\sum_{p,q} v_{ap}^{qj}\lambda_{qb}^{ip} = -\delta_B^I\frac{1}{\bar{n}_b}\left(\frac{c^i}{n^i}c_b v_{ai}^{bj} - n_b v_{ab}^{ij}\right) \ ,$$

and other similar contributions. This term may grow very large for intrageminal excitations if ϕ^i is weakly occupied and ϕ^b is strongly occupied. Note that the spectator term corresponding to $i = b$ is just $-v_{ai}^{ij}$, and is well-behaving.

Looking at Eq. (6.1.1), we can see that initial value of the amplitudes assigned to these excited states could thus take huge values, which usually cannot be countered by the control parameters; as a simple example, the analysis of Appendix E confirms the finiteness of the MBPT-like energy denominator $D_{ab}^{ij} = -(\epsilon_a^i + \epsilon_b^j)$.

In principle, such very small-norm excitations should not pose a problem for the general theory, since the effect of overly large dressed vertices is always dampened by the small occupation factors in the amplitude equations $n^i\bar{n}_a\omega_a^i = 0$, $n^i n^j\bar{n}_a\bar{n}_b\omega_{ab}^{ij} = 0$. However, the above considerations show that in practice, these states may introduce serious numerical

instabilities during the CC iteration, endangering the convergence of the procedure. In order to avoid this, we discard these excited states from the ic-MRCC Ansatz. Note that the ill-effect of excitations changes with the change of occupation factors along the PES: during the dissociation of a covalent bond, strongly and weakly occupied orbitals of a GVB geminal become quasi-degenerate in occupation, and the problem associated with e.g. Eq. (6.1.2) diminishes. This means that in the dissociation limit, certain excitations could be allowed which are forbidden around equilibrium geometry. But this would potentially introduce discontinuities along the PES in points where the number of excitations changes; therefore, it is best to discard problematic excitations uniformly along the dissociation curve.

By examining the other terms of one- and two-body dressed vertices similary to the above, it is not hard to identify potentially dangerous excited states, hence define excitation selection schemes for our ic-MRCC approximations.
The rCCD(UGA) calculations are carried out with the following restrictions on the excitation manifold $\{:E_{ij}^{ab}: |\Phi_{GVB}\rangle\}$:

- $n_i \bar{n}^a > 1/4$ if $i \neq a$ and $I = A$,
- $n_i \bar{n}^b > 1/4$ if $i \neq b$ and $I = B$,
- $n_j \bar{n}^a > 1/4$ if $j \neq a$ and $J = A$,
- $n_j \bar{n}^b > 1/4$ if $j \neq b$ and $J = B$,
- $n_i > 1/2$ if $i = j$,
- $\bar{n}_a > 1/2$ if $a = b$.

The excitation manifold $\{:E_k^c: |\Phi_{GVB}\rangle\} \cup \{:E_{ij}^{ab}: |\Phi_{GVB}\rangle\}$ is somewhat more restricted for the idCCSD, idCCSD+λF and rmCCD methods, spectator-type excitations (see Sec. 5.2) being omitted:

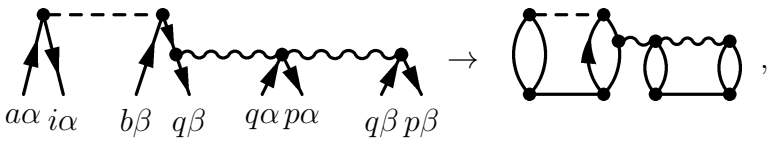
- $n_k > 1/2$,
- $\bar{n}_c > 1/2$,
- $n_i \bar{n}^a > 1/4$ if $I = A$,
- $n_i \bar{n}^b > 1/4$ if $I = B$,
- $n_j \bar{n}^a > 1/4$ if $J = A$,
- $n_j \bar{n}^b > 1/4$ if $J = B$,
- $n_i > 1/2$ if $i = j$,
- $\bar{n}_a > 1/2$ if $a = b$.

Of course, the first two conditions (restricting single excitations) do not have any effect on rmCCD. Note that the above restriction of excitations does not violate size consistency. Also note that linear dependence of excited states persists even with such a strict selection: states $:E^{qr}_{pp}: |\Phi_{GVB}\rangle$ and $:E^{ss}_{qr}: |\Phi_{GVB}\rangle$ (with $p,q \in \mathbb{G}_P$, $r,s \in \mathbb{G}_R$, $P \neq R$, $n_p \geq n_q$, $n_r \geq n_s$) form redundant two-by-two blocks, and need to be treated with the procedure of Sec. 5.2.

6.1.2 High-rank dressed vertices in GVB

In Sec. 3.3.3, we argued that the inclusion of high-rank dressed vertices in the energy formula might not lead to a balanced MRCCSD formalism. Nevertheless, it would be important to have at least an estimation about the effect of these neglected dressed vertices on the energy. The simple structure of GVB cumulants makes it possible to carry out such an analysis in a special case.
In this section, spin orbital dressed vertices $\tilde{W}_K$ are used instead of spin free vertices W_K. Spin indices are shown explicitily.

Let us consider the GVB description of a system with only two active spatial orbitals (that is, four spin orbitals with indices $p\alpha, p\beta, q\alpha, q\beta$; let us choose $n_p \geq n_q$). A basic example would be a molecule with only one correlated bond, all other electron pairs being treated on the HF level (that is, their geminals contain only one spatial orbital). Furthermore, let us apply the stricter excitation selection scheme devised for idCCSD, idCCSD+λF and rmCCD.
It is not hard to see that almost all energy contributions associated with high-rank vertices vanish in this setting. The reasons can be summed up as follows. Cumulant terms of the dressed vertex can only take values from the one geminal with active orbitals, that is, the only labels that can be assigned to cumulant legs are $p\alpha, p\beta, q\alpha, q\beta$. This means that vertex diagrams may have at most four incoming and four outgoing cumulant legs, otherwise they vanish by antisymmetry. Finally, only those vertex diagrams can contribute to the energy which can be connected to amplitudes of allowed excitations (that is, excitations not forbidden by the selection scheme of the previous section). Vertices of rank five or higher (having at least seven cumulant legs) cannot conform to all of these conditions. In the four-body vertex, one may find a diagram that satisfies all the above criteria:



and another one obtained by reversing the orientation of fermion lines. However, this diagram is zero in GVB, since $\lambda^{q\alpha q\beta p\beta}_{p\alpha p\beta q\beta} = 0$ is the only cumulant contributing to it. The vanishing of this cumulant can be easily seen: based on Appendix C, it can be written as a linear combination of spin-summed cumulants Λ^{qqp}_{ppq} and Λ^{qqp}_{pqp}, both of which are zero based on Eq. (2.2.11).
The three-body vertex may contribute to the energy through the fourth and fifth skeletons of Eq. (3.2.9). However, as discussed in Sec. 3.3.3, its effect on the energy is limited, as it must be contracted to at least one t_1 (which is expected to be small).

The above considerations were based on an overly simple model system with only two active orbitals. Despite this, the examples might serve as an indicator that high-rank vertices can be omitted in more realistic systems as well.

6.2 Some numerical results

The quality of our proposed ic-MRCC approximations is tested on PES calculations. The PES is calculated by adding the nuclear repulsion term to the electronic energy at each nuclear configuration $\boldsymbol{R}$:

$$E_{PES}(\boldsymbol{R}) = E(\boldsymbol{R}) + \sum_{A<B} \frac{Z_A Z_B}{|\boldsymbol{R}_A - \boldsymbol{R}_B|} \ ,$$

electronic energy $E(\boldsymbol{R}) = E_{ref}(\boldsymbol{R}) + \Delta E(\boldsymbol{R})$ being made up of reference energy $E_{ref} = \langle H \rangle$ and correction ΔE calculated with one of the methods in question.

The primary test cases include systems undergoing covalent bond dissociations: the single and double bond dissociation of H_2O and the triple bond dissociation of N_2. The BeH_2 $\rightarrow$Be+H_2 reaction pathway provides an example for a process involving qualitative change of molecular structure. Another process involving change of structure, the square-to-rectangle transformation of H_4 is also examined. In the case of these strongly correlated or structure changing systems, Full Configuration Interaction (FCI) serves as benchmark, calculated with an in-house implementation of Olsen's algorithm [161]. The one exception is the case of N_2 bond dissociation, where the semistochastic heatbath CI (shCI) of Sharma et al. [162] is used for this purpose.
The performance of GVB-rCCD(UGA) is also assessed on weakly correlated systems, involving calculation of conformational barriers for NH_3, H_2O_2 and H_2O, deprotonation energy for H_2O and CH_3OH and dispersion interactions of He and Ne dimers. These results are tested against HF-CCSD(T) energies calculated with the Gaussian 09 software [163].

For the sake of comparison, results obtained with another MR correction method, the GVB-based Extended RPA (GVB-ERPA) of Pernal et al. [55, 56] are also presented.

All calculations based on the GVB reference were performed with the Budapest version of the MUNGAUSS program package [164]. General details of the calculations were discussed in Sec. 6.1.
The rCCD(UGA) calculations were presented in Ref. [138], except for the dispersion energy curves which appear in Ref. [147]. All other calculations involving idCCSD, idCCSD+λF and rmCCD were presented in Ref. [137].

6.2.1 Covalent bond dissociations

Single bond dissociation of H_2O

As discussed in Sec. 2.2.1, strongly orthogonal geminal wave functions are ideal choices for a qualitatively correct description of a single dissociating covalent bond. The single bond

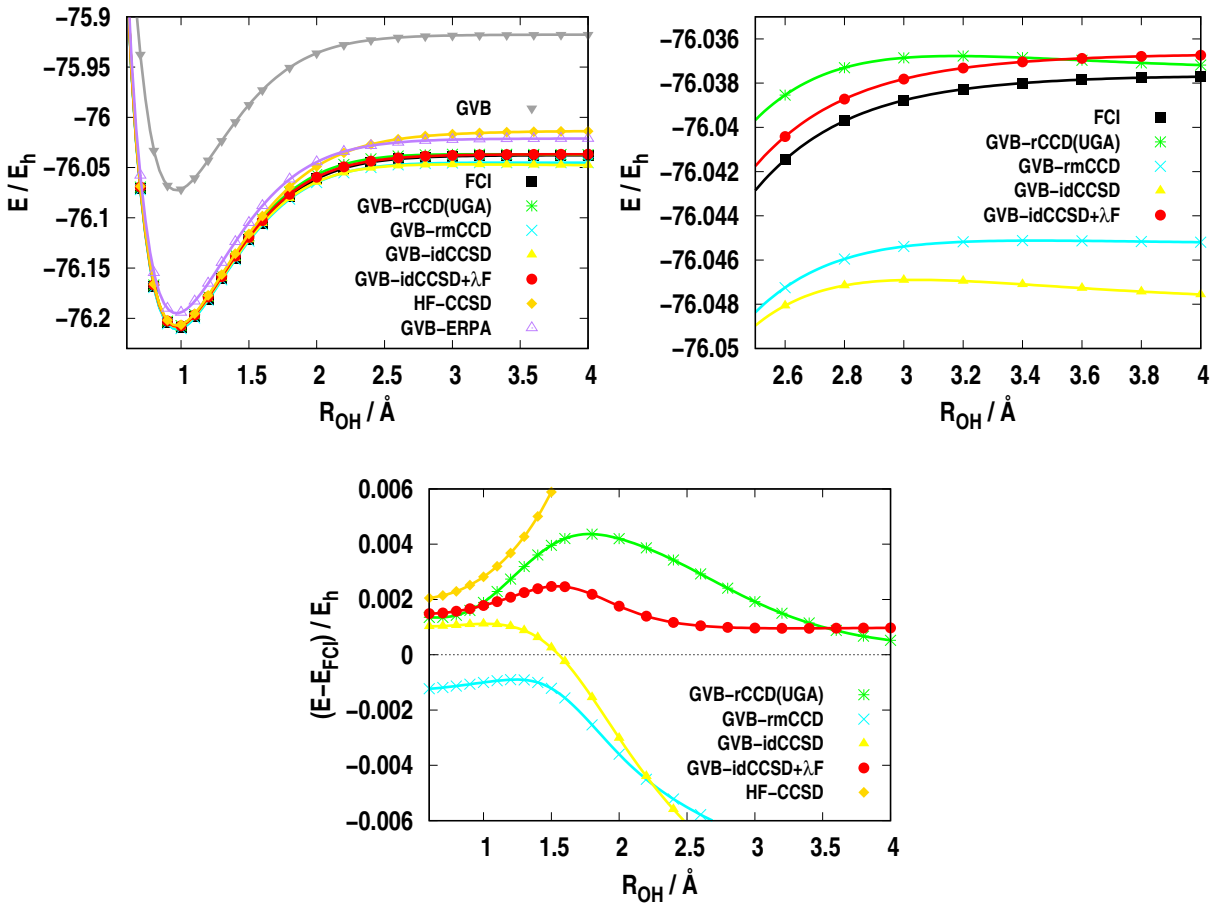


Figure 6.1: Single bond dissociation of the H_2O molecule in 6-31G* basis set, at $\alpha_{OHO} = 104.5°$ bond angle and $R_{OH'} = 1.000$ Å bond length for the non-dissociating bond. Total energy and energy difference computed as $\Delta E = E - E_{\mathrm{FCI}}$ are displayed.

dissociation of H_2O, depicted in Fig. 6.1 thus illustrates the capabilities of the dressed interaction vertex schemes in the most ideal situation, from the point of view of reliability of the reference. A bird's eye view on the PES, shown in the left panel indicates the success of correction methods in reducing the ∼100 mE_h overall error of GVB. It is also apparent in the left panel, that the proposed MRCC methods outperform both HF-CCSD and GVB-ERPA in the dissociated regime (the latter two giving rather similar results).
The right and bottom panels serve for comparing the various MR-CC approaches. Difference curves taken with FCI indicate that in spite of idCCSD being more elaborate than rCCD(UGA), it brings a deterioration in numerical terms. The absolute error of idCCSD builds up from ∼2 mE_h at around equilibrium, to ∼10 mE_h upon dissociation. This is to be compared with the largest error of rCCD(UGA), that is around 4 mE_h. The rmCCD model, which in a sense involves less than idCCSD but more than rCCD(UGA), performs in between. It represents an underestimation of the total energy similarly to the observation made in single determinantal studies [70], but its error is smaller and varies on a narrower

range compared with idCCSD. It is important to see that these three methods (idCCSD, rmCCD, rCCD(UGA)) share a common problem: they all predict a qualitatively incorrect shape for the dissociation curve displaying an unphysical local maximum around 3 Å (cf. the right and bottom panels of Fig. 6.1). GVB-ERPA performs better in this respect, as it does not display such local maxima.
The best performing dressed vertex scheme is the one including not only intravertex, but intervertex correlation, denoted idCCSD+λF. Besides reducing the error to the $\sim$2 mE_h regime along the entire curve, it does not show any local maximum.
Among the MR-CC schemes, the performance of idCCSD varies on the broadest scale, as it provides the smallest error in absolute value at around equilibrium distance, but also the worse estimate in the dissociated regime. We remark that the phenomenon is not accompanied by idCCSD amplitudes growing unduly large upon bond breaking.
The above results hint that a more carefully designed inclusion of terms involving cumulant elements possibly growing large along the PES may represent a cure to the dissociation problem whilst staying within an MRCCSD framework.

Double bond dissociation of H_2O

The double bond dissociation of H_2O, depicted in Fig. 6.2 shows a behaviour of idCCSD similar to the previous example. It outperforms others at around equilibrium, providing an error on the order of 1 mE_h until 1.2 Å bond length, but the error grows into the regime of 50 mE_h as the bonds are stretched further. As the bottom panel of Fig. 6.2 reflects, both idCCSD+λF and rmCCD outperform idCCSD in terms of overall error along the bond breaking process. The error curve of idCCSD+λF in the bottom panel is of similar quality as that of rCCD(UGA), starting with a couple of mE_h around equilibrium and extending to $\sim$15 mE_h upon dissociation. Lack of a local maximum is, however, again an obvious advantage of idCCSD+λF over rCCD(UGA). The rmCCD model is extremely successful here in keeping the absolute error in the couple of mE_h regime along the entire process. A slight hump, hardly discernible in the right panel is present though on the rmCCD curve at around 3 Å. Hump of rCCD(UGA) is somewhat more pronounced.
The ERPA correction shows a similar behaviour to the case of single bond dissociation: its error grows quite large, however, no local maximum is observed on the dissociation curve.

In order to demonstrate the effect of cumulants involved in dressed one- and two-body vertices, a systematic study is carried out on the GVB-rCCD(UGA) description of the H_2O double dissociation, the results being shown in Fig. 6.3. In this study, the rCCD(UGA) method is further approximated by neglecting various rank K cumulant contributions in the dressed one- and two-body vertices. Retained cumulant ranks are shown in parentheses; the acronym UGA is neglected for brevity.
Neglecting all cumulants of rank two and higher defines GVB-rCCD(Λ_1), essentially a single determinantal rCCD with fractional occupation numbers. This HF-like approximation apparently inherits the divergent behaviour of restricted single determinantal rCCD [70], as no converged GVB-rCCD(Λ_1) solution can be found beyond $\sim$1.8 Å.
Since cumulants decay fast with increasing rank around equilibrium, it is understandable

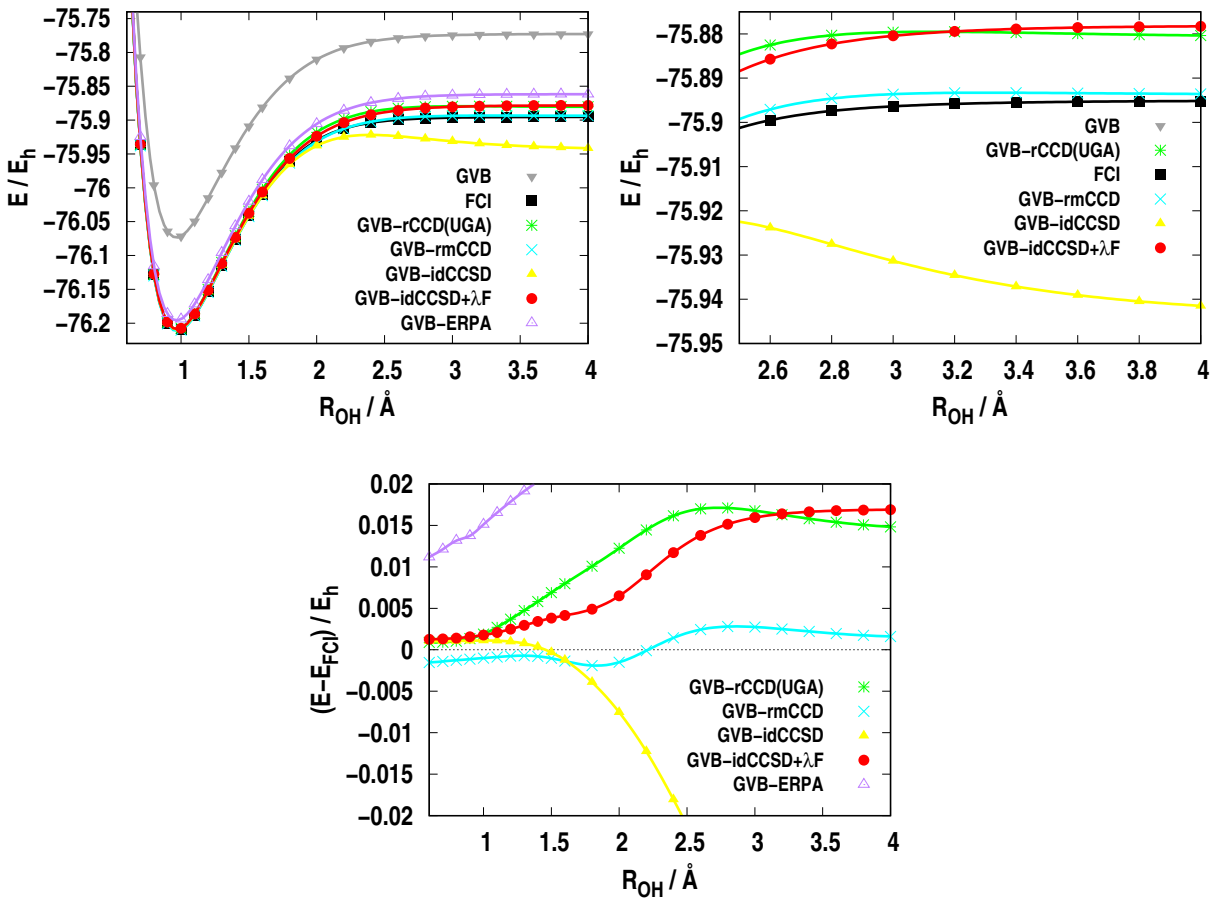


Figure 6.2: Double bond dissociation of H_2O in 6-31G* basis. Bond angle is fixed at α_{OHO}=104.5°. Total energy and energy difference computed as $\Delta E = E - E_{\mathrm{FCI}}$ are displayed.

that inclusion of two-body cumulants already yields quite good results around 1 Å. This scheme, denoted GVB-rCCD(Λ_1, Λ_2) gives a finite but rather disappointing solution in the dissociating regime, where higher rank cumulants are in general not negligible. Accordingly, GVB-rCCD(Λ_1, Λ_2) produces a large hump in the spin recoupling region and tends to an energy value rather close to idCCSD, lying off from FCI by some −50 mEh in the dissociation limit.

The scheme GVB-rCCD($\Lambda_1, \Lambda_2, \Lambda_3$) produces an even more expressed hump and tends essentially to the same limit as GVB-rCCD(Λ_1, Λ_2). This is not surprising: all components of Λ_3 belonging to the dissociating geminal vanish, as was discussed in Secs. 2.1.4 and 2.2.2. Allowing Λ_4 to enter, but discarding Λ_3 defines GVB-rCCD($\Lambda_1, \Lambda_2, \Lambda_4$) giving acceptable results. Curiously, GVB-rCCD($\Lambda_1, \Lambda_2, \Lambda_4$) stays somewhat more close to FCI than GVB-rCCD(UGA) with full inclusion of cumulants (denoted GVB-rCCD($\Lambda_1, \Lambda_2, \Lambda_3, \Lambda_4$) in Fig. 6.3).

The above results confirm what one would expect based on Fig. 2.1 of Sec. 2.2.2, and one can generally conclude that retaining Λ_4 in the dressed two-body vertex is

essential to obtain correct PES for covalent bond dissociation. We remark that the success of GVB-rCCD($\Lambda_1, \Lambda_2, \Lambda_4$) originates from a fortuitous rather than systematic error compensation taking place in this example. Scheme GVB-rCCD($\Lambda_1, \Lambda_2, \Lambda_4$) outperforming GVB-rCCD(UGA) does not occur in general.

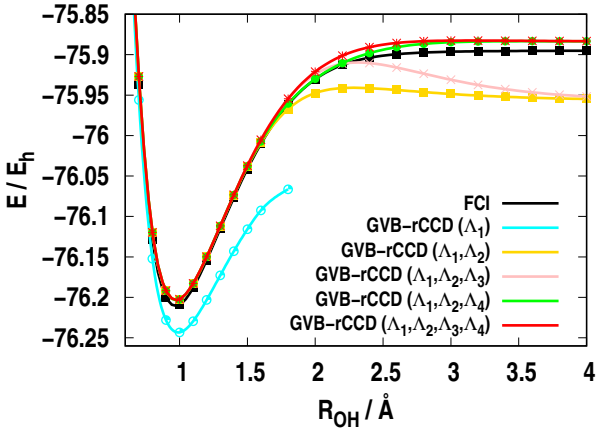


Figure 6.3: Total energy by various cumulant rank based cutoff schemes in GVB-rCCD(UGA) for the double dissociation process shown in Fig. 6.2. Cumulants retained in the one- and two-body dressed vertices are indicated in parenthesis.

Symmetric dissociation of the H_2O molecule serves for one further test, once again demonstrated on rCCD(UGA): examination of the numerical effect of various excitation types as well as the role of excitations affected by redundancy. Results are presented at two geometries in Table 6.1, around equilibrium and in the dissociated regime. The number of excitations retained in a given scheme (n_{exc}), collected in Table 6.1, has to be contested with 10107, the number of original excitations when following the pruning described in Sec. 6.1.1. Note that individual excitations or blocks of excitations with all overlap eigenvalues below the numerical threshold are already excluded in the original scheme. The total number of excitations as well as the structure of the redundant blocks remain unchanged when stepping from $R_{OH} = 1.0$ Å to $R_{OH} = 4.0$ Å.

Excitation types omitted in the schemes examined in Table 6.1 involve exchange and direct spectators ($:E^{pa}_{ip}:$ and $:E^{ap}_{ip}:$) abbreviated as 'exc.spec' and 'dir.spec', respectively. Deexcitation type transitions, i.e. $:E^{ab}_{ij}:$ where $n_i < n_a$ with $I \neq A$ or $n_j < n_b$ with $J \neq B$, are abbreviated as 'deexc'.
In addition, complete elimination of excitations affected by redundancy (i.e. the presence of at least one overlap eigenvalue below the threshold Σ_{min} in a block of $\boldsymbol{S}$) is also examined, abbreviated as 'red'. Note that these excitation types do not define disjoint sets. Some transitions, like $:E^{ai}_{ii}:$ may be considered both exchange and direct spectators; moreover, spectators are heavily affected by redundancy.

As Table 6.1 indicates, roughly 20% of the amplitudes appear in redundant blocks of matrix $\boldsymbol{S}$ in this example. Attempts to converge the amplitude equations without redundancy

excitation type omitted	n_{exc}	$(E - E_{\mathrm{rCCD(UGA)}})$ / E_h	
		$R_{OH} = 1.0$ Å	$R_{OH} = 4.0$ Å
red	8079	$9.562 \cdot 10^{-4}$	$8.284 \cdot 10^{-4}$
deexc	7779	$1.365 \cdot 10^{-4}$	$1.507 \cdot 10^{-4}$
exc.spec	9067	$7.764 \cdot 10^{-5}$	$6.154 \cdot 10^{-5}$
dir.spec	9067	$1.976 \cdot 10^{-4}$	$5.560 \cdot 10^{-4}$
exc.spec, dir.spec	8127	$3.216 \cdot 10^{-4}$	$2.429 \cdot 10^{-4}$
exc.spec, dir.spec, deexc	6415	$4.055 \cdot 10^{-4}$	$3.624 \cdot 10^{-4}$

Table 6.1: Number of retained excitations n_{exc} and difference in total energy compared to GVB-rCCD(UGA) when neglecting excitation types from T. Example is provided by the symmetric dissociation of the H_2O molecule. Number of excitations are to be compared to the total number 10107. See text for abbreviations.

treatment have often run into serious numerical instabilities. Eliminating redundancy affected amplitudes altogether introduces a change in energy essentially on the mE_h level. Other schemes examined in Table 6.1 are accompanied by an energy change that is about an order of magnitude smaller. Interestingly, the role of exchange spectators appears somewhat smaller than direct spectators. Spectators altogether amount to roughly 20% of the amplitudes. Deexcitation type transitions are somewhat more numerous, around 23% of the total amplitudes. The energetic role of deexcitations, however, appears smaller than the effect of all spectators. Variation of the energy change with geometry is most pronounced for direct spectators, but even for this excitation type it falls in the same order of magnitude for $R_{OH} = 1.0$ Å and 4.0 Å.
These results might also help to estimate the loss of energy correction stemming from the stricter excitation selection scheme (applied to methods idCCSD, idCCSD+λF and rmCCD), that completely excludes spectators.

Bond dissociation of N_2

The example of N_2 dissociation (depicted in Fig. 6.4) reflects a further deterioration of the results in the dissociated region, in parallel with the GVB reference becoming gradually less reliable in this sequence of test cases. The idCCSD method is acceptable at around equilibrium, but the dissociation curve shape is deficient, exhibiting a hump at around 2 Å, reminiscent of single determinantal CC methods. Errors taken with shCI, shown in the bottom panel of Fig. 6.4 allow to draw a conclusion similar to the H_2O double dissociation case, with the extent of discrepancies increased. The error of both rCCD(UGA) and idCCSD+λF amount to 50-70 mE_h in the dissociated regime. At variance with previous observations, rCCD(UGA) deviates significantly from other CC methods at around equilibrium. The rmCCD model is again the most successful method regarding overall error. While the negative shift of rmCCD with respect to other CC approximations persists, its error remains in the positive domain. Albeit the error of rmCCD is the smallest among the MR-CC methods tested, it is markedly larger when compared with the previous example

once the bond is stretched. Local maxima in the spin recoupling region of GVB, observable in the right panel of Fig. 6.4, are most expressed among all for this test system. It is only idCCSD+λF that is free of this shortcoming among the CC methods investigated. The ERPA-GVB curve is similary correct in qualitative terms, but has a larger error than idCCSD+λF.

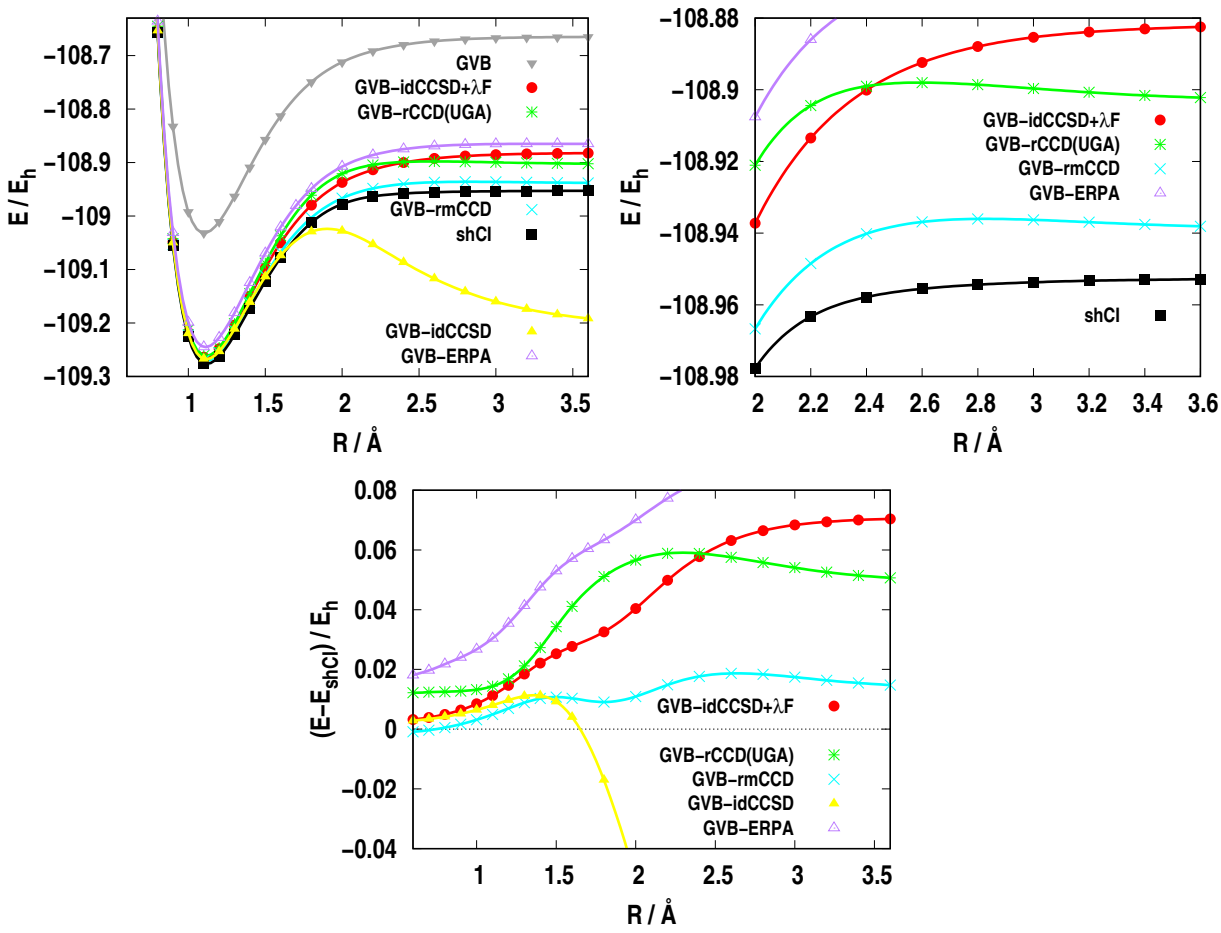


Figure 6.4: Dissociation potential curve of the N_2 molecule in 6-31G* basis. Total energy and energy difference computed as $\Delta E = E - E_{\text{shCI}}$ are displayed.

6.2.2 Structural changes

The BeH_2 system

The BeH_2 →Be+H_2 reaction pathway, as computed by Purvis and Bartlett [165], has become a widely used test of performance in the presence of strong correlation. A further numerical study for our correction methods is carried out on this classical MR test case, using the same double-ζ quality basis set and the same geometry points proposed by the aforementioned authors. In particular, beryllium is placed in the origin of the coordinate system and the

position of the two hydrogen atoms is given by $(d_{\text{Be-H}_2}, \pm d_{\text{H}})$, with $d_{\text{H}} = 1.344\,\text{Å} - 0.46\,d_{\text{Be-H}_2}$. Values for $d_{\text{Be-H}_2}$ in Å (with corresponding points of Purvis et al. indicated in parenthesis) are as follows: 0.000 (A), 0.529 (B), 1.058 (C), 1.322 (D), 1.455 (E), 1.588 (F), 1.852 (G), 2.117 (H), 3.175 (I).
The difficulty of the BeH_2 example lies with a change in the Lewis structure along the reaction pathway: in geometry points A-E (till about $d_{\text{Be-H}_2} = 1.5$ Å) the two valence geminals can be assigned to the two Be-H bonds; however, starting at point F, geminals of the the lowest energy GVB solution can rather be identified as one residing on atom Be and another describing the H-H bond. The change in the nature of the reference is necessarily accompanied by a discontinuity on potential curves.

A pair of single excitations with imaginary excitation energy are excluded from the ERPA correction at geometry points E and I. The rCCD(UGA) calculation is also tweaked at point E: excited states $:E^{rr}_{pq}:\,|\Phi_{GVB}\rangle$ and $:E^{ss}_{pq}:\,|\Phi_{GVB}\rangle$ (orbitals ϕ^p, ϕ^r and ϕ^q, ϕ^s constituting the two BeH geminals, respectively) had to be omitted from T in order to achieve convergence. No such tinkering was required for the convergence of the other three CC approaches.

Fig. 6.5 shows the results. Comparison with the FCI benchmark shows that GVB-ERPA describes a larger portion of the correlation than GVB-rCCD(UGA) in this example. Fig. 6.5. also reflects that parallelity of GVB-rCCD is slightly better: difference from FCI lies in the range 3-29 mE_h for GVB-ERPA (peaking at point E), and 13-30 mE_h for GVB-rCCD. On the other hand, idCCSD, idCCSD+λF and rmCCD all perform much better on this system, their error being in the range of a few mE_h-s at most (also see Table 6.2).

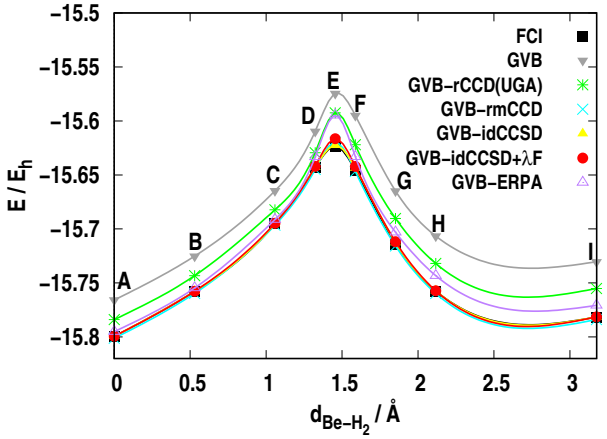


Figure 6.5: Total energy of the BeH_2 system starting from a linear molecule (point A) and ending in a distant Be atom and a H_2 molecule (point I). Geometry and the double-ζ quality basis set taken from Ref. [165].

The BeH_2 system is considered a standard test, offering the possibility of comparison with some of the MRCC methods mentioned in Sec. 2.3.4. Table 6.2 shows the energy differences compared to FCI in points C-G, as obtained with our methods and with three known MRCC approaches from the literature (results taken from Refs. [100, 136]). Among these,

$d_{\text{Be-H}_2}$ / Å		1.058	1.322	1.455	1.588	1.852
ΔE / mE$_h$	GVB-rCCD(UGA)	13.013	13.811	29.772	26.611	24.675
	GVB-rmCCD	-1.187	-1.716	3.772	-0.946	-1.710
	GVB-idCCSD	0.306	0.278	0.928	2.229	1.321
	GVB-idCCSD+λF	0.466	1.097	6.965	3.708	2.178
	CASCCSD(sw)	0.024	0.044	0.072	0.402	0.206
	Mk-MRCCSD	0.609	0.320	-0.613	-1.140	0.257
	ic-MRCCSD	0.239	0.385	1.043	0.534	0.222

Table 6.2: Energy difference computed as $\Delta E = E - E_{\text{FCI}}$ for the BeH_2 molecule. The CASCCSD(sw) results are quoted from Ref. [136], while the Mk-MRCCSD and ic-MRCCSD results are quoted from Ref. [100]. Reference function underlying CASCCSD(sw), Mk-MRCCSD and ic-MRCCSD is CAS(2,2).

CASCCSD(sw) is a single determinantal-like formulation due to Lyakh and coworkers [166], adapted for the MR situation following the basic idea of Ivanov and Adamowicz [134]; abbreviation "sw" in the acronym refers to a swap of the largest weight determinant around $d_{\text{Be-H}_2} = 1.322$ Å. The Mk-MRCCSD of Mukherjee et al. [112] and the ic-MRCCSD of Evangelista and Gauss [117] both fall in the category of genuine Hilbert-space MRCC techniques, harnessing a JM-type and an internally contracted parametrization, respectively. As an ic-MRCC model, the latter bears the most kinship with our approach. The reference underlying these three methods is a CAS(2,2) function (CAS standing for Complete Active Space).

The energy differences of Table 6.2 reflect that the performance of GVB-rCCD(UGA) lags behind MR-based CCSD schemes by roughly two orders of magnitude. This holds true regarding either of the benchmark methods. The other GVB-based methods fare better: in particular, idCCSD proves to be the best dressed vertex-based method for this system. The fact that rmCCD and idCCSD+λF are somewhat less successful in this example, points to the need of careful investigation of beyond-idCCSD terms in various strong correlation situations. Swap in the GVB description of the system occurring between 1.455 and 1.588 Å for $d_{\text{Be-H}_2}$ is accompanied by a slight increase in the error of idCCSD. Notably, idCCSD errors are in the range of those by ic-MRCCSD up to $d_{\text{Be-H}_2} = 1.5$ Å. Errors of internally contracted methods fall behind CASCCSD(sw), compelling further improvement of ic-MRCC approaches.

The H_4 system

Transformation of the H_4 system from rectangle to square geometry is an other classic example of change in the geminal structure along the PES. This change leads to a cusp for GVB at $\alpha_{HXH} = 90°$, as displayed in Fig. 6.6. The cusp shape is roughly conserved by ERPA as well as by rCCD(UGA), making a striking difference with the zero derivative at the square arrangement of the correct FCI curve. While the incorrect curve shape is not alleviated much by the correction methods, the error of GVB is reduced considerably by

rCCD(UGA) as shown by Fig. 6.6. Similarly to the previous examples, rCCD(UGA) and ERPA are comparable, the latter lagging behind rCCD(UGA) by a couple of mE_h-s.

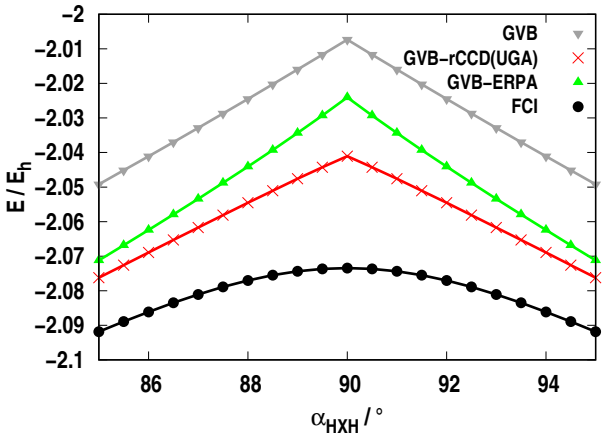


Figure 6.6: Rectangle to square transformation of H_4 in 6-31G** basis. A dummy atom at the center of mass is denoted by X, distance XH is fixed at $R_{XH} = 0.75$ Å.

6.2.3 Conformational barriers

The GVB model accounts for the interaction of chemical bonds only at the SCF level. The interest in torsional barriers described by geminal based methods [56, 167] derives from this fact, since covalent bonds are essentially conserved but their interaction changes during such processes. The three examples picked here are the bending motion of H_2O and the umbrella inversion of NH_3 in the left and right panel of Fig. 6.7, as well as the torsional motion of H_2O_2 in the bottom panel of Fig. 6.7.

Fig. 6.7 uniformly shows a considerable improvement over GVB both by rCCD(UGA) and ERPA. The rCCD(UGA) correction is closer to the CCSD(T) than ERPA for H_2O bending and H_2O_2 torsion. The quality of rCCD(UGA) and ERPA are also comparable when tested against CCSD(T) on the example of NH_3 inversion, the error curves in the bottom right panel of Fig. 6.7 falling in the kcal mol^{-1} range. According to the top right panel, rCCD(UGA) barriers are slightly better in this example.

6.2.4 Deprotonation energies

Deprotonation energy of the H_2O and CH_3OH molecules is calculated as the adiabatic energy difference between the neutral molecule and the anion. Geometry for both structures is optimized at the B3LYP/6-31G* level. Comparing the results collected in Table 6.3 with the CCSD(T) benchmark we see that the error of the dissociation energy is on the order of 10 kcal mol^{-1} by GVB. This is reduced to a couple of kcal mol^{-1} both by GVB-ERPA

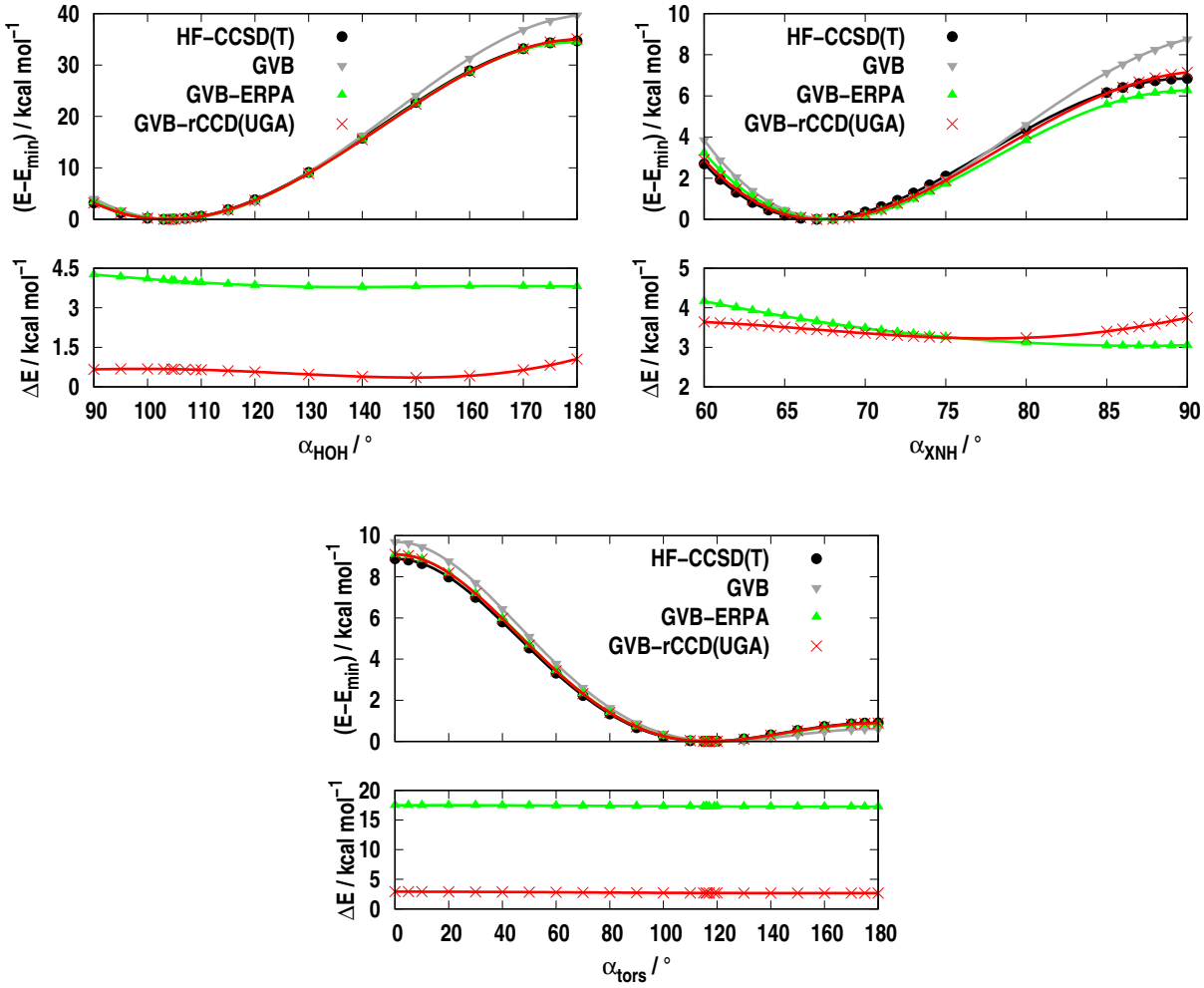


Figure 6.7: Bending motion of H_2O (left panel) and umbrella inversion of NH_3 (right panel) in Dunning-type cc-pVTZ basis set [168]. O-H bond lengths and N-H bond lengths are fixed at $R_{OH} = 0.96$ Å and $R_{NH} = 1.01$ Å, respectively. In the case of NH_3, X denotes a dummy atom in the center of mass.
Bottom panel: torsional motion of H_2O_2 in Dunning-type cc-pVDZ basis set [168]. Bond lengths are fixed at $R_{OH} = 0.97$ Å and $R_{OO} = 1.45$ Å, the value of the bond angle is $\alpha_{OHO} = 99.5°$. The dihedral angle is labeled α_{tors} on axis x.
Energy difference is measured from the minimum of the respective curve in the top halves of each panel, while it is computed as $\Delta E = E - E_{\text{CCSD(T)}}$ in the bottom halves.

and GVB-rCCD(UGA). Accuracy of the two correction schemes does not appear remarkably different in these test cases.

		E_{neutral} / E_h	E_{anion} / E_h	ΔE / kcal mol^{-1}
H_2O	GVB	-76.1060	-75.4627	403.62
	GVB-rCCD(UGA)	-76.2719	-75.6453	393.18
	GVB-ERPA	-76.2580	-75.6245	397.51
	HF-CCSD(T)	-76.2739	-75.6438	395.42
CH_3OH	GVB	-115.1717	-114.5318	401.58
	GVB-rCCD(UGA)	-115.4435	-114.8234	389.12
	GVB-ERPA	-115.4269	-114.7986	394.25
	HF-CCSD(T)	-115.4556	-114.8336	390.29

Table 6.3: Deprotonation energies of the H_2O and CH_3OH molecules in aug-cc-pVDZ basis set [168].

6.2.5 Dispersion energies

Here dispersion energy calculations are presented for dimers He_2 and Ne_2. Both calculations were carried out using the Dunning aug-cc-pVDZ atomic basis set [168] (with five d-orbitals). In reporting the results, E_{disp} stands for the total energy of the supersystem minus total energies of the subsystems, computed according to the Boys-Bernardi scheme [169].
Equilibrium atomic distances and the corresponding dispersion energies can be seen in Table 6.4, while Fig. 6.8 shows the dispersion energy curves for He_2 and Ne_2, respectively. Just like HF, GVB cannot describe dispersion interactions, resulting in non-bonding energy curves. GVB-ERPA and GVB-rCCD(UGA) improve significantly on this situation, although the results are far from perfect. While GVB-ERPA underestimates the dispersion energy in both cases, GVB-rCCD(UGA) overestimates it, although to a somewhat lesser extent. A more significant difference shows up in the equilibrium atomic distances: GVB-ERPA overshoots by 0.4 bohr in each case, while GVB-rCCD(UGA) reproduces the benchmark CCSD(T) value to the digits displayed in Table 6.4.
For the sake of fairness, we remark that deficiencies of GVB-ERPA have been noted before in describing weak interaction, and electron group embedding has been suggested as a cure (GVB-EERPA) [57, 58]. It has also been demonstrated that GVB-EERPA represents a significant improvement over GVB-ERPA in such situations.

The algebraic ring approximation rCCD(UGA) as applied to a HF reference is also presented in Fig. 6.8 and Table 6.4. The results of HF-rCCD(UGA) and GVB-rCCD(UGA) are very similar. This is not surprising: the electron correlation is so weak, that the problem is essentially of single determinantal nature, and the dressed interaction vertices of GVB-rCCD(UGA) barely differ from their bare counterparts. However, the small influence of cumulant dressing still changes the results in the right direction, as both equilibrium energies and distances are improved in GVB-rCCD(UGA) compared to HF-rCCD(UGA).

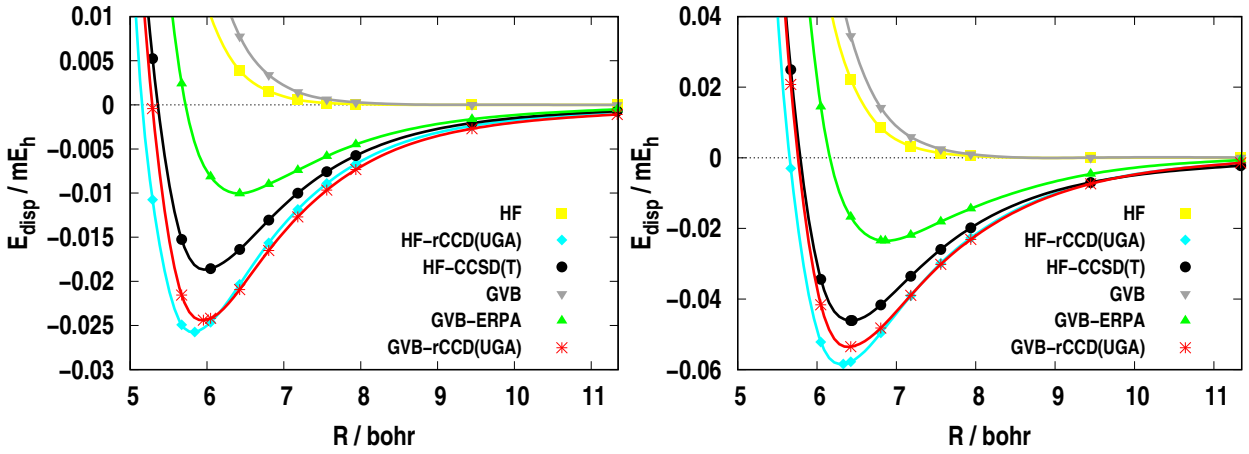


Figure 6.8: Dispersion energy of noble gas dimers calculated in the aug-cc-pVDZ basis set [168] (left panel: He_2; right panel: Ne_2)

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	He_2		Ne_2	
	R_{min} / bohr	E_{int} / mE$_h$	R_{min} / bohr	E_{int} / mE$_h$
HF-CCSD(T)	6.0	-0.0186	6.4	-0.0461
HF-rCCD(UGA)	5.8	-0.0257	6.3	-0.0584
GVB-rCCD(UGA)	6.0	-0.0244	6.4	-0.0529
GVB-ERPA	6.4	-0.0101	6.8	-0.0248

Table 6.4: Equilibrium atomic distances and interaction energies $E_{int} = E_{disp}(R_{min})$ for He_2 and Ne_2, calculated in the aug-cc-pVDZ basis set [168].

Chapter 7

Conclusion and outlook

In the previous chapters, a novel formulation of the ic-MRCC theory has been worked out, based on the notion of cumulant-dependent dressed interaction vertices. Many appealing features of the formalism have been demonstrated, such as a straightforward diagrammatic construction of equations, clear interpretation based on an expansion formally resembling MBPT, size extensivity and consistency over singlet subsystems and invariance under unitary transformations in the subspace of core or virtual orbitals.

Four simple approximations of the general theory have been proposed based on diagrammatic (idCCSD, rmCCD, idCCSD+λF), or algebraic (rCCD(UGA)) arguments.
Performance of these methods has been assessed in Chapter 6 on strongly and weakly correlated molecular systems using a GVB reference. The obtained numerical results indicate that treating only the intravertex part of electron correlation is in general not sufficient to properly describe strongly correlated systems. The idCCSD approximation (which may have appeared to be quite a natural approximation) turned out to perform poorly for bond dissociation processes, both in quantitative and qualitative terms. While rmCCD and rCCD(UGA) typically produced results much closer to benchmark values, both methods lead to qualitatively wrong dissociation curves displaying unphysical local maxima. The only approach that does not showcase such qualitative errors is idCCSD+λF incorporating some portion of intervertex correlation.

The above numerical results naturally raise the question of how to find manageable yet accurate approximations in the general ic-MRCC theory. Estimating the magnitude of individual diagrams (or diagram classes) poses the main difficulty in this respect. The results clearly suggest that intuition gained from the diagrammatic analysis of single determinantal theories may not lead to the correct conclusions in the strongly correlated regime.
Furthermore, one cannot neglect the possibility of compensation effects between diagrams: it is by no means obvious whether blindly adding further explicitly cumulant-dependent diagrams to our equations would improve the results.
Dependence on the reference is manifest in the appearance of cumulants in diagrams. In order to assess the importance of certain diagrams, one should analyse which elements of cumulants grow large in the presence of strong correlation, and how these large elements are connected to interaction vertices in different terms.

Such an analysis should be carried out in the future for GVB to find out which diagrammatic contributions should be further included (or conversely, which contributions should be omitted) to arrive at a well-balanced ic-MRCC method. Given the conceptual advantages of the outlined theory, it would definitely be worth taking this extra effort.

On a more speculative note, let us briefly mention two further possibilities offered by this general framework.

As another route to find simple ic-MRCC models, one may consider drastically reducing the number of excited states in the parametrization of T. We define orbitals ϕ^i with $n_i \geq 1/2$ and orbitals ϕ^a with $n_a < 1/2$ (that is, $\bar{n}_a > 1/2$) to represent hole and particle states, respectively. This is a stricter categorization than the one given in Sec. 2.3.4, as it makes the sets of hole and particle states disjoint. Only excitations from hole states to particle states are allowed; while this would lead to the loss of various excited states of $|\Phi\rangle$ (such as spectators and deexcitations; cf. Sec. 6.2.1), the computational advantages would be prominent. With the disjoint hole-particle sets, one does not need to keep different arrays for the different variants of dressed vertices (corresponding to the different leg orientations): one could store e.g. all elements of F and $\tilde{V}$ in single arrays of size M^2 and M^4, respectively. Also, this categorization would naturally rule out the problematic excitations of Sec. 6.1.1, as well as the redundancy of the excitation manifold (the latter being a consequence of having formally the same excitations as in a single determinantal theory).
From a more conceptual point of view, this separation would allow us to define a dressed MRGNO Hamiltonian in terms of dressed interaction vertices:

$$\begin{aligned}\bar{\mathcal{H}} &= \sum_K \frac{1}{(K!)^2} \tilde{W}^{q_1 \dots q_K}_{p_1 \dots p_K} \, :a^{p_1 \dots p_K}_{q_1 \dots q_K}: \\ &= \sum_{p,q} F^q_p \, :a^p_q: + \frac{1}{4} \sum_{p,q,r,s} \tilde{V}^{qs}_{pr} \, :a^{pr}_{qs}: + \frac{1}{36} \sum_{p,q,r,s,t,u} \tilde{W}^{qsu}_{prt} \, :a^{prt}_{qsu}: + \dots \, ,\end{aligned}$$

indices $p, q, r, s, \dots$ running over all states. Using this Hamiltonian, the general ic-MRCC equations of Chapter 3 can be compactly written as

$$\Delta E = \langle \bar{\mathcal{H}} \, :\exp(T):\rangle_\lambda \quad , \quad 0 = \langle :a^\dagger_K: \, \bar{\mathcal{H}} \, :\exp(T):\rangle_\lambda \, .$$

Subscript λ refers to a connected expectation value where $\bar{\mathcal{H}}$ cannot be contracted with cumulants, only via pair-contractions ($:a^\dagger_K:$ and $:\exp(T):$ can be contracted with cumulants). Of course, the same diagrams are contained in these equations as in Sec. 3.3. However, assuming a GVB reference, the strict hole-particle separation combined with the sparse structure of GVB cumulants makes many classes of diagrams vanish (e.g. the second, seventh, twelfth, and thirteenth diagrams of Eq. (3.3.7)), leading to a more transparent theory. In this simplified framework, one could possibly have a clearer view on which diagrams are important in the presence of strong correlation.

Another interesting possibility is the calculation of various molecular properties with the help of ic-MRCC density matrices. Starting from the definition of Eq. (2.1.12) and essentially following the single determinantal treatment of Ref. [26], one could find

$$\bar{\gamma}_K = \frac{\langle :\exp(T^\dagger): \, a_K \, :\exp(T):\rangle}{\langle :\exp(T^\dagger)::\exp(T):\rangle} = \langle :\exp(T^\dagger): \, a_K \, :\exp(T):\rangle_L$$

for the K-RDM calculated with the ic-MRCC wave function. Subscript L denotes the linked expectation value, as in Sec. 3.1. Applying the MRGNO and the cumulant expansion of density matrices to this formula helps us to identify ic-MRCC density cumulants as

$$\bar{\lambda}_K = \lambda_K + \langle :\exp(T^\dagger)::a_K::\exp(T):\rangle_C \ ,$$

where λ_K is just the cumulant calculated with our starting MR function. This formally exact expression gives ic-MRCC cumulants as an expansion of connected terms. However, simplicity of the above formula is deceptive, as evaluation of the right hand side with the MRGWT becomes extremely complicated even for small powers of T, $T^\dagger$. Diagrammatic techniques (similar to the single determinantal treatment of Ref. [170]) might facilitate this process as well. Finding reasonable (e.g. N-representable) approximations for $\bar{\lambda}_K$ could provide us with ways to accurately calculate various properties of strongly correlated systems.

Addressing these questions remains to be a project for the future.

Appendix A

The relationship between first and second quantization

First and second quantization are equivalent formulations of the same theory, and in this appendix we outline how one can easily switch between these representations. We start by introducing the so-called field operators

$$\psi(x) = \sum_p \phi^p(x) a_p \quad , \quad \psi^\dagger(x) = \sum_p \phi_p(x) a^p \ . \tag{A.0.1}$$

These have a similar interpretation to that of creation-annihilation operators: $\psi^\dagger(x)$ creates an electron at x (i.e. at $\boldsymbol{r}$ with spin σ), while $\psi(x)$ deletes it. The above definitions can be inverted by using the orthonormality of $\{\phi^p(x)\}$:

$$a_p = \int \mathrm{d}x \phi_p(x)\psi(x) \quad , \quad a^p = \int \mathrm{d}x \phi^p(x)\psi^\dagger(x) \ .$$

In this sense, a_p and $\psi(x)$ form Fourier pairs. The anticommutation relations of $\psi(x)$ and $\psi^\dagger(x)$ are easily found using Eq. (2.1.2) and the completeness of $\{\phi^p(x)\}$:

$$\{\psi(x), \psi(x')\} = \{\psi^\dagger(x), \psi^\dagger(x')\} = 0 \quad , \quad \{\psi(x), \psi^\dagger(x')\} = \delta(x, x') \ , \tag{A.0.2}$$

where $\delta(x, x') = \delta(\boldsymbol{r} - \boldsymbol{r}')\delta_{\sigma\sigma'}$. The N-coordinate eigenstate is given by

$$|\boldsymbol{x}_N\rangle \equiv |x_1, ..., x_N\rangle = \frac{1}{\sqrt{N!}} \psi^\dagger(x_1)...\psi^\dagger(x_N)|0\rangle \ . \tag{A.0.3}$$

Using these, Slater determinants of the abstract space (c.f. Eq. (2.1.3)) can be expressed as

$$\begin{aligned} |\Phi^{p_1 \cdots p_N}\rangle &= \int \mathrm{d}x_{\{N\}} \phi^{p_1}(x_1)...\phi^{p_N}(x_N)\psi^\dagger(x_1)...\psi^\dagger(x_N)|0\rangle \\ &= \frac{1}{\sqrt{N!}} \int \mathrm{d}x_{\{N\}} \Phi^{p_1 \cdots p_N}(\boldsymbol{x}_N)\psi^\dagger(x_1)...\psi^\dagger(x_N)|0\rangle \\ &= \int \mathrm{d}x_{\{N\}} \Phi^{p_1 \cdots p_N}(\boldsymbol{x}_N)|\boldsymbol{x}_N\rangle \ . \end{aligned}$$

Here $\int \mathrm{d}x_{\{N\}} = \int \mathrm{d}x_1 ... \int \mathrm{d}x_N$. In the second line we antisymmetrized the product of spin orbitals (exploiting the fact that the integrand was already antisymmetric with respect to the interchange of coordinates), and recognized the real space representation of the Slater determinant:

$$\Phi^{p_1 \dots p_N}(\boldsymbol{x}_N) = \frac{1}{\sqrt{N!}} \mathcal{A}^{p_1 \dots p_N} \left(\prod_{k=1}^{N} \phi^{p_k}(x_k) \right) \ .$$

Non-idempotent operator $\mathcal{A}$ antisymmetrizes the indices in a product of antisymmetric tensors, but only between different tensors. Antisymmetrization within a K-particle tensor would only contribute a factor of $K!$. Simple examples include

$$\mathcal{A}^{pr} \mathcal{A}_{qs} \left(\chi^p_q \xi^r_s \right) = \chi^p_q \xi^r_s - \chi^p_s \xi^r_q - \chi^r_q \xi^p_s + \chi^r_s \xi^p_q \ ,$$

$$\begin{aligned} \mathcal{A}^{prt} \mathcal{A}_{qsu} \left(\chi^p_q \xi^{rt}_{su} \right) = {} & \chi^p_q \xi^{rt}_{su} - \chi^p_s \xi^{rt}_{qu} - \chi^p_u \xi^{rt}_{sq} - \chi^r_q \xi^{pt}_{su} + \chi^r_s \xi^{pt}_{qu} + \chi^r_u \xi^{pt}_{sq} \\ & - \chi^t_q \xi^{rp}_{su} + \chi^t_s \xi^{rp}_{qu} + \chi^t_u \xi^{rp}_{sq} \ . \end{aligned}$$

Using Eq. (2.1.1), a general state of the abstract space can be expressed as

$$|\Psi_N\rangle = \int \mathrm{d}x_{\{N\}} \Psi_N(\boldsymbol{x}_N) |\boldsymbol{x}_N\rangle \ , \tag{A.0.4}$$

where

$$\Psi_N(\boldsymbol{x}_N) = \langle \boldsymbol{x}_N | \Psi_N \rangle = \frac{1}{N!} \sum_{p_1, \dots, p_N} C_{p_1 \dots p_N} \Phi^{p_1 \dots p_N}(\boldsymbol{x}_N) \ .$$

We will show how the – possibly more familiar – first quantized form of the K-particle density matrix can be obtained from Eq. (2.1.12) (with $\mathcal{N}_K = 1$). First we transform γ_K to real space by $\{\phi^p(x)\}$, then use Eqs. (A.0.1) and (A.0.4):

$$\begin{aligned} \gamma(\boldsymbol{x}_K; \boldsymbol{x}'_K) &= \sum_{\substack{p_1, \dots, p_K \\ q_1, \dots, q_K}} \gamma^{p_1 \dots p_K}_{q_1 \dots q_K} \phi_{p_1}(x'_1) ... \phi_{p_K}(x'_K) \phi^{q_1}(x_1) ... \phi^{q_K}(x_K) \\ &= \langle \psi^\dagger(x'_1) ... \psi^\dagger(x'_K) \psi(x_K) ... \psi(x_1) \rangle \\ &= \int \mathrm{d}y_{\{N\}} \int \mathrm{d}y'_{\{N\}} \Psi_N(\boldsymbol{y}_N) \Psi^*_N(\boldsymbol{y}'_N) \Delta(\boldsymbol{x}_K; \boldsymbol{x}'_K; \boldsymbol{y}_N; \boldsymbol{y}'_N) \ , \end{aligned}$$

where $\langle (...) \rangle$ stands for $\langle \Psi_N | (...) | \Psi_N \rangle$. We also introduced

$$\Delta(\boldsymbol{x}_K; \boldsymbol{x}'_K; \boldsymbol{y}_N; \boldsymbol{y}'_N) = \langle \boldsymbol{y}'_N | \psi^\dagger(x'_1) ... \psi^\dagger(x'_K) \psi(x_K) ... \psi(x_1) | \boldsymbol{y}_N \rangle \ .$$

This term can be evaluated using Eqs. (A.0.2) and (A.0.3), or equivalently, with the Wick theorem (cf. Sec. 2.1.5), by considering all possible contractions (pairings) between the various ψ-s and $\psi^\dagger$-s, each contraction contributing a delta function. Note that $\psi(x_i)$ and $\psi^\dagger(x'_j)$ can only be contracted to operators of $|\boldsymbol{y}_N\rangle$ and $\langle \boldsymbol{y}'_N|$, respectively. Because of the antisymmetry, each term in Δ gives the same contribution to $\gamma(\boldsymbol{x}_K; \boldsymbol{x}'_K)$ after performing the integrations; this leads to

$$\gamma(\boldsymbol{x}_K; \boldsymbol{x}'_K) = \Omega_{N,K} \int \mathrm{d}y_{\{N-K\}} \Psi_N(\boldsymbol{x}_K, \boldsymbol{y}_{N-K}) \Psi^*_N(\boldsymbol{x}'_K, \boldsymbol{y}_{N-K}) \ ,$$

where the factor $\Omega_{N,K}$ is just the number of all possible contractions in Δ. Operators $\psi(x_K)...\psi(x_1)$ can be contracted to $|\boldsymbol{y}_N\rangle$ in $\binom{N}{K}K! = N!/(N-K)!$ ways; the same holds for $\psi^\dagger(x'_1)...\psi^\dagger(x'_K)$ and $\langle \boldsymbol{y}'_N|$. The remaining operators of $\langle \boldsymbol{y}'_N|$ and $|\boldsymbol{y}_N\rangle$ can be contracted in $(N-K)!$ ways. Furthermore, a factor of $1/N!$ comes from the normalization in Eq. (A.0.3). Putting these together, we obtain

$$\Omega_{N,K} = \left(\frac{N!}{(N-K)!}\right)^2 \frac{(N-K)!}{N!} = \frac{N!}{(N-K)!} ,$$

giving the final expression for the first quantized $\gamma(\boldsymbol{x}_K; \boldsymbol{x}'_K)$:

$$\gamma(\boldsymbol{x}_K; \boldsymbol{x}'_K) = \frac{N!}{(N-K)!} \int \mathrm{d}y_{\{N-K\}} \Psi_N(\boldsymbol{x}_K, \boldsymbol{y}_{N-K}) \Psi^*_N(\boldsymbol{x}'_K, \boldsymbol{y}_{N-K}) . \tag{A.0.5}$$

One could similarly find the real space representation of "vacancy density matrices" defined in Eq. (2.1.20). For η_1 we have

$$\eta(x; x') = \sum_{p,q} \phi^q(x)\phi_p(x')\eta^p_q = \langle \psi(x)\psi^\dagger(x')\rangle = \delta(x,x') - \gamma(x;x') .$$

From Eq. (A.0.5), one can immediately deduce the real space partial trace relation of the K-RDM (to be compared with Eq. (2.1.17)):

$$\int \mathrm{d}x_K \gamma(\boldsymbol{x}_K; \boldsymbol{x}'_K) = (N-K+1)\gamma(\boldsymbol{x}_{K-1}; \boldsymbol{x}'_{K-1}) .$$

Spin-free density matrices in real space are defined in analogy with Eq. (2.1.56):

$$\Gamma(\boldsymbol{r}_1, ..., \boldsymbol{r}_K; \boldsymbol{r}'_1, ..., \boldsymbol{r}'_K) = \int \mathrm{d}\sigma_1 ... \int \mathrm{d}\sigma_K \gamma(\boldsymbol{r}_1\sigma_1, ..., \boldsymbol{r}_K\sigma_K; \boldsymbol{r}'_1\sigma_1, ..., \boldsymbol{r}'_K\sigma_K) . \tag{A.0.6}$$

Appendix B

The role of cumulants – a minimal example

In order to discuss the importance and interpretation of cumulants, it is instructive to look at a simple model system with a well-known exact solution; for this reason, let us consider the minimal LCAO description of a dissociated H_2 molecule. The discussion is mostly based on Ref. [171].

We have a bonding ("gerade") and an unbonding ("ungerade") spatial orbital:

$$\phi_g(\boldsymbol{r}) = \frac{1}{\sqrt{2}}[\chi_A(\boldsymbol{r}) + \chi_B(\boldsymbol{r})] \quad , \quad \phi_u(\boldsymbol{r}) = \frac{1}{\sqrt{2}}[\chi_A(\boldsymbol{r}) - \chi_B(\boldsymbol{r})] \ ,$$

$\chi_A(\boldsymbol{r})$ and $\chi_B(\boldsymbol{r})$ being hydrogenic $1s$ functions localized on infinitely separated nuclei. Because of this infinite separation, terms like $\chi_A(\boldsymbol{r})\chi_B(\boldsymbol{r})$ always vanish.
Two determinants contribute to the exact wave function: the usual HF determinant

$$\Phi_{g\bar{g}}(\boldsymbol{r}_1\sigma_1, \boldsymbol{r}_2\sigma_2) = \phi_g(\boldsymbol{r}_1)\phi_g(\boldsymbol{r}_2)\Theta(\sigma_1, \sigma_2) \ ,$$

and the doubly excited determinant

$$\Phi_{u\bar{u}}(\boldsymbol{r}_1\sigma_1, \boldsymbol{r}_2\sigma_2) = \phi_u(\boldsymbol{r}_1)\phi_u(\boldsymbol{r}_2)\Theta(\sigma_1, \sigma_2) \ .$$

Here $\Theta(\sigma_1, \sigma_2)$ is a singlet spin function:

$$\Theta(\sigma_1, \sigma_2) = \frac{1}{\sqrt{2}}[\alpha(\sigma_1)\beta(\sigma_2) - \alpha(\sigma_2)\beta(\sigma_1)] \ .$$

The exact wave function can be written as

$$\begin{aligned}
\Psi(\boldsymbol{r}_1\sigma_1, \boldsymbol{r}_2\sigma_2) &= \frac{1}{\sqrt{2}}[\Phi_{g\bar{g}}(\boldsymbol{r}_1\sigma_1, \boldsymbol{r}_2\sigma_2) - \Phi_{u\bar{u}}(\boldsymbol{r}_1\sigma_1, \boldsymbol{r}_2\sigma_2)] \\
&= \frac{1}{\sqrt{2}}[\phi_g(\boldsymbol{r}_1)\phi_g(\boldsymbol{r}_2) - \phi_u(\boldsymbol{r}_1)\phi_u(\boldsymbol{r}_2)]\Theta(\sigma_1, \sigma_2) \\
&= \frac{1}{\sqrt{2}}[\chi_A(\boldsymbol{r}_1)\chi_B(\boldsymbol{r}_2) + \chi_B(\boldsymbol{r}_1)\chi_A(\boldsymbol{r}_2)]\Theta(\sigma_1, \sigma_2) \ .
\end{aligned}$$

The last equality shows that Ψ describes a purely covalent system with exactly one electron localized around each nuclei. Using Ψ, it is easy to calculate the spin-free density matrices as (cf. Eq. (A.0.6)). For the 1-RDM we find

$$\Gamma(\boldsymbol{r}_1;\boldsymbol{r}_1') = \chi_A(\boldsymbol{r}_1)\chi_A(\boldsymbol{r}_1') + \chi_B(\boldsymbol{r}_1)\chi_B(\boldsymbol{r}_1') \ .$$

One then immediately has the electron density $\rho(\boldsymbol{r}_1) = \Gamma(\boldsymbol{r}_1;\boldsymbol{r}_1)$ too:

$$\rho(\boldsymbol{r}_1) = \chi_A^2(\boldsymbol{r}_1) + \chi_B^2(\boldsymbol{r}_1) \ .$$

It is also easy to find the 2-RDM. For the sake of brevity, we only quote its diagonal part, the pair density:

$$\rho(\boldsymbol{r}_1,\boldsymbol{r}_2) = \chi_A^2(\boldsymbol{r}_1)\chi_B^2(\boldsymbol{r}_2) + \chi_B^2(\boldsymbol{r}_1)\chi_A^2(\boldsymbol{r}_2) = \rho_{cov}(\boldsymbol{r}_1,\boldsymbol{r}_2)$$

Again, the pair density formula mirrors the localized and covalent nature of our system: the only event with a non-negligible probability is finding electron 1 around nucleus A while electron 2 is around nucleus B (or the other way around).

To highlight the importance of cumulants, we decompose the pair density as (cf. Eq. (2.1.30))

$$\rho(\boldsymbol{r}_1,\boldsymbol{r}_2) = \Lambda(\boldsymbol{r}_1,\boldsymbol{r}_2) + \rho(\boldsymbol{r}_1)\rho(\boldsymbol{r}_2) - \frac{1}{2}|\Gamma(\boldsymbol{r}_1;\boldsymbol{r}_2)|^2$$

The first term of the expansion is the cumulant part; the second and third terms constitute the wedge product part. The wedge product part simply reads

$$\begin{aligned}\rho(\boldsymbol{r}_1)\rho(\boldsymbol{r}_2) - \frac{1}{2}|\Gamma(\boldsymbol{r}_1;\boldsymbol{r}_2)|^2 =& \frac{1}{2}[\chi_A^2(\boldsymbol{r}_1)\chi_A^2(\boldsymbol{r}_2) + \chi_B^2(\boldsymbol{r}_1)\chi_B^2(\boldsymbol{r}_2)] \\ &+ \chi_A^2(\boldsymbol{r}_1)\chi_B^2(\boldsymbol{r}_2) + \chi_B^2(\boldsymbol{r}_1)\chi_A^2(\boldsymbol{r}_2) \\ =& \frac{1}{2}\rho_{ion}(\boldsymbol{r}_1,\boldsymbol{r}_2) + \rho_{cov}(\boldsymbol{r}_1,\boldsymbol{r}_2) \ ,\end{aligned}$$

hence the cumulant part must be

$$\Lambda(\boldsymbol{r}_1,\boldsymbol{r}_2) = -\frac{1}{2}\rho_{ion}(\boldsymbol{r}_1,\boldsymbol{r}_2) \ .$$

This is a negative contribution to the pair density with a simple interpretation. Let us compare our results with the analogous expressions in HF approximation (i.e. $\Psi \approx \Phi_{g\bar{g}}$):

$$\begin{aligned}\Gamma_{HF}(\boldsymbol{r}_1;\boldsymbol{r}_1') =& \chi_A(\boldsymbol{r}_1)\chi_A(\boldsymbol{r}_1') + \chi_A(\boldsymbol{r}_1)\chi_B(\boldsymbol{r}_1') \\ &+ \chi_B(\boldsymbol{r}_1)\chi_A(\boldsymbol{r}_1') + \chi_B(\boldsymbol{r}_1)\chi_B(\boldsymbol{r}_1') \ , \\ \rho_{HF}(\boldsymbol{r}_1) =& \chi_A^2(\boldsymbol{r}_1) + \chi_B^2(\boldsymbol{r}_1) = \rho(\boldsymbol{r}_1) \ ,\end{aligned}$$

and

$$\begin{aligned}\rho_{HF}(\boldsymbol{r}_1,\boldsymbol{r}_2) &= \rho_{HF}(\boldsymbol{r}_1)\rho_{HF}(\boldsymbol{r}_2) - \frac{1}{2}|\Gamma_{HF}(\boldsymbol{r}_1;\boldsymbol{r}_2)|^2 \\ &= \frac{1}{2}\rho_{HF}(\boldsymbol{r}_1)\rho_{HF}(\boldsymbol{r}_2) \\ &= \frac{1}{2}[\rho_{ion}(\boldsymbol{r}_1,\boldsymbol{r}_2) + \rho_{cov}(\boldsymbol{r}_1,\boldsymbol{r}_2)]\end{aligned}$$

The pair density in HF approximation is a mixture of ionic and covalent contributions. This is just the manifestation of the well-known deficiency of a single determinantal description, in which electrons with different spins are uncorrelated, and therefore can be easily localized on the same atomic site. The wedge product terms of the exact pair density have a very similar structure: although the covalent contribution appears with a larger weight, there are still remaining ionic terms. Calculating the pair density simply as a wedge product would then lead to a fundamentally flawed description of the system. The role of the cumulant term is to cancel the erroneous ionic terms and decrease the probability of finding electrons near each other, thus forming the correct covalent configuration. In this sense, cumulants are responsible for the left-right electron correlation.

Admittedly, the above simple example is somewhat artificial: when knowing the exact pair density, there is no reason to try a wedge product approximation. Still, the model showcases the importance of cumulants in forming the correct probabilistic description of strongly correlated systems by cancelling spurious contributions coming from the factorizable part of the density.

Appendix C

Spin-summation of density matrices and cumulants

Evaluating the spin-summed form of a product of spin orbital quantities (like density matrices or cumulants) is highly non-trivial. However, as it was pointed out first by Shamasundar [38], the summation can − at least in principle − always be performed by first considering the "inverse problem", that is, expressing the spin orbital quantities in terms of spin-free ones. The validity of these inversion formulas can be proven by studying permutation groups [172]. However, in this appendix we present a more straightforward derivation for them, generalizing the concepts of Ref [22]. We shall consider singlet states, although the results remain valid for spin-averaged ensembles of high-multiplicity states as well. Latin indices now denote spatial orbitals.

We discuss spin-summation of density matrices throughout this appendix. However, in the end, we will generalize the results to other quantities (cumulants, interaction vertices, CC amplitudes) as well.

Spin symmetry imposes serious constraints on elements of density matrices. An obvious constraint is posed by M_S conservation: elements such as $\gamma^{p\alpha}_{q\beta}$ or $\gamma^{p\alpha r\alpha}_{q\beta s\alpha}$ (having different number of α spins in the upper and lower indices, thus not conserving M_S) must vanish by virtue of spin orthogonality.
Let us now consider a simple two dimensional rotation, that is, an element of $SO(2) \subset SU(2)$:

$$\begin{bmatrix}\alpha\\ \beta\end{bmatrix} \rightarrow \begin{bmatrix}\tilde{\alpha}\\ \tilde{\beta}\end{bmatrix} = \begin{bmatrix}\cos(\theta) & \sin(\theta)\\ -\sin(\theta) & \cos(\theta)\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}.$$

This rotation transforms every spin index of a tensor, for example:

$$\chi^{p\alpha}_{q\alpha} \;\rightarrow\; \chi^{p\tilde{\alpha}}_{q\tilde{\alpha}} = \cos^2(\theta)\chi^{p\alpha}_{q\alpha} + \sin^2(\theta)\chi^{p\beta}_{q\beta} + \cos(\theta)\sin(\theta)\left(\chi^{p\alpha}_{q\beta} + \chi^{p\beta}_{q\alpha}\right) .$$

However, singlet states, and therefore the corresponding density matrices are invariant under the action of spin rotation group $SU(2)$:

$$\gamma^{p_1\sigma_1\ldots p_K\sigma_K}_{q_1\sigma'_1\ldots q_K\sigma'_K} \;\rightarrow\; \gamma^{p_1\tilde{\sigma}_1\ldots p_K\tilde{\sigma}_K}_{q_1\tilde{\sigma}'_1\ldots q_K\tilde{\sigma}'_K} = \gamma^{p_1\sigma_1\ldots p_K\sigma_K}_{q_1\sigma'_1\ldots q_K\sigma'_K} .$$

As a consequence of this invariance, density matrix elements are not linearly independent; relations can be derived between them that can be used to perform difficult spin-summations. The simplest relation – spin-flip symmetry – is obtained by setting $\theta = \pi/2$: density matrices are invariant under interchanging spin indices, for example $\gamma^{p\alpha}_{q\alpha} = \gamma^{p\beta}_{q\beta}$, or $\gamma^{p\alpha r\beta}_{q\alpha s\beta} = \gamma^{p\beta r\alpha}_{q\beta s\alpha}$.
Invariance under $SU(2)$, however, also leads to much less trivial relations; before tackling the general problem, it is instructive to start with a simple example. Consider the case of $\gamma^{p\alpha r\alpha}_{q\alpha s\alpha}$:

$$\gamma^{p\alpha r\alpha}_{q\alpha s\alpha} = \left(\cos^4(\theta) + \sin^4(\theta)\right)\gamma^{p\alpha r\alpha}_{q\alpha s\alpha} + 2\cos^2(\theta)\sin^2(\theta)\left(\gamma^{p\alpha r\beta}_{q\alpha s\beta} + \gamma^{p\alpha r\beta}_{q\beta s\alpha}\right) \ ,$$

where we exploited the spin-flip symmetry and only wrote out the non-vanishing M_S-conserving terms. After a simple rearranging, we have

$$2\cos^2(\theta)\sin^2(\theta)\left(\gamma^{p\alpha r\beta}_{q\alpha s\beta} + \gamma^{p\alpha r\beta}_{q\beta s\alpha} - \gamma^{p\alpha r\alpha}_{q\alpha s\alpha}\right) = 0 \ ,$$

which can be true for all values of θ only if

$$\gamma^{p\alpha r\alpha}_{q\alpha s\alpha} = \gamma^{p\alpha r\beta}_{q\alpha s\beta} + \gamma^{p\alpha r\beta}_{q\beta s\alpha} \ . \tag{C.0.1}$$

We have thus found a non-trivial relation between elements of the two-particle density matrix.

Turning to the general K-particle case, it is useful to introduce the notation $(G_{K,K'})^{p_1\ldots p_K}_{q_1\ldots q_K}$ for the sum of all density matrix elements having K' β spin idices in both the upper and the lower indices (the order of spatial indices being unchanged). In the case of $K' = K/2$ only one term is taken into account from each spin-flip pair, respectively. Two examples are provided by

$$(G_{2,1})^{pr}_{qs} = \gamma^{p\alpha r\beta}_{q\alpha s\beta} + \gamma^{p\alpha r\beta}_{q\beta s\alpha} \ ,$$

and

$$\begin{aligned}(G_{3,1})^{prt}_{qsu} = {} & \gamma^{p\alpha r\alpha t\beta}_{q\alpha s\alpha u\beta} + \gamma^{p\alpha r\alpha t\beta}_{q\alpha s\beta u\alpha} + \gamma^{p\alpha r\alpha t\beta}_{q\beta s\alpha u\alpha} + \gamma^{p\alpha r\beta t\alpha}_{q\alpha s\alpha u\beta} + \gamma^{p\alpha r\beta t\alpha}_{q\alpha s\beta u\alpha} + \gamma^{p\alpha r\beta t\alpha}_{q\beta s\alpha u\alpha} \\ & + \gamma^{p\beta r\alpha t\alpha}_{q\alpha s\alpha u\beta} + \gamma^{p\beta r\alpha t\alpha}_{q\alpha s\beta u\alpha} + \gamma^{p\beta r\alpha t\alpha}_{q\beta s\alpha u\alpha}\end{aligned}$$

A density matrix element with only α spins is just $(G_{K,0})^{p_1\ldots p_K}_{q_1\ldots q_K} = \gamma^{p_1\alpha\ldots p_K\alpha}_{q_1\alpha\ldots q_K\alpha}$. Obviously, $G_{K,K'} = G_{K,K-K'}$ because of the spin-flip symmetry. Using this notation, it is easy to write the invariance condition for the all-α density matrix element:

$$\begin{aligned}G_{K,0} = {} & \left(\cos^{2K}(\theta) + \sin^{2K}(\theta)\right)G_{K,0} \\ & + \sum_{K'=1}^{\lfloor \frac{K}{2} \rfloor}\left(\cos^{2(K-K')}(\theta)\sin^{2K'}(\theta) + \cos^{2K'}(\theta)\sin^{2(K-K')}(\theta)\right)G_{K,K'} \ ,\end{aligned}$$

where $\lfloor ... \rfloor$ is the floor function. Spatial indices have been suppressed for the sake of brevity. Introducing notation $\xi = \sin^2(\theta)$, the above equation can be recast in a simpler form:

$$C_{K,0}(\xi)G_{K,0} + \sum_{K'=1}^{\lfloor \frac{K}{2} \rfloor} C_{K,K'}(\xi)G_{K,K'} = 0 \ , \tag{C.0.2}$$

where

$$C_{K,0}(\xi) = (1-\xi)^K + \xi^K - 1 \quad , \quad C_{K,K'}(\xi) = (1-\xi)^{K-K'}\xi^{K'} + (1-\xi)^{K'}\xi^{K-K'} \; . \tag{C.0.3}$$

Coefficients $\{C_{K,K'}\}$ (for $K' = 1, ..., \lfloor \frac{K}{2} \rfloor$) are clearly linearly independent, since the lowest power of ξ involved is different for all of them. There is, however, a relation between $C_{K,0}$ and the other coefficients. For odd K we have

$$C_{K,0}(\xi) + \sum_{K'=1}^{\frac{K-1}{2}} \binom{K}{K'} C_{K,K'}(\xi) = 0 \; ,$$

while for even K we have

$$C_{K,0}(\xi) + \sum_{K'=1}^{\frac{K}{2}-1} \binom{K}{K'} C_{K,K'}(\xi) + \frac{1}{2}\binom{K}{\frac{K}{2}} C_{K,\frac{K}{2}}(\xi) = 0 \; .$$

These relations can be verified by substituting Eq. (C.0.3). We can now substitute $C_{K,0}$ back to Eq. (C.0.2). For odd K we find

$$\sum_{K'=1}^{\frac{K-1}{2}} C_{K,K'}(\xi) \left[G_{K,K'} - \binom{K}{K'} G_{K,0} \right] = 0 \; ,$$

and for even K:

$$\sum_{K'=1}^{\frac{K}{2}-1} C_{K,K'}(\xi) \left[G_{K,K'} - \binom{K}{K'} G_{K,0} \right] + C_{K,\frac{K}{2}}(\xi) \left[G_{K,\frac{K}{2}} - \frac{1}{2}\binom{K}{\frac{K}{2}} G_{K,0} \right] = 0 \; .$$

Since coefficients $\{C_{K,K'}\}$ are linearly independent, the quantities in brackets must vanish separately. We have (for $K' \neq K/2$)

$$G_{K,K'} = \binom{K}{K'} G_{K,0} \quad , \quad G_{K,\frac{K}{2}} = \frac{1}{2}\binom{K}{\frac{K}{2}} G_{K,0} \; . \tag{C.0.4}$$

Eq. (C.0.4) provides general relations between elements of spin orbital K-particle density matrices. Note that the case of $G_{2,1}$ just gives back Eq. (C.0.1).

Now we use Eqs. (C.0.4) to find the relation between spin orbital and spin-summed density matrices. We define an antisymmetrized version of the spin-summed density matrix as

$$\Gamma^{p_1 \; \cdots \; p_K}_{[q_1, \; \cdots \; ,q_K]} = \frac{1}{K!} \sum_{\hat{P} \in S_K} (-1)^\pi \Gamma^{p_1 \cdots p_K}_{\hat{P}(q_1 \ldots q_K)} \; .$$

Using the definition of spin-summation, as well as spin-flip and permutation symmetries, this can be rewritten in terms of $\{G_{K,K'}\}$:

$$\Gamma^{p_1 \; \cdots \; p_K}_{[q_1, \; \cdots \; ,q_K]} = 2 \sum_{K'=0}^{\lfloor \frac{K}{2} \rfloor} \frac{1}{\binom{K}{K'}} (G_{K,K'})^{p_1 \ldots p_K}_{q_1 \ldots q_K} \; . \tag{C.0.5}$$

For example,

$$\Gamma^{p\ r}_{[q,s]} = \frac{1}{2}\left(\Gamma^{pr}_{qs} - \Gamma^{pr}_{sq}\right) = 2\gamma^{p\alpha r\alpha}_{q\alpha s\alpha} + \gamma^{p\alpha r\beta}_{q\alpha s\beta} + \gamma^{p\alpha r\beta}_{q\beta s\alpha} \ ,$$

as can be easily seen by explicit calculation. On the other hand, rearranging and summing up Eq. (C.0.4) for various K' yields

$$(G_{K,0})^{p_1\dots p_K}_{q_1\dots q_K} = \frac{2}{K+1}\sum_{K'=0}^{\lfloor\frac{K}{2}\rfloor}\frac{1}{\binom{K}{K'}}(G_{K,K'})^{p_1\dots p_K}_{q_1\dots q_K} \tag{C.0.6}$$

Comparing Eqs. (C.0.5) and (C.0.6), we find

$$\gamma^{p_1\alpha\dots p_K\alpha}_{q_1\alpha\dots q_K\alpha} = \frac{1}{K+1}\Gamma^{p_1\ \dots\ p_K}_{[q_1,\ \dots\ ,q_K]} \ . \tag{C.0.7}$$

At this stage of the derivation, the all-α density matrix is expressed in terms of spin-summed density matrices. Note, however, that this decomposition is not unique for $K \geq 3$. To see this, let us define a symmetrization as

$$\Gamma^{p_1\ \dots\ p_K}_{\{q_1,\ \dots\ ,q_K\}} = \frac{1}{K!}\sum_{\hat{P}\in S_K}\Gamma^{p_1\dots p_K}_{\hat{P}(q_1\dots q_K)} \ .$$

As a consequence of the Pauli principle, $E^{p\ r\ t}_{\{q,s,u\}} = 0$, hence

$$\Gamma^{p\ r\ t}_{\{q,s,u\}} = 0 \ . \tag{C.0.8}$$

More generally, symmetrizing any three indices of Γ_K yields zero; this gives additional equations that can be used to simplify Eq. (C.0.7), although at the cost of antisymmetry not being manifest anymore. The simplest example is for $K = 3$:

$$\gamma^{p\alpha r\alpha t\alpha}_{q\alpha s\alpha u\alpha} = \frac{1}{4}\Gamma^{p\ r\ t}_{[q,s,u]} = \frac{1}{4}\left(\Gamma^{p\ r\ t}_{[q,s,u]} + \Gamma^{p\ r\ t}_{\{q,s,u\}}\right) = \frac{1}{12}\left(\Gamma^{prt}_{qsu} + \Gamma^{prt}_{uqs} + \Gamma^{prt}_{suq}\right) \ .$$

Using all of the above, we can now complete the relationship between spin orbital and spin-free density matrices. A trivial but important property of spin-summation is that summed up spin indices are unaffected by rotations:

$$\sum_{\sigma_{K'+1},\dots,\sigma_K}\gamma^{p_1\sigma_1\dots p_{K'}\sigma_{K'}p_{K'+1}\sigma_{K'+1}\dots p_K\sigma_K}_{q_1\sigma'_1\dots q_{K'}\sigma'_{K'}q_{K'+1}\sigma_{K'+1}\dots q_K\sigma_K} \rightarrow \sum_{\sigma_{K'+1},\dots,\sigma_K}\gamma^{p_1\tilde{\sigma}_1\dots p_{K'}\tilde{\sigma}_{K'}p_{K'+1}\sigma_{K'+1}\dots p_K\sigma_K}_{q_1\tilde{\sigma}'_1\dots q_{K'}\tilde{\sigma}'_{K'}q_{K'+1}\sigma_{K'+1}\dots q_K\sigma_K} \ .$$

Only "free" spin indices $\sigma_1, \sigma'_1, \dots, \sigma_{K'}, \sigma'_{K'}$ are transformed. This means that the relations between density matrix elements must also hold as partial spin trace relations. For example (cf. Eq. (C.0.1)):

$$\sum_{\sigma_3,\dots,\sigma_K}\gamma^{p_1\alpha p_2\alpha p_3\sigma_3\dots p_K\sigma_K}_{q_1\alpha q_2\alpha q_3\sigma_3\dots q_K\sigma_K} = \sum_{\sigma_3,\dots,\sigma_K}\gamma^{p_1\alpha p_2\beta p_3\sigma_3\dots p_K\sigma_K}_{q_1\alpha q_2\beta q_3\sigma_3\dots q_K\sigma_K} + \sum_{\sigma_3,\dots,\sigma_K}\gamma^{p_1\alpha p_2\beta p_3\sigma_3\dots p_K\sigma_K}_{q_1\beta q_2\alpha q_3\sigma_3\dots q_K\sigma_K} \ ,$$

and (cf. Eq. (C.0.7))

$$\sum_{\sigma_{K'+1},\dots,\sigma_K}\gamma^{p_1\alpha\dots p_{K'}\alpha p_{K'+1}\sigma_{K'+1}\dots p_K\sigma_K}_{q_1\alpha\dots q_{K'}\alpha q_{K'+1}\sigma_{K'+1}\dots q_K\sigma_K} = \frac{1}{K'+1}\Gamma^{p_1\ \dots\ p_{K'}p_{K'+1}\dots p_K}_{[q_1,\dots,q_{K'}]q_{K'+1}\dots q_K} \ .$$

With the help of these two relations, we can easily express other spin orbital density matrices different from the all-α type too. For example:

$$\begin{aligned}\gamma^{p_1\alpha\ldots p_{K-1}\alpha p_K\beta}_{q_1\alpha\ldots q_{K-1}\alpha q_K\beta} &= \sum_\sigma \gamma^{p_1\alpha\ldots p_{K-1}\alpha p_K\sigma}_{q_1\alpha\ldots q_{K-1}\alpha q_K\sigma} - \gamma^{p_1\alpha\ldots p_{K-1}\alpha p_K\alpha}_{q_1\alpha\ldots q_{K-1}\alpha q_K\alpha} \\ &= \frac{1}{K}\Gamma^{p_1\ \ldots\ p_{K-1}p_K}_{[q_1,\ \ldots\ ,q_{K-1}]q_K} - \frac{1}{K+1}\Gamma^{p_1\ \ldots\ p_K}_{[q_1,\ \ldots\ ,q_K]} \ . \end{aligned} \tag{C.0.9}$$

By using Eqs. (C.0.7) and (C.0.9) recursively (and exploiting the usual spin-flip and permutation symmetries), all spin orbital density matrix elements can be expressed in terms of spin-summed ones. We refer to the formulae obtained as Shamasundar identities. The first few of these identities read

$$\gamma^{p\alpha}_{q\alpha} = \frac{1}{2}\Gamma^p_q \ ,$$

$$\gamma^{p\alpha r\alpha}_{q\alpha s\alpha} = \frac{1}{3}\Gamma^{p\ r}_{[q,s]} = \frac{1}{6}\left(\Gamma^{pr}_{qs} - \Gamma^{pr}_{sq}\right) \ , \tag{C.0.10}$$

$$\gamma^{p\alpha r\beta}_{q\alpha s\beta} = \frac{1}{2}\Gamma^{pr}_{qs} - \frac{1}{3}\Gamma^{p\ r}_{[q,s]} = \frac{1}{6}\left(2\Gamma^{pr}_{qs} + \Gamma^{pr}_{sq}\right) \ . \tag{C.0.11}$$

Spin-summation of arbitary products of density matrices is now (in principle) straightforward. Two examples which are most important in the thesis are (using Eqs. (C.0.10) and (C.0.11))

$$\sum_{\sigma_1,\ldots,\sigma_4} \gamma^{p\sigma_1 r\sigma_2}_{q\sigma_1 s\sigma_3}\gamma^{t\sigma_3 v\sigma_4}_{u\sigma_2 w\sigma_4} = \frac{1}{2}\Gamma^{pr}_{qs}\Gamma^{tv}_{uw} \ , \tag{C.0.12}$$

and

$$\sum_{\sigma_1,\ldots,\sigma_4} \gamma^{p\sigma_1 r\sigma_2}_{q\sigma_3 s\sigma_4}\gamma^{t\sigma_3 v\sigma_4}_{u\sigma_1 w\sigma_2} = \frac{1}{6}\left(2\Gamma^{pr}_{qs}\Gamma^{tv}_{uw} + \Gamma^{pr}_{sq}\Gamma^{tv}_{uw} + \Gamma^{pr}_{qs}\Gamma^{tv}_{wu} + 2\Gamma^{pr}_{sq}\Gamma^{tv}_{wu}\right) \ . \tag{C.0.13}$$

We have derived the above results for density matrices. However, we have never really used any specific property of density matrices; we only requiered M_S-conservation, antisymmetry and $SU(2)$ invariance. This means that all the above formulae stand for any antisymmetric K-particle tensor fulfilling these criteria. Density cumulants, for example, satisfy these conditions by construction, as well as Coupled Cluster amplitudes and matrix elements of second quantized operators (particularly, the Fockian and the antisymmetrized two-electron integral in the Hamiltonian). Furthermore, the results remain naturally valid for tensors obtained by contraction of two or more tensors: the most important examples are the dressed K-particle interaction vertices introduced in Secs. 3.1 and 3.2.

For one- and two-particle quantities (density matrices, cumulants, amplitudes, bare or dressed vertices) all of these results can be summarized in a particularly simple form:

$$\chi^{p\sigma_1}_{q\sigma_2} = \delta^{\sigma_1}_{\sigma_2}\chi^p_q \quad , \quad \tilde{\chi}^{p\sigma_1 r\sigma_3}_{q\sigma_2 s\sigma_4} = \delta^{\sigma_1}_{\sigma_2}\delta^{\sigma_3}_{\sigma_4}\chi^{pr}_{qs} - \delta^{\sigma_1}_{\sigma_4}\delta^{\sigma_3}_{\sigma_2}\chi^{pr}_{sq} \ , \tag{C.0.14}$$

where $\chi^p_q = \chi^{p\alpha}_{q\alpha}$ and $\chi^{pr}_{qs} = \tilde{\chi}^{p\alpha r\beta}_{q\alpha s\beta}$.

Appendix D

Interpretation of dressed vertices in real space

Dressed vertices in the orbital space representation are thoroughly discussed in Chapter 3. While this representation is the most useful in developing theories, the real space representation is easier to interpret. This representation will be briefly reviewed here.

Previously, dressed vertices were considered in the basis of natural orbitals, but the concept remains meaningful in any orbital basis. The only difference is that legs in Eq. (3.2.11) cannot be added to the vertex by simple multiplicative factors n_i, $\bar{n}^a$, they have to be introduced by contraction with $\gamma_i^{i'}$, $\eta_{a'}^a$. A few examples are

$$\sum_{\substack{i' \\ a'}} \gamma_i^{i'} \eta_{a'}^a F_{i'}^{a'} = \langle \mathcal{H} :a_i^a: \rangle \quad , \tag{D.0.1}$$

$$\sum_{\substack{i',j' \\ a',b'}} \gamma_i^{i'} \gamma_j^{j'} \eta_{a'}^a \eta_{b'}^b \tilde{V}_{i'j'}^{a'b'} = \langle \mathcal{H} :a_{ij}^{ab}: \rangle \ , \tag{D.0.2}$$

and

$$\sum_{a',b',c',d'} \eta_{a'}^a \eta_{b'}^b \eta_{c'}^c \eta_{d'}^d \tilde{V}_{c'd'}^{a'b'} = \langle :a_{cd}: \mathcal{H} :a^{ab}: \rangle_C \ .$$

Of course, these quantities could also be represented by diagrams, after introducing a suitable graphical notation for γ_1 and η_1, each with a single pair of (down- or upgoing) legs.
We can always transform such orbital space formulae to real space with $\{\phi^p(x)\}$. Eq. (D.0.1) is readily transformed to

$$\int \mathrm{d}y \int \mathrm{d}y' \gamma(x;y) F_{p;h}(y;y') \eta(y';x') = \langle \mathcal{H} :\psi^\dagger(x')\psi(x): \rangle \ , \tag{D.0.3}$$

with the vertex part being an integral kernel:

$$F_{p;h}(x;x') = \sum_{\substack{i' \\ a'}} F_{i'}^{a'} \phi^{i'}(x) \phi_{a'}(x') \ .$$

The transformation can be carried out, since both sides of Eq. (D.0.1) vanish identically whenever i or a refer to virtual or core orbitals, respectively, hence summations can run over all orbitals. Real space MRGWT expansion of the right hand side of Eq. (D.0.3) mirrors that of orbital space (cf. Eq. (3.2.4)) and reflects the diagrammatic structure of Eq. (3.2.6):

$$\begin{aligned}\int \mathrm{d}y \int \mathrm{d}y' \gamma(x;y) F_{p;h}(y;y') \eta(y';x') = &\int \mathrm{d}y \int \mathrm{d}y' \gamma(x;y) f(y;y') \eta(y';x') \\ &+ \int \mathrm{d}y \int \mathrm{d}y' f(y;y') \lambda(y',x;y,x') \\ &+ \frac{1}{2} \int \mathrm{d}\boldsymbol{y}_{\{2\}} \int \mathrm{d}\boldsymbol{y}'_{\{2\}} \lambda(x,y'_2;y_1,y_2) \tilde{v}(y_1,y_2;y'_1,y'_2) \eta(y'_1;x') \\ &- \frac{1}{2} \int \mathrm{d}\boldsymbol{y}_{\{2\}} \int \mathrm{d}\boldsymbol{y}'_{\{2\}} \lambda(y_1,y_2;x',y'_2) \tilde{v}(y'_1,y'_2;y_1,y_2) \gamma(x;y'_1) \\ &+ \frac{1}{4} \int \mathrm{d}\boldsymbol{y}_{\{2\}} \int \mathrm{d}\boldsymbol{y}'_{\{2\}} \tilde{v}(y_1,y_2;y'_1,y'_2) \lambda(y'_1,y'_2,x;y_1,y_2,x') \ ,\end{aligned} \tag{D.0.4}$$

where $f(x;x')$ and $\tilde{v}(x_1,x_2;x'_1,x'_2)$ are the usual Fockian and electron-electron vertices:

$$\begin{aligned} f(x;x') &= \sum_{p,q} f^q_p \phi^p(x) \phi_q(x') \ , \\ \tilde{v}(x_1,x_2;x'_1,x'_2) &= \sum_{p,q,r,s} \tilde{v}^{qs}_{pr} \phi^p(x_1) \phi^r(x_2) \phi_q(x'_1) \phi_s(x'_2) \ . \end{aligned} \tag{D.0.5}$$

Eq. (D.0.4) is interpreted as a scattering process between a particle at x' and a hole at x, mediated by a potential represented by $F_{p;h}$. This includes scattering on the bare f as well as internal processes involving other vertices and density disturbances described by cumulants. All possible scattering scenarios belonging to a specific cumulant are summed up by integration over internal points.
Eq (D.0.2) can be similarly transformed to give

$$\begin{aligned}&\int \mathrm{d}\boldsymbol{y}_{\{2\}} \int \mathrm{d}\boldsymbol{y}'_{\{2\}} \gamma(x_1;y_1) \gamma(x_2;y_2) \tilde{V}_{pp;hh}(y_1,y_2;y'_1,y'_2) \eta(y'_1;x'_1) \eta(y'_2;x'_2) \\ =&\langle \mathcal{H} :\psi^\dagger(x'_1) \psi^\dagger(x'_2) \psi(x_2) \psi(x_1): \rangle \ ,\end{aligned} \tag{D.0.6}$$

the vertex part being

$$\tilde{V}_{pp;hh}(x_1,x_2;x'_1,x'_2) = \sum_{\substack{i',j' \\ a',b'}} \tilde{V}^{a'b'}_{i'j'} \phi^{i'}(x_1) \phi^{j'}(x_2) \phi_{a'}(x'_1) \phi_{b'}(x'_2) \ .$$

Upon expanding the right hand side of Eq. (D.0.6), a propagation pattern similar to Eq. (D.0.4) would be found, but now between two particles at x'_1, x'_2 and two holes at x_1, x_2 scattering on $\tilde{V}_{pp;hh}$, with all possible internal processes (represented by cumulants) and exchange processes taken into account.
Of course, the same interpretation stands for any dressed vertex variant as well. For example,

$$\int \mathrm{d}y \int \mathrm{d}y' \gamma(x;y) F_{h;h}(y;y') \gamma(y';x') = -\langle :\psi^\dagger(x'): \mathcal{H} :\psi(x): \rangle$$

is associated with one-body hole-hole scattering in real space by

$$F_{h;h}(x;x') = \sum_{i',k'} F_{i'}^{k'} \phi^{i'}(x)\phi_{k'}(x') \ .$$

Different variants of dressed vertices are represented formally by the same amputated diagrams (see e.g. Eqs. (3.2.15) and (3.2.16)).

Let us finally comment on the non-locality of dressed vertices. A vertex is called local, if it can only affect the propagation of each particle or hole in a single point of space, respectively. In this sense, the bare electron-electron interaction is local:

$$\tilde{v}(x_1,x_2;x_1',x_2') = \hat{v}(x_1,x_2)[\delta(x_1,x_1')\delta(x_2,x_2') - \delta(x_1,x_2')\delta(x_2,x_1')] \ ,$$

where

$$\hat{v}(x_1,x_2) = \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \ .$$

This can be seen by using Eq. (2.1.8) and the completeness of $\{\phi^p(x)\}$ in Eq. (D.0.5). On the other hand, the folded electron-electron interaction gives rise to a non-local exchange term in the Fockian:

$$f(x;x') = \left[-\frac{1}{2}\nabla^2 - \sum_A \frac{Z_A}{|\boldsymbol{r} - \boldsymbol{R}_A|} + \int \mathrm{d}y \hat{v}(x,y)\rho(y)\right]\delta(x,x') - \hat{v}(x,x')\gamma(x;x') \ .$$

Hence, during the real space propagation of particles and holes, the Fockian can connect distant points x and x'.
Besides containing these Fockian parts, a further layer of non-locality is added to dressed vertices in the form of cumulant-involving internal processes. Repeated scattering on cumulants (all possible points taken into account by integration) can make dressed vertices affect incoming and outgoing particles or holes in different points of space. This is not surprising: non-locality arises whenever many-body effects are folded into effective fewer-body quantities (a possible example given by CC amplitudes).
While strictly speaking, dressed vertices are non-local, their range is limited to certain parts of space if orbitals and cumulants are localized. For example, in the case of strongly orthogonal geminal wave functions like APSG, cumulants are centered around localized electron pairs (e.g. covalent bonds) and decay fast for large distances (see Sec. 2.2.2). In such a case, due to connectedness, non-local dressing effects are dominant only along the spatial extent of bonds.

Appendix E

On the positivity of ϵ_i^a

In Chapter 4, during the evaluation of second-order idCCSD dispersion energy, positivity of formal energy denominator

$$\epsilon_a^i = \bar{n}_a F_a^a - n^i F_i^i$$

has been assumed. This statement is supported by numerical results for GVB, tested in various (e.g. bond breaking) scenarios. While we cannot prove this statement in an exact way, some formal rationalization can still be provided based on reasonable assumptions about the diagonal elements of the GVB Fockian. These are the following: f_p^p is assumed to be negative if spatial orbital p is strongly occupied ($n_p \lesssim 1$), and positive (or negative with a very small absolute value) if p is weakly occupied ($n_p \gtrsim 0$). We shall also exploit that GVB geminals consist of two orbitals, here denoted as $i, i' \in \mathbb{G}_I$ and $a, a' \in \mathbb{G}_A$ with $n_i + n_{i'} = n_a + n_{a'} = 1$.
Although in Chapter 4 only atoms were considered as subsystems, this short analysis also covers the possibility of dissociating geminals within the subsystem.

The spin orbital expressions for the hole-hole and particle-particle variant of the one-body vertex (cf. Eqs. (3.2.19) and (3.2.20)) can be easily converted to spatial orbitals with the results of Appendix C:

$$\begin{aligned} n_i n^k F_i^k &= n_i n^k F_{i\alpha}^{k\alpha} \\ &= n_i n^k f_i^k - \frac{1}{2}\sum_{p,q} f_p^q \ \Lambda_{qi}^{pk} + \frac{1}{2}\sum_{p,q,r}\left(n^k v_{rp}^{kq} \ \Lambda_{iq}^{rp} + n_i v_{iq}^{rp} \ \Lambda_{rp}^{kq}\right) - \frac{1}{4}\sum_{p,q,r,s} v_{pr}^{qs} \ \Lambda_{qsi}^{prk} \ , \end{aligned}$$

$$\begin{aligned} \bar{n}^a \bar{n}_c F_c^a &= \bar{n}^a \bar{n}_c F_{c\alpha}^{a\alpha} \\ &= \bar{n}^a \bar{n}_c f_c^a - \frac{1}{2}\sum_{p,q} f_p^q \ \Lambda_{qc}^{pa} - \frac{1}{2}\sum_{p,q,r}\left(\bar{n}^a v_{rp}^{aq} \ \Lambda_{cq}^{rp} + \bar{n}_c v_{cq}^{rp} \ \Lambda_{rp}^{aq}\right) - \frac{1}{4}\sum_{p,q,r,s} v_{pr}^{qs} \ \Lambda_{qsc}^{pra} \ . \end{aligned}$$

Using the explicit GVB form of Λ_2 and Λ_3 given by Eqs. (2.2.8) and (2.2.11), we can express the two terms of the denominator as

$$\begin{aligned} n_i F_i^i = &+ (2n_i - 1) f_i^i + 2\bar{n}_i f_{i'}^{i'} + 2n_i \bar{n}_i v_{ii}^{ii} - 2\sqrt{n_i \bar{n}_i} v_{i'i'}^{ii} \\ &- 2n_i \bar{n}_i (2v_{ii'}^{ii'} - v_{i'i}^{ii'}) - \bar{n}_i (2n_i - 1)(2v_{ii'}^{ii'} - v_{i'i}^{ii'} - v_{i'i'}^{i'i'}) \ , \end{aligned}$$

$$\bar{n}^a F^a_a = -(2n^a - 1) f^a_a + 2n_a f^{a'}_{a'} - 2n_a \bar{n}_a v^{aa}_{aa} + 2\sqrt{n_a \bar{n}_a} v^{aa}_{a'a'} + 2n_a \bar{n}_a (2v^{aa'}_{aa'} - v^{aa'}_{a'a}) - n_a(2n_a - 1)(2v^{aa'}_{aa'} - v^{aa'}_{a'a} - v^{a'a'}_{a'a'}) \ .$$

Above we made use of the fact that geminal coefficients c_i and $c_{i'}$ have opposite signs (the same applying to a and a'); see Sec. 2.2.1 and Appendix B. Also, real orbitals are assumed. We shall analyze these expressions in two limiting cases: the mean-field-like case (one orbital in the geminal is strongly occupied, the other is weakly occupied) and the case of the dissociating geminal (both orbitals in the geminal having the same occupation). We then have:

$$n_i \approx 1 \ \rightarrow \ n_i F^i_i \approx f^i_i \ , \tag{E.0.1}$$

$$n_i \approx 0 \ \rightarrow \ n_i F^i_i \approx -f^i_i + 2f^{i'}_{i'} - \left(v^{i'i'}_{i'i'} + v^{ii}_{i'i'} - 2v^{ii'}_{ii'}\right) \ , \tag{E.0.2}$$

$$n_i \approx \frac{1}{2} \ \rightarrow \ n_i F^i_i \approx f^{i'}_{i'} + \frac{1}{2}\left(v^{ii}_{ii} - v^{ii}_{i'i'} - 2v^{ii'}_{ii'}\right) \approx f^i_i - \frac{1}{2}\left(v^{ii}_{ii} + v^{ii}_{i'i'}\right) \ , \tag{E.0.3}$$

$$\bar{n}^a \approx 1 \ \rightarrow \ \bar{n}^a F^a_a \approx f^a_a \ , \tag{E.0.4}$$

$$\bar{n}^a \approx 0 \ \rightarrow \ \bar{n}^a F^a_a \approx -f^a_a + 2f^{a'}_{a'} + \left(v^{a'a'}_{a'a'} + v^{aa}_{a'a'} - 2v^{aa'}_{aa'}\right) \ , \tag{E.0.5}$$

$$\bar{n}^a \approx \frac{1}{2} \ \rightarrow \ \bar{n}^a F^a_a \approx f^{a'}_{a'} - \frac{1}{2}\left(v^{aa}_{aa} - v^{aa}_{a'a'} - 2v^{aa'}_{aa'}\right) \approx f^a_a + \frac{1}{2}\left(v^{aa}_{aa} + v^{aa}_{a'a'}\right) \ . \tag{E.0.6}$$

We have used the fact that in the dissociating regime $f^i_i \approx f^{i'}_{i'}$ and $v^{ii}_{ii} \approx v^{i'i'}_{i'i'} \approx v^{ii'}_{ii'}$ (and similar relations for a), since in this limit both orbitals are roughly symmetric-antisymmetric combinations of atomic orbitals, similarly to the case of H_2 (see Appendix B).
All two-electron integrals appearing in (E.0.1)-(E.0.6) are positive. The two-electron contribution is small compared to the Fockian part around equilibrium geometry, while it is the other way around in the dissociating regime. Based on these equations and our initial assumption concerning the sign of f^p_p, one can argue the positivity of ϵ^a_i for all the different pairings of $-n_i F^i_i$ and $\bar{n}^a F^a_a$. In e.g. (E.0.2) $2f^{i'}_{i'}$ is expected to give the (large negative) dominant contribution. The positivity of the dominant contribution in (E.0.5) can be argued similarly, leading to the positivity of ϵ^a_i, despite ϕ^i and ϕ^a being weakly and strongly occupied, respectively. As a further example, let us assume that orbitals ϕ^i and ϕ^a belong to the same dissociating geminal, that is, $I = A$, $i \neq a$ and $n_i \approx n_a \approx 1/2$. Then $i' = a$ and $a' = i$. Using (E.0.3) and (E.0.6), ϵ^a_i reads:

$$\epsilon^a_i \approx v^{ii}_{ii} + v^{aa}_{ii} \geq 0 \ .$$

We investigated only the two extreme cases of equilibrium and dissociated bonds. However, intermediate cases (corresponding to somewhat elongated bonds) are expected to behave similarly, following from the monotonicity of occupation numbers and of the magnitude of Λ_2 (see Fig. 2.1 of Sec. 2.2.2). While the magnitude of Λ_3 does not change monotonically, its effect is small compared to Λ_2.

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Abstract

Perturbative or Coupled Cluster-type correction methods built on a single Slater-determinant tend to perform poorly in the presence of strong electron correlation (a common example being molecules with dissociating covalent bonds), raising the need for an MR treatment employing a linear combination of determinants as reference function. This dissertation focuses on analytical and numerical studies concerning the so-called internally contracted Multi-Reference Coupled Cluster (ic-MRCC) model, a general theory aiming to accurately describe strongly correlated many-electron systems.

The starting point of the investigation is the normal ordered ic-MRCC parametrization of Mukherjee et al. Derivations are carried out with the help of the MR generalization of operator normal ordering and the corresponding Wick theorem, as developed by Kutzelnigg and Mukherjee. The novelty of the proposed approach lies in the introduction of the so-called dressed interaction vertices. Dressed one- and two-body interaction vertices formally resemble the Fockian and the electron-electron interaction vertices of a single determinantal theory, and one can use them in an analogous way to derive the working equations of the formalism. However, dressed vertices are functions of so-called density cumulants (which carry the correlation content of the reference), and for this reason, they are more suitable for strongly correlated scenarios than their "bare" counterparts. Working with dressed vertices makes the otherwise complicated derivations of the ic-MRCC equations considerably easier. Construction and plausible interpretation of dressed vertices are facilitated by diagrammatic techniques.

In this thesis, a completely general ic-MRCC theory based on the concept of dressed vertices is developed. Algebraic and diagrammatic techniques are worked out in detail for the construction of these vertices, and similar techniques are used when applying them in the derivation of ic-MRCC formulae. We present a diagrammatic analysis of CC amplitudes and the energy correction formula in order to highlight the role of dressed vertices in electron correlation. We also discuss diagrammatic approaches for the singlet spin-adaptation of the theory.

The manifold of elementary excitations used in ic-MRCC is typically linearly dependent, and one must treat this redundancy in order to avoid possible numerical instabilities. A novel approach is discussed on the treatment of this problem.

As an interesting sidenote, we investigate typically weak-correlation phenomena, dispersion interactions in the ic-MRCC framework. We demonstrate the emergence of the London dispersion formula for an approximation of the general theory.

Based on diagrammatic and algebraic arguments, we propose simple approximations to the general ic-MRCC theory. These include a model formally resembling traditional CCSD (with bare vertices replaced by dressed ones), ring-like approximations to the former, as well as a more involved scheme incorporating some of the so-called intervertex electron correlation.

Applying these approximate correction schemes to a simple geminal wave function, we test the quality of the proposed models on different, strongly or weakly correlated molecular systems. The examples involve the breaking of covalent bonds and structural rearrangements, as well as the calculation of conformational barriers and dispersion interactions. We discuss both strengths and weaknesses of the proposed approximations.

Összefoglaló

Az ún. erős elektronkorreláció (pl. disszociáló kovalens kötéseket tartalmazó molekulák esetén) jelentősen rontja a Slater-determinánsra épülő perturbatív vagy Coupled Cluster korrekciók megbízhatóságát. Ilyen esetekben több determináns lineáris kombinációjaként előálló Multi-Referencia (MR) hullámfüggvények alkalmazása szükséges. Jelen disszertáció az ún. "internally contracted Multi-Reference Coupled Cluster" (ic-MRCC) elmélet képletszintű és numerikus vizsgálatát mutatja be.
A modell kiindulópontját a Mukherjee és munkatársai által javasolt normálrendezett ic-MRCC parametrizáció adja. A levezetéseket a Kutzelnigg és Mukherjee által bevezetett, MR-hullámfüggvényekre általánosított normálrendezés és Wick-tétel segítségével végezzük.
A megközelítésünk újdonsága az ún. "öltöztetett" (dressed) kölcsönhatási vertexek bevezetésében áll. A dressed egy- és kétrészecske vertexek formálisan a Fock- és az elektron-elektron kölcsönhatási vertexekre emlékeztetnek, és ezekkel analóg szerepet játszanak a CC egyenletek levezetésében. Az elektronkorrelációt kódoló ún. sűrűségkumulánsoktól való függésük révén azonban alkalmasabbak erősen korrelált rendszerek leírására, mint "csupasz" megfelelőik. A dressed vertexek jelentősen megkönnyítik az ic-MRCC egyenletek (máskülönben kezelhetetlenül bonyolulttá váló) levezetését; maguk a dressed vertexek egyszerűen megszerkeszthetők és interpretálhatók Feynman-diagramos módszerek segítségével.
A disszertációban egy általános ic-MRCC elmélet kiépítését mutatjuk be dressed vertexek segítségével. Algebrai és diagramos módszerekkel is tárgyaljuk ezen vertexek, majd a segítségükkel felépülő ic-MRCC egyenletek levezetését. Diagram alapú módszerekkel vizsgáljuk a dressed vertexek hatását a CC amplitudókra, valamint a számolt energiakorrekcióra. Hasonló módon mutatjuk be az elmélet szinglet rendszerekre történő spinadaptálását is.
Az ic-MRCC számításokhoz felhasznált elemi gerjesztések lineárisan összefüggő rendszert alkotnak: ezen redundanciát kezelni kell, az esetleges numerikus problémákat elkerülendő. Ennek kapcsán egy új redundanciakezelő eljárás is bemutatásra kerül.
Egy érdekes mellékágként a London-féle diszperziós kölcsönhatások leírását is vizsgáljuk az ic-MRCC formalizmus egy közelítésének keretében.
Diagramos és algebrai érvekre támaszkodva, egyszerű közelítéseket javaslunk az általános ic-MRCC elméletre: egy formálisan a hagyományos CCSD-re emlékeztető modellt (dressed vertexekkel helyettesítve az eredeti, csupasz vertexeket), ún. ring-diagram alapú közelítő modelleket, valamint egy bonyolultabb módszert, ami az elektronkorreláció ún. intervertex részét is figyelembe tudja venni.
Egy egyszerű geminál hullámfüggvényre alkalmazva ezeket a módszereket, különböző erősen, vagy gyengén korrelált rendszereken teszteljük a hatékonyságukat. A tesztek során disszociáló kovalens kötések, szerkezetátrendeződések, konformációs energiagátak, valamint diszperziós kölcsönhatások leírását vizsgáljuk. A módszerek erősségeit és gyengeségeit egyformán diszkutáljuk.